



Supporting Information

A Spiro Phosphamide Catalyzed Enantioselective Proton Transfer of Ylides in a Free Carbene Insertion into N–H Bonds

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1. Method

All reactions and manipulations were performed using standard Schlenk techniques. All solvents were purified and dried using standard procedure.^[1] The ammonia was pumped into the toluene to get $\text{NH}_3 \cdot \text{Toluene}$ solution (0.27 mol/L). NMR spectra were recorded with a Bruker AV 400 and 600 spectrometer at 400 or 600 MHz (^1H NMR), 101 or 151 MHz (^{13}C NMR). Chemical shifts (δ values) were reported in ppm down field from internal Me_4Si . High Resolution Mass Spectra (HRMS) were recorded on an IonSpec FT-ICR mass spectrometer with Electron Spray Ionization (ESI) resource. HPLC analysis was performed on a Hewlett Packard Model HP 1100 Series chromatography. Optical rotations were determined by a Perkin Elmer 341 MC polarimeter. The in situ IR spectra were recorded using a React IR 15 from Mettler-Toledo AutoChem fitted with a Silicon-tipped probe.

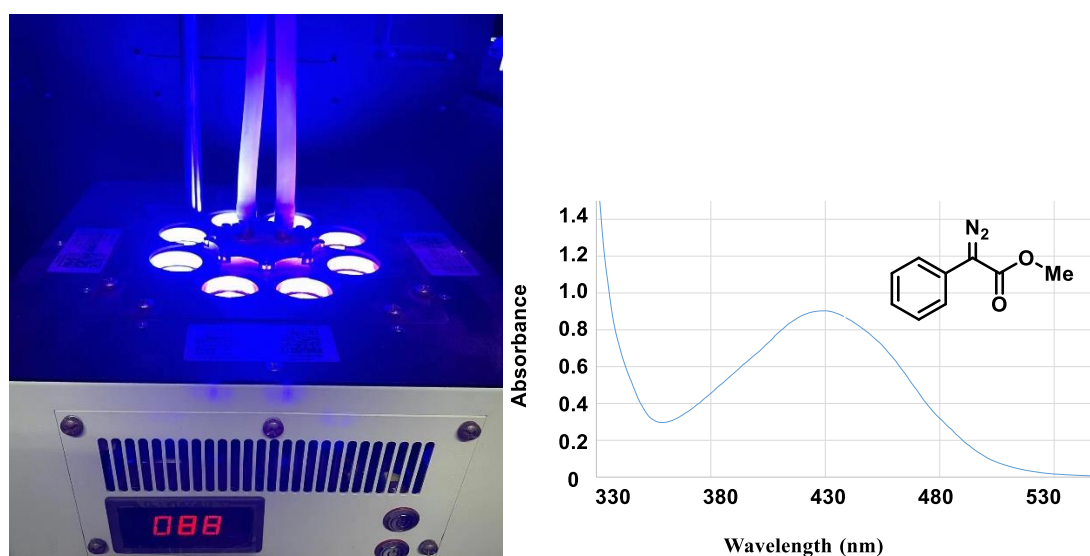


Figure S1. Blue LEDs photo reactor and the absorption spectrum of the substrate.

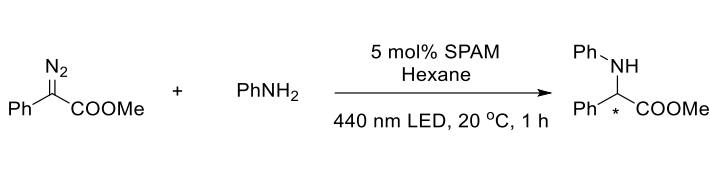
Table S1. Enantioselective N–H insertion of α -diazoester with aniline: optimization of the solvent.

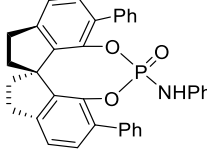
$ \begin{array}{c} \text{N}_2 \\ \\ \text{Ph}-\text{C}=\text{C}-\text{COOMe} \end{array} + \text{PhNH}_2 \xrightarrow[440 \text{ nm LED, } 20^\circ\text{C, } 1 \text{ h}]{5 \text{ mol\% SPAM, Solvent}} \begin{array}{c} \text{Ph}-\text{NH} \\ \\ \text{Ph}-\text{C}-\text{COOMe} \\ \\ \text{*} \end{array} $			
SPAM-1			
Entry*	Solvent	Yield (%) [†]	ee (%) [‡]
1	DCM	76	12
2	DCE	76	12
3	Tol	68	32
4	Benzene	67	43
5	PhCF_3	67	28

6	Et ₂ O	49	31
7	MTBE	58	35
8	THF	23	0
9	<i>n</i> -hexane	67	56
10	cyclohexane	62	54
11	<i>n</i> -pentane	60	54
12	cyclopentane	61	52
13	MeCN	57	0

*Reaction conditions: α -diazoester (0.2 mmol), PhNH₂ (0.2 mmol), SPAM-1 (5 mol%), 2 mL solvent, 20 °C, 440 nm LEDs, 1 h. †Isolated yield was given. ‡The ee values were determined by high performance liquid chromatography.

Table S2. Enantioselective N–H insertion of α -diazoester with aniline: optimization of the concentration.



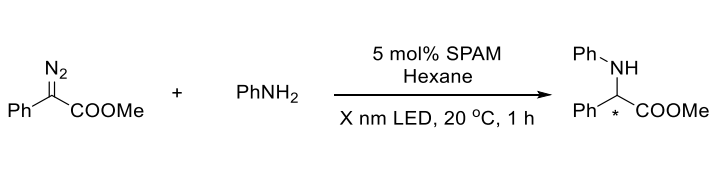


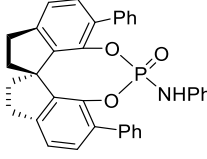
SPAM-1

Entry*	Hexane	Yield (%) [†]	ee (%) [‡]
1	1 mL	66	33
2	2 mL	67	56
3	3 mL	66	61
4	4 mL	67	63

*Reaction conditions: α -diazoester (0.2 mmol), PhNH₂ (0.2 mmol), SPAM-1 (5 mol%), 20 °C, 440 nm LEDs, 1 h. †Isolated yield was given. ‡The ee values were determined by high performance liquid chromatography.

Table S3. Enantioselective N–H insertion of α -diazoester with aniline: optimization of the LEDs light.



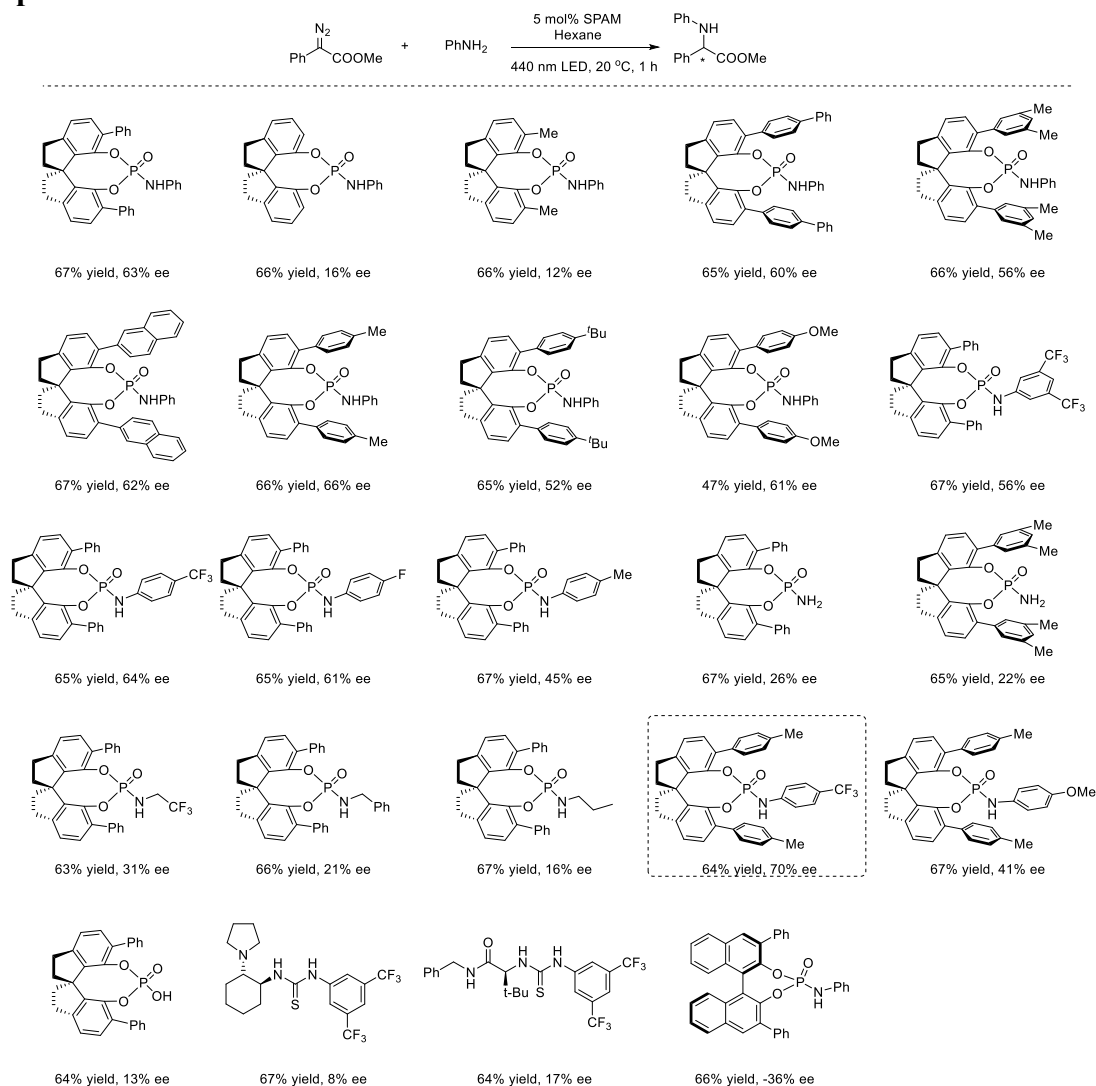


SPAM-1

Entry*	X nm	Yield (%) [†]	ee (%) [‡]
1	390 nm	66	63
2	405 nm	67	62
3	420 nm	66	61
4	440 nm	66	63
5	460 nm	67	62
6	470 nm	66	62
7	490 nm	65	63

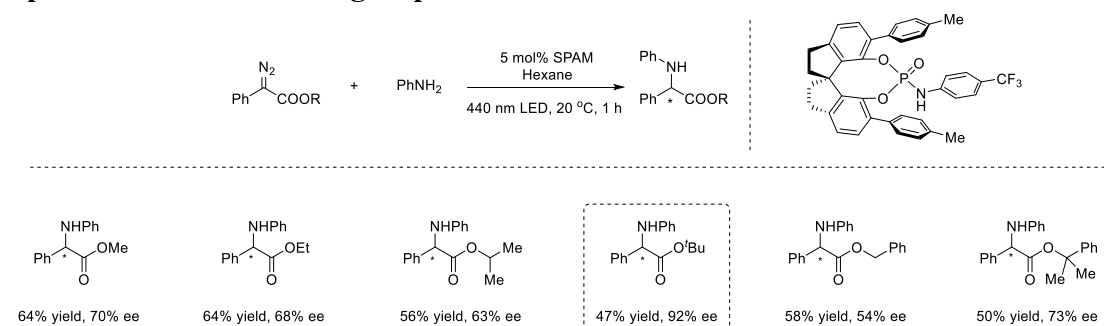
*Reaction conditions: α -diazooester (0.2 mmol), PhNH₂ (0.2 mmol), SPAM-1 (5 mol%), 4 mL hexane, 20 °C, 440 nm LEDs, 1 h. †Isolated yield was given. ‡The ee values were determined by high performance liquid chromatography.

Table S4. Enantioselective N–H insertion of α -diazooester with aniline: optimization of the SPAM.



Reaction conditions: α -diazooester (0.2 mmol), PhNH₂ (0.2 mmol), SPAM (5 mol%), 4 mL Hexane, 20 °C, 440 nm LEDs, 1 h. Isolated yield was given. The ee values were determined by high performance liquid chromatography.

Table S5. Enantioselective N–H insertion of α -diazoester with aniline: optimization of the ester group of substrates.



Reaction conditions: α -diazoester (0.2 mmol), PhNH₂ (0.2 mmol), SPAM (5 mol%), 4 mL hexane, 20 °C, 440 nm LEDs, 1 h. Isolated yield was given. The ee values were determined by high performance liquid chromatography.

Table S6. Enantioselective N–H insertion of α -diazoester with aniline: optimization of temperature and catalyst loading.

X mol% SPAM
Hexane
440 nm LED, T, 1 h

Entry*	X	T (°C)	Yield (%) [†]	ee (%) [‡]
1	1	20	46	92
2	2	20	47	92
3	5	20	46	91
4	10	20	46	92
5	2	30	44	91
6	2	0	47	92

*Reaction conditions: α -diazoester (0.2 mmol), PhNH₂ (0.2 mmol), 4 mL hexane, 440 nm LEDs, 1 h. [†]Isolated yield was given. [‡]The ee values were determined by high performance liquid chromatography.

Table S7. Enantioselective N–H insertion of α -diazoester with aniline: optimization of substrate equivalents.

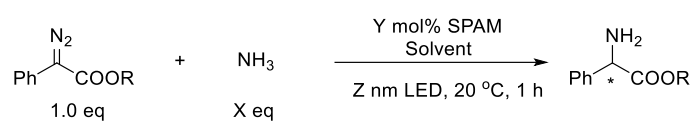
2 mol% SPAM
Hexane
440 nm LED, 20 °C, 1 h

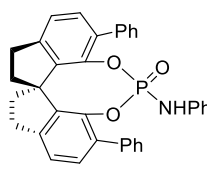
Entry*	X	Y	Hexane (mL)	Yield (%) [†]	ee (%) [‡]
1	0.2	0.2	4	46	92
2	0.2	0.4	4	71	81

3	0.2	0.6	4	82	69
4	0.1	0.1	4	42	94
5	0.1	0.1	8	37	96
6	0.1	0.2	8	56	94
7	0.1	0.3	8	69	92
8 ^a	0.1	0.3	8	0	-
9 ^b	0.1	0.3	8	67	0

*Reaction conditions: SPAM (2 mol%), 20 °C, 440 nm LEDs, 1 h. [†]Isolated yield was given. [‡]The ee values were determined by high performance liquid chromatography. ^a In dark. ^b Without SPAM.

Table S8. Enantioselective N–H insertion of α -diazoester with ammonia: optimization.



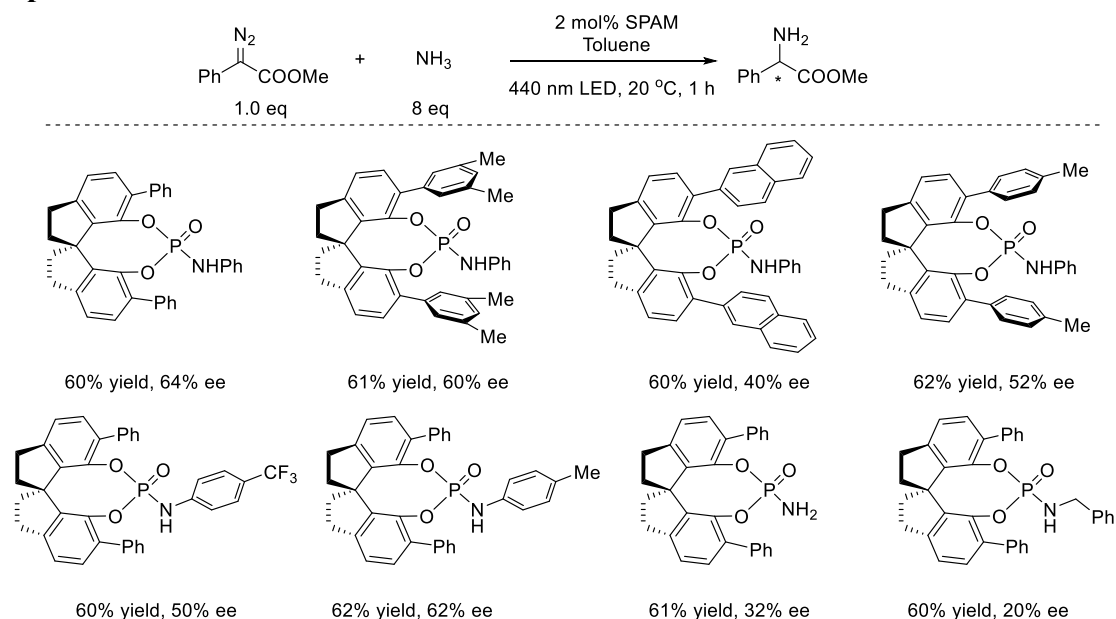


SPAM-1

Entry*	R	Solvent	X	Y	Z	Yield (%) [†]	ee (%) [‡]
1	Me	DCM	8	5	440	66	40
2	Me	Toluene	8	5	440	62	64
3	Me	THF	8	5	440	11	8
4	Me	MTBE	8	5	440	22	66
5	Me	Toluene	4	5	440	37	66
6	Me	Toluene	6	5	440	49	65
7	Me	Toluene	8	2	440	63	64
8	Me	Toluene	8	2	405	62	62
9	Me	Toluene	8	2	470	62	64
10	Et	Toluene	8	2	440	58	54
11	ⁱ Pr	Toluene	8	2	440	47	58
12	^t Bu	Toluene	8	2	440	31	50

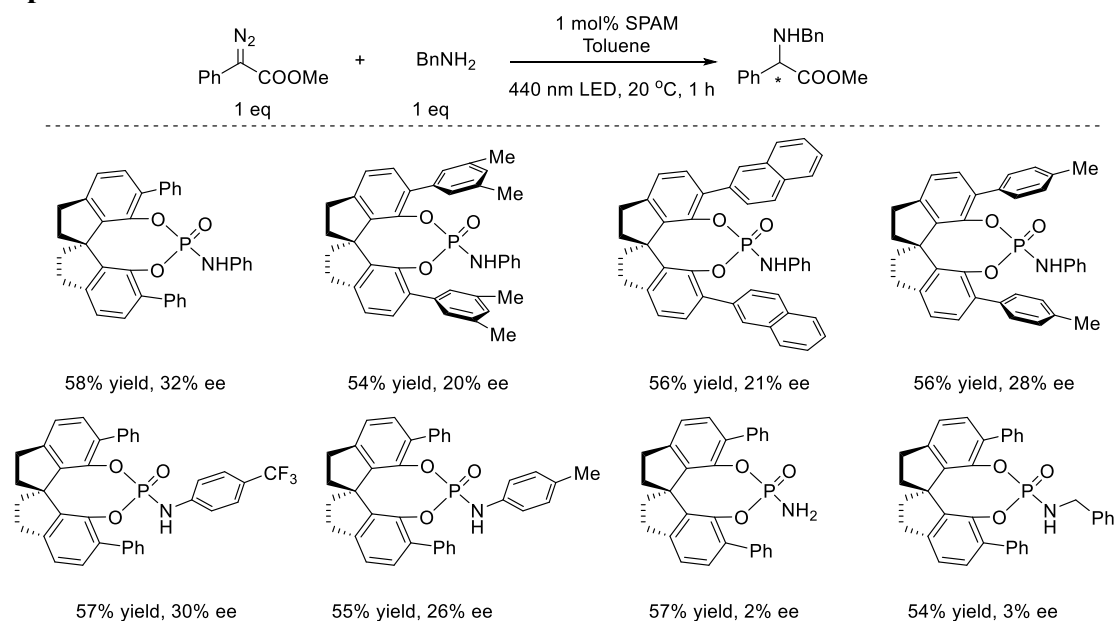
*Reaction conditions: diazoester (0.2 mmol), 1 h. [†]Isolated yield was given. [‡]The ee values were determined by high performance liquid chromatography.

Table S9. Enantioselective N–H insertion of α -diazooester with ammonia: optimization of SPAM.



*Reaction conditions: α -diazooester (0.2 mmol), ammonia (1.6 mmol, 0.27M in Toluene), SPAM (2 mol%), 6 mL toluene, 440 nm LEDs, 20 °C, 1 h. †Isolated yield was given. ‡The ee values were determined by high performance liquid chromatography.

Table S10. Enantioselective N–H insertion of α -diazooester with benzylamine: optimization of SPAM.



*Reaction conditions: SPAM (2 mol%), 20 °C, 2 mL Toluene, 440 nm LEDs, 1 h. †Isolated yield was given. ‡The ee values were determined by high performance liquid chromatography.

Table S11. Enantioselective N–H insertion of α -diazoester with benzylamine: optimization.

Entry*	R	Solvent	X	Y	Z	Yield (%) [†]	ee (%) [‡]
1	Me	Toluene	0.2	0.2	2	58	32
2	Me	DCM	0.2	0.2	2	76	14
3	Me	n-Hexane	0.2	0.2	2	66	27
4	Me	Benzene	0.2	0.2	2	65	26
5	Et	Toluene	0.2	0.2	2	54	33
6	<i>i</i> Pr	Toluene	0.2	0.2	2	49	38
7	<i>t</i> Bu	Toluene	0.2	0.2	2	33	48
8	<i>t</i> Bu	Toluene	0.2	0.4	2	43	40
9	<i>t</i> Bu	Toluene	0.4	0.2	2	45	48
10	<i>t</i> Bu	Toluene	0.6	0.2	2	55	48
11	<i>t</i> Bu	Toluene	0.6	0.2	5	54	48
12 ^a	<i>t</i> Bu	Toluene	0.6	0.2	2	41	54

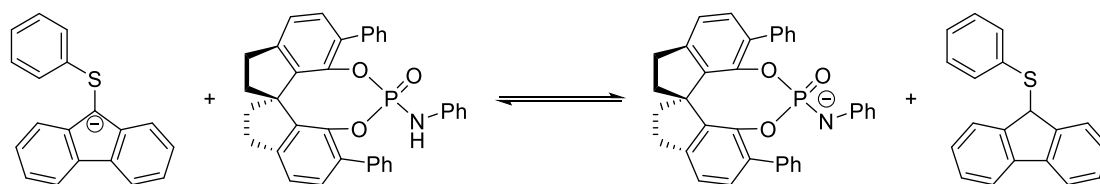
*Reaction conditions: Solvent (2 mL), 1 h. [†]Isolated yield was given. [‡]The ee values were determined by high performance liquid chromatography. ^a4 mL toluene was used.

2. Procedure for the Determination of pK_a

Overlapping indicator method

We used the overlapping indicator method for pK_a determination.^[2] The method consists essentially in relating the acidity of two acids by titrating a solution of one acid, containing a known concentration of its colored anion, with a solution of a second acid, the anion of which is either colorless or absorbs at a different wavelength.

In this study, the UV spectrum measurements were performed on Agilent Cary 60 UV/vis spectrometer. The procedures for purification and preparation of the DMSO solvent, the potassium dimsyl base, the solutions of the indicator were reported in the literature.^[3-4] We choose (9H-fluoren-9-yl)(phenyl)sulfane as the indicator ($pK_a = 15.4$ in DMSO, $\lambda_{max} = 454$ nm).



The exact amount of dry DMSO (usually 10–25 mL) and indicator (usually 100 mg) were determined gravimetrically to prepare the solution of indicator whose

concentration was about 50 mmol/L. All the operations were performed in the glove box.

Titration and evaluation

A spectrophotometric cell ($d = 1$ cm, quartz) was equipped with a stirring bar, sealed with a septum cap, and weighted. The cell was charged with freshly distilled DMSO and weighted again. The first UV spectrum was recorded to determine the base line, and then the titration was started. Several aliquots (15–20 μL) of the indicator stock solution was added, the weight of the cell and the UV spectrum recorded after each step, until no further increase of the absorbance value at the chosen wavelength was observed. The absorbance value and the concentration of the indicator in the cell corresponded according to Lambert-Beer's law (Figure S2). The slope of the linear fit gave the molar extinction coefficient, and was determined in every titration. At this stage the K-dimsyl base in the cell was consumed completely and the solution contained only the two species of the indicator. The concentration of the deprotonated form could be calculated directly from the absorbance value. The overall amount of the two species was known, which led to the concentration of the protonated form. (We assumed that the dilute stock solutions had the same density as that of the pure DMSO).

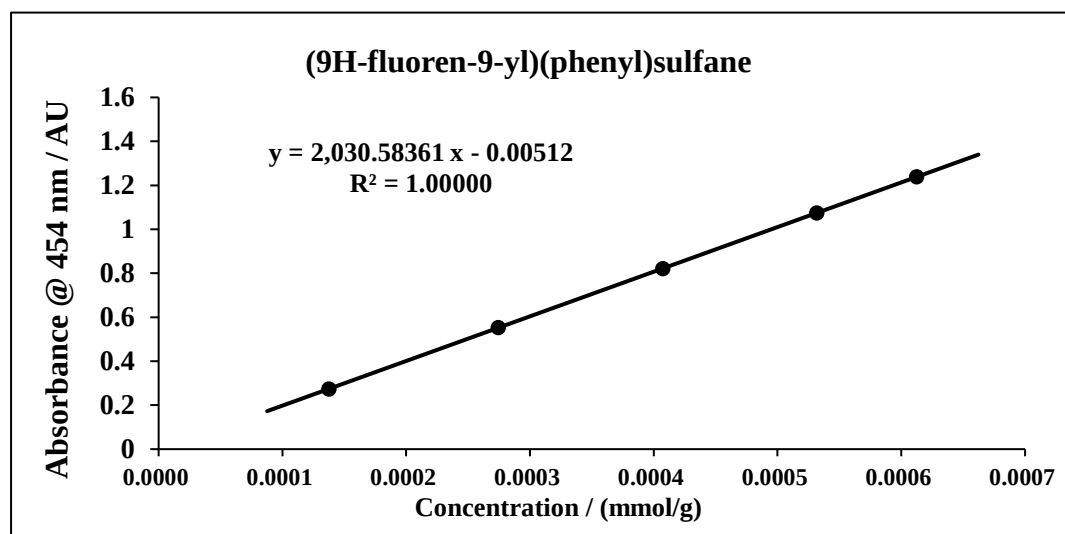


Figure S2. The Lambert-Beer plot for (9H-fluoren-9-yl)(phenyl)sulfane.

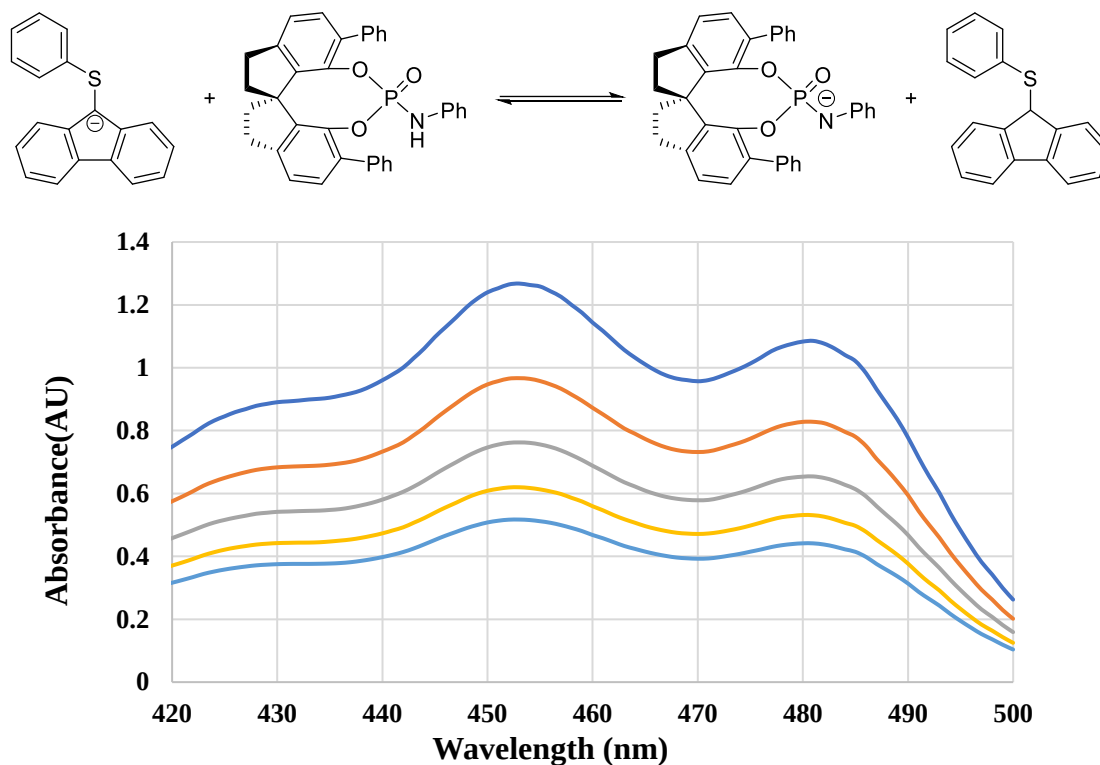
Then, four increments of the catalyst solution were added, the weight of the cell and the absorbance value recorded after each step. With an appropriate indicator a new acid-base equilibrium was established with each addition, causing a loss in the absorbance value at a chosen wavelength, as the indicator anion was reprotonated. The change could only be implemented by the catalyst (HA) as proton source, so the following equations applied:



With the necessary concentration values in hand, the equilibrium constant could be calculated, which led to the desired pK_a value.

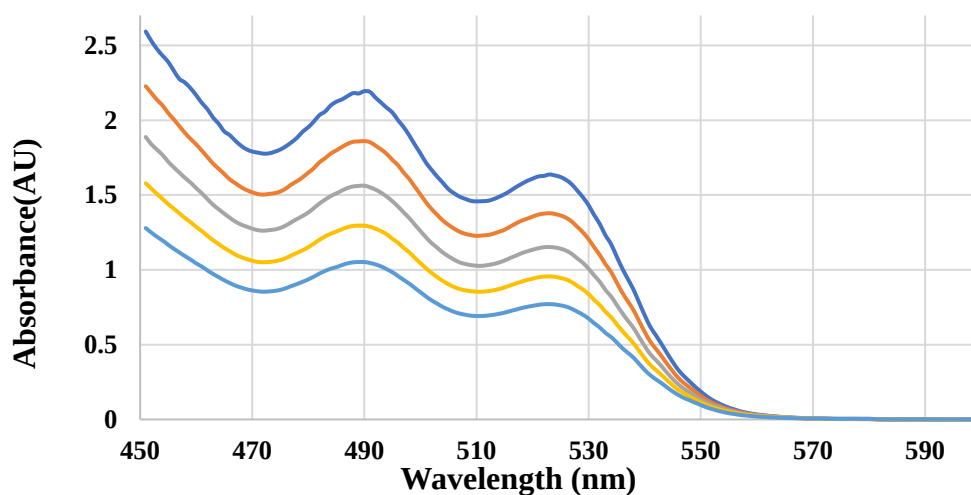
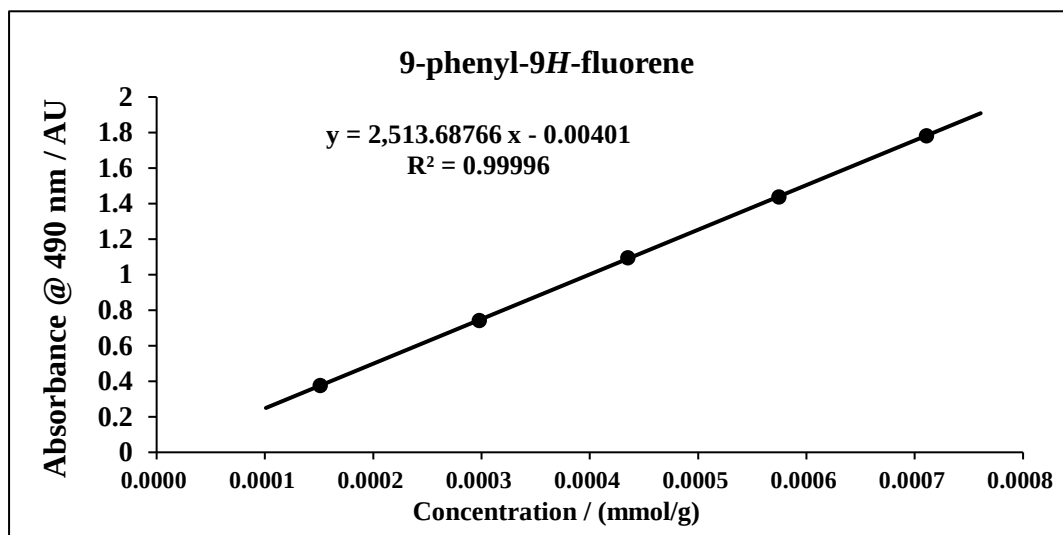
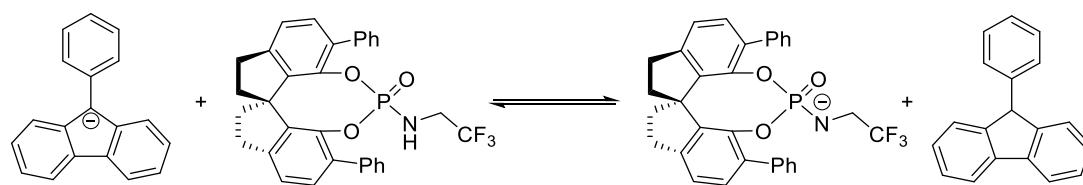
$$pK_{HA} = pK_{HIn} - \log K_{eq} = 15.4 - \log \left(\frac{[HIn][A^-]}{[HA][In^-]} \right) \quad (2)$$

The titration provided 4 pK_a values, the average of which is taken as the result. In the titration for SPAM-1 (Figure S3), the following values were obtained:



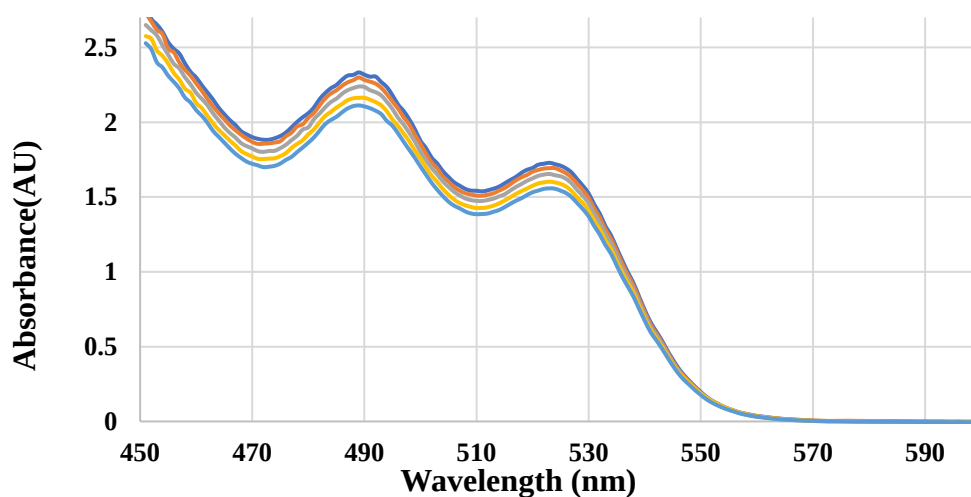
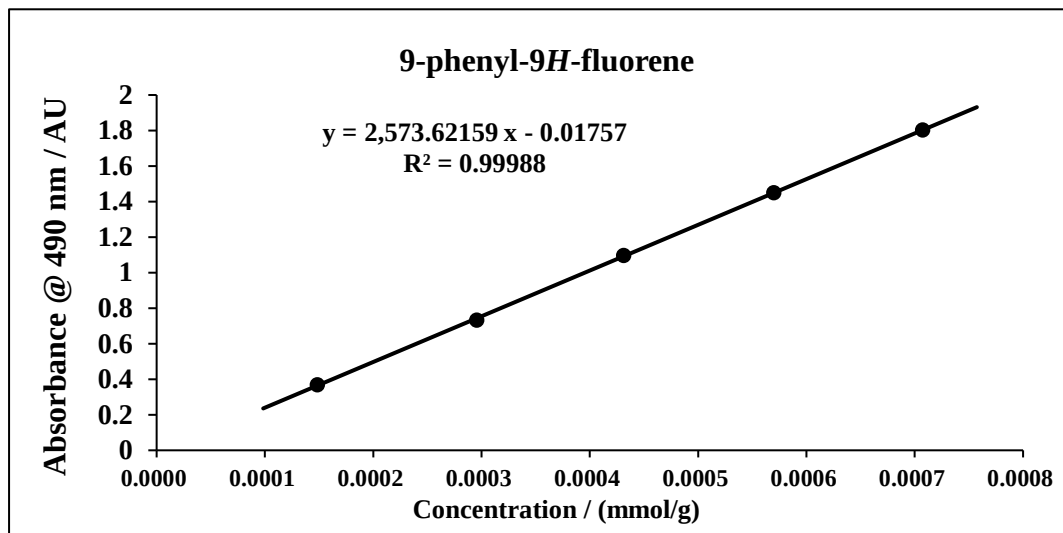
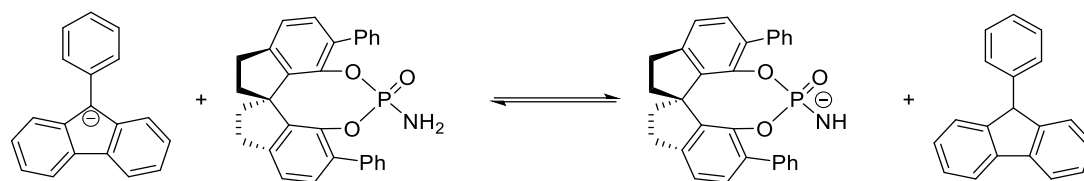
HIn	A ⁻	HA	In ⁻	K_{eq}	pK_a
0.0011050	0.0001424	0.0002325	0.0004779	1.416290257	15.24884773
0.0011932	0.0002382	0.0005059	0.0003773	1.488929543	15.22712585
0.0012509	0.0003033	0.0008026	0.0003074	1.537360033	15.21322441
0.0012897	0.0003492	0.0011016	0.0002569	1.591024167	15.19832322
Concentration / (mmol/g)				Mean value	15.22188031
				St. dev.	0.018605231

Figure S3. Absorption spectra of (9H-fluoren-9-yl)(phenyl)sulfane obtained during the second part of the titration for SPAM-1.



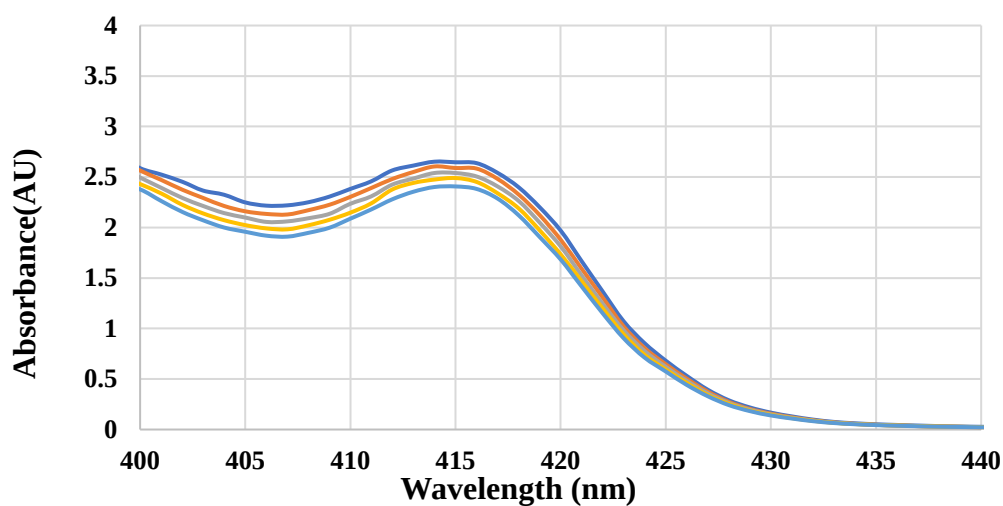
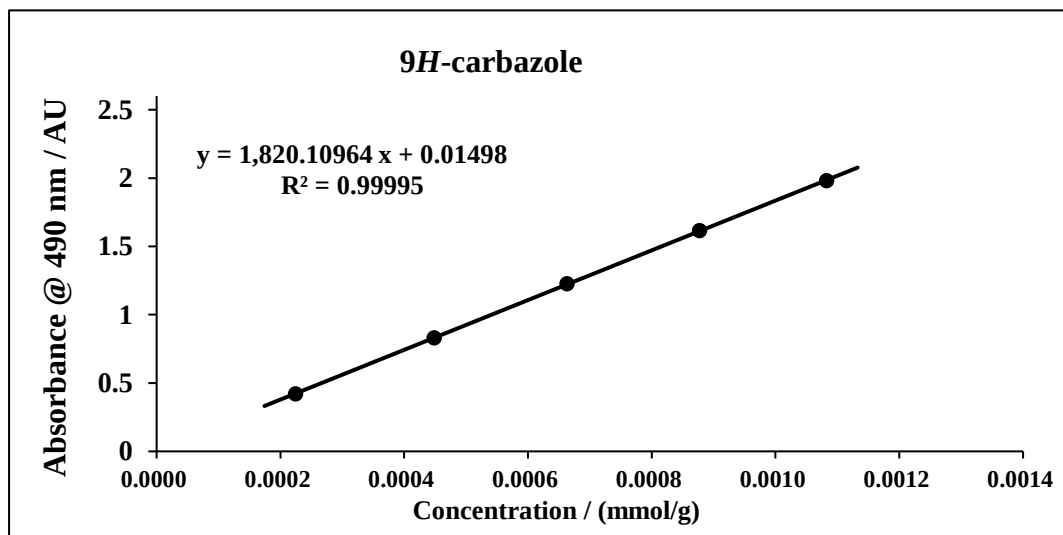
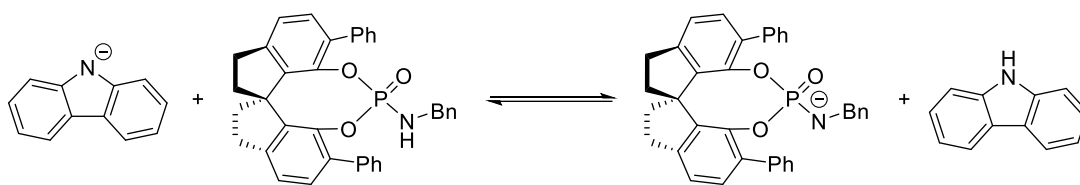
HIn	A ⁻	HA	In ⁻	K_{eq}	p <i>K</i> _a
0.00087156	0.0001241	4.118E-05	0.00074068	3.547393997	17.35009057
0.00098409	0.0002393	7.442E-05	0.00062241	5.084940274	17.19371414
0.00108399	0.0003419	0.00012202	0.00051671	5.878668762	17.13072101
0.00117464	0.0004353	0.00018057	0.00042019	6.739402985	17.07137857
Concentration / (mmol/g)				Mean value	17.13193791
				St. dev.	0.049950699

Figure S4. The overlapping indicator method to determine the p*K*_a.



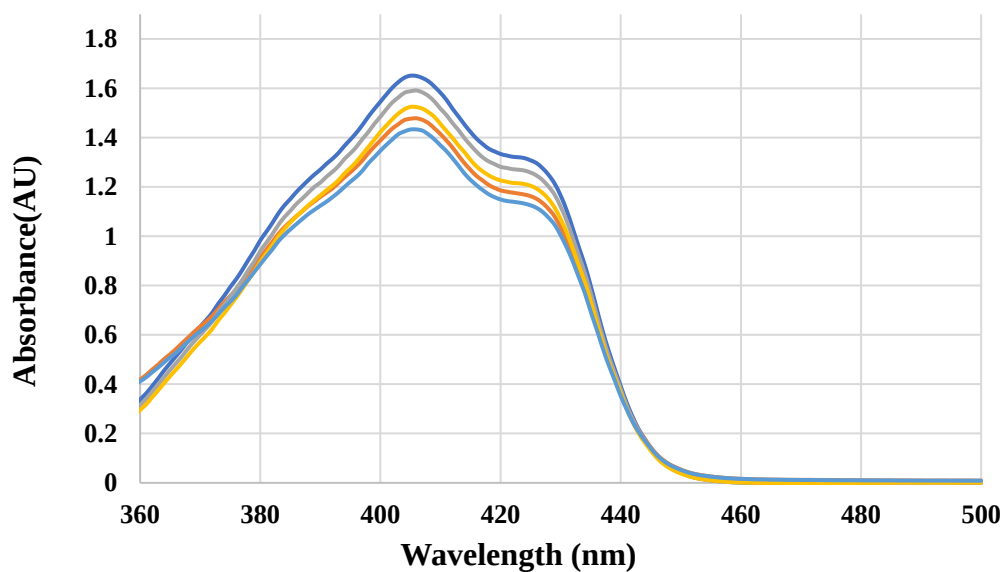
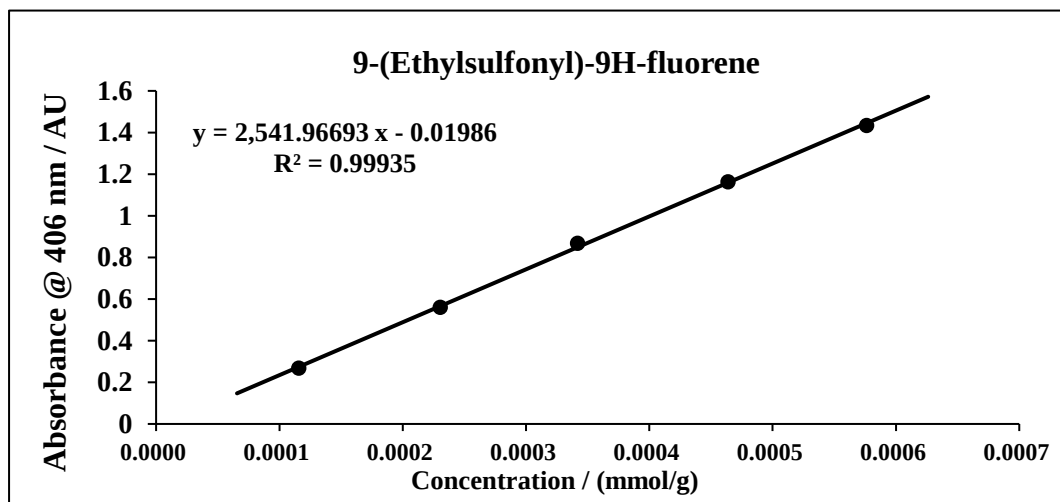
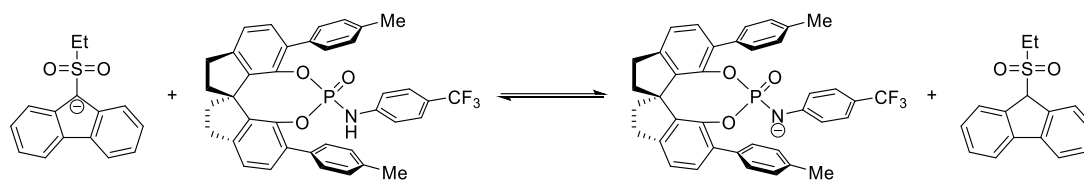
HIn	A ⁻	HA	In ⁻	<i>K</i> _{eq}	p <i>K</i> _a
0.00067837	7.539E-06	0.00028599	0.0009001	0.01986728	19.6018614
0.00068930	2.383E-05	0.00065726	0.0008765	0.02851741	19.4448898
0.00070696	4.651E-05	0.00099650	0.0008471	0.03895787	19.3094047
0.00071459	5.923E-05	0.00135149	0.0008274	0.03784988	19.3219354
Concentration / (mmol/g)				Mean value	19.41952288
				St. dev.	0.11783479

Figure S5. The overlapping indicator method to determine the p*K*_a.



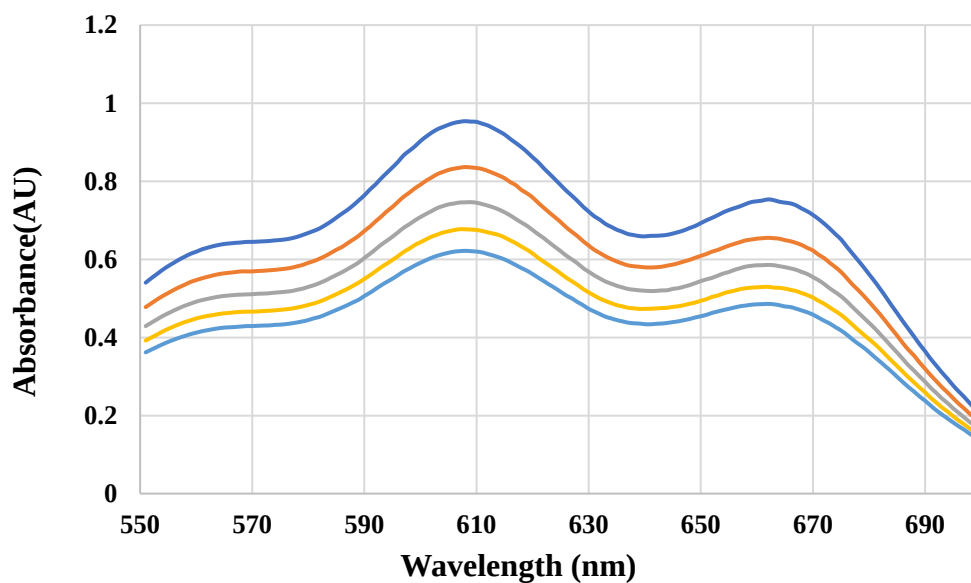
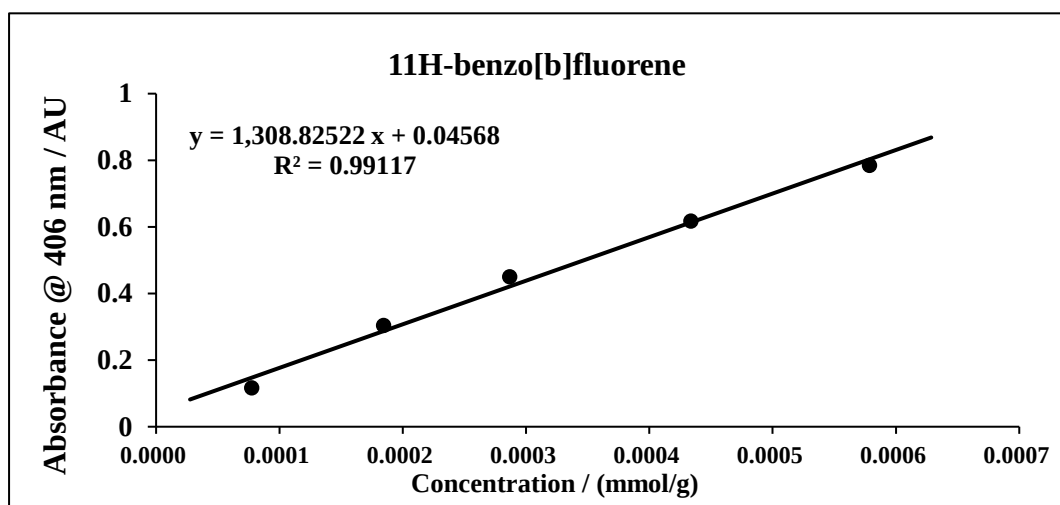
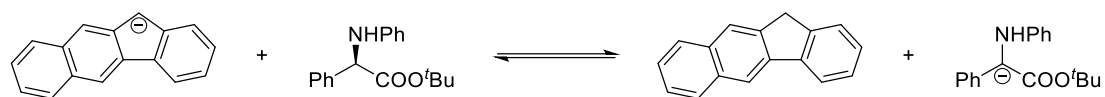
HIn	A ⁻	HA	In ⁻	K_{eq}	p <i>K</i> _a
0.00125784	3.503E-05	0.00028453	0.00116465	0.132978446	20.77621875
0.00128163	6.878E-05	0.00056658	0.00112113	0.138788761	20.7576457
0.00129771	9.457E-05	0.00084871	0.00108581	0.133177378	20.77556954
0.00131805	0.0001246	0.00112581	0.00104628	0.139425626	20.7556574
Concentration / (mmol/g)				Mean value	20.76295755
				St. dev.	0.008954893

Figure S6. The overlapping indicator method to determine the p*K*_a.



HIn	A ⁻	HA	In ⁻	K_{eq}	pK_a
0.00056778	8.494E-06	8.7506E-05	0.00063565	0.069955165	13.45518021
0.00057353	1.705E-05	0.0001528	0.00062386	0.08302474	13.38079248
0.00058237	3.326E-05	0.00033102	0.00059916	0.079591539	13.3991331
0.00059171	4.895E-05	0.00048260	0.00057617	0.085386942	13.36860854
Concentration / (mmol/g)				Mean value	13.40092858
				St. dev.	0.033153096

Figure S7. The overlapping indicator method to determine the pK_a .



HIn	A ⁻	HA	In ⁻	<i>K</i> _{eq}	p <i>K</i> _a
0.00099746	3.500E-05	0.00011818	0.00026236	1.126233057	23.04837173
0.00101620	6.041E-05	0.00023331	0.00023488	1.120420724	23.05061887
0.00102960	8.060E-05	0.00035584	0.00021261	1.096861142	23.05984835
0.00103856	9.547E-05	0.00047001	0.00019562	1.080885445	23.06622033
Concentration / (mmol/g)				Mean value	23.05626482
				St. dev.	0.007178641

Figure S8. The overlapping indicator method to determine the p*K*_a.

3. Computational Data

DFT Methods for the proton transfer catalyzed by the SPAM.

We calculated the proton transfer modes involving the spiro phosphamide catalyst. The 1,2-proton transfer of ylide and 1,3-proton transfer of enol were explored with (*R*)-SPAM (Figure S9). **TSRa-I** has the lowest energy among six different transition states (Table S12).

DFT method

DFT calculations have been carried out using the Gaussian 16 program package.^[5] The B3LYP method and the dispersion energy corrections by D3 version of Grimme's dispersion with the original D3BJ damping function were used.^[6-8] The def2svp basis set of Ahlrichs and coworkers for all the atoms has been selected for geometry optimizations and calculation of Gibbs energy corrections at 298 K temperature.^[9-10] Final energies were retrieved from single-point calculations at the B3LYP level of theory with def2tzvpp basis set for all atoms. All structures have been optimized considering solvent effects using the SMD model for *n*-hexane.^[11] Reaction paths were traced by the intrinsic reaction coordinate method for all transition states. All energetics reported throughout the text are in kcal/mol. Structures were generated using CYLview.^[12]

Table S12. Calculated Gibbs energy for the transition states of the product.

	Gibbs energy corrections	Single-point energies	Gibbs energy	ΔG
TSRa-I	0.862012	-3293.456422	-3292.594410	0.0
TSRa-II	0.862304	-3293.456216	-3292.593912	0.3
TSRb-I	0.85874	-3293.441224	-3292.582484	7.5
TSSa-I	0.859478	-3293.44764	-3292.588166	3.9
TSSa-II	0.861918	-3293.448702	-3292.586784	4.8
TSSa-III	0.861912	-3293.447543	-3292.585631	5.5
TSSa-IV	0.860905	-3293.44636	-3292.585452	5.6
TSSa-V	0.860548	-3293.44285	-3292.582304	7.6
TSSb-I	0.857363	-3293.433899	-3292.576536	11.2

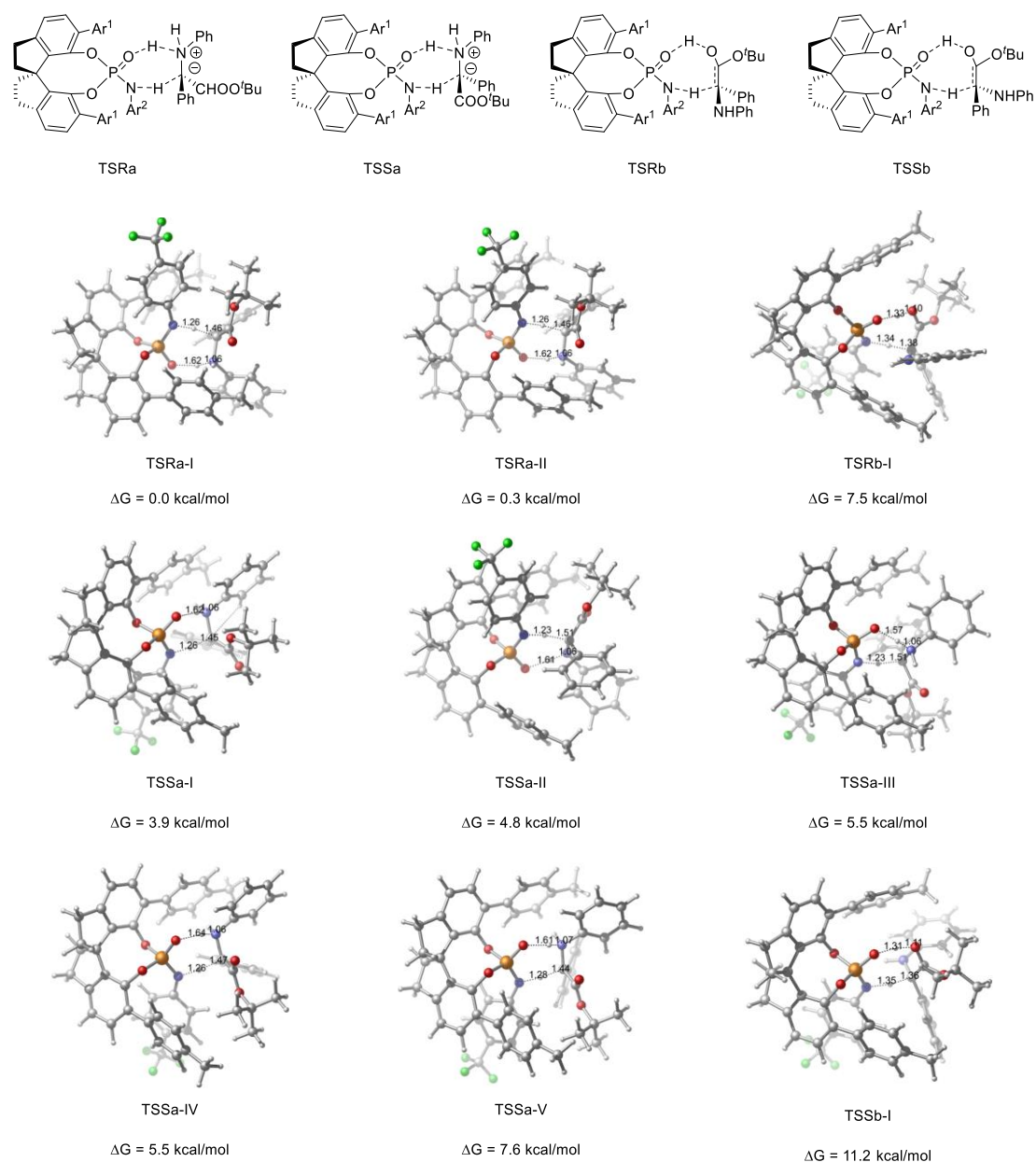


Figure S9. The proton transfers of ylide and enol to the new formed stereogenic center at product.

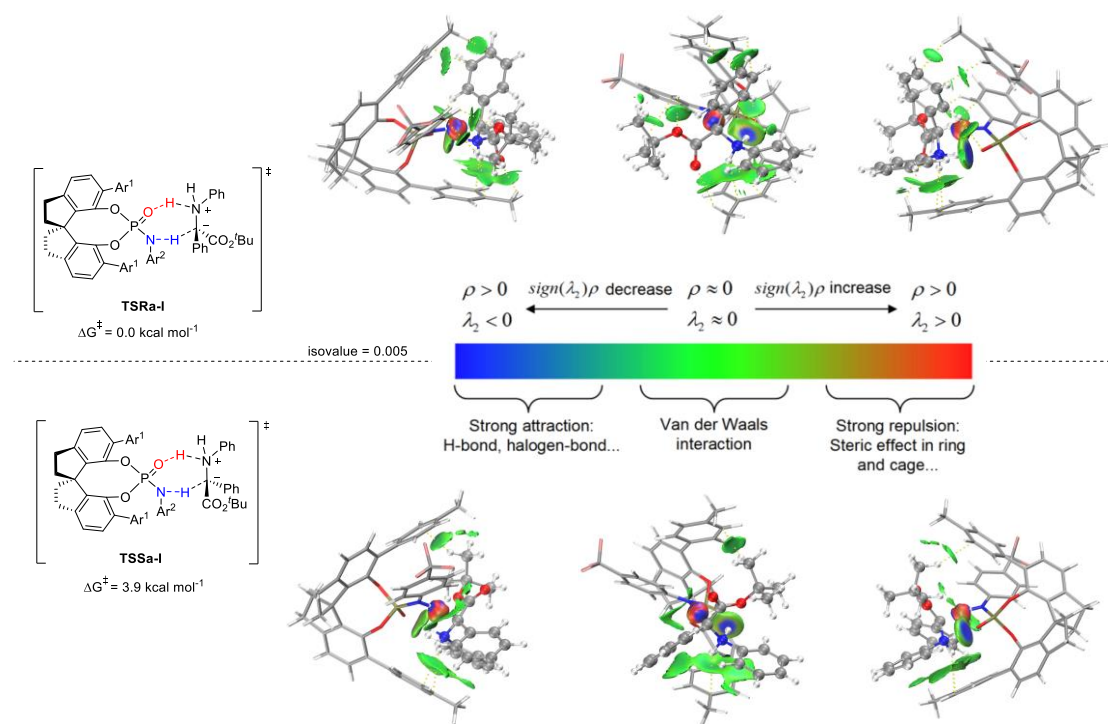


Figure S10. The independent gradient model analysis for TSRa-I and TSSa-I.

Relative pK_a calculations

The overlapping indicator method for pK_a determination can not be applied to the **ylide intermediates** and the determination of pK_a of this species is rarely reported in the literatures. Therefore, we use the relative pK_a method based on the thermodynamic cycles to caculate the pK_a of the **ylide intermediates** (Figure S11). With this cycle, the exchange free energy in solution is:

$$\Delta G_{\text{exchange}} = \Delta G_{\text{gas, exchange}} + \Delta G_{\text{solv}}(A^-) + \Delta G_{\text{solv}}(H^+) - \Delta G_{\text{solv}}(HA) \quad (3)$$

$\Delta G_{\text{exchange}}$ represents the overall change in Gibbs free energy of this reaction in solution. $\Delta G_{\text{gas, exchange}}$ is the change in the gas-phase Gibbs free energy, and ΔG_{solv} is the change in Gibbs free energy of solvation. Then, the pK_a is:

$$pK_a = \frac{\Delta G_{\text{exchange}}}{RT \ln(10)} \quad (4)$$

Relative pK_a calculations can be highly accurate, within a few tenths of a kcal/mol, as errors in both the gas-phase and solution-phase values tend to cancel. Numerous highly accurate relative pK_a values have been reported in the literature.^[13]

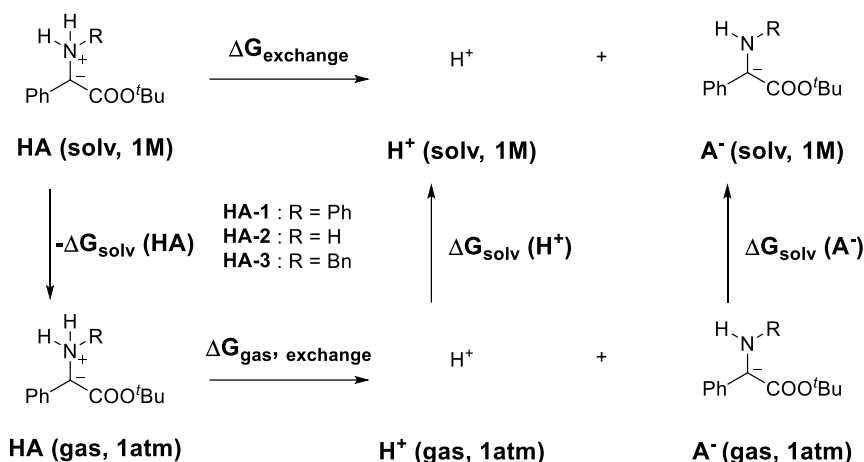


Figure S11. Thermodynamic cycles employed in this study

DFT Methods for pK_a calculations^[14-15]

All DFT calculations have been carried out using the Gaussian 16 program package.

The change in the Gibbs free energy

The B3LYP method and the dispersion energy corrections by D3 version of Grimme's dispersion with the original D3BJ damping function were used. The 6-31+G(d) basis set for all the atoms has been selected for geometry optimizations and calculation of Gibbs energy corrections at 298 K temperature in the gas phase. Final energies were retrieved from single-point calculations at the M062X-D3 level of theory with 6-311++G(2df,2p) for all atoms in the gas phase.^[16-17] The free energies of solvation in DMSO were calculated using SMD model.

Table S13. Calculated Gibbs energy profile for the compounds.

	Gibbs energy corrections	Single-point energies (gas)	Single-point energies (dmsol)	ΔG_{solv}
H^+	-0.010000	0.000000	---	-0.427633 ^[18]
HA-1	0.304625	-903.653365	-903.679834	-0.026469
A ⁻ -1	0.289629	-903.141327	-903.218241	-0.076914
HA-2	0.23139	-672.631096	-672.659144	-0.028048
A ⁻ -2	0.215796	-672.100779	-672.179429	-0.078650
HA-3	0.331821	-942.964708	-942.992927	-0.028219
A ⁻ -3	0.315180	-942.433373	-942.511637	-0.078264
SPAM-2	0.529284	-2388.064404	-2388.100619	-0.036215
SPAM-2 ⁻	0.515317	-2387.529728	-2387.620388	-0.090660

$$\text{HA-1} \quad pK_a = \frac{\Delta G_{\text{exchange}}}{RT \ln(10)} = 4.13 \quad (5)$$

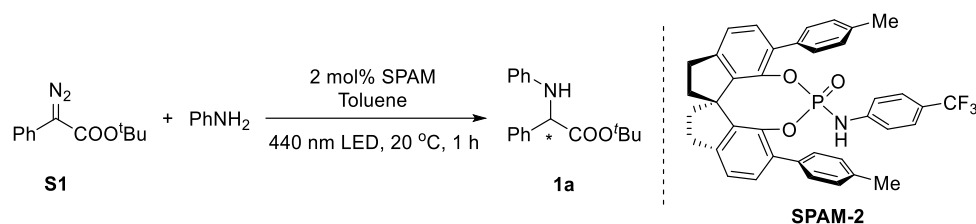
$$\text{HA-2} \quad pK_a = \frac{\Delta G_{\text{exchange}}}{RT \ln(10)} = 12.20 \quad (6)$$

$$\text{HA-3} \quad pK_a = \frac{\Delta G_{\text{exchange}}}{RT \ln(10)} = 12.45 \quad (7)$$

To check the correctness of the calculation method, we applied the method to the pK_a calculation of the catalyst. We calculated the pK_a value of the optimal catalyst **SPAM-2**. The calculated value of pK_a is 13.19, the experimental value is 13.40, and the error between these is only 0.2 unit. Therefore, we think this method is reliable.

$$\text{SPAM-2} \quad pK_a = \frac{\Delta G_{\text{exchange}}}{RT \ln(10)} = 13.19 \quad (8)$$

4. Kinetic Experiments



A three necked reaction vessel was fitted with a magnetic stirring bar. The IR probe was inserted through an adapter into the middle neck; the other two necks were capped by septa for reagent injections and a N₂ line. To this vessel were added **SPAM-2**. This vessel was evacuated and purged with N₂ three times. To this vessel was added PhNH₂ (0.2 mmol) and Toluene (2 mL) by a syringe. The mixture was stirred at 20 °C in a water bath meanwhile turn on the LED light. Then the IR data collection was started, followed by addition of diazoester **S1**. In situ IR spectra were recorded over the course of the reaction. ConcIRT Spectrum of the **1a** with peak at 1741 cm⁻¹ was used for the generation of the kinetic profiles.

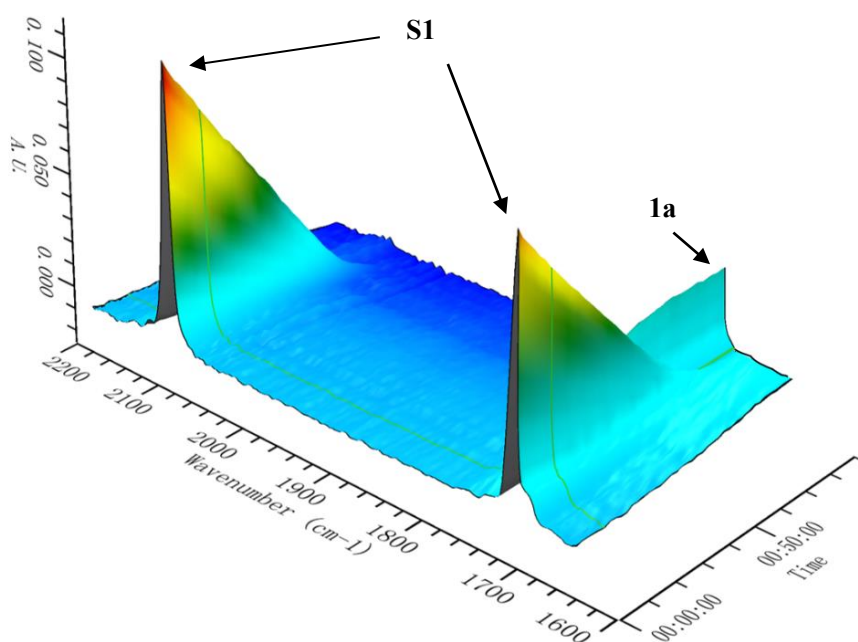
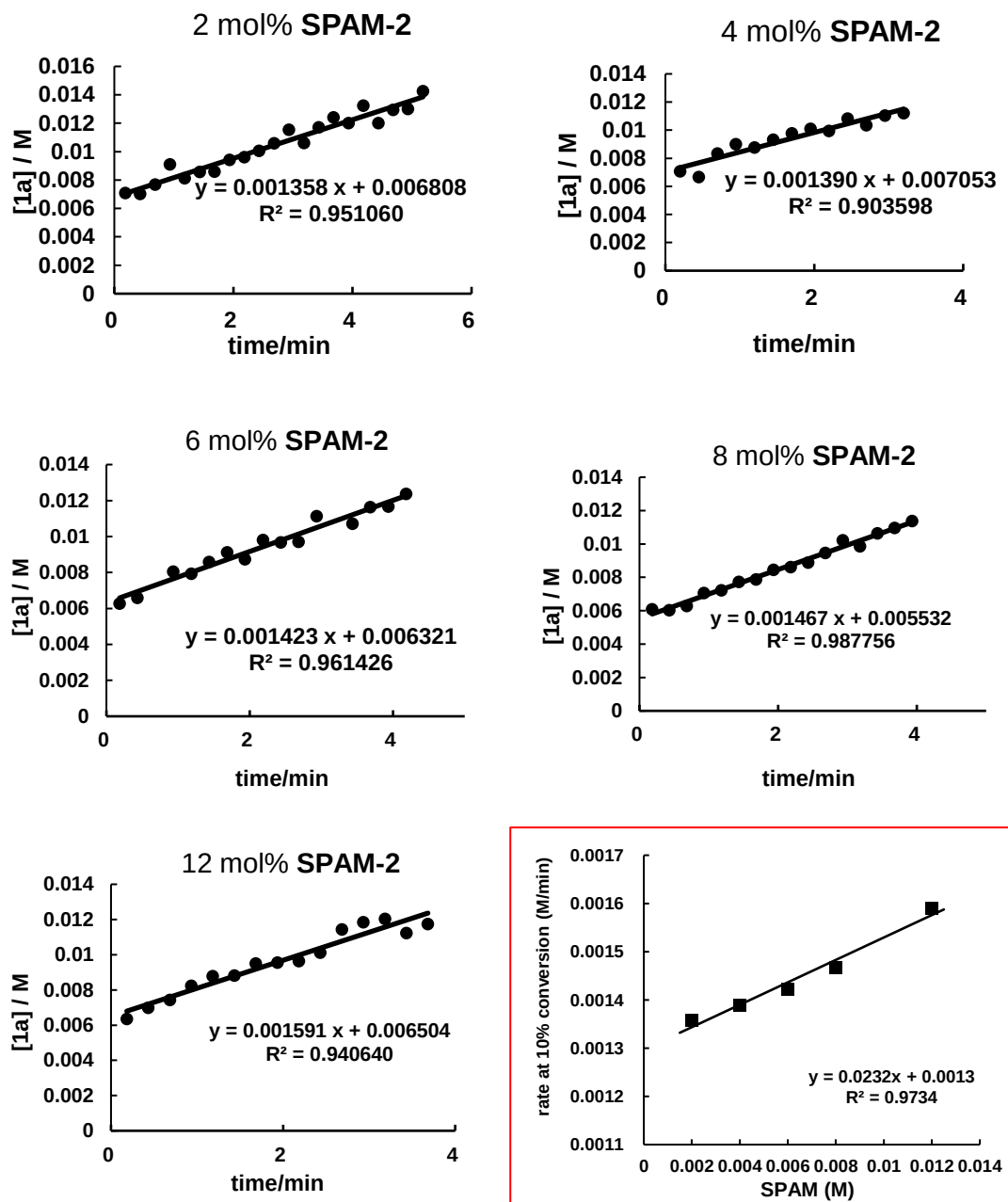


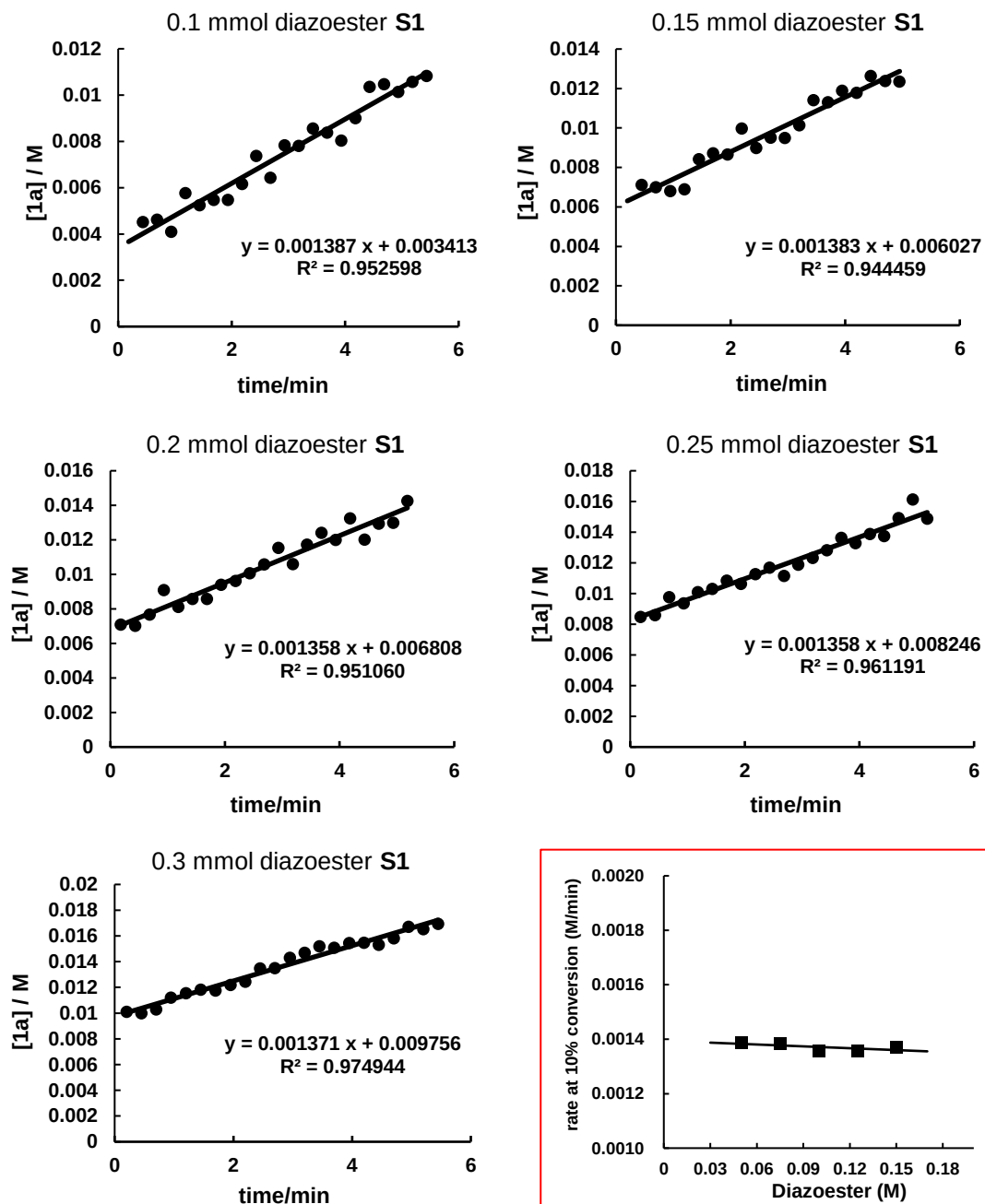
Figure S12. Kinetic profiles of Photo-catalyzed N–H insertion reaction of S1 and PhNH₂. In situ IR spectroscopy of temporal changes in the stretching bands of S1 (2080 cm⁻¹ and 1703 cm⁻¹) and 1a (1741 cm⁻¹).



SPAM-2 (M)	0.002	0.004	0.006	0.008	0.012
k_{obs} (M/min)	0.001358	0.001390	0.001423	0.001467	0.001591

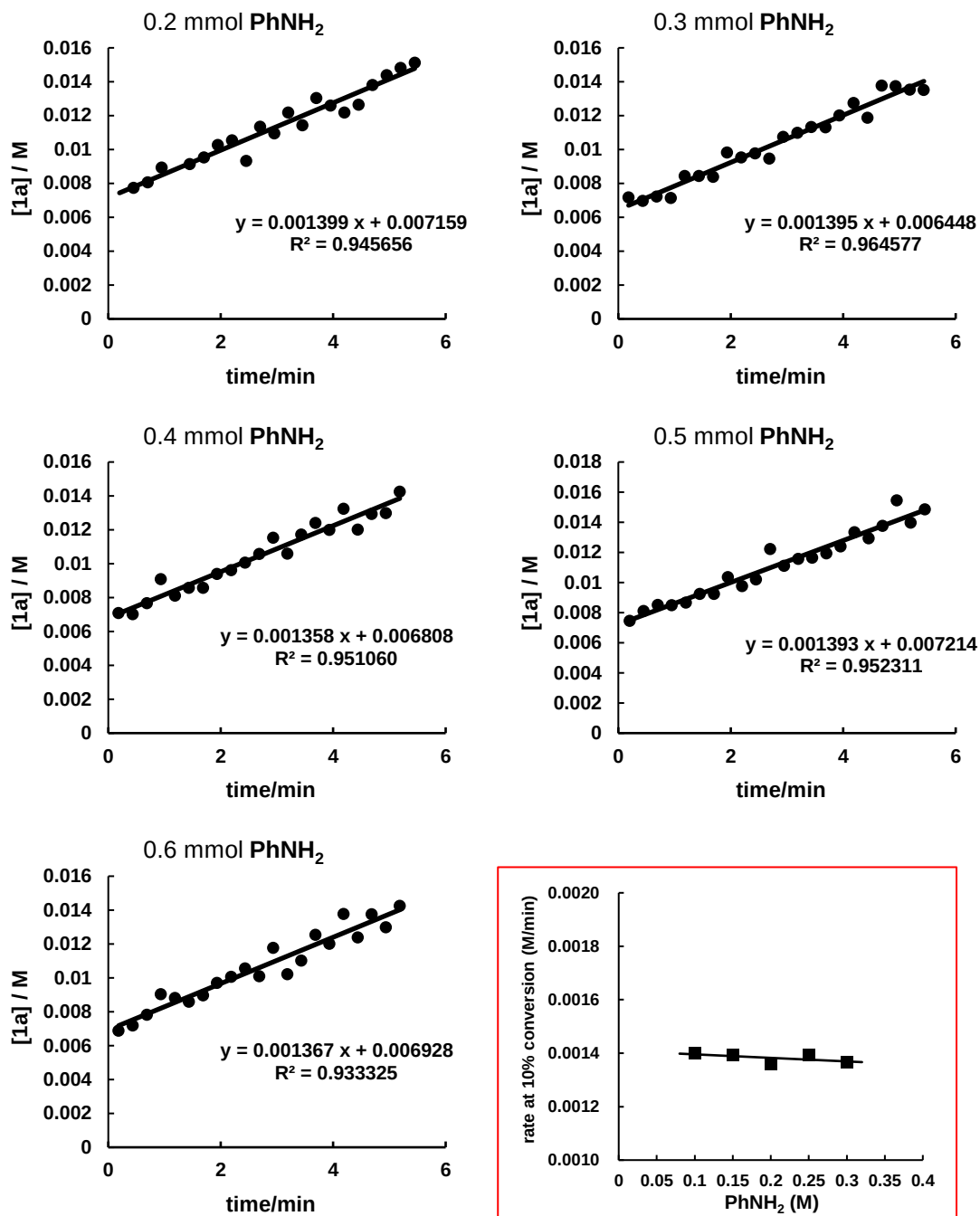
Figure S13. Dependence of the reaction rate on concentration of SPAM-2.

Kinetic profiles of different initial concentrations of SPAM-2 (from 2 mol% to 12 mol%). Reaction conditions: S1 (0.1 M), PhNH₂ (0.2 M), carried out in toluene (2 mL) at 20 °C under 440 nm light irradiation.



Diazoester S1 (M)	0.05	0.075	0.1	0.125	0.15
k_{obs} (M/min)	0.001387	0.001383	0.001358	0.001358	0.001371

Figure S14. Dependence of the reaction rate on concentration of diazoester S1. Kinetic profiles of different initial concentrations of diazoester S1 (from 0.1 mmol to 0.3 mmol). Reaction conditions: SPAM-2 (0.002 M), PhNH₂ (0.2 M), carried out in toluene (2 mL) at 20 °C under 440 nm light irradiation.



PhNH_2 (M)	0.1	0.015	0.2	0.25	0.3
k_{obs} (M/min)	0.001399	0.001395	0.001358	0.001393	0.001367

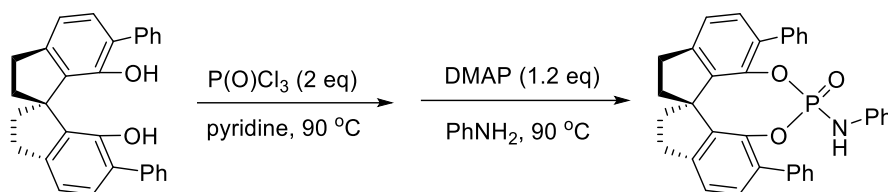
Figure S15. Dependence of the reaction rate on concentration of PhNH_2 .

Kinetic profiles of different initial concentrations of PhNH_2 (from 0.2 mmol to 0.6 mmol). Reaction conditions: SPAM-2 (0.002 M), S1 (0.1 M), carried out in toluene (2 mL) at 20 °C under 440 nm light irradiation.

5. Synthesis and Analytical Data

Table S14. Synthesis and Analytical Data of Spiro Phosphamide Catalysts

(*R*)-1,10-diphenyl-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



According to the literature,^[19] we can obtain (*R*)-SPINOL containing various substituted arenes.

To a 25 mL oven-dried Schlenk flask containing substituted (*R*)-SPINOL (404 mg, 1 mmol) was added 3 mL anhydrous pyridine and freshly distilled P(O)Cl₃ (302 mg, 2 mmol) under a nitrogen atmosphere. The mixture was stirred for 12 h under 90 °C. After cooling to room temperature, excess hydrochloric acid was added. The mixture was extracted by DCM (20 mL × 3). The combined organic layer was concentrated to give a white solid compound.

The dried white solid compound was added DMAP (147 mg, 1.2 mmol), PhNH₂ (3 mL) was added. The mixture was stirred at 90 °C for 12 h, and then cooled by an ice-bath, excess hydrochloric acid was added. The mixture was extracted by DCM (20 mL × 3). The combined organic layer was concentrated and the residue was purified by chromatography on silica gel (petroleum ether/ethyl acetate = 5:1) to give 487 mg white solid product.

White solid, 485 mg, 90% yield, m.p. 118–120 °C. TLC: *R_f* = 0.27 (PE/EA = 3:1, v/v). $[\alpha]_D^{25} +420$ (c 0.2, CHCl₃).

¹H NMR (400 MHz, CDCl₃)

δ 7.66–7.55 (m, 2H), 7.52–7.45 (m, 2H), 7.43–7.27 (m, 7H), 7.24–7.14 (m, 3H), 6.93–6.59 (m, 3H), 6.02–5.72 (m, 2H), 4.25 (d, *J* = 10.0 Hz, 1H), 3.39–3.05 (m, 2H), 3.06–2.79 (m, 2H), 2.52–2.07 (m, 4H).

¹³C NMR (151 MHz, CDCl₃)

δ 146.7, 145.0, 143.1 (d, *J* = 10.2 Hz), 141.4 (d, *J* = 7.5 Hz), 140.7 (d, *J* = 2.3 Hz), 140.6 (d, *J* = 3.8 Hz), 138.3, 138.2, 136.4, 136.1 (d, *J* = 3.7 Hz), 132.6 (d, *J* = 4.0 Hz), 130.8, 130.4, 130.3, 129.8, 129.1, 128.9, 128.3, 128.2, 127.1, 123.2, 122.8, 121.5, 117.4 (d, *J* = 7.4 Hz), 60.2, 39.3, 38.3, 30.4.

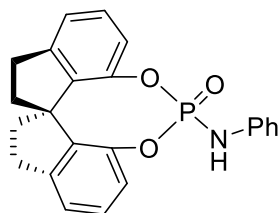
³¹P NMR (162 MHz, CDCl₃)

δ –7.9.

HRMS (ESI)

Calcd for [C₃₅H₂₈NO₃P, M+H]⁺: 542.1880; Found: 542.1885.

(*R*)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 75 mg, 94% yield, m.p. 129–131 °C. TLC: R_f = 0.33 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +150 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.30 – 7.22 (m, 3H), 7.17 (d, J = 7.4 Hz, 1H), 7.15 – 7.09 (m, 2H), 7.09 – 7.04 (m, 3H), 7.02 (t, J = 7.5 Hz, 1H), 6.67 (d, J = 7.9 Hz, 1H), 4.88 (d, J = 8.2 Hz, 1H), 3.11 (td, J = 11.0, 5.8 Hz, 2H), 2.85 (dd, J = 16.3, 7.8 Hz, 2H), 2.32 (dd, J = 12.1, 6.3 Hz, 1H), 2.26 (dd, J = 12.0, 6.3 Hz, 1H), 2.11 – 1.97 (m, 2H).

¹³C NMR (151 MHz, CDCl₃)

δ 146.8 (d, J = 1.7 Hz), 146.6 (d, J = 1.5 Hz), 146.3 (d, J = 9.5 Hz), 143.5 (d, J = 7.4 Hz), 139.7 (d, J = 3.2 Hz), 139.6 (d, J = 4.0 Hz), 138.5 (d, J = 3.3 Hz), 129.4, 128.9, 128.7, 123.2, 122.7, 122.6, 121.9 (d, J = 2.7 Hz), 120.9 (d, J = 4.0 Hz), 118.3 (d, J = 7.2 Hz), 59.2, 38.4, 38.3, 30.6.

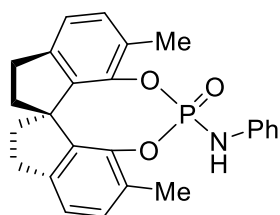
³¹P NMR (162 MHz, CDCl₃)

δ –5.9.

HRMS (ESI)

Calcd for [C₂₃H₂₀NO₃P, M+Na]⁺: 412.1073; Found: 412.1078.

(*R*)-1,10-dimethyl-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 70 mg, 82% yield, m.p. 115–117 °C. TLC: R_f = 0.40 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +304 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.15 (t, J = 7.8 Hz, 2H), 7.11 (d, J = 7.6 Hz, 1H), 7.03 (d, J = 7.5 Hz, 1H), 6.95 (d, J = 8.6 Hz, 3H), 6.91 (t, J = 8.6 Hz, 2H), 5.15 (d, J = 7.2 Hz, 1H), 3.04 (qd, J = 15.0, 14.6, 6.5 Hz, 2H), 2.80 (td, J = 15.9, 7.9 Hz, 2H), 2.44 (s, 3H), 2.26 (dd, J = 12.0, 6.3 Hz, 1H), 2.18 (dd, J = 12.0, 6.3 Hz, 1H), 2.10 (s, 3H), 2.08 – 1.96 (m, 2H).

¹³C NMR (151 MHz, CDCl₃)

δ 144.1 (d, $J = 10.3$ Hz), 143.7 (d, $J = 1.8$ Hz), 143.5, 142.5 (d, $J = 8.1$ Hz), 139.9 (d, $J = 3.2$ Hz), 139.5 (d, $J = 3.3$ Hz), 138.9, 130.7, 130.5, 130.2 (d, $J = 3.9$ Hz), 129.4 (d, $J = 4.1$ Hz), 129.1, 122.5, 122.2, 118.1, 118.0, 59.6, 38.7, 38.4, 30.1, 30.0, 16.7, 16.1.

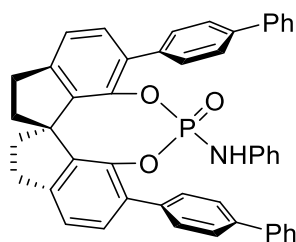
^{31}P NMR (162 MHz, CDCl_3)

$\delta -7.2$.

HRMS (ESI)

Calcd for $[\text{C}_{25}\text{H}_{24}\text{NO}_3\text{P}, \text{M}+\text{Na}]^+$: 440.1386; Found: 440.1390.

(*R*)-1,10-di([1,1'-biphenyl]-4-yl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 118 mg, 85% yield, m.p. 159 – 161 °C. TLC: $R_f = 0.28$ (PE/EA = 3:1, v/v). $[\alpha]_D^{25} +454$ (c 0.2, CHCl_3).

^1H NMR (600 MHz, CDCl_3)

δ 7.66 – 7.51 (m, 12H), 7.42 (t, $J = 7.6$ Hz, 4H), 7.38 – 7.29 (m, 4H), 7.28 – 7.21 (m, 2H), 6.69 (t, $J = 7.7$ Hz, 2H), 6.60 (t, $J = 7.4$ Hz, 1H), 5.97 (d, $J = 7.8$ Hz, 2H), 4.37 (d, $J = 9.7$ Hz, 1H), 3.17 (ddd, $J = 17.2, 9.9, 5.2$ Hz, 2H), 3.03 – 2.87 (m, 2H), 2.50 – 2.27 (m, 3H), 2.23 (td, $J = 11.8, 8.2$ Hz, 1H).

^{13}C NMR (151 MHz, CDCl_3)

δ 146.7, 145.2, 143.1 (d, $J = 10.2$ Hz), 141.3 (d, $J = 8.1$ Hz), 140.9 (d, $J = 4.5$ Hz), 140.8 (d, $J = 2.7$ Hz), 140.7 (d, $J = 3.1$ Hz), 140.3, 139.8, 138.1, 137.4, 135.7 (d, $J = 3.0$ Hz), 135.3, 132.4 (d, $J = 4.1$ Hz), 130.7, 130.6, 130.3, 130.1, 128.8, 128.7, 128.6, 127.7, 127.5, 127.2, 127.1 (d, $J = 3.9$ Hz), 126.8, 123.2, 122.9, 121.6, 117.4 (d, $J = 7.6$ Hz), 60.3, 39.3, 38.4, 30.4.

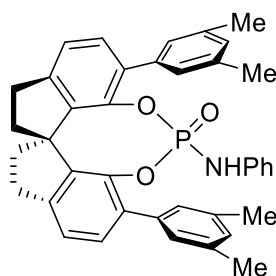
^{31}P NMR (162 MHz, CDCl_3)

$\delta -7.8$.

HRMS (ESI)

Calcd for $[\text{C}_{47}\text{H}_{36}\text{NO}_3\text{P}, \text{M}+\text{Na}]^+$: 716.2325; Found: 716.2330.

(*R*)-1,10-bis(3,5-dimethylphenyl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 97 mg, 81% yield, m.p. 187–189 °C. TLC: R_f = 0.32 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +385 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.36 – 7.31 (m, 1H), 7.28 – 7.23 (m, 3H), 7.18 (dd, J = 18.5, 7.5 Hz, 2H), 7.11 (d, J = 6.2 Hz, 2H), 7.02 (d, J = 5.4 Hz, 1H), 6.97 (d, J = 5.0 Hz, 1H), 6.81 (q, J = 7.2 Hz, 2H), 6.73 (q, J = 7.3, 6.8 Hz, 1H), 5.84 (t, J = 6.8 Hz, 2H), 4.46 (t, J = 8.9 Hz, 1H), 3.23 – 3.07 (m, 2H), 2.92 (dt, J = 16.1, 8.2 Hz, 2H), 2.39 – 2.29 (m, 9H), 2.29 – 2.24 (m, 6H), 2.22 – 2.15 (m, 1H).

¹³C NMR (151 MHz, CDCl₃)

δ 146.7, 144.7, 143.2 (d, J = 10.0 Hz), 141.5 (d, J = 7.7 Hz), 140.6 (d, J = 13.3 Hz), 138.9, 138.3 (d, J = 20.0 Hz), 137.6, 136.5, 136.3, 132.6 (d, J = 4.2 Hz), 130.6, 130.3, 130.0, 128.9, 128.8, 128.2, 127.7, 123.0, 122.7, 121.5, 116.9 (d, J = 7.9 Hz), 60.2, 39.5, 38.1, 30.3, 21.4, 21.3.

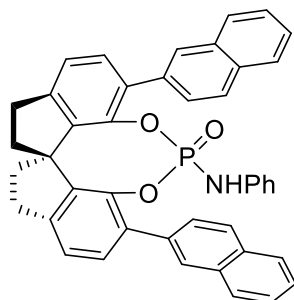
³¹P NMR (162 MHz, CDCl₃)

δ –7.6.

HRMS (ESI)

Calcd for [C₃₉H₃₆NO₃P, M+Na]⁺: 620.2325; Found: 620.2330.

(*R*)-1,10-di(naphthalen-2-yl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 118 mg, 92% yield, m.p. 158–160 °C. TLC: R_f = 0.30 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +456 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 8.01 (s, 1H), 7.93 (s, 1H), 7.87 – 7.79 (m, 3H), 7.76 (dd, J = 6.4, 3.7 Hz, 2H), 7.71 (dd, J = 13.0, 8.4 Hz, 2H), 7.64 – 7.58 (m, 1H), 7.48 – 7.38 (m, 5H), 7.37 (d, J = 7.6 Hz, 1H), 7.30 – 7.21 (m, 2H), 6.34 (t, J = 7.4 Hz, 1H), 6.08 (t, J = 7.7 Hz, 2H), 5.52 (d, J = 7.8 Hz, 2H), 4.25 (d, J = 10.0 Hz, 1H), 3.20 (dtd, J = 17.6, 10.8, 6.6 Hz, 2H), 2.98 (dt, J = 16.1, 7.0 Hz, 2H), 2.40 (tdd, J = 21.9, 15.8, 9.1 Hz, 3H), 2.27 (td, J = 11.7, 8.0 Hz, 1H).

¹³C NMR (151 MHz, CDCl₃)

δ 146.9, 145.2, 143.3 (d, J = 10.2 Hz), 141.6 (d, J = 8.1 Hz), 140.8 (d, J = 3.0 Hz), 140.7 (d, J = 3.1 Hz), 137.6, 136.1 (d, J = 2.8 Hz), 135.8, 133.9, 133.4, 133.2, 132.7, 132.6 (d, J = 4.0 Hz), 132.5, 130.9, 130.6, 129.1, 128.8, 128.6, 128.4, 128.3, 128.1, 127.7 (d, J = 4.0 Hz), 127.6, 127.3, 126.6 (d, J = 9.5 Hz), 125.9, 123.2, 122.9, 121.3, 116.8 (d, J = 7.6 Hz), 60.3, 39.4, 38.4, 30.4.

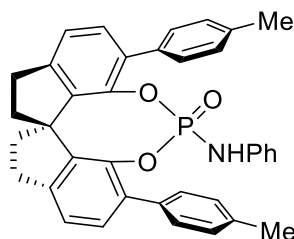
³¹P NMR (162 MHz, CDCl₃)

δ -7.7.

HRMS (ESI)

Calcd for [C₄₃H₃₂NO₃P, M+H]⁺: 642.2193; Found: 642.2198.

(*R*)-12-(phenylamino)-1,10-di-*p*-tolyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 108 mg, 95% yield, m.p. 208–210 °C. TLC: *R*_f = 0.33 (PE/EA = 3:1, v/v). [α]_D²⁵ +368 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.48 (dd, *J* = 7.9, 4.3 Hz, 2H), 7.37 (t, *J* = 6.6 Hz, 2H), 7.31 (dd, *J* = 7.7, 3.5 Hz, 1H), 7.27 – 7.23 (m, 1H), 7.22 – 7.11 (m, 6H), 6.80 (q, *J* = 6.7 Hz, 2H), 6.73 (t, *J* = 6.2 Hz, 1H), 5.91 (t, *J* = 6.5 Hz, 2H), 4.33 (t, *J* = 7.9 Hz, 1H), 3.26 – 3.06 (m, 2H), 2.92 (dt, *J* = 16.2, 8.0 Hz, 2H), 2.48 – 2.22 (m, 9H), 2.18 (q, *J* = 11.0 Hz, 1H).

¹³C NMR (151 MHz, CDCl₃)

δ 146.4, 144.8, 143.1 (d, *J* = 9.9 Hz), 141.5 (d, *J* = 8.0 Hz), 140.6 (t, *J* = 3.8 Hz), 138.3, 138.2, 136.5, 136.0 (d, *J* = 3.2 Hz), 135.5, 133.4, 132.6 (d, *J* = 4.1 Hz), 130.7, 130.2, 129.8, 129.6, 128.9, 128.6, 123.0, 122.7, 121.4, 117.4 (d, *J* = 7.5 Hz), 60.2, 39.3, 38.3, 30.3, 21.3, 21.2.

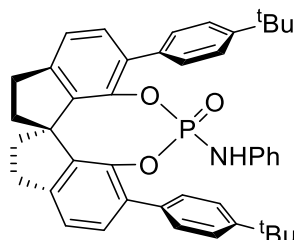
³¹P NMR (162 MHz, CDCl₃)

δ -7.9.

HRMS (ESI)

Calcd for [C₃₇H₃₂NO₃P, M+H]⁺: 570.2193; Found: 570.2196.

(*R*)-1,10-bis(4-(tert-butyl)phenyl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 103 mg, 79% yield, m.p. 245–247 °C. TLC: *R*_f = 0.38 (PE/EA = 3:1, v/v). [α]_D²⁵ +364 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.57 (d, J = 8.0 Hz, 2H), 7.43 (d, J = 8.0 Hz, 4H), 7.38 (d, J = 8.2 Hz, 2H), 7.35 (d, J = 7.7 Hz, 1H), 7.25 (d, J = 10.9 Hz, 1H), 7.19 (dd, J = 14.5, 7.6 Hz, 2H), 6.81 (t, J = 7.7 Hz, 2H), 6.71 (t, J = 7.4 Hz, 1H), 5.94 (d, J = 7.8 Hz, 2H), 4.26 (d, J = 9.5 Hz, 1H), 3.14 (ddt, J = 17.6, 13.0, 6.7 Hz, 2H), 2.92 (ddd, J = 16.3, 7.9, 2.5 Hz, 2H), 2.36 (ddd, J = 18.1, 12.1, 6.4 Hz, 2H), 2.28 (q, J = 11.5 Hz, 1H), 2.20 (td, J = 11.5, 8.4 Hz, 1H), 1.34 (s, 9H), 1.29 (s, 9H).

^{13}C NMR (151 MHz, CDCl_3)

δ 151.4, 149.6, 146.4, 144.8, 143.2 (d, J = 9.6 Hz), 141.6 (d, J = 8.3 Hz), 140.7 (d, J = 2.6 Hz), 140.5 (d, J = 3.4 Hz), 138.2, 135.9 (d, J = 3.3 Hz), 135.6, 133.4, 132.3 (d, J = 3.8 Hz), 131.2, 130.2, 129.9, 129.5, 128.7, 126.1, 124.9, 123.1, 122.8, 121.6, 117.8 (d, J = 7.3 Hz), 60.3, 39.2, 38.4, 34.6 (d, J = 11.4 Hz), 31.5, 31.3, 30.3.

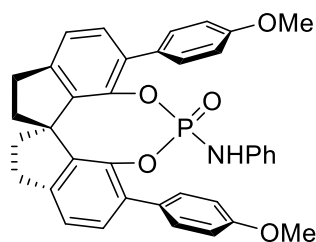
^{31}P NMR (162 MHz, CDCl_3)

δ -7.5.

HRMS (ESI)

Calcd for $[\text{C}_{43}\text{H}_{44}\text{NO}_3\text{P}, \text{M}+\text{H}]^+$: 654.3132; Found: 654.3134.

(*R*)-1,10-bis(4-methoxyphenyl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 108 mg, 90% yield, m.p. 139–141 °C. TLC: R_f = 0.31 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +334 (c 0.2, CHCl_3).

^1H NMR (600 MHz, CDCl_3)

δ 7.52 (d, J = 8.4 Hz, 2H), 7.39 (d, J = 8.3 Hz, 2H), 7.29 (d, J = 7.6 Hz, 1H), 7.25 (d, J = 10.8 Hz, 1H), 7.18 (t, J = 7.4 Hz, 2H), 6.90 (d, J = 7.2 Hz, 4H), 6.83 (t, J = 7.7 Hz, 2H), 6.73 (t, J = 7.4 Hz, 1H), 5.93 (d, J = 7.8 Hz, 2H), 4.36 (d, J = 10.0 Hz, 1H), 3.83 (s, 3H), 3.78 (s, 3H), 3.14 (dtd, J = 18.7, 12.6, 12.1, 6.4 Hz, 2H), 2.92 (dt, J = 16.7, 8.7 Hz, 2H), 2.35 (ddd, J = 14.7, 12.1, 6.4 Hz, 2H), 2.28 (q, J = 11.6 Hz, 1H), 2.18 (tt, J = 11.4, 5.0 Hz, 1H).

^{13}C NMR (151 MHz, CDCl_3)

δ 159.6, 158.8, 146.1, 144.7, 143.1 (d, J = 10.3 Hz), 141.5 (d, J = 8.1 Hz), 140.6, 138.2, 135.7 (d, J = 3.4 Hz), 132.2 (d, J = 4.1 Hz), 131.4, 130.9, 130.8, 130.7, 130.0, 128.7, 128.6, 123.1, 122.7, 121.5, 117.3 (d, J = 7.4 Hz), 114.5, 113.6, 60.3, 55.3 (d, J = 15.8 Hz), 39.3, 38.2, 31.6, 30.3 (d, J = 3.7 Hz), 22.7, 14.1.

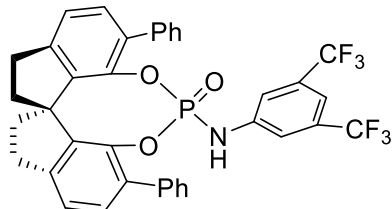
^{31}P NMR (162 MHz, CDCl_3)

δ -7.9.

HRMS (ESI)

Calcd for $[C_{37}H_{32}NO_5P, M+H]^+$: 602.2091; Found: 602.2096.

(*R*)-12-((3,5-bis(trifluoromethyl)phenyl)amino)-1,10-diphenyl-4,5,6,7-tetrahydroindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 99 mg, 73% yield, m.p. 144 – 146 °C. TLC: R_f = 0.20 (PE/EA = 3:1, v/v). $[\alpha]_D^{25} +400$ (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.45 (d, J = 7.6 Hz, 2H), 7.32 (d, J = 7.7 Hz, 1H), 7.28 – 7.22 (m, 6H), 7.20 (t, J = 7.4 Hz, 1H), 7.17 – 7.07 (m, 5H), 6.67 (s, 2H), 4.77 (d, J = 6.5 Hz, 1H), 3.16 (dtd, J = 17.7, 11.4, 6.4 Hz, 2H), 2.96 (dd, J = 16.3, 7.9 Hz, 2H), 2.43 (dd, J = 12.2, 6.4 Hz, 1H), 2.37 (dd, J = 12.2, 6.4 Hz, 1H), 2.25 (tt, J = 20.5, 10.6 Hz, 2H).

^{13}C NMR (151 MHz, $CDCl_3$)

δ 146.4, 146.3, 145.6, 142.1 (d, J = 10.4 Hz), 140.8 (d, J = 3.9 Hz), 140.4 (d, J = 3.0 Hz), 140.2 (d, J = 8.0 Hz), 140.0, 137.6, 136.1, 135.5 (d, J = 3.1 Hz), 133.5 (d, J = 4.0 Hz), 131.7 (q, J = 33.2 Hz), 130.8, 130.2, 129.5, 129.1, 128.3, 128.0, 127.7, 127.2, 123.5, 123.0 (q, J = 272.8 Hz), 117.4, 115.3, 60.2, 39.1, 38.6, 30.3 (d, J = 4.0 Hz).

^{31}P NMR (162 MHz, $CDCl_3$)

δ -10.0.

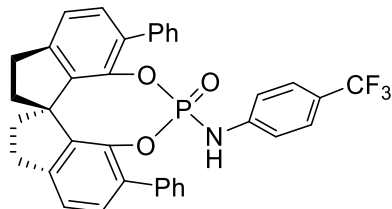
^{19}F NMR (376 MHz, $CDCl_3$)

δ -62.9.

HRMS (ESI)

Calcd for $[C_{37}H_{26}F_6NO_3P, M+Na]^+$: 700.1447; Found: 700.1450.

(*R*)-1,10-diphenyl-12-((4-(trifluoromethyl)phenyl)amino)-4,5,6,7-tetrahydroindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 97 mg, 80% yield, m.p. 134–136 °C. TLC: R_f = 0.23 (PE/EA = 3:1, v/v). $[\alpha]_D^{25} +300$ (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.53 (d, J = 7.5 Hz, 2H), 7.47 (dd, J = 6.5, 2.9 Hz, 2H), 7.40 – 7.35 (m, 3H), 7.33 (t, J = 7.3 Hz, 3H), 7.28 (dd, J = 7.6, 3.3 Hz, 2H), 7.23 (t, J = 8.5

Hz, 2H), 7.03 (d, $J = 8.2$ Hz, 2H), 5.97 (d, $J = 8.2$ Hz, 2H), 4.52 (d, $J = 9.4$ Hz, 1H), 3.16 (ddt, $J = 16.1, 10.4, 5.2$ Hz, 2H), 2.95 (dt, $J = 15.8, 7.5$ Hz, 2H), 2.38 (ddd, $J = 19.8, 12.1, 6.5$ Hz, 2H), 2.29 (q, $J = 11.6$ Hz, 1H), 2.19 (td, $J = 11.9, 8.4$ Hz, 1H).

^{13}C NMR (101 MHz, CDCl_3)

δ 145.7, 144.1, 141.8 (d, $J = 10.2$ Hz), 140.5, 139.8 (d, $J = 7.6$ Hz), 139.5 (d, $J = 7.5$ Hz), 137.1, 135.1, 134.9, 131.6 (d, $J = 3.8$ Hz), 129.8, 129.5, 129.1, 128.6, 128.0, 127.2, 126.1, 125.0 (d, $J = 4.0$ Hz), 123.2 (q, $J = 271.5$ Hz), 122.3 (q, $J = 32.8$ Hz), 122.2 (d, $J = 22.8$ Hz), 115.8 (d, $J = 8.0$ Hz), 59.2, 38.3, 37.3, 29.3.

^{31}P NMR (162 MHz, CDCl_3)

δ -8.9.

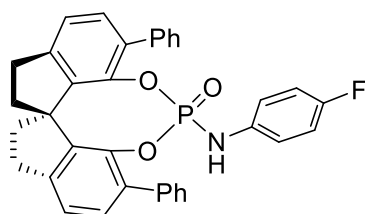
^{19}F NMR (376 MHz, CDCl_3)

δ -62.0.

HRMS (ESI)

Calcd for $[\text{C}_{36}\text{H}_{27}\text{F}_3\text{NO}_3\text{P}, \text{M}+\text{H}]^+$: 610.1753; Found: 610.1758.

(*R*)-12-((4-fluorophenyl)amino)-1,10-diphenyl-4,5,6,7-tetrahydroindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 95 mg, 85% yield, m.p. 134–136 °C. TLC: $R_f = 0.26$ (PE/EA = 3:1, v/v).

$[\alpha]_{\text{D}}^{25} +398$ (c 0.2, CHCl_3).

^1H NMR (600 MHz, CDCl_3)

δ 7.59 (d, $J = 7.6$ Hz, 2H), 7.46 (dd, $J = 6.6, 2.9$ Hz, 2H), 7.39 (t, $J = 7.6$ Hz, 2H), 7.37 – 7.29 (m, 5H), 7.26 (d, $J = 7.9$ Hz, 1H), 7.22 (dd, $J = 15.5, 7.7$ Hz, 2H), 6.50 (t, $J = 8.5$ Hz, 2H), 5.80 (dd, $J = 8.9, 4.5$ Hz, 2H), 4.20 (d, $J = 9.7$ Hz, 1H), 3.16 (dtd, $J = 16.8, 10.4, 6.9$ Hz, 2H), 2.94 (dt, $J = 15.8, 7.6$ Hz, 2H), 2.37 (ddd, $J = 18.0, 12.1, 6.4$ Hz, 2H), 2.29 (q, $J = 11.5$ Hz, 1H), 2.19 (td, $J = 11.9, 8.6$ Hz, 1H).

^{13}C NMR (151 MHz, CDCl_3)

δ 158.0 (d, $J = 240.1$ Hz), 146.8, 145.1, 143.0 (d, $J = 10.4$ Hz), 141.3 (d, $J = 7.4$ Hz), 140.6 (d, $J = 2.6$ Hz), 140.5 (d, $J = 3.1$ Hz), 138.3, 136.5, 136.0 (d, $J = 3.3$ Hz), 134.1, 132.5 (d, $J = 4.0$ Hz), 130.7, 130.4, 130.3, 129.7, 129.1, 128.3, 128.2, 127.1, 123.2, 122.9, 119.0 (t, $J = 7.7$ Hz), 115.4 (d, $J = 22.6$ Hz), 60.2, 39.4, 38.2, 30.3.

^{31}P NMR (162 MHz, CDCl_3)

δ -7.7.

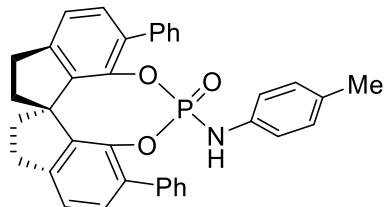
^{19}F NMR (162 MHz, CDCl_3)

δ -122.4.

HRMS (ESI)

Calcd for $[C_{35}H_{27}FNO_3P, M+H]^+$: 560.1785; Found: 560.1789.

(*R*)-1,10-diphenyl-12-(*p*-tolylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 90 mg, 81% yield, m.p. 121–123 °C. TLC: R_f = 0.36 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +328 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.59 (d, J = 7.6 Hz, 2H), 7.47 (dd, J = 6.6, 2.9 Hz, 2H), 7.40 – 7.27 (m, 8H), 7.20 (dd, J = 12.8, 7.7 Hz, 2H), 6.61 (d, J = 7.9 Hz, 2H), 5.75 (d, J = 7.9 Hz, 2H), 4.15 (d, J = 9.9 Hz, 1H), 3.23 – 3.06 (m, 2H), 2.93 (ddd, J = 16.3, 8.0, 3.1 Hz, 2H), 2.37 (ddd, J = 17.7, 12.1, 6.5 Hz, 2H), 2.29 (dd, J = 11.5, 8.3 Hz, 1H), 2.19 (q, J = 11.4 Hz, 1H), 2.13 (s, 3H).

^{13}C NMR (151 MHz, $CDCl_3$)

δ 146.7, 145.0, 143.1 (d, J = 9.9 Hz), 141.5 (d, J = 7.4 Hz), 140.7 (d, J = 2.9 Hz), 140.6 (d, J = 2.7 Hz), 138.3, 136.5, 136.1 (d, J = 3.2 Hz), 135.4, 132.7 (d, J = 3.8 Hz), 131.0, 130.7, 130.3, 130.3, 129.8, 129.3, 129.1, 128.2, 128.1, 127.0, 123.0, 122.7, 117.6 (d, J = 7.3 Hz), 60.2, 39.3, 38.3, 30.3, 20.5.

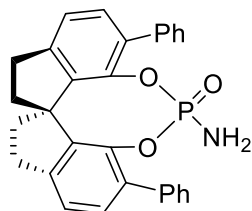
^{31}P NMR (162 MHz, $CDCl_3$)

δ -7.5.

HRMS (ESI)

Calcd for $[C_{36}H_{30}NO_3P, M+H]^+$: 556.2036; Found: 556.2039.

(*R*)-12-amino-1,10-diphenyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 83 mg, 89% yield, m.p. 154–156 °C. TLC: R_f = 0.30 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +470 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.53 (d, J = 7.6 Hz, 2H), 7.48 (d, J = 7.6 Hz, 2H), 7.33 (dt, J = 16.0, 7.5 Hz, 5H), 7.30 – 7.23 (m, 3H), 7.21 (d, J = 7.7 Hz, 1H), 7.18 (d, J = 7.7 Hz, 1H), 3.14 (ddt, J = 16.9, 11.6, 6.1 Hz, 2H), 2.91 (ddd, J = 16.1, 7.9, 5.8 Hz,

2H), 2.36 (ddd, $J = 23.7, 12.1, 6.5$ Hz, 2H), 2.24 (td, $J = 11.3, 7.7$ Hz, 1H), 2.16 (td, $J = 11.5, 11.0, 8.0$ Hz, 1H), 1.98 (d, $J = 5.6$ Hz, 2H).

¹³C NMR (151 MHz, CDCl₃)

δ 146.1, 145.3, 143.2 (d, $J = 10.2$ Hz), 141.3 (d, $J = 3.7$ Hz), 141.0 (d, $J = 7.5$ Hz), 140.7 (d, $J = 3.9$ Hz), 138.1, 137.5, 135.5 (d, $J = 3.4$ Hz), 133.8 (d, $J = 4.0$ Hz), 130.4, 129.9, 129.8, 129.5, 128.6, 127.8, 127.5, 127.0, 122.8 (d, $J = 17.5$ Hz), 60.1, 38.9, 38.5, 30.4.

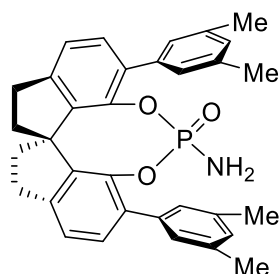
³¹P NMR (162 MHz, CDCl₃)

δ -0.2.

HRMS (ESI)

Calcd for [C₂₉H₂₄NO₃P, M+H]⁺: 466.1567; Found: 466.1570.

(*R*)-12-amino-1,10-bis(3,5-dimethylphenyl)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 77 mg, 74% yield, m.p. 164–166 °C. TLC: $R_f = 0.39$ (PE/EA = 3:1, v/v).

$[\alpha]_D^{25} +398$ (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.34 (d, $J = 7.6$ Hz, 1H), 7.28 (d, $J = 7.7$ Hz, 1H), 7.21 – 7.14 (m, 4H), 7.12 (s, 2H), 6.89 (d, $J = 8.7$ Hz, 2H), 3.13 (ddt, $J = 16.8, 11.2, 5.6$ Hz, 2H), 2.90 (ddd, $J = 16.1, 7.9, 5.7$ Hz, 2H), 2.41 – 2.10 (m, 16H), 2.01 (d, $J = 5.0$ Hz, 2H).

¹³C NMR (151 MHz, CDCl₃)

δ 145.8, 145.0, 143.3 (d, $J = 10.5$ Hz), 141.2 (d, $J = 3.4$ Hz), 141.0 (d, $J = 8.1$ Hz), 140.7 (d, $J = 2.7$ Hz), 138.2, 137.9, 137.3, 137.1, 135.7 (d, $J = 2.9$ Hz), 133.8 (d, $J = 3.7$ Hz), 130.3, 129.9, 129.1, 128.6, 127.6, 127.3, 122.6 (d, $J = 20.5$ Hz), 60.1, 38.9, 38.4, 30.4, 21.3 (d, $J = 7.9$ Hz).

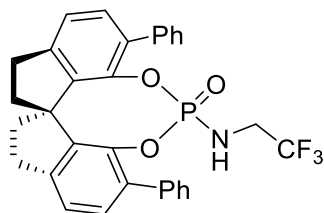
³¹P NMR (162 MHz, CDCl₃)

δ 0.0.

HRMS (ESI)

Calcd for [C₃₃H₃₂NO₃P, M+H]⁺: 522.2193; Found: 522.2197.

(*R*)-1,10-diphenyl-12-((2,2,2-trifluoroethyl)amino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 86 mg, 79% yield, m.p. 157–159 °C. TLC: R_f = 0.19 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +396 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.58 (d, J = 7.7 Hz, 2H), 7.47 (d, J = 7.6 Hz, 2H), 7.39 (t, J = 7.4 Hz, 3H), 7.35 – 7.27 (m, 4H), 7.25 – 7.19 (m, 3H), 3.15 (ddd, J = 17.4, 11.4, 6.5 Hz, 2H), 2.92 (dt, J = 16.7, 8.7 Hz, 2H), 2.81 – 2.60 (m, 1H), 2.36 (ddd, J = 28.3, 12.2, 6.4 Hz, 2H), 2.28 – 2.12 (m, 3H), 2.05 (ddt, J = 15.0, 9.1, 4.5 Hz, 1H).

¹³C NMR (101 MHz, CDCl₃)

δ 145.4 (d, J = 2.6 Hz), 144.2 (d, J = 2.5 Hz), 141.6 (d, J = 10.1 Hz), 140.2 (d, J = 3.4 Hz), 139.9 (d, J = 7.5 Hz), 139.4 (d, J = 3.6 Hz), 136.6, 136.0, 134.5 (d, J = 3.2 Hz), 132.2 (d, J = 3.8 Hz), 129.3 (d, J = 2.5 Hz), 128.9 (d, J = 2.4 Hz), 128.6, 128.2, 127.7, 126.8, 126.6, 126.0, 122.6 (qd, J = 278.4, 6.7 Hz), 122.0 (d, J = 2.3 Hz), 59.0 (d, J = 1.8 Hz), 40.9 (qd, J = 35.0, 1.8 Hz), 37.9, 37.4, 29.3 (d, J = 2.6 Hz).

³¹P NMR (162 MHz, CDCl₃)

δ –3.2.

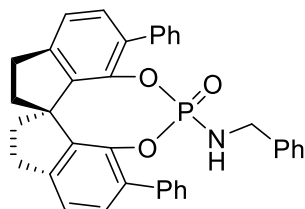
¹⁹F NMR (376 MHz, CDCl₃)

δ –74.6.

HRMS (ESI)

Calcd for [C₃₁H₂₅F₃NO₃P, M+H]⁺: 548.1597; Found: 548.1601.

(*R*)-12-(benzylamino)-1,10-diphenyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 103 mg, 93% yield, m.p. 102–104 °C. TLC: R_f = 0.34 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +364 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.61 (d, J = 7.6 Hz, 2H), 7.49 (d, J = 7.4 Hz, 2H), 7.41 – 7.33 (m, 5H), 7.32 – 7.25 (m, 3H), 7.20 (dd, J = 12.5, 7.7 Hz, 2H), 7.17 – 7.11 (m, 3H), 6.74 (t, J = 3.7 Hz, 2H), 3.36 (ddd, J = 15.1, 10.6, 8.3 Hz, 1H), 3.14 (ddd, J

= 17.1, 11.2, 6.4 Hz, 2H), 2.91 (ddd, $J = 16.2, 7.9, 3.2$ Hz, 2H), 2.77 (ddd, $J = 15.3, 9.8, 5.7$ Hz, 1H), 2.35 (ddd, $J = 23.1, 12.1, 6.5$ Hz, 2H), 2.30 – 2.20 (m, 1H), 2.18 – 2.10 (m, 1H), 2.06 (dq, $J = 13.5, 6.8, 6.2$ Hz, 1H).

¹³C NMR (151 MHz, CDCl₃)

δ 146.0 (d, $J = 1.7$ Hz), 145.3, 143.1 (d, $J = 9.5$ Hz), 141.4 (d, $J = 3.3$ Hz), 141.3, 141.2, 140.7 (d, $J = 3.3$ Hz), 139.4 (d, $J = 5.3$ Hz), 138.2, 137.4, 135.7 (d, $J = 3.5$ Hz), 133.6 (d, $J = 4.2$ Hz), 130.2, 130.0, 129.8, 129.4, 128.6, 128.2, 127.8, 127.4, 127.1, 127.0, 126.9, 122.7 (d, $J = 15.0$ Hz), 60.0, 44.1, 38.9, 38.5, 30.4 (d, $J = 8.6$ Hz).

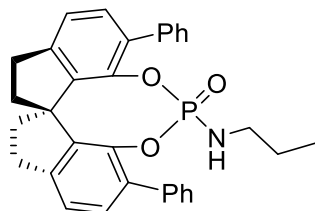
³¹P NMR (162 MHz, CDCl₃)

δ –1.8.

HRMS (ESI)

Calcd for [C₃₆H₃₀NO₃P, M+H]⁺: 556.2036; Found: 556.2038.

(*R*)-1,10-diphenyl-12-(propylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 86 mg, 85% yield, m.p. 112–114 °C. TLC: $R_f = 0.35$ (PE/EA = 3:1, v/v). $[\alpha]_D^{25} +464$ (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.58 (d, $J = 7.6$ Hz, 2H), 7.48 (d, $J = 7.6$ Hz, 2H), 7.35 (t, $J = 7.7$ Hz, 3H), 7.33 – 7.27 (m, 3H), 7.27 – 7.21 (m, 2H), 7.18 (dd, $J = 13.0, 7.7$ Hz, 2H), 3.13 (ddd, $J = 17.0, 11.2, 6.4$ Hz, 2H), 2.90 (dt, $J = 15.4, 7.2$ Hz, 2H), 2.37 (dd, $J = 12.1, 6.4$ Hz, 1H), 2.32 (dd, $J = 12.1, 6.4$ Hz, 1H), 2.27 – 2.20 (m, 1H), 2.19 – 2.11 (m, 1H), 2.11 – 2.03 (m, 1H), 1.79 (ddt, $J = 17.4, 13.3, 6.7$ Hz, 1H), 1.65 (dt, $J = 13.2, 6.9$ Hz, 1H), 0.93 (qd, $J = 7.1, 2.9$ Hz, 2H), 0.50 (t, $J = 7.4$ Hz, 3H).

¹³C NMR (151 MHz, CDCl₃)

δ 146.0 (d, $J = 1.5$ Hz), 145.2, 143.2 (d, $J = 9.5$ Hz), 141.5 (d, $J = 7.4$ Hz), 141.2 (d, $J = 3.2$ Hz), 140.7 (d, $J = 3.5$ Hz), 138.2, 137.4, 135.6 (d, $J = 3.0$ Hz), 133.6 (d, $J = 4.0$ Hz), 130.2, 129.9, 129.8, 129.4, 128.5, 127.7, 127.4, 126.3, 122.5 (d, $J = 14.9$ Hz), 60.0, 42.5, 39.0, 38.4, 30.4 (d, $J = 3.6$ Hz), 24.6 (d, $J = 5.4$ Hz), 10.6.

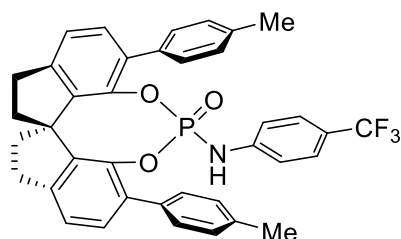
³¹P NMR (162 MHz, CDCl₃)

δ –1.0.

HRMS (ESI)

Calcd for [C₃₂H₃₀NO₃P, M+Na]⁺: 530.1856; Found: 530.1858.

(*R*)-1,10-di-*p*-tolyl-12-((4-(trifluoromethyl)phenyl)amino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 103 mg, 81% yield, m.p. 251–253 °C. TLC: R_f = 0.25 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +348 (c 0.2, CHCl₃).

¹H NMR (400 MHz, CDCl₃)

δ 7.38 (dd, J = 15.0, 7.8 Hz, 4H), 7.28 (dd, J = 13.3, 7.6 Hz, 2H), 7.23 – 7.14 (m, 4H), 7.10 (d, J = 7.9 Hz, 2H), 7.02 (d, J = 8.3 Hz, 2H), 6.03 (d, J = 8.4 Hz, 2H), 4.59 (d, J = 9.6 Hz, 1H), 3.14 (tt, J = 16.8, 7.2 Hz, 2H), 2.91 (ddd, J = 16.7, 9.9, 7.8 Hz, 2H), 2.45 – 2.21 (m, 9H), 2.16 (td, J = 11.5, 7.8 Hz, 1H).

¹³C NMR (101 MHz, CDCl₃)

δ 146.4 (d, J = 2.4 Hz), 144.9 (d, J = 2.7 Hz), 142.9 (d, J = 10.4 Hz), 141.7, 141.0 (d, J = 8.0 Hz), 140.6 (d, J = 3.5 Hz), 138.2, 136.8, 135.9 (d, J = 3.7 Hz), 135.3, 133.3, 132.7 (d, J = 4.1 Hz), 130.8, 130.3, 130.1, 129.7, 129.5, 129.0, 125.9 (q, J = 3.9 Hz), 124.3 (q, J = 271.2 Hz), 123.3 (q, J = 32.7 Hz), δ 123.2 (d, J = 2.3 Hz), 123.0 (d, J = 2.4 Hz), 116.9 (t, J = 8.6 Hz), 60.3, 39.4, 38.3, 30.4, 21.1 (d, J = 10.0 Hz).

³¹P NMR (162 MHz, CDCl₃)

δ –8.9.

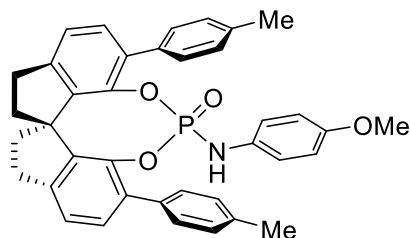
¹⁹F NMR (376 MHz, CDCl₃)

δ –61.9.

HRMS (ESI)

Calcd for [C₃₈H₃₁F₃NO₃P, M+H]⁺: 638.2066; Found: 638.2071.

(*R*)-12-((4-methoxyphenyl)amino)-1,10-di-*p*-tolyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



White solid, 88 mg, 73% yield, m.p. 130–133 °C. TLC: R_f = 0.24 (PE/EA = 3:1, v/v). $[\alpha]_D^{25}$ +328 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.48 (dd, J = 7.7, 2.7 Hz, 2H), 7.35 (dd, J = 8.0, 2.7 Hz, 2H), 7.31 (d, J = 7.7 Hz, 1H), 7.24 (d, J = 7.8 Hz, 1H), 7.21 – 7.12 (m, 6H), 6.37 (d, J = 8.9 Hz, 2H), 5.85 (d, J = 9.0 Hz, 2H), 4.10 (dd, J = 10.2, 3.7 Hz, 1H), 3.67 (s,

3H), 3.21 – 3.03 (m, 2H), 2.99 – 2.86 (m, 2H), 2.49 – 2.24 (m, 9H), 2.17 (q, $J = 11.6, 11.2$ Hz, 1H).

¹³C NMR (151 MHz, CDCl₃)

δ 154.6, 146.4, 144.8, 143.2 (d, $J = 10.1$ Hz), 141.5 (d, $J = 7.6$ Hz), 140.6 (d, $J = 2.8$ Hz), 140.5 (d, $J = 3.9$ Hz), 138.1, 136.4, 136.0 (d, $J = 3.1$ Hz), 135.5, 133.6, 132.6 (d, $J = 4.1$ Hz), 131.4, 130.6, 130.2 (d, $J = 4.6$ Hz), 129.8, 129.6, 128.8, 123.0, 122.7, 119.3 (d, $J = 7.2$ Hz), 114.0, 60.2, 55.4, 39.3, 38.2, 31.6, 30.3, 22.7, 21.3, 21.2, 14.2.

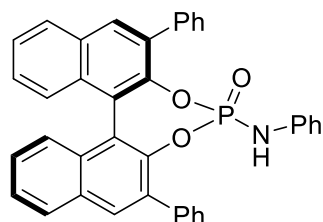
³¹P NMR (162 MHz, CDCl₃)

δ -7.2.

HRMS (ESI)

Calcd for [C₃₈H₃₄NO₄P, M+H]⁺: 600.2298; Found: 600.2301.

(*R*)-2,6-diphenyl-4-(phenylamino)dinaphtho[2,1-d:1',2'-f][1,3,2]dioxaphosphepine 4-oxide



White solid, 290 mg, 87% yield, m.p. 170–174 °C. TLC: $R_f = 0.36$ (PE/EA = 5:1, v/v). $[\alpha]_D^{25} -238$ (c 0.1, CHCl₃).

¹H NMR (400 MHz, CDCl₃)

δ 8.07 (d, $J = 2.8$ Hz, 1H), 8.02 – 7.90 (m, 3H), 7.80 (t, $J = 5.9$ Hz, 2H), δ 7.51 (q, $J = 7.6$ Hz, 2H), 7.44 (d, $J = 7.6$ Hz, 5H), 7.38 (d, $J = 7.4$ Hz, 2H), 7.33 (q, $J = 7.3$ Hz, 2H), 7.29 – 7.21 (m, 3H), 6.85 (t, $J = 7.6$ Hz, 2H), 6.78 (dd, $J = 7.3, 2.7$ Hz, 1H), 6.33 (dd, $J = 8.1, 2.7$ Hz, 2H), 4.78 (d, $J = 9.0$ Hz, 1H).

¹³C NMR (101 MHz, CDCl₃)

δ 145.1 (d, $J = 7.8$ Hz), 143.9, 137.8, 136.9, 136.1, 134.4, 134.0, 132.1 (d, $J = 4.4$ Hz), 131.8, 131.7 (d, $J = 4.0$ Hz), 131.2, 130.1, 129.8, 128.8, 128.5, 128.4, 128.3, 127.9, 127.8, 127.1, 126.8 (d, $J = 3.5$ Hz), 126.2 (d, $J = 3.3$ Hz), 122.9 (d, $J = 2.2$ Hz), 122.7, 121.9, 117.8 (t, $J = 6.5$ Hz).

³¹P NMR (162 MHz, CDCl₃)

δ 4.9.

HRMS (ESI)

Calcd for [C₃₈H₂₆NO₃P, M+H]⁺: 576.1723; Found: 576.1728.

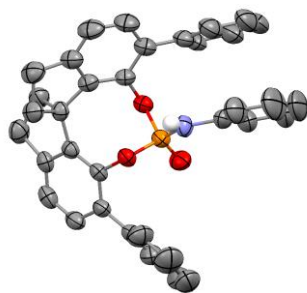
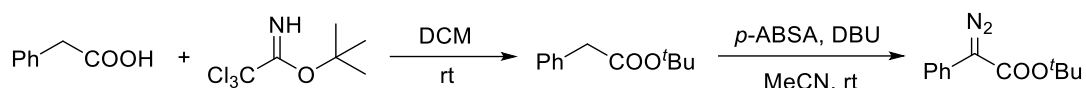


Figure S16. The X-ray crystal structure of SPAM-1.

Table S15. Crystal data and structure refinement for SPAM-1.

Compound	SPAM-1
CCDC deposition No.	2181590
Empirical formula	C ₃₅ H ₃₀ NO ₄ P
Formula weight	559.57
Temperature/K	294.15
Crystal system	monoclinic
Space group	I2
a/Å	18.0229(14)
b/Å	7.2760(5)
c/Å	23.0585(17)
α /°	90
β /°	102.453(7)
γ /°	90
Volume/Å ³	2952.6(4)
Z	4
$\rho_{\text{calc}}/\text{cm}^3$	1.259
μ/mm^{-1}	0.133
F(000)	1176.0
Crystal size/mm ³	0.25 × 0.16 × 0.13
Radiation	Mo K α (λ = 0.71073)
2 θ range for data collection/°	6.344 to 64.672
Index ranges	-25 ≤ h ≤ 25, -10 ≤ k ≤ 10, -33 ≤ l ≤ 30
Reflections collected	15819
Independent reflections	8681 [R_{int} = 0.0631, R_{sigma} = 0.1341]
Data/restraints/parameters	8681/3/378
Goodness-of-fit on F ²	1.008
Final R indexes [$I \geq 2\sigma(I)$]	R_1 = 0.0668, wR_2 = 0.0984
Final R indexes [all data]	R_1 = 0.1768, wR_2 = 0.1374
Largest diff. peak/hole / e Å ⁻³	0.21/-0.16
Flack parameter	0.14(11)

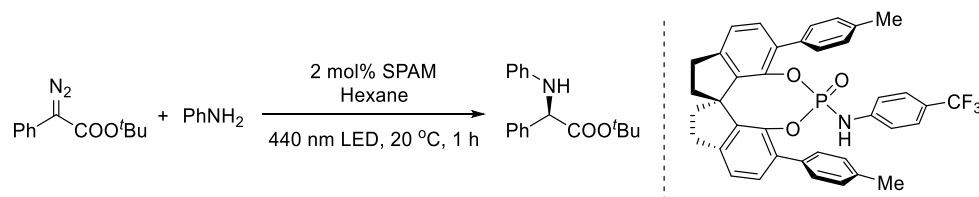
Synthesis of α -Diazoesters



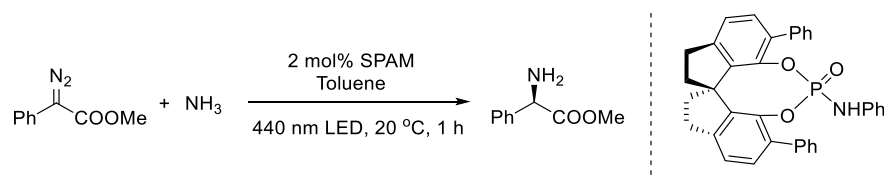
The imidates (9.59 g, 44 mmol) were added to 2-phenylacetic acid (5.44 g, 40.00 mmol) in dry DCM (20 mL) under Ar atmosphere at room temperature. The mixture was stirred for 12 h and filtered to remove trichloroacetamide. The filter cake was washed with DCM. The combined organic layer was concentrated. The residue was purified by a short silica gel column to remove the unreacted raw materials to obtain target product as an oil.

To a stirred suspension of tert-butyl 2-phenylacetate (3.84 g, 20 mmol) and 4-acetoamidobenzenesulfonyl azide (*p*-ABSA) (7.22 g, 30 mmol) in MeCN (40 mL) at room temperature, DBU (4.56g, 30 mmol) in MeCN (10 mL) was added under Ar atmosphere. After stirring for 24 h, the reaction mixture was quenched with saturated NH_4Cl solution and extracted with Et_2O . The organic phase was dried over Na_2SO_4 and concentrated to give crude product. Purification of crude product on column chromatography gave α -phenyl diazoesters as yellow oil.

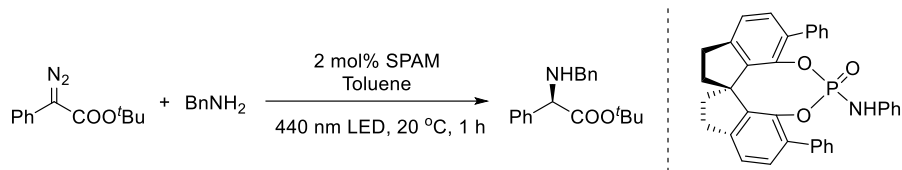
Typical Procedure for Enantioselective N–H Insertion and Analytical Data



SPAM (1.3 mg, 0.002 mmol, 2 mol%) and PhNH_2 (28.0 mg, 0.3 mmol, 3 eq) were introduced into an oven-dried Schlenk flask under a nitrogen atmosphere. The *tert*-butyl 2-diazo-2-phenylacetate (21.8 mg, 0.1 mmol, 1.0 eq.) was added. Hexane (8 mL) was injected into the Schlenk flask. The resulting mixture was stirred at 20 °C for 1 h under 440 nm light irradiation. When the reaction is over, the system was transferred directly to the round bottom flask and concentrated in vacuo. The product was purified by chromatography through wet loading (PE/EA = 20:1, v/v).



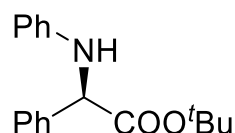
The SPAM (2.2 mg, 0.004 mmol, 2 mol%) and diazoester (35.4 mg, 0.2 mmol, 1.0 eq) were introduced into an oven-dried Sealing pipe in an argon-filled glovebox. A solution of ammonia in toluene (0.27 M, 6 mL, 1.6 mmol, 8.0 eq.) was injected into the Sealing pipe. The resulting mixture was stirred for 1 h at 20 °C under 440 nm light irradiation. When the reaction is over, the system was transferred directly to the round bottom flask and concentrated in vacuo. The product was purified by chromatography through wet loading (DCM/MeOH = 20:1, v/v).



The SPAM (2.2 mg, 0.004 mmol, 2 mol%) and BnNH₂ (21.4 mg, 0.2 mmol, 1 eq) were introduced into an oven-dried Schlenk flask under a nitrogen atmosphere. Diazoester was added (105.4 mg, 0.6 mmol, 3.0 eq.). Toluene (4 mL) was injected into the Schlenk flask. The resulting mixture was stirred for 1 h at 20 °C under 440 nm light irradiation. When the reaction is over, the system was transferred directly to the round bottom flask and concentrated in vacuo. The product was purified by chromatography through wet loading (PE/EA = 20:1, v/v).

Supplementary notes: Racemic samples of N–H insertion products can be prepared without chiral catalyst.

(*R*)-tert-butyl 2-phenyl-2-(phenylamino)acetate (1a)

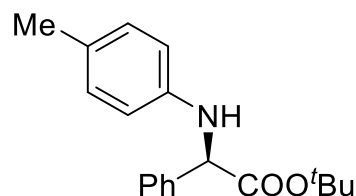


White solid, TLC: R_f = 0.78 (PE/EA = 10:1, v/v), m.p. 76–78 °C, 19 mg, 69% yield, 91% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 5.54 min (major) and t_R = 5.96 min (minor). $[\alpha]_D^{25}$ –75.0 (c 0.7, CHCl₃). By comparing the specific rotation, $[\alpha]_D^{25}$ –63.4 (c 0.71, CHCl₃) for *R* enantiomer in literature,^[20] the configuration of the insertion product **1** was determined to be *R*.

¹H NMR (400 MHz, CDCl₃)

δ 7.48 (d, J = 7.5 Hz, 2H), 7.33 (t, J = 7.5 Hz, 2H), 7.27 (t, J = 7.3 Hz, 1H), 7.10 (t, J = 7.7 Hz, 2H), 6.67 (t, J = 7.3 Hz, 1H), 6.54 (d, J = 7.9 Hz, 2H), 4.95 (s, 2H), 1.38 (s, 9H).

(-)-tert-butyl 2-phenyl-2-(p-tolylamino)acetate (1b)



White solid, TLC: R_f = 0.82 (PE/EA = 10:1, v/v), m.p. 102–104 °C, 21 mg, 71% yield, 86% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 5.65 min (major) and t_R = 6.05 min (minor). $[\alpha]_D^{25}$ –61.0 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.47 (d, J = 7.6 Hz, 2H), 7.32 (t, J = 7.5 Hz, 2H), 7.27 (t, J = 7.3 Hz, 1H), 6.91 (d, J = 8.0 Hz, 2H), 6.47 (d, J = 8.0 Hz, 2H), 4.92 (s, 1H), 4.81 (s, 1H), 2.19 (s, 3H), 1.37 (s, 9H).

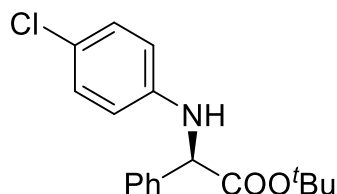
¹³C NMR (151 MHz, CDCl₃)

δ 171.0, 143.8, 138.3, 129.7, 128.6, 127.9, 127.1, 126.9, 113.4, 82.2, 61.5, 27.8, 20.3.

HRMS (ESI)

Calcd for [C₁₉H₂₃NO₂, M+Na]⁺: 320.1621; Found: 320.1626.

(-)-tert-butyl 2-((4-chlorophenyl)amino)-2-phenylacetate (1c)



White solid, TLC: R_f = 0.74 (PE/EA = 10:1, v/v), m.p. 122–124 °C, 24 mg, 75% yield, 84% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 6.50 min (major) and t_R = 7.86 min (minor). $[\alpha]_D^{25}$ -43.0 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.44 (d, J = 7.5 Hz, 2H), 7.34 (t, J = 7.4 Hz, 2H), 7.29 (t, J = 7.2 Hz, 1H), 7.04 (d, J = 8.4 Hz, 2H), 6.45 (d, J = 8.4 Hz, 2H), 4.99 (d, J = 6.0 Hz, 1H), 4.90 (d, J = 5.9 Hz, 1H), 1.38 (s, 9H).

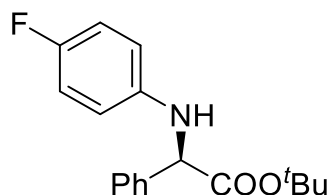
¹³C NMR (151 MHz, CDCl₃)

δ 170.6, 144.6, 137.7, 129.0, 128.7, 128.1, 127.0, 122.4, 114.4, 82.6, 61.1, 27.8.

HRMS (ESI)

Calcd for [C₁₈H₂₀ClNO₂, M+Na]⁺: 340.1075; Found: 340.1082.

(-)-tert-butyl 2-((4-fluorophenyl)amino)-2-phenylacetate (1d)



White solid, TLC: R_f = 0.74 (PE/EA = 10:1, v/v), m.p. 75–77 °C, 21 mg, 73% yield, 92% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 6.06 min (major) and t_R = 7.54 min (minor). $[\alpha]_D^{25}$ -71.0 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.46 (d, J = 7.5 Hz, 2H), 7.34 (t, J = 7.5 Hz, 2H), 7.29 (t, J = 7.3 Hz, 1H), 6.81 (t, J = 8.6 Hz, 2H), 6.47 (dd, J = 9.0, 4.3 Hz, 2H), 4.89 (s, 1H), 4.83 (s, 1H), 1.37 (s, 9H).

¹³C NMR (151 MHz, CDCl₃)

δ 170.8, 156.0 (d, *J* = 235.2 Hz), 142.5, 138.0, 128.7, 128.0, 127.0, 115.6 (d, *J* = 22.6 Hz), 114.2 (d, *J* = 7.4 Hz), 82.5, 61.8, 27.8.

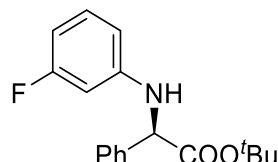
¹⁹F NMR (376 MHz, CDCl₃)

δ -127.6.

HRMS (ESI)

Calcd for [C₁₈H₂₀FNO₂, M+Na]⁺: 324.1370; Found: 324.1373.

(-)-*tert*-butyl 2-((3-fluorophenyl)amino)-2-phenylacetate (1e)



White solid, TLC: *R*_f = 0.69 (PE/EA = 10:1, v/v), m.p. 55–57 °C, 20 mg, 67% yield, 84% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak OD-H column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, *t*_R = 3.73 min (major) and *t*_R = 4.24 min (minor). [α]_D²⁵ -72.0 (*c* 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.45 (d, *J* = 7.5 Hz, 2H), 7.34 (t, *J* = 7.5 Hz, 2H), 7.29 (t, *J* = 7.3 Hz, 1H), 7.03 (q, *J* = 7.7 Hz, 1H), 6.40 – 6.26 (m, 2H), 6.20 (dt, *J* = 11.6, 2.4 Hz, 1H), 5.10 (d, *J* = 5.6 Hz, 1H), 4.91 (d, *J* = 5.6 Hz, 1H), 1.38 (s, 9H).

¹³C NMR (151 MHz, CDCl₃)

δ 170.5, 163.9 (d, *J* = 243.1 Hz), 147.8 (d, *J* = 10.6 Hz), 137.7, 130.3 (d, *J* = 10.3 Hz), 128.7, 128.1, 127.0, 109.2, 104.2 (d, *J* = 21.3 Hz), 100.1 (d, *J* = 25.2 Hz), 82.6, 61.1, 27.8.

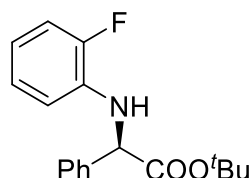
¹⁹F NMR (376 MHz, CDCl₃)

δ -112.8.

HRMS (ESI)

Calcd for [C₁₈H₂₀FNO₂, M+Na]⁺: 324.1370; Found: 324.1372.

(-)-*tert*-butyl 2-((2-fluorophenyl)amino)-2-phenylacetate (1f)



White solid, TLC: *R*_f = 0.83 (PE/EA = 10:1, v/v), m.p. 53–55 °C, 18 mg, 62% yield, 82% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 99:1, flow rate = 0.5 mL/min, 220 nm UV detector, *t*_R = 11.51 min (minor) and *t*_R = 12.52 min (major). [α]_D²⁵ -60.0 (*c* 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.47 (d, J = 7.5 Hz, 2H), 7.34 (t, J = 7.5 Hz, 2H), 7.29 (t, J = 7.3 Hz, 1H), 6.97 (dd, J = 11.7, 8.0 Hz, 1H), 6.84 (t, J = 7.7 Hz, 1H), 6.67 – 6.53 (m, 1H), 6.42 (t, J = 8.4 Hz, 1H), 5.21 (s, 1H), 4.96 (d, J = 6.0 Hz, 1H), 1.38 (s, 10H).

^{13}C NMR (151 MHz, CDCl_3)

δ 170.4, 151.6 (d, J = 239.3 Hz), 137.7, 134.7 (d, J = 11.5 Hz), 128.7, 128.1, 127.0, 124.4 (d, J = 3.3 Hz), 117.3 (d, J = 7.1 Hz), 114.6 (d, J = 18.5 Hz), 112.9 (d, J = 3.0 Hz), 82.5, 61.0, 27.8.

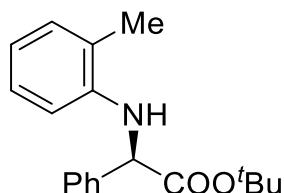
^{19}F NMR (376 MHz, CDCl_3)

δ -135.7.

HRMS (ESI)

Calcd for $[\text{C}_{18}\text{H}_{20}\text{FNO}_2, \text{M}+\text{Na}]^+$: 324.1370; Found: 324.1372.

(-)-*tert*-butyl 2-phenyl-2-(*o*-tolylamino)acetate (1g)



White solid, TLC: R_f = 0.74 (PE/EA = 10:1, v/v), m.p. 48–50 °C, 19 mg, 64% yield, 93% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak OD-H column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 4.40 min (minor) and t_R = 5.21 min (major). $[\alpha]_D^{25}$ -41.0 (c 0.2, CHCl_3).

^1H NMR (600 MHz, CDCl_3)

δ 7.48 (d, J = 7.6 Hz, 2H), 7.33 (t, J = 7.5 Hz, 2H), 7.28 (t, J = 7.3 Hz, 1H), 7.06 (d, J = 7.3 Hz, 1H), 6.96 (t, J = 7.8 Hz, 1H), 6.62 (t, J = 7.3 Hz, 1H), 6.32 (d, J = 8.0 Hz, 1H), 4.99 (d, J = 5.0 Hz, 1H), 4.91 (s, 1H), 2.28 (s, 3H), 1.38 (s, 9H).

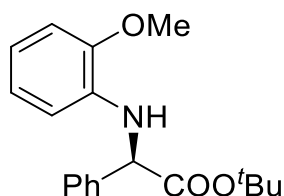
^{13}C NMR (151 MHz, CDCl_3)

δ 171.1, 144.2, 138.3, 130.1, 128.6, 127.9, 127.1, 126.9, 122.3, 117.4, 110.7, 82.4, 61.2, 27.9, 17.6.

HRMS (ESI)

Calcd for $[\text{C}_{19}\text{H}_{23}\text{NO}_2, \text{M}+\text{Na}]^+$: 320.1621; Found: 320.1625.

(-)-*tert*-butyl 2-((2-methoxyphenyl)amino)-2-phenylacetate (1h)



White solid, TLC: R_f = 0.53 (PE/EA = 10:1, v/v), m.p. 62–64 °C, 21 mg, 68% yield, 87% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak OD-H column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol =

90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 6.70 min (minor) and t_R = 14.21 min (major). $[\alpha]_D^{25}$ -47.0 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.34 (d, J = 6.3 Hz, 1H), 7.26 – 7.21 (m, 1H), 7.11 (t, J = 7.7 Hz, 2H), 6.91 (dd, J = 8.0, 5.1 Hz, 2H), 6.66 (t, J = 7.3 Hz, 1H), 6.61 (d, J = 8.0 Hz, 2H), 5.39 (s, 1H), 4.86 (s, 1H), 3.90 (s, 3H), 1.36 (s, 9H).

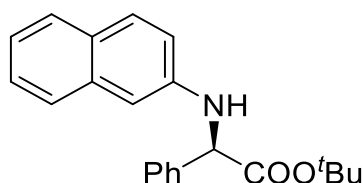
¹³C NMR (151 MHz, CDCl₃)

δ 171.4, 157.2, 146.6, 129.1, 129.0, 127.8, 127.1, 120.8, 117.6, 113.4, 110.9, 81.6, 55.6, 55.3, 27.8.

HRMS (ESI)

Calcd for [C₁₉H₂₃NO₃, M+Na]⁺: 336.1570; Found: 336.1575.

(-)-*tert*-butyl 2-(naphthalen-2-ylamino)-2-phenylacetate (1i)



White solid, TLC: R_f = 0.73 (PE/EA = 10:1, v/v), m.p. 93–95 °C, 24 mg, 74% yield, 91% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 6.52 min (major) and t_R = 8.55 min (minor). $[\alpha]_D^{25}$ -92.0 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.95 (s, 1H), 7.89 – 7.71 (m, 3H), 7.60 (dd, J = 8.5, 1.8 Hz, 1H), 7.50 – 7.37 (m, 2H), 7.09 (t, J = 7.8 Hz, 2H), 6.66 (t, J = 7.3 Hz, 1H), 6.58 (d, J = 8.0 Hz, 2H), 5.24 – 4.97 (m, 2H), 1.37 (s, 9H).

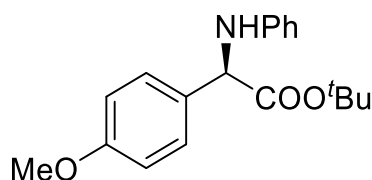
¹³C NMR (151 MHz, CDCl₃)

δ 170.8, 146.1, 135.8, 133.4, 133.2, 129.2, 128.4, 128.1, 127.7, 126.1, 126.0, 125.1, 117.8, 113.4, 82.5, 61.4, 27.8.

HRMS (ESI)

Calcd for [C₂₂H₂₃NO₂, M+Na]⁺: 356.1621; Found: 356.1626.

(-)-*tert*-butyl 2-(4-methoxyphenyl)-2-(phenylamino)acetate (1j)



White solid, TLC: R_f = 0.61 (PE/EA = 10:1, v/v), m.p. 50–52 °C, 23 mg, 75% yield, 90% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 7.65 min (major) and t_R = 8.48 min (minor). $[\alpha]_D^{25}$ -55.0 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.38 (d, J = 8.2 Hz, 2H), 7.10 (t, J = 7.7 Hz, 2H), 6.86 (d, J = 8.2 Hz, 2H), 6.67 (t, J = 7.3 Hz, 1H), 6.54 (d, J = 7.9 Hz, 2H), 4.90 (s, 2H), 3.77 (s, 3H), 1.38 (s, 9H).

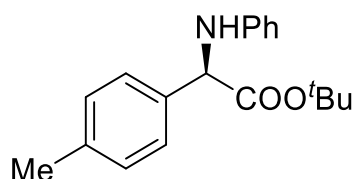
¹³C NMR (151 MHz, CDCl₃)

δ 171.2, 159.3, 146.2, 130.1, 129.1, 128.2, 117.7, 114.0, 113.3, 82.2, 60.6, 55.2, 27.8.

HRMS (ESI)

Calcd for [C₁₉H₂₃NO₃, M+Na]⁺: 336.1570; Found: 336.1574.

(-)-*tert*-butyl 2-(phenylamino)-2-(*p*-tolyl)acetate (1k)



White solid, TLC: R_f = 0.76 (PE/EA = 10:1, v/v), m.p. 63–65 °C, 21 mg, 72% yield, 90% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 5.81 min (major) and t_R = 6.19 min (minor). $[\alpha]_D^{25}$ -62.0 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.35 (d, J = 7.8 Hz, 2H), 7.16 – 7.07 (m, 4H), 6.67 (t, J = 7.3 Hz, 1H), 6.54 (d, J = 7.9 Hz, 2H), 4.96 – 4.83 (m, 2H), 2.32 (s, 3H), 1.38 (s, 9H).

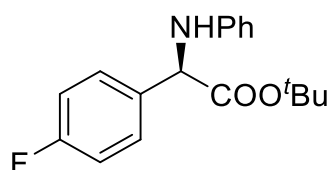
¹³C NMR (151 MHz, CDCl₃)

δ 171.1, 146.2, 137.6, 135.1, 129.3, 129.1, 126.9, 117.7, 113.3, 82.2, 60.9, 27.8, 21.1.

HRMS (ESI)

Calcd for [C₁₉H₂₃NO₂, M+Na]⁺: 320.1621; Found: 320.1627.

(-)-*tert*-butyl 2-(4-fluorophenyl)-2-(phenylamino)acetate (1l)



White solid, TLC: R_f = 0.77 (PE/EA = 10:1, v/v), m.p. 44–46 °C, 20 mg, 66% yield, 82% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak OD-H column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 4.08 min (major) and t_R = 4.78 min (minor). $[\alpha]_D^{25}$ -60.0 (c 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.45 (dd, J = 8.5, 5.3 Hz, 2H), 7.11 (t, J = 7.7 Hz, 2H), 7.02 (t, J = 8.5 Hz, 2H), 6.68 (t, J = 7.4 Hz, 1H), 6.52 (d, J = 8.0 Hz, 2H), 5.03 – 4.75 (m, 2H), 1.38 (s, 9H).

¹³C NMR (151 MHz, CDCl₃)

δ 170.6, 162.5 (d, *J* = 245.8 Hz), 145.9, 134.0 (d, *J* = 3.1 Hz), 129.2, 128.7 (d, *J* = 8.1 Hz), 117.9, 115.6 (d, *J* = 21.6 Hz), 113.4, 82.6, 60.5, 27.8.

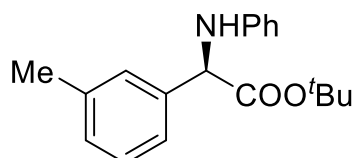
¹⁹F NMR (376 MHz, CDCl₃)

δ -114.5.

HRMS (ESI)

Calcd for [C₁₈H₂₀FNO₂, M+Na]⁺: 324.1370; Found: 324.1371.

(-)-*tert*-butyl 2-(phenylamino)-2-(*m*-tolyl)acetate (1m)



White solid, TLC: *R_f* = 0.81 (PE/EA = 10:1, v/v), m.p. 65–67 °C, 22 mg, 73% yield, 91% ee. The enantiomeric ratio was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, *t_R* = 5.11 min (major) and *t_R* = 5.58 min (minor). [α]_D²⁵ -66.0 (*c* 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.34 – 7.17 (m, 3H), 7.15 – 7.04 (m, 3H), 6.67 (t, *J* = 7.3 Hz, 1H), 6.55 (d, *J* = 8.0 Hz, 2H), 4.91 (s, 2H), 2.33 (s, 3H), 1.38 (s, 9H).

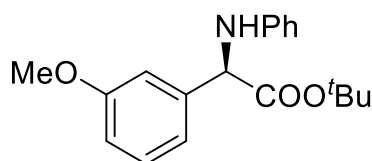
¹³C NMR (151 MHz, CDCl₃)

δ 171.1, 146.3, 138.3, 138.1, 129.2, 128.7, 128.5, 127.7, 124.2, 117.7, 113.3, 82.2, 61.3, 27.8, 21.5.

HRMS (ESI)

Calcd for [C₁₉H₂₃NO₂, M+Na]⁺: 320.1621; Found: 320.1626.

(-)-*tert*-butyl 2-(3-methoxyphenyl)-2-(phenylamino)acetate (1n)



White solid, TLC: *R_f* = 0.60 (PE/EA = 10:1, v/v), m.p. 62–64 °C, 22 mg, 72% yield, 90% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, *t_R* = 7.15 min (major) and *t_R* = 8.65 min (minor). [α]_D²⁵ -67.0 (*c* 0.2, CHCl₃).

¹H NMR (600 MHz, CDCl₃)

δ 7.24 (t, *J* = 7.9 Hz, 1H), 7.11 (t, *J* = 7.7 Hz, 2H), 7.07 (d, *J* = 7.7 Hz, 1H), 7.03 (s, 1H), 6.82 (dd, *J* = 8.1, 2.6 Hz, 1H), 6.68 (t, *J* = 7.3 Hz, 1H), 6.55 (d, *J* = 7.9 Hz, 2H), 4.92 (s, 2H), 3.78 (s, 3H), 1.39 (s, 9H).

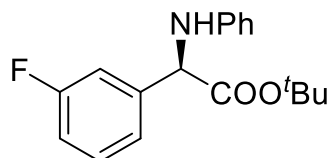
¹³C NMR (151 MHz, CDCl₃)

δ 170.8, 159.8, 146.1, 139.8, 129.6, 129.1, 119.5, 117.8, 113.5, 113.3, 112.5, 82.3, 61.2, 55.2, 27.8.

HRMS (ESI)

Calcd for $[C_{19}H_{23}NO_3, M+Na]^+$: 336.1570; Found: 336.1572.

(-)-*tert*-butyl 2-(3-fluorophenyl)-2-(phenylamino)acetate (1o)



White solid, TLC: R_f = 0.69 (PE/EA = 10:1, v/v), m.p. 74–76 °C, 21 mg, 70% yield, 86% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 5.04 min (major) and t_R = 5.85 min (minor). $[\alpha]_D^{25}$ –58.0 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.35 – 7.26 (m, 2H), 7.21 (d, J = 7.8 Hz, 1H), 7.11 (t, J = 7.7 Hz, 2H), 6.98 (dt, J = 9.4, 4.6 Hz, 1H), 6.69 (t, J = 7.3 Hz, 1H), 6.52 (d, J = 7.9 Hz, 2H), 4.99 (s, 1H), 4.94 (d, J = 4.6 Hz, 1H), 1.39 (s, 9H).

^{13}C NMR (151 MHz, $CDCl_3$)

δ 170.2, 163.0 (d, J = 246.2 Hz), 145.8, 141.1 (d, J = 7.1 Hz), 130.1 (d, J = 7.7 Hz), 129.2, 122.8 (d, J = 2.5 Hz), 118.0, 114.9 (d, J = 21.5 Hz), 114.1 (d, J = 22.5 Hz), 113.4, 82.8, 60.8, 27.8.

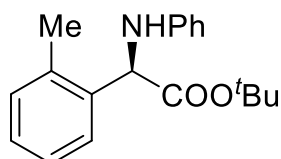
^{19}F NMR (376 MHz, $CDCl_3$)

δ –112.7.

HRMS (ESI)

Calcd for $[C_{18}H_{20}FNO_2, M+H]^+$: 302.1551; Found: 302.1559.

(-)-*tert*-butyl 2-(phenylamino)-2-(*o*-tolyl)acetate (1p)



White solid, TLC: R_f = 0.78 (PE/EA = 10:1, v/v), m.p. 70–72 °C, 19 mg, 65% yield, 90% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak OD-H column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 3.64 min (major) and t_R = 4.43 min (minor). $[\alpha]_D^{25}$ –64.0 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.39 (d, J = 7.5 Hz, 1H), 7.21 – 7.13 (m, 3H), 7.10 (t, J = 7.7 Hz, 2H), 6.67 (t, J = 7.3 Hz, 1H), 6.50 (d, J = 7.9 Hz, 2H), 5.16 (s, 1H), 4.87 (s, 1H), 2.54 (s, 3H), 1.37 (s, 9H).

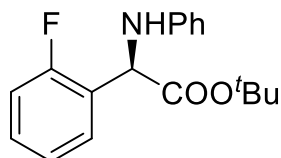
^{13}C NMR (151 MHz, $CDCl_3$)

δ 171.3, 146.3, 136.6, 136.5, 130.7, 129.2, 127.8, 126.5, 126.1, 117.7, 113.1, 82.2, 57.7, 27.8, 19.5.

HRMS (ESI)

Calcd for $[C_{19}H_{23}NO_2, M+Na]^+$: 320.1621; Found: 320.1627.

(-)-tert-butyl 2-(2-fluorophenyl)-2-(phenylamino)acetate (1q)



White solid, TLC: R_f = 0.73 (PE/EA = 10:1, v/v), m.p. 59–61 °C, 19 mg, 64% yield, 81% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 5.13 min (major) and t_R = 5.92 min (minor). $[\alpha]_D^{25}$ -62.0 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.41 (t, J = 7.0 Hz, 1H), 7.29 – 7.17 (m, 1H), 7.10 (dt, J = 18.8, 7.9 Hz, 4H), 6.68 (t, J = 7.3 Hz, 1H), 6.57 (d, J = 7.9 Hz, 2H), 5.32 (d, J = 5.0 Hz, 1H), 4.98 (s, 1H), 1.37 (s, 9H).

^{13}C NMR (151 MHz, $CDCl_3$)

δ 170.3, 160.9 (d, J = 247.6 Hz), 145.8, 129.6 (d, J = 7.9 Hz), 129.2, 127.9 (d, J = 3.6 Hz), 125.9 (d, J = 13.8 Hz), 124.5 (d, J = 3.8 Hz), 118.0, 115.6 (d, J = 22.0 Hz), 113.3, 82.6, 54.3, 27.8.

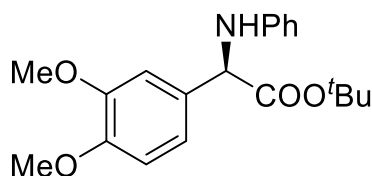
^{19}F NMR (376 MHz, $CDCl_3$)

δ -118.7.

HRMS (ESI)

Calcd for $[C_{18}H_{20}FNO_2, M+Na]^+$: 324.1370; Found: 324.1375.

(-)-tert-butyl 2-(3,4-dimethoxyphenyl)-2-(phenylamino)acetate (1r)



White solid, TLC: R_f = 0.25 (PE/EA = 10:1, v/v), m.p. 111–113 °C, 25 mg, 74% yield, 91% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 9.15 min (major) and t_R = 13.71 min (minor). $[\alpha]_D^{25}$ -77.0 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.11 (t, J = 7.8 Hz, 2H), 7.05 – 6.97 (m, 2H), 6.82 (d, J = 8.2 Hz, 1H), 6.68 (t, J = 7.4 Hz, 1H), 6.56 (d, J = 7.9 Hz, 2H), 4.89 (s, 2H), 3.86 (s, 3H), 3.85 (s, 3H), 1.39 (s, 9H).

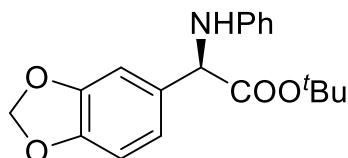
^{13}C NMR (151 MHz, $CDCl_3$)

δ 171.1, 149.0, 148.7, 146.2, 130.6, 129.1, 119.4, 117.8, 113.4, 111.1, 110.0, 82.2, 60.9, 55.9, 55.8, 27.8.

HRMS (ESI)

Calcd for $[C_{20}H_{25}NO_4, M+Na]^+$: 366.1676; Found: 366.1680.

(-)-tert-butyl 2-(benzo[d][1,3]dioxol-5-yl)-2-(phenylamino)acetate (1s)



White solid, TLC: R_f = 0.62 (PE/EA = 10:1, v/v), m.p. 82–84 °C, 22 mg, 68% yield, 90% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 8.43 min (major) and t_R = 9.83 min (minor). $[\alpha]_D^{25}$ –42.0 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.11 (t, J = 7.7 Hz, 2H), 6.99 – 6.92 (m, 2H), 6.76 (d, J = 8.4 Hz, 1H), 6.68 (t, J = 7.3 Hz, 1H), 6.54 (d, J = 7.9 Hz, 2H), 5.93 (d, J = 5.5 Hz, 2H), 4.92 (s, 1H), 4.85 (s, 1H), 1.39 (s, 9H).

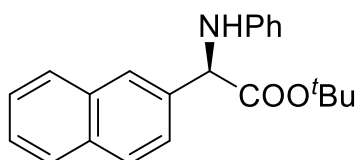
^{13}C NMR (151 MHz, $CDCl_3$)

δ 170.8, 147.9, 147.3, 146.0, 132.2, 129.2, 120.6, 117.8, 113.3, 108.3, 107.3, 101.1, 82.4, 60.8, 27.8.

HRMS (ESI)

Calcd for $[C_{19}H_{21}NO_4, M+Na]^+$: 350.1363; Found: 350.1368.

(-)-tert-butyl 2-(naphthalen-2-yl)-2-(phenylamino)acetate (1t)



White solid, TLC: R_f = 0.72 (PE/EA = 10:1, v/v), m.p. 120–122 °C, 24 mg, 73% yield, 90% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak OD-H column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 5.36 min (major) and t_R = 5.79 min (minor). $[\alpha]_D^{25}$ –92.0 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.95 (s, 1H), 7.88 – 7.76 (m, 3H), 7.61 (d, J = 8.5 Hz, 1H), 7.49 – 7.37 (m, 2H), 7.09 (t, J = 7.8 Hz, 2H), 6.67 (t, J = 7.3 Hz, 1H), 6.58 (d, J = 7.9 Hz, 2H), 5.15 – 4.98 (m, 2H), 1.37 (s, 9H).

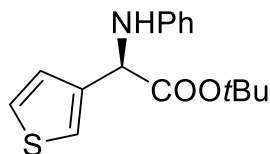
^{13}C NMR (151 MHz, $CDCl_3$)

δ 170.8, 146.1, 135.7, 133.4, 133.1, 129.2, 128.4, 128.1, 127.6, 126.1, 126.0, 125.0, 117.8, 113.4, 82.5, 61.4, 27.8.

HRMS (ESI)

Calcd for $[C_{22}H_{23}NO_2, M+Na]^+$: 356.1621; Found: 356.1625.

(-)-tert-butyl 2-(phenylamino)-2-(thiophen-3-yl)acetate (1u)



White solid, TLC: R_f = 0.74 (PE/EA = 10:1, v/v), m.p. 60–62 °C, 18 mg, 63% yield, 80% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 6.75 min (major) and t_R = 7.51 min (minor). $[\alpha]_D^{25}$ –57.0 (c 0.2, $CHCl_3$).

1H NMR (600 MHz, $CDCl_3$)

δ 7.38 – 7.26 (m, 2H), 7.21 – 7.07 (m, 3H), 6.71 (t, J = 7.3 Hz, 1H), 6.59 (d, J = 8.0 Hz, 2H), 5.07 (d, J = 4.5 Hz, 1H), 4.80 (s, 1H), 1.42 (s, 9H).

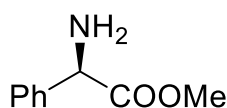
^{13}C NMR (151 MHz, $CDCl_3$)

δ 170.6, 146.3, 138.8, 129.2, 126.5, 126.1, 122.5, 118.1, 113.4, 82.5, 57.6, 27.9.

HRMS (ESI)

Calcd for $[C_{16}H_{19}NO_2S, M+Na]^+$: 312.1029; Found: 312.1032.

(R)-methyl 2-amino-2-phenylacetate (1v)

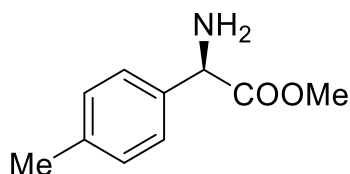


Colorless oil, TLC: R_f = 0.28 (DCM/MeOH = 20:1, v/v), 21 mg, 63% yield, 64% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak IA-3 column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 8.38 min (minor) and t_R = 8.96 min (major). $[\alpha]_D^{25}$ –88.0 (c 1.0, $CHCl_3$). By comparing the specific rotation, $[\alpha]_D^{25}$ +139 (c 1.0, $CHCl_3$) for S enantiomer in the literature,^[21] the configuration of the insertion product **22** was determined to be R.

1H NMR (400 MHz, $CDCl_3$)

δ 7.44 – 7.33 (m, 4H), 7.30 (t, J = 6.7 Hz, 1H), 4.62 (s, 1H), 3.70 (s, 3H), 2.00 (s, 2H).

(-)-methyl 2-amino-2-(p-tolyl)acetate (1w)



Colorless oil, TLC: R_f = 0.25 (DCM/MeOH = 20:1, v/v), 22 mg, 62% yield, 66% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak IA-3 column (25 cm $\times$ 0.46 cm ID), hexane/2-propanol = 90:10,

flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 8.39 min (minor) and t_R = 9.12 min (major). $[\alpha]_D^{25}$ -117 (c 0.2, CHCl₃).

¹H NMR (400 MHz, CDCl₃)

δ 7.30 – 7.22 (m, 2H), 7.20 – 7.12 (m, 2H), 4.58 (s, 1H), 3.69 (s, 3H), 2.34 (s, 3H), 1.94 (s, 2H).

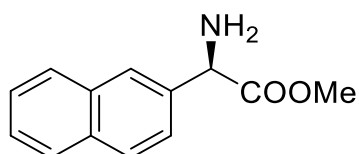
¹³C NMR (101 MHz, CDCl₃)

δ 174.6, 137.8, 137.4, 129.5, 126.7, 58.5, 52.4, 21.1.

HRMS (ESI)

Calcd for [C₁₀H₁₃NO₂, M+H]⁺: 180.1019; Found: 180.1021.

(-)-methyl 2-amino-2-(naphthalen-2-yl)acetate (1x)



Colorless liquid, TLC: R_f = 0.30 (DCM/MeOH = 20:1, v/v), 24 mg, 55% yield, 66% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak IA-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 11.53 min (minor) and t_R = 12.93 min (major). $[\alpha]_D^{25}$ -115 (c 0.2, CHCl₃).

¹H NMR (400 MHz, CDCl₃)

δ 7.89 – 7.64 (m, 4H), 7.56 – 7.34 (m, 3H), 4.78 (s, 1H), 3.69 (s, 3H), 2.05 (s, 2H).

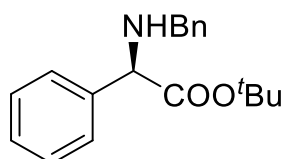
¹³C NMR (101 MHz, CDCl₃)

δ 174.4, 137.7, 133.4, 133.1, 128.7, 128.0, 127.7, 126.4, 126.2, 125.9, 124.7, 58.9, 52.5.

HRMS (ESI)

Calcd for [C₁₃H₁₃NO₂, M+H]⁺: 216.1019; Found: 216.1023.

(-)-tert-butyl 2-(benzylamino)-2-phenylacetate (1y)



Colorless liquid, TLC: R_f = 0.45 (PE/EA = 10:1, v/v), 24 mg, 41% yield, 54% ee. The ee value was determined by high performance liquid chromatography. HPLC condition: Chiralpak AD-3 column (25 cm × 0.46 cm ID), hexane/2-propanol = 90:10, flow rate = 1.0 mL/min, 220 nm UV detector, t_R = 4.99 min (minor) and t_R = 6.19 min (major). $[\alpha]_D^{25}$ -55.0 (c 0.2, CHCl₃).

¹H NMR (400 MHz, CDCl₃)

δ 7.41 – 7.26 (m, 10H), 4.27 (s, 1H), 3.73 (s, 2H), 2.08 (s, 1H), 1.39 (s, 9H).

¹³C NMR (101 MHz, CDCl₃)

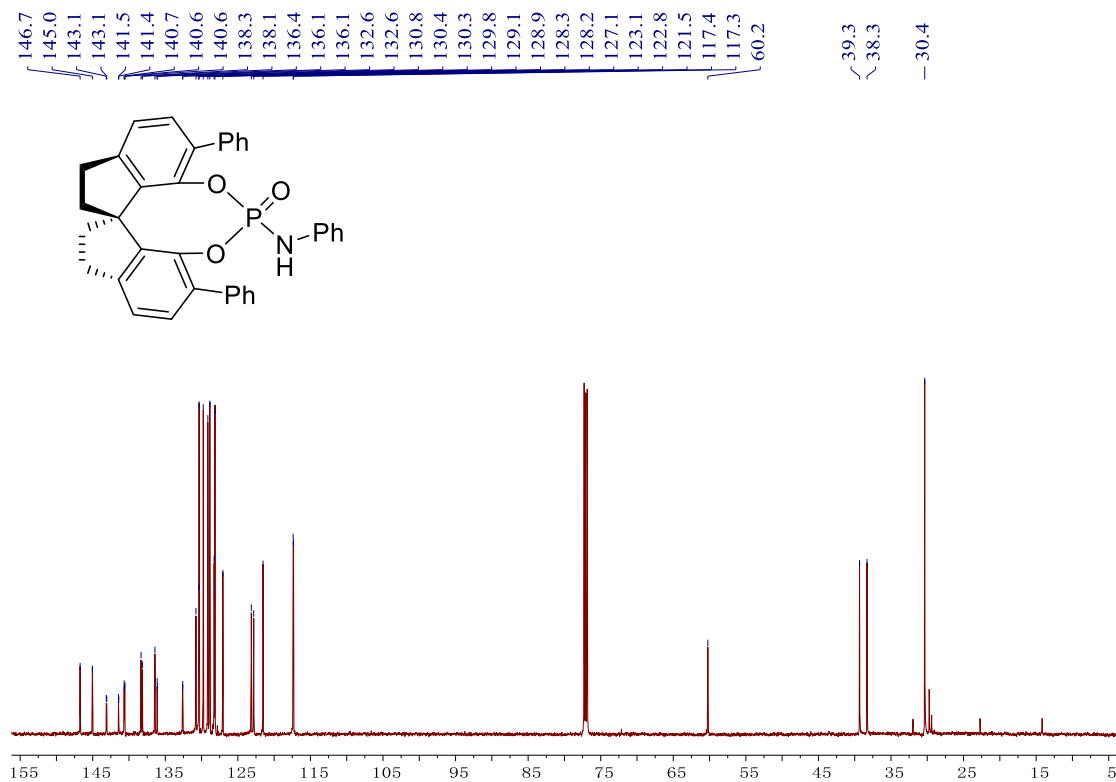
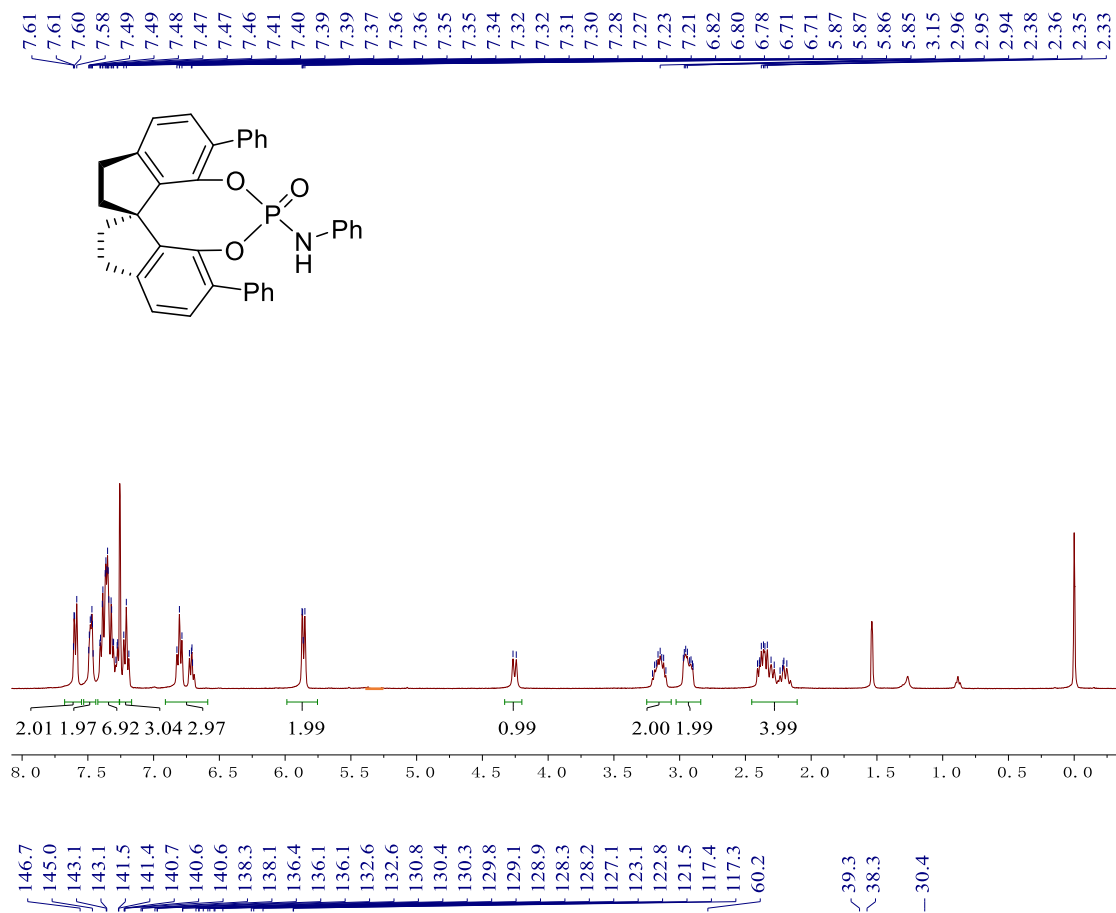
δ 172.2, 139.7, 138.7, 128.5, 128.4, 128.3, 127.7, 127.4, 127.1, 81.5, 65.1, 51.4, 27.9.

HRMS (ESI)

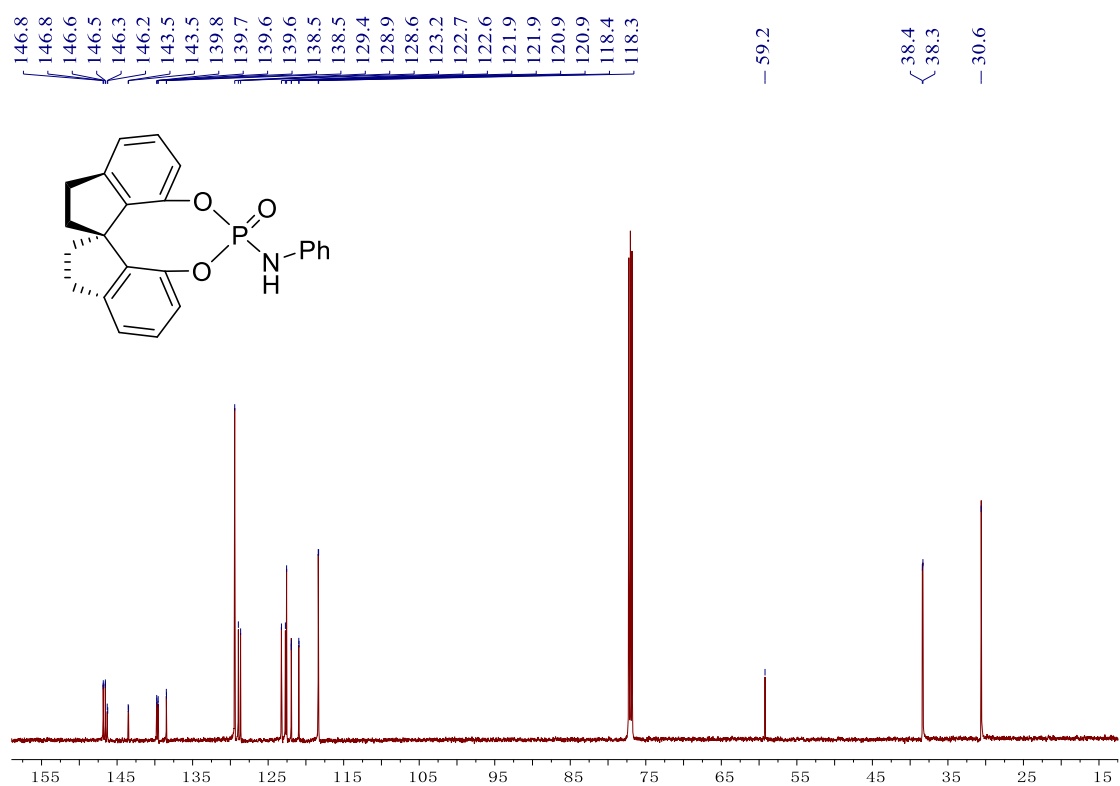
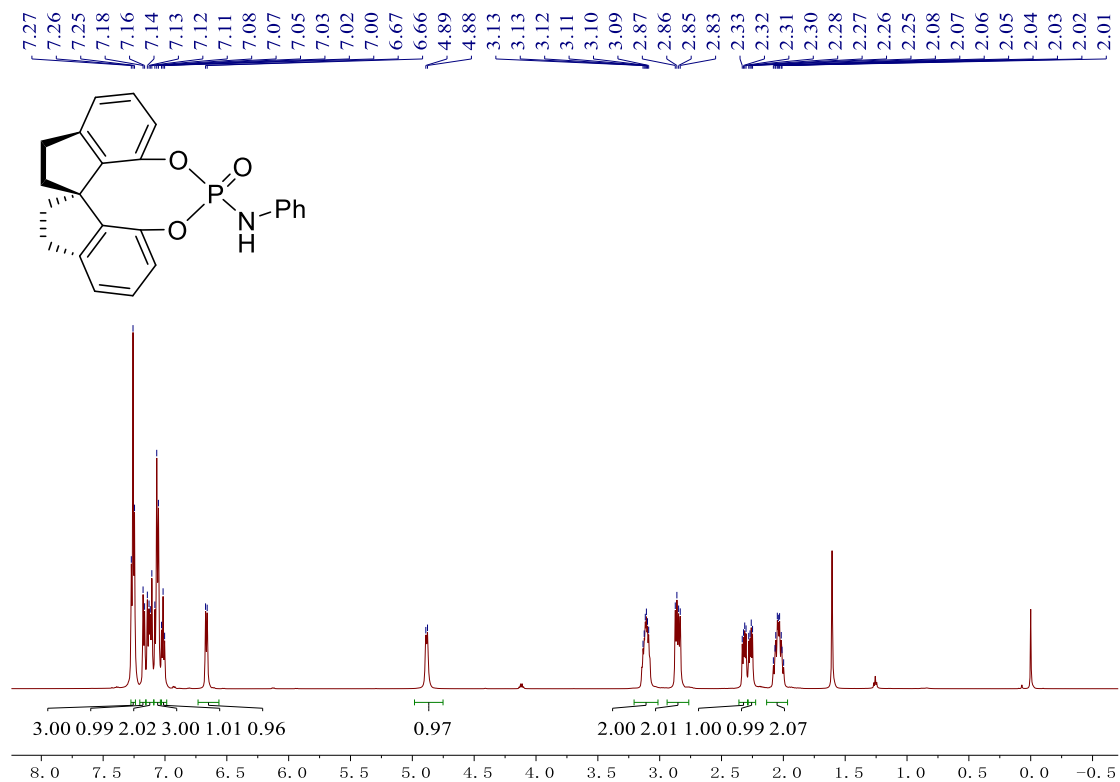
Calcd for $[\text{C}_{19}\text{H}_{23}\text{NO}_2, \text{M}+\text{H}]^+$: 298.1802; Found: 298.1806.

6. NMR Spectra for New Compounds

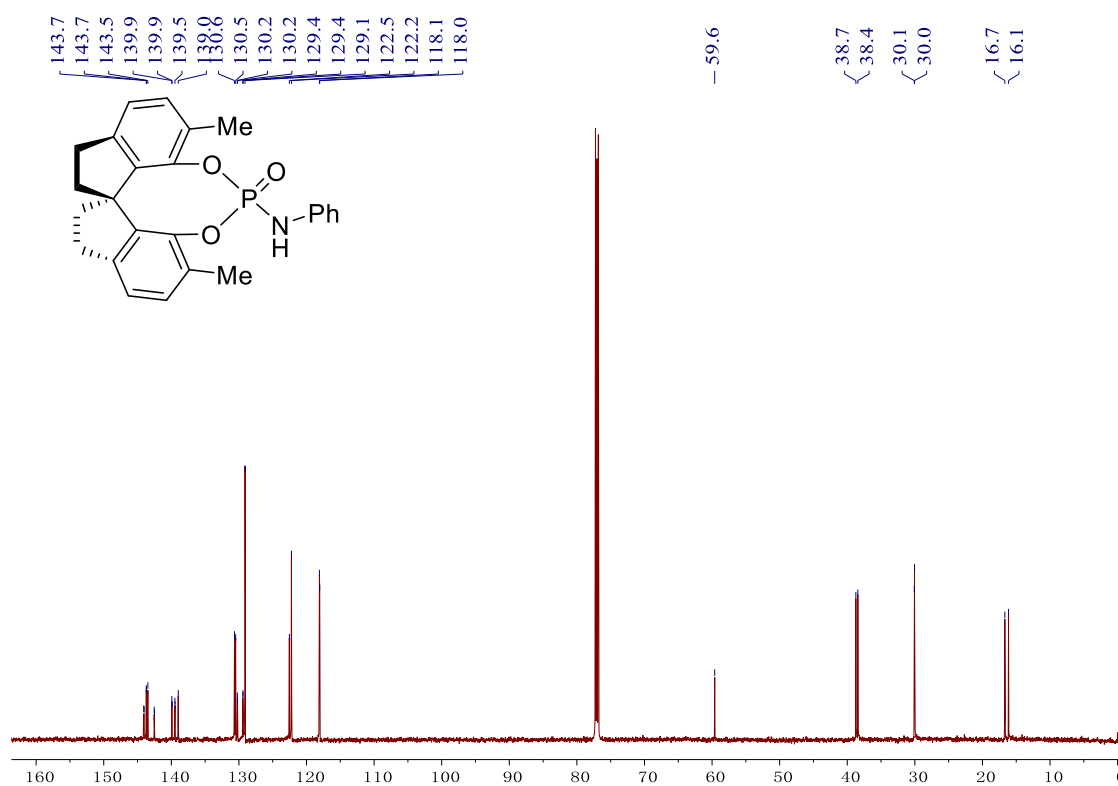
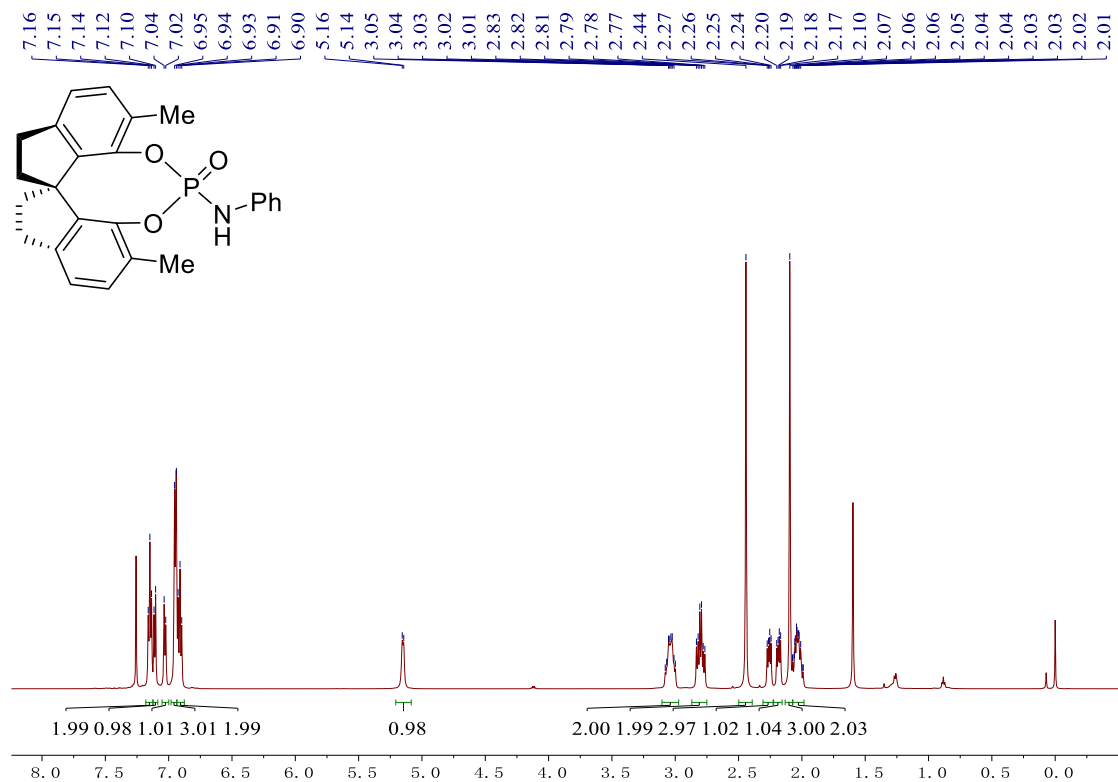
(*R*)-1,10-diphenyl-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



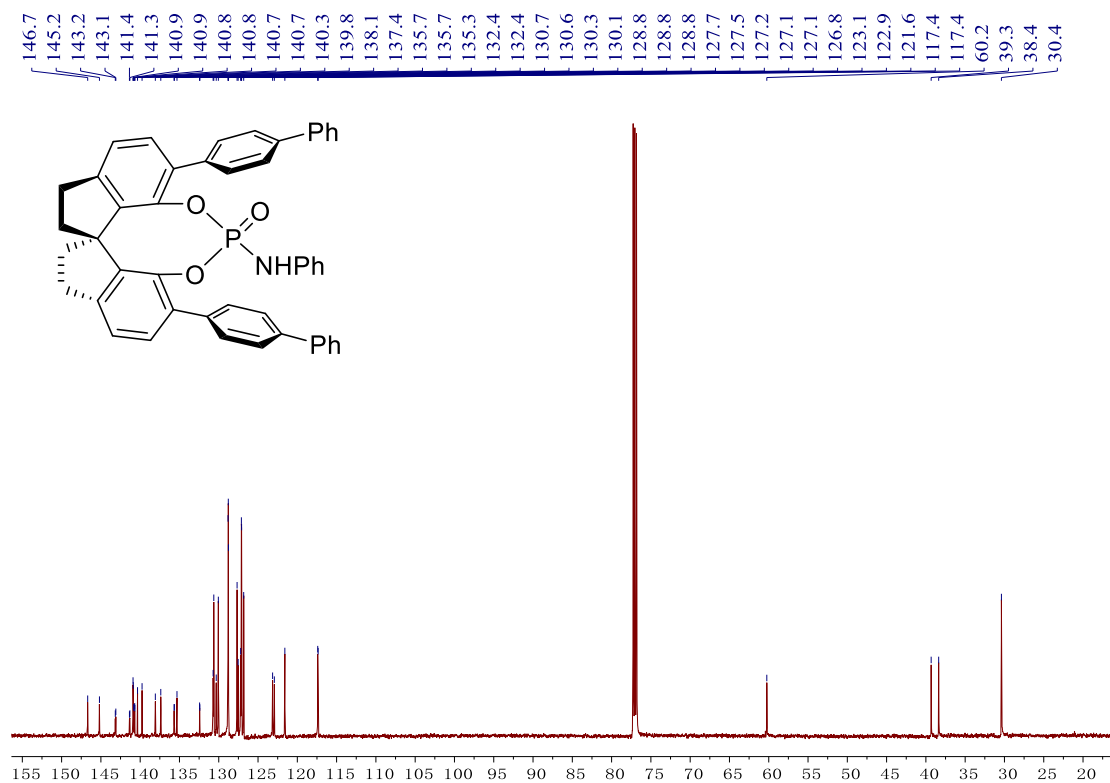
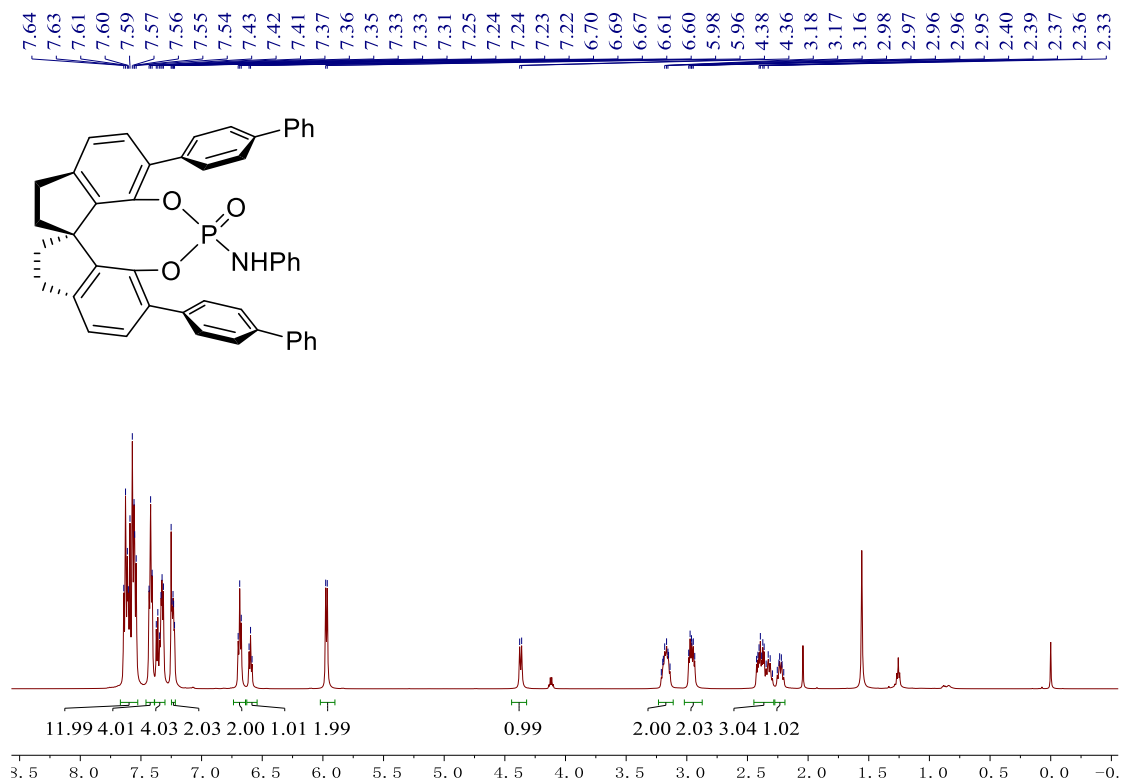
(R)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



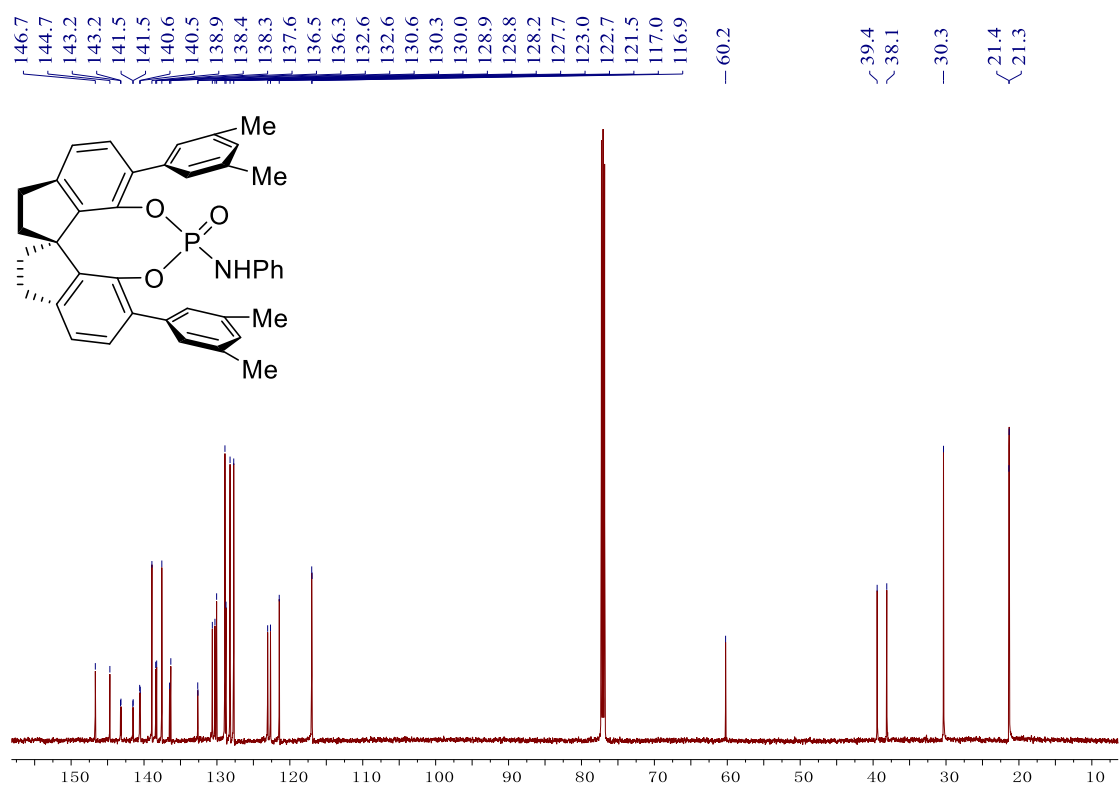
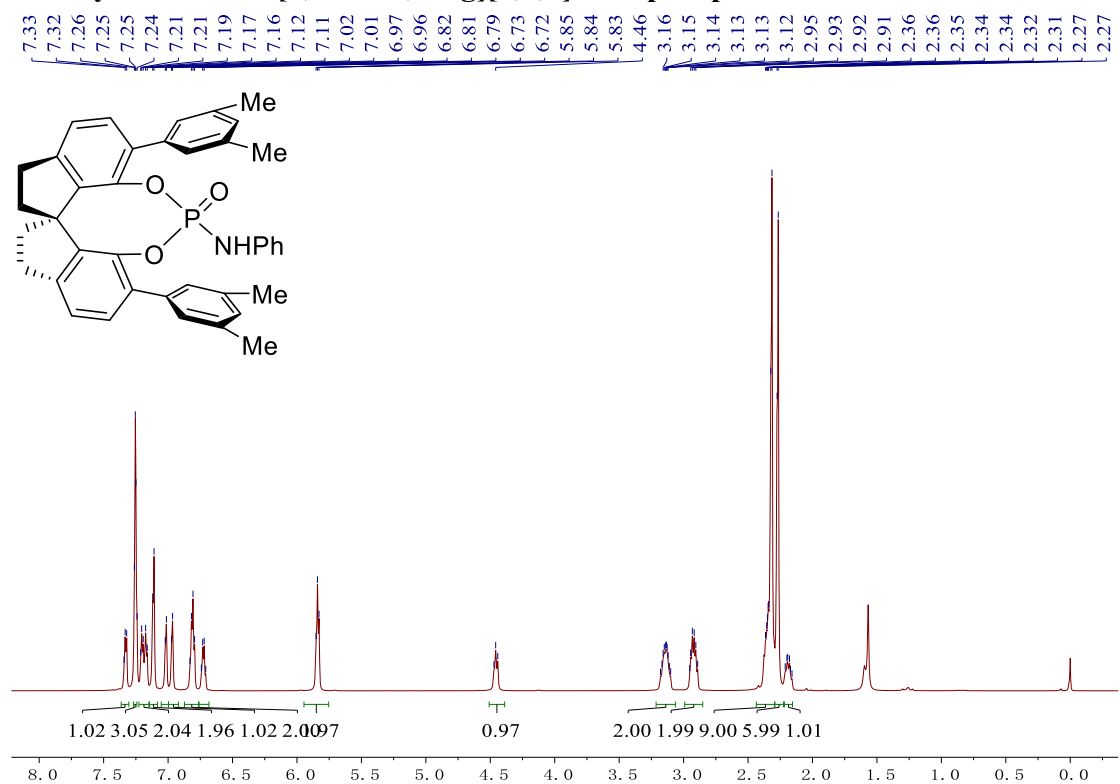
(*R*)-1,10-dimethyl-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



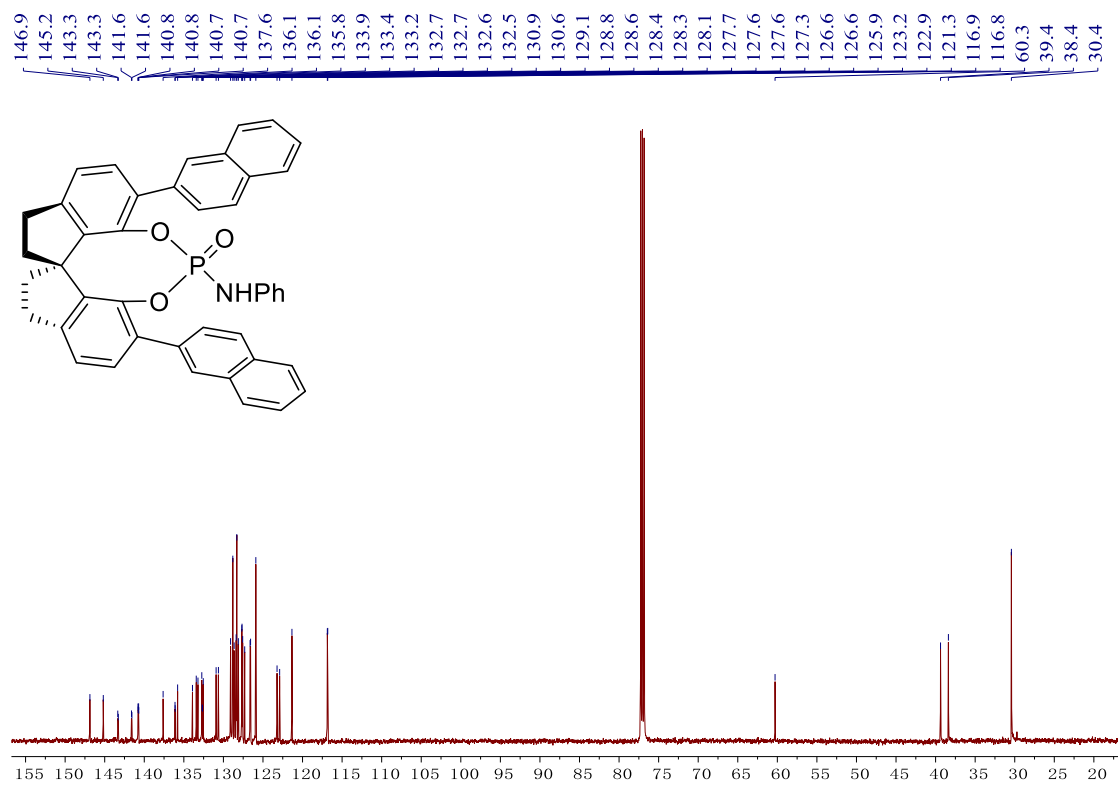
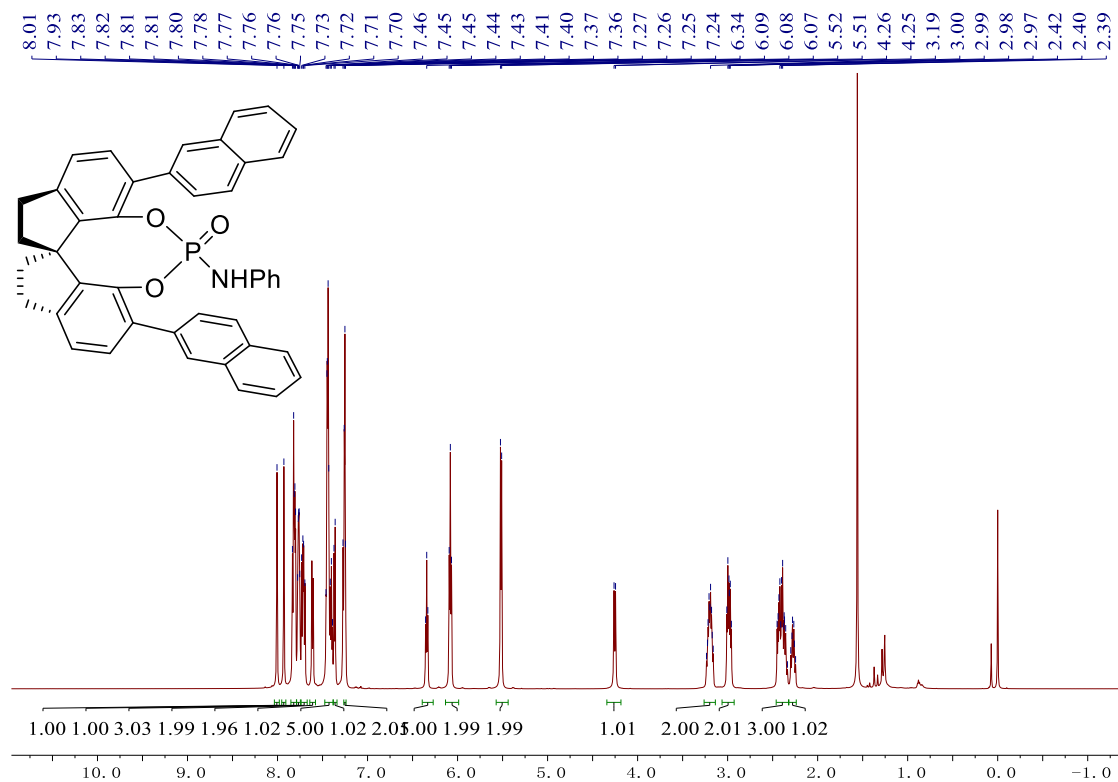
(R)-1,10-di([1,1'-biphenyl]-4-yl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



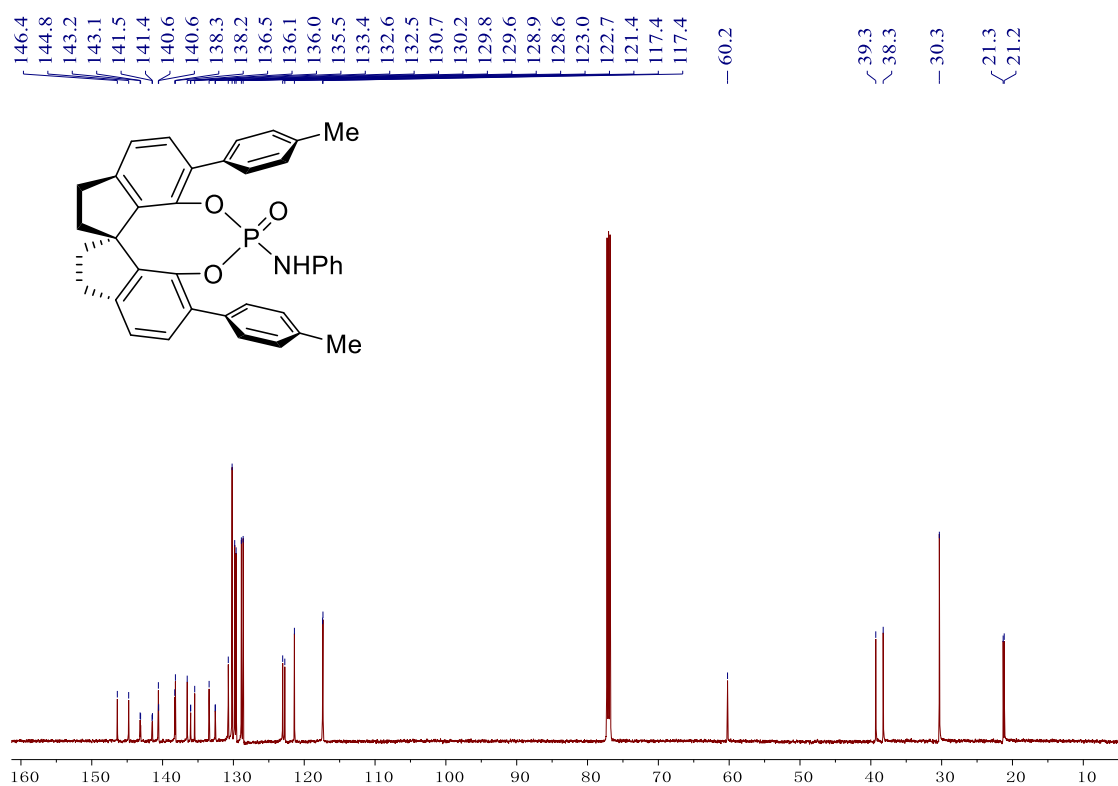
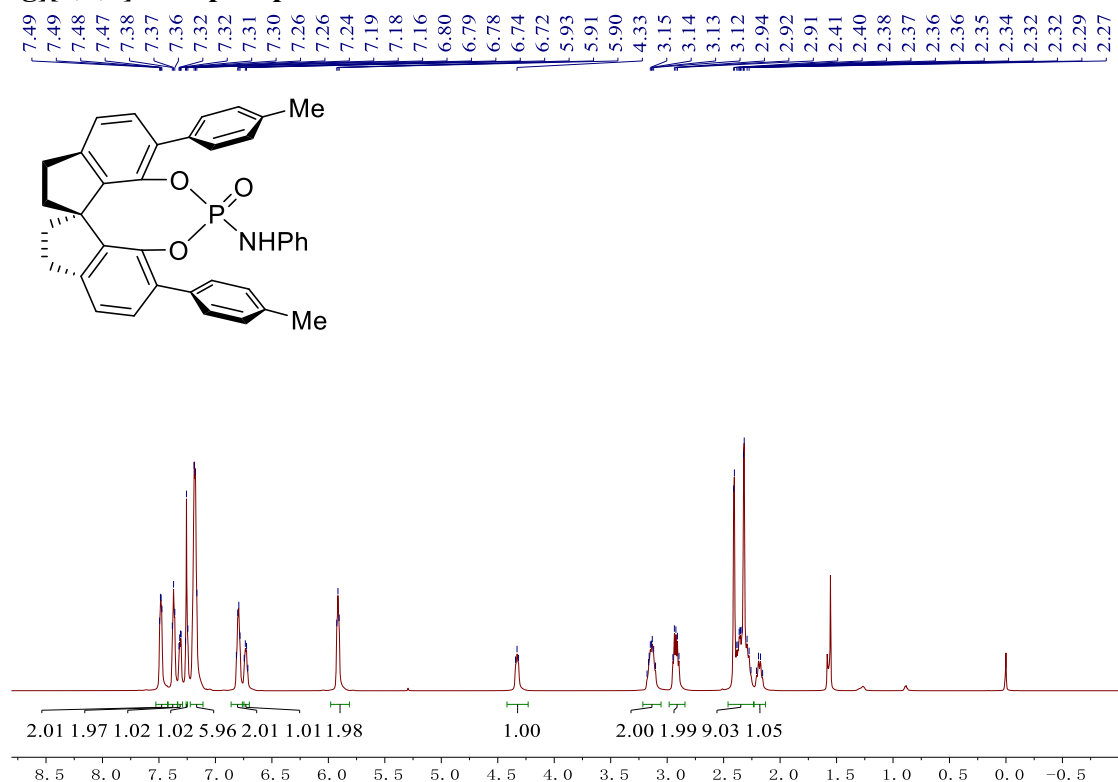
(*R*)-1,10-bis(3,5-dimethylphenyl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



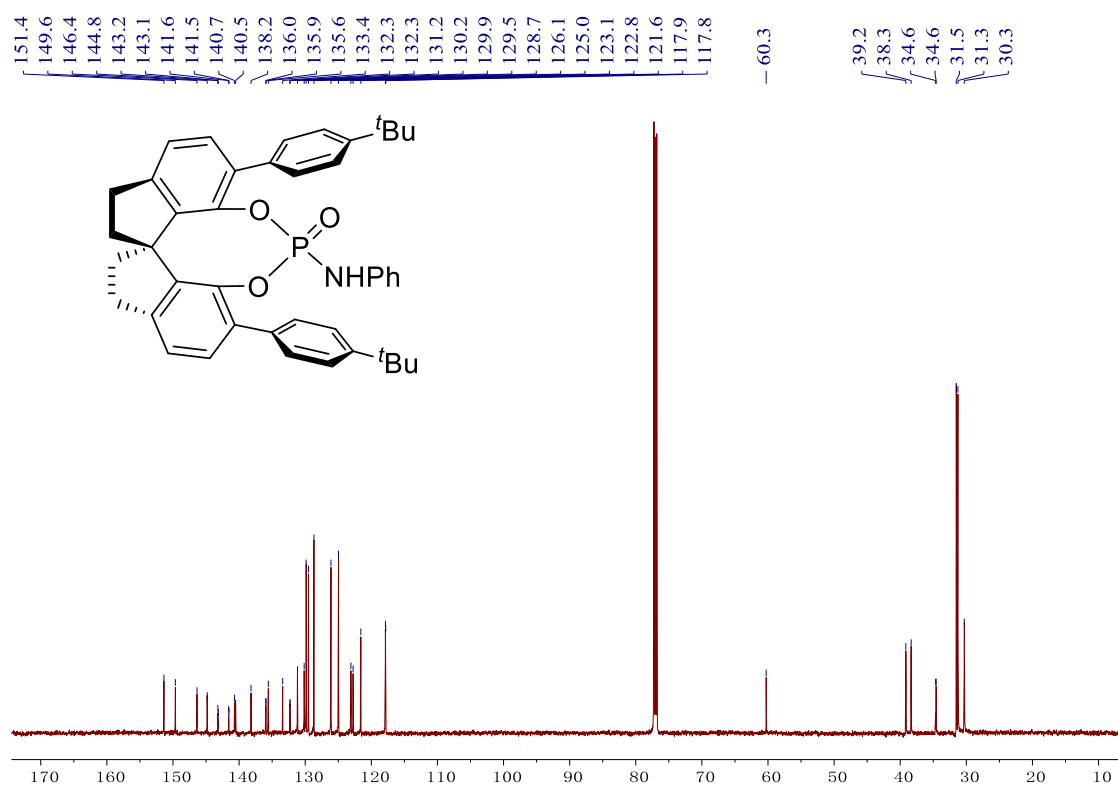
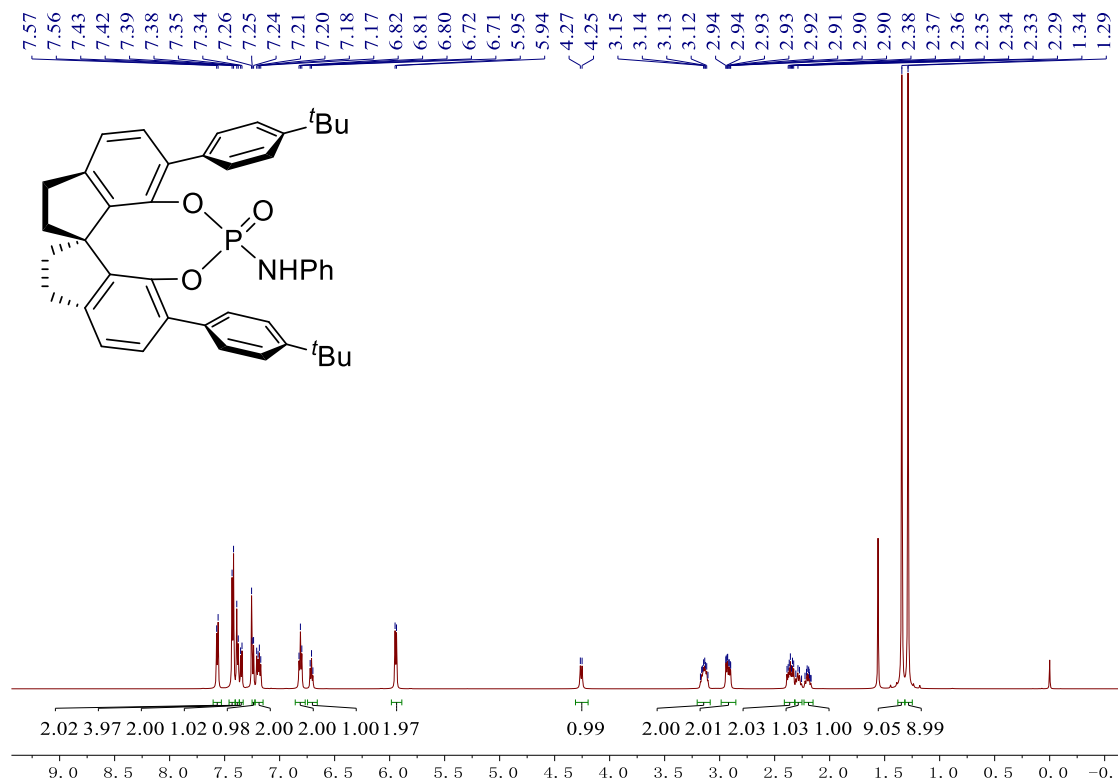
(*R*)-1,10-di(naphthalen-2-yl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



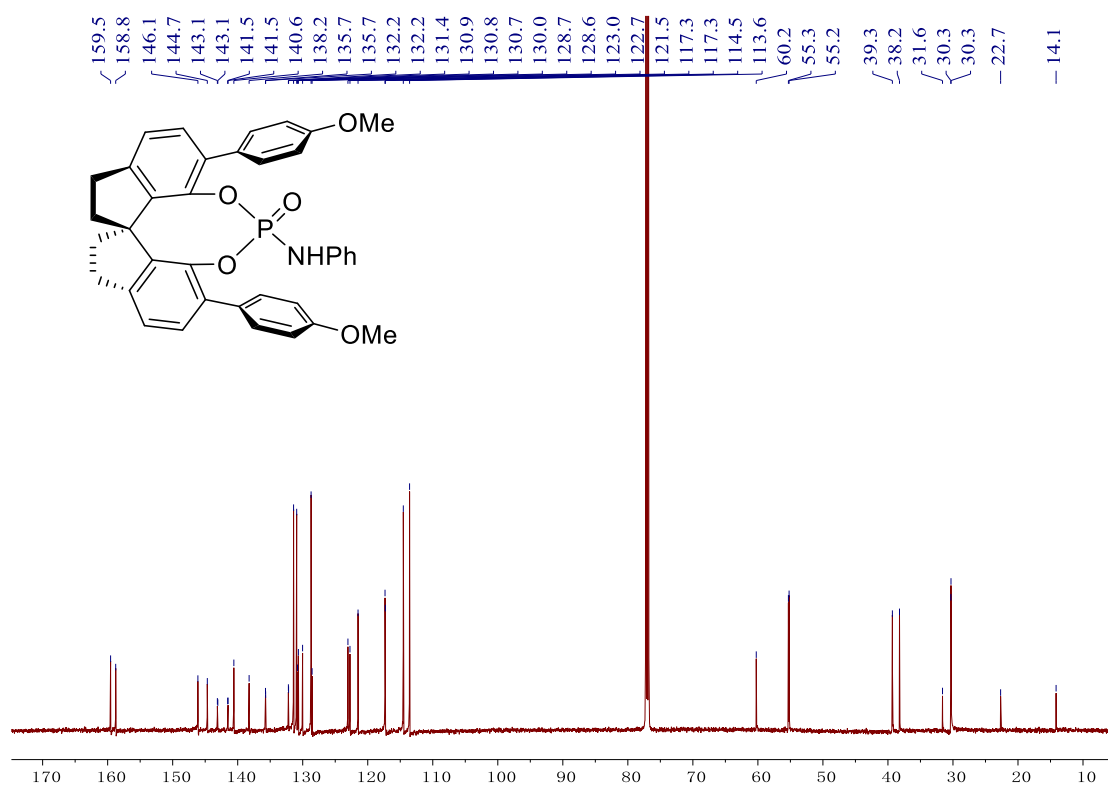
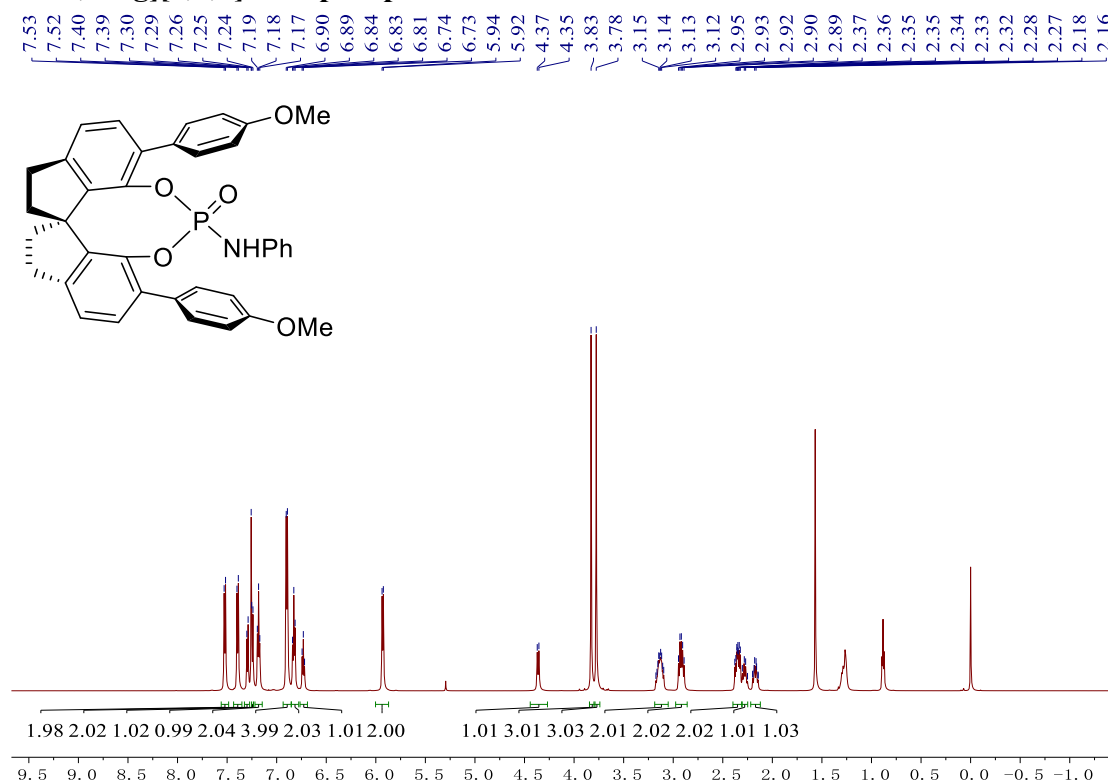
(R)-12-(phenylamino)-1,10-di-p-tolyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



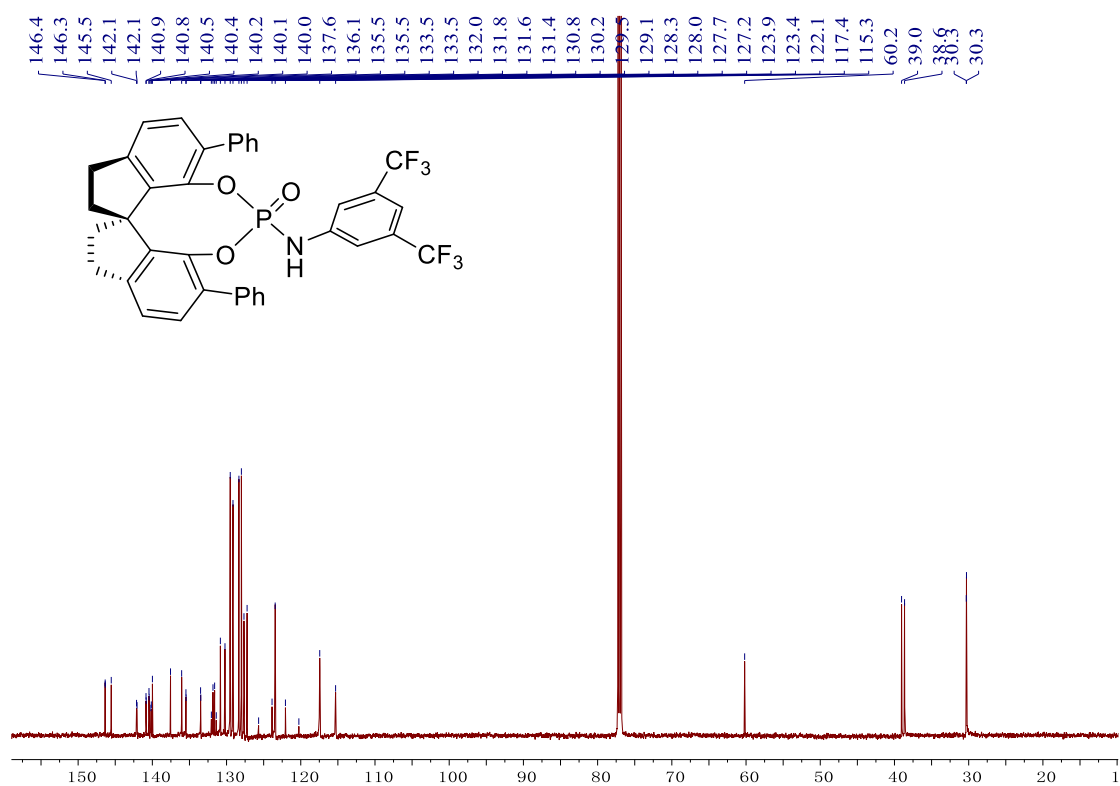
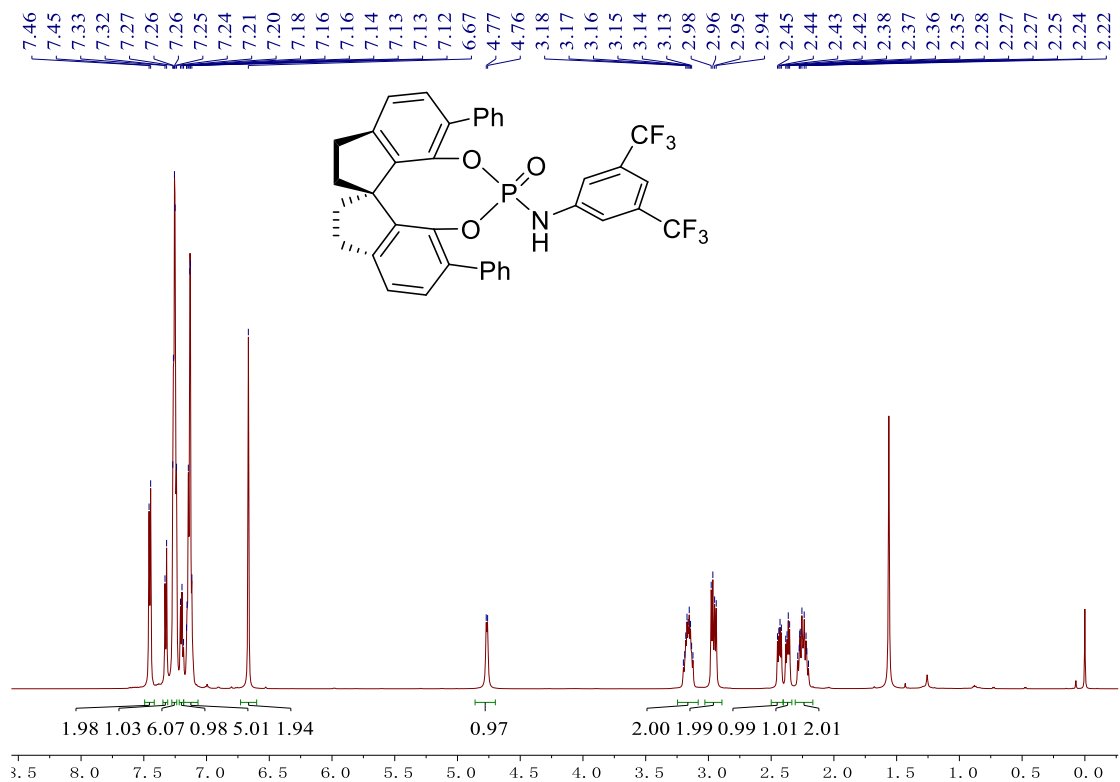
(*R*)-1,10-bis(4-(tert-butyl)phenyl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



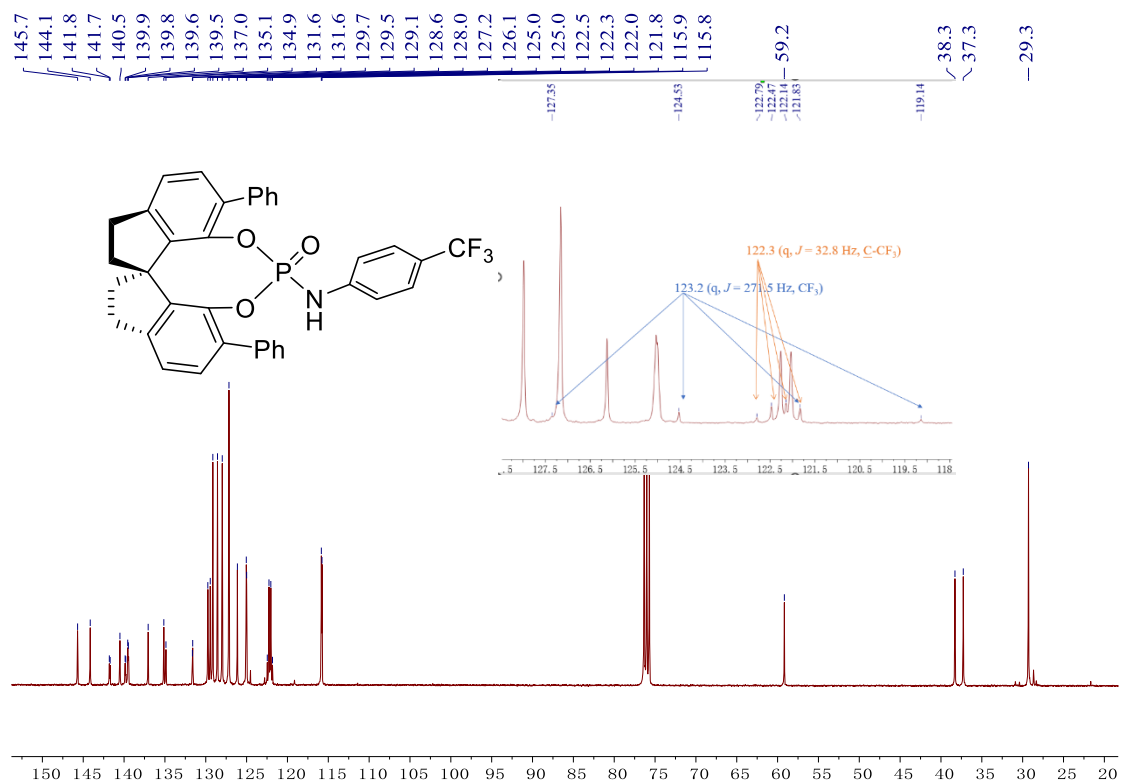
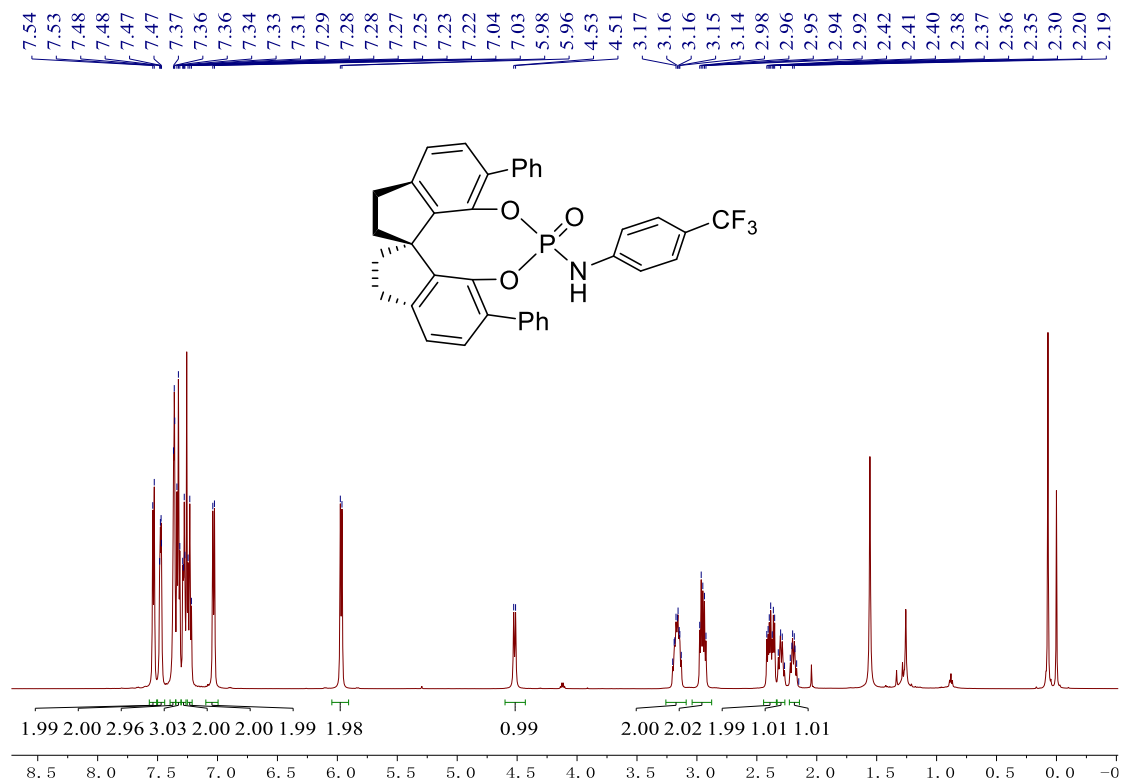
(*R*)-1,10-bis(4-methoxyphenyl)-12-(phenylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



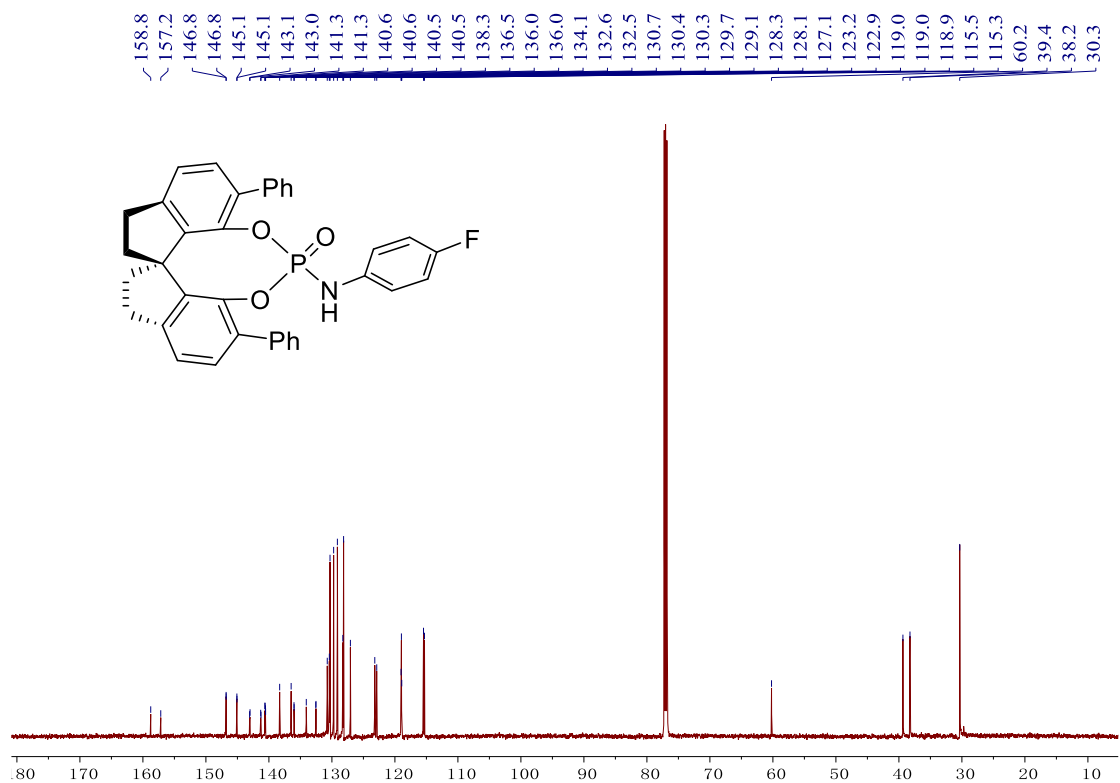
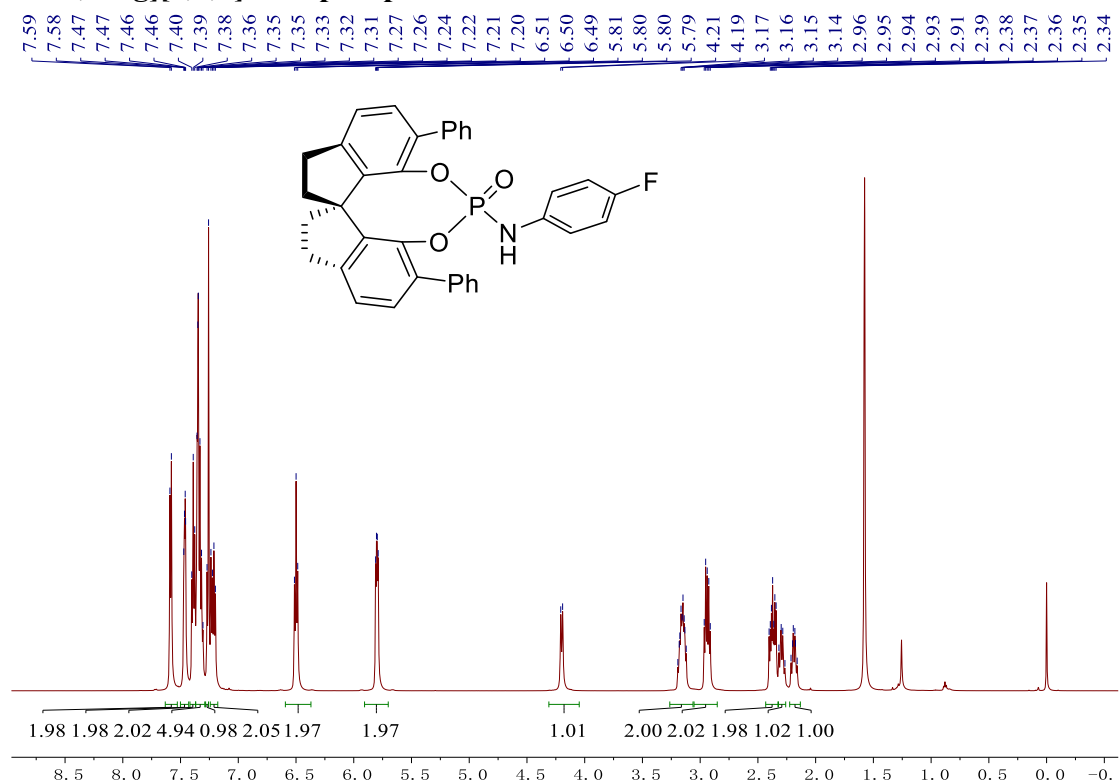
(*R*)-12-((3,5-bis(trifluoromethyl)phenyl)amino)-1,10-diphenyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



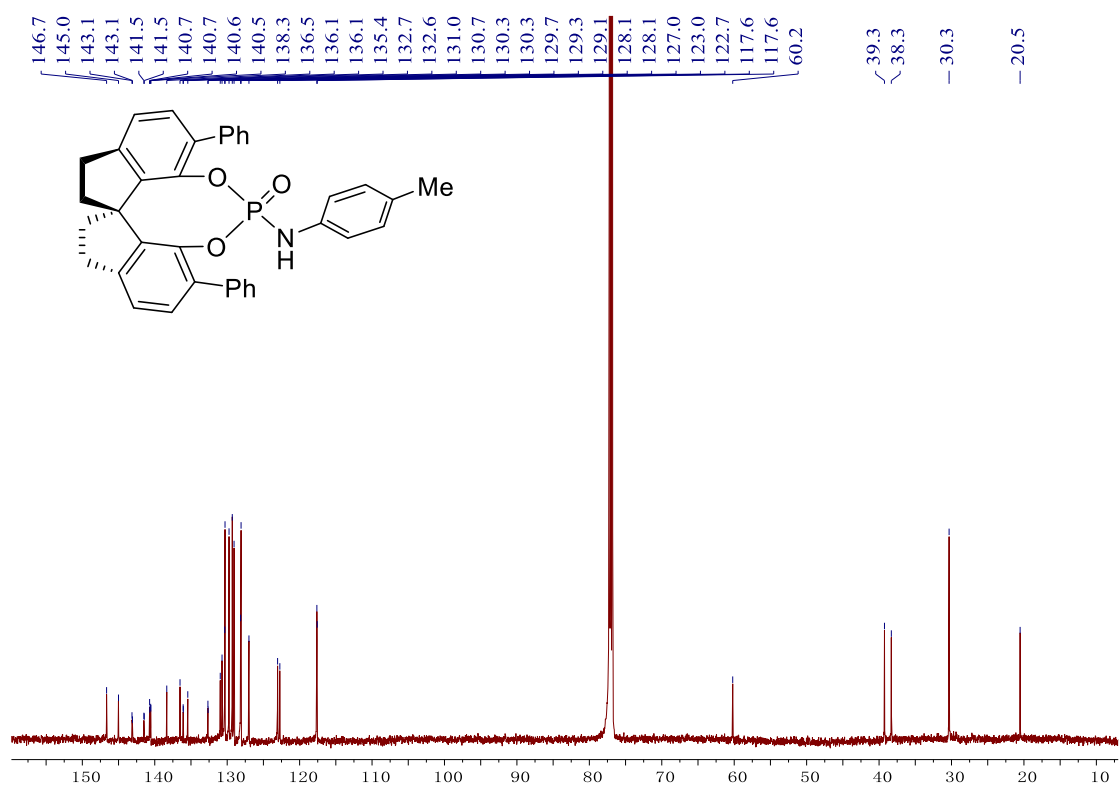
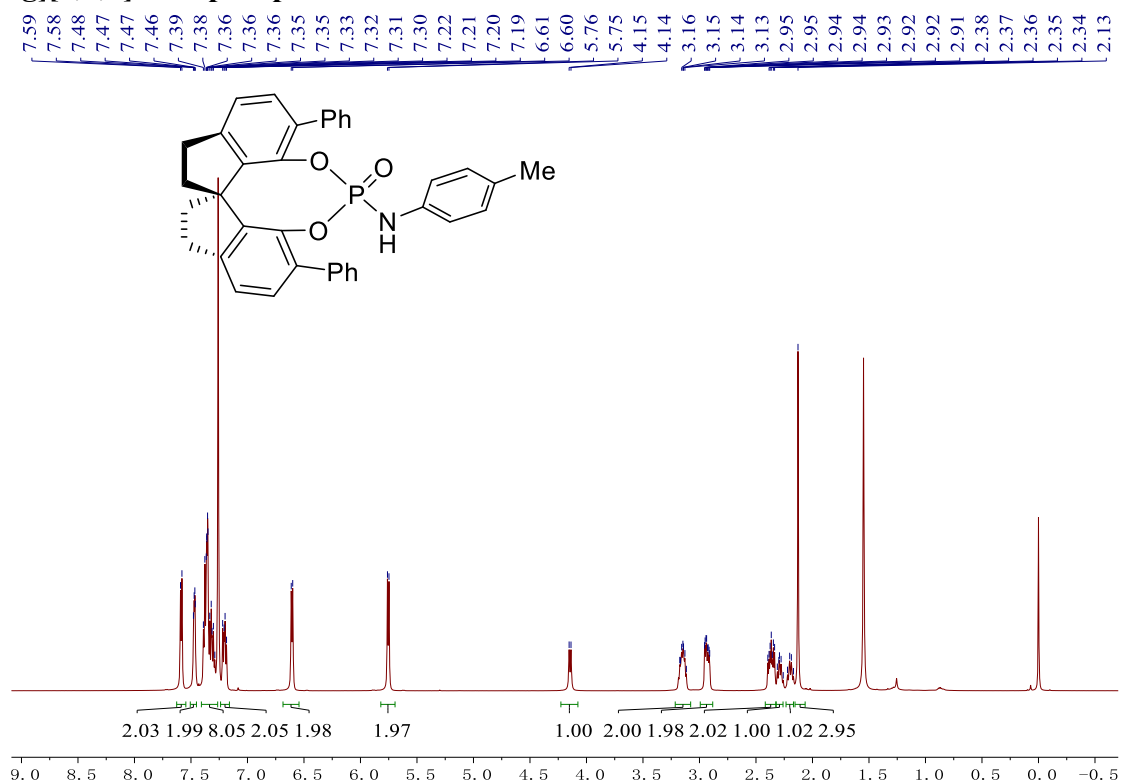
(*R*)-1,10-diphenyl-12-((4-(trifluoromethyl)phenyl)amino)-4,5,6,7-tetrahydroindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



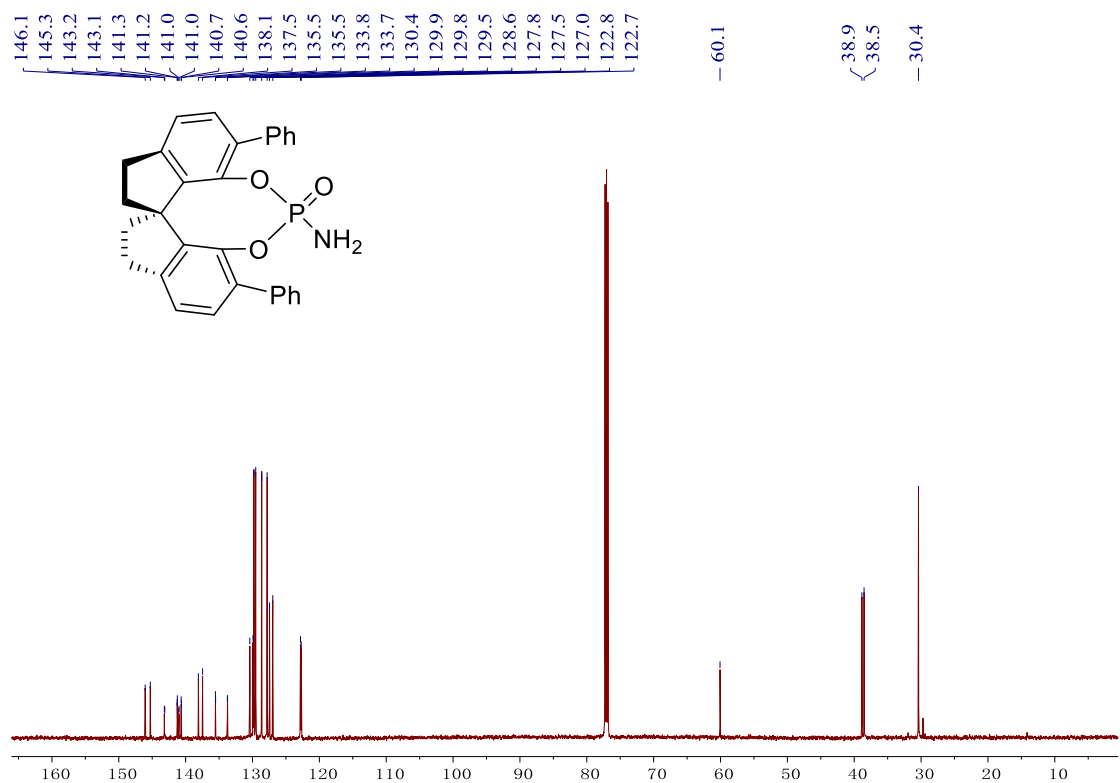
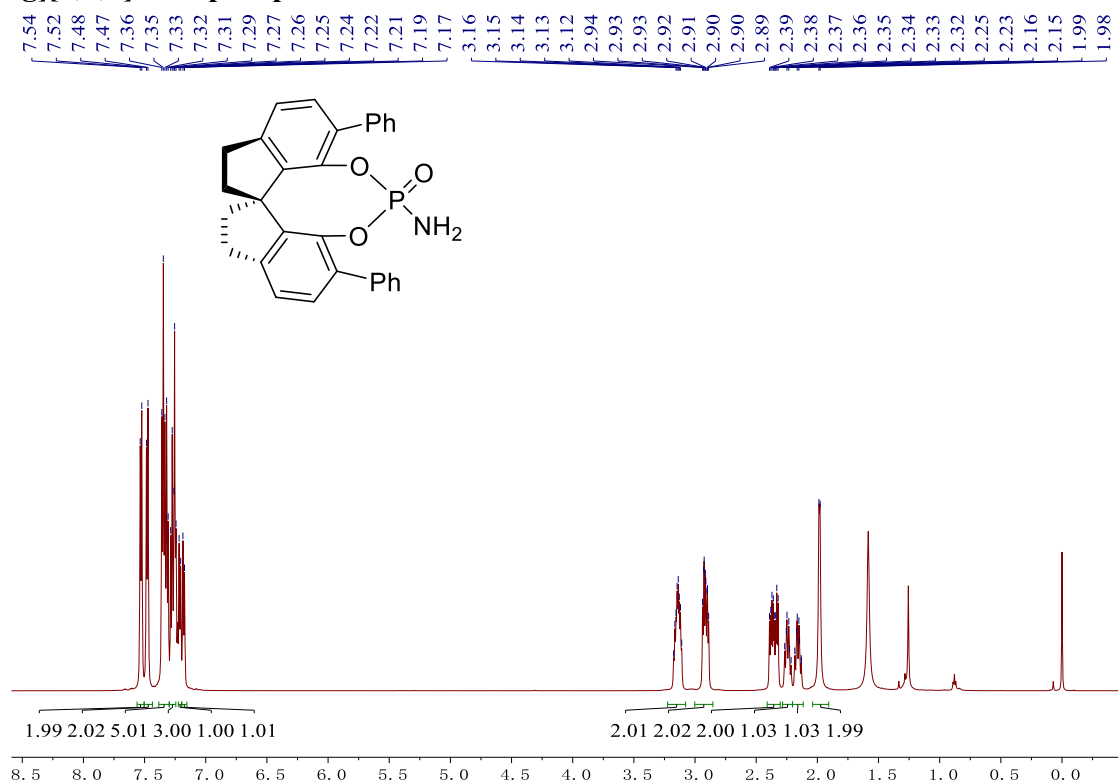
(*R*)-12-((4-fluorophenyl)amino)-1,10-diphenyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



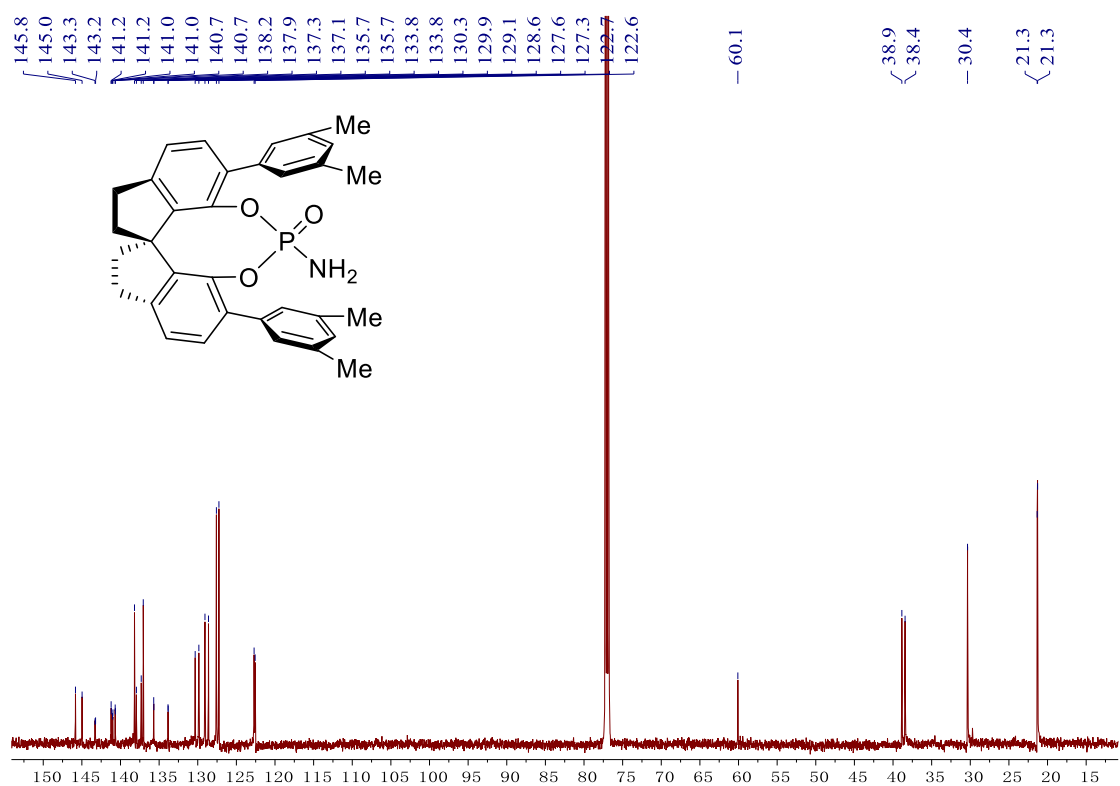
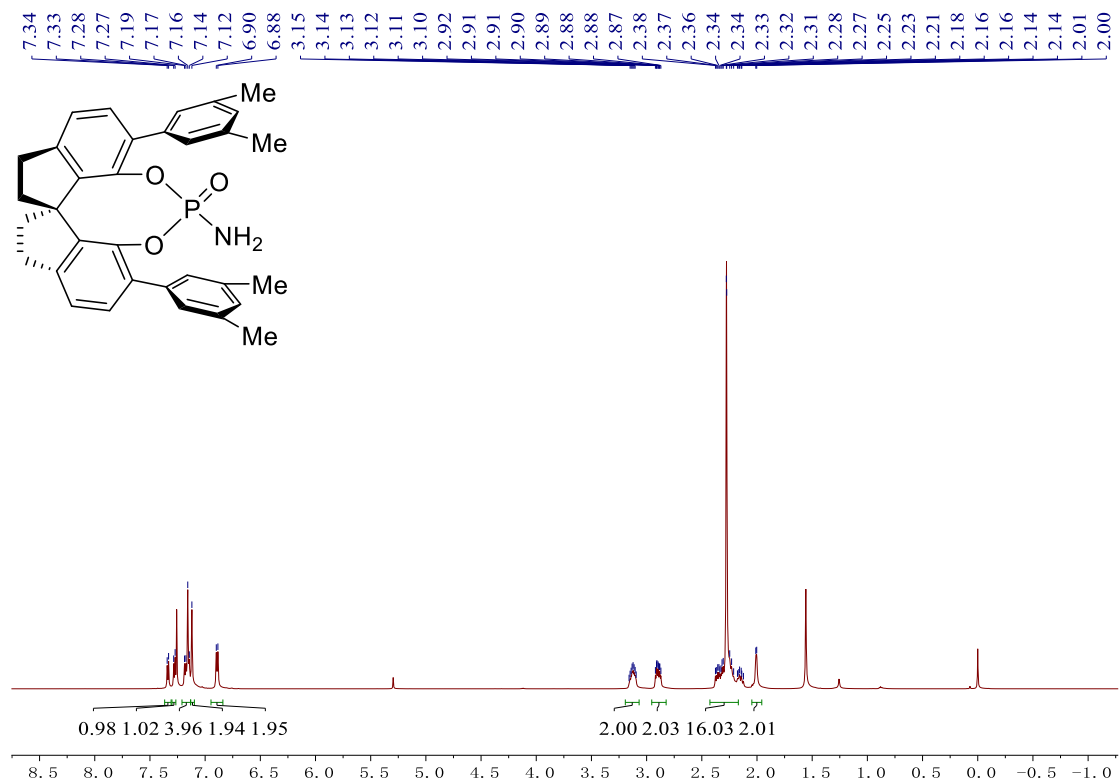
(*R*)-1,10-diphenyl-12-(*p*-tolylamino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



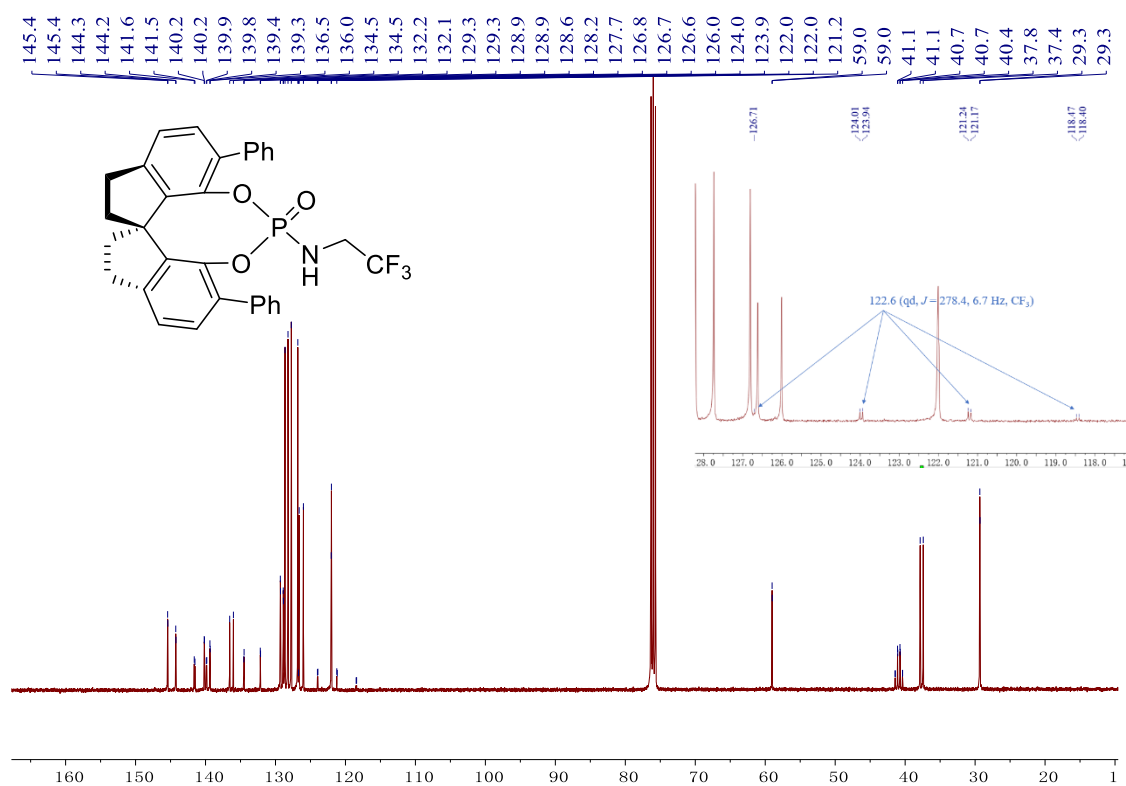
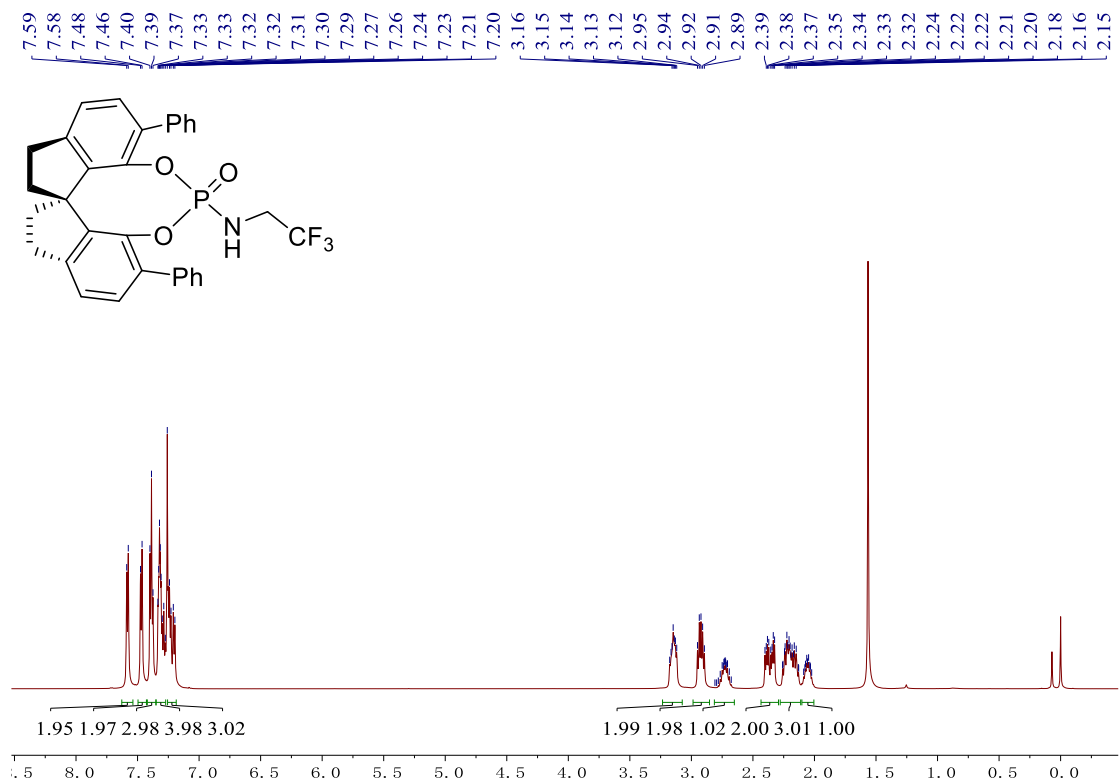
(R)-12-amino-1,10-diphenyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



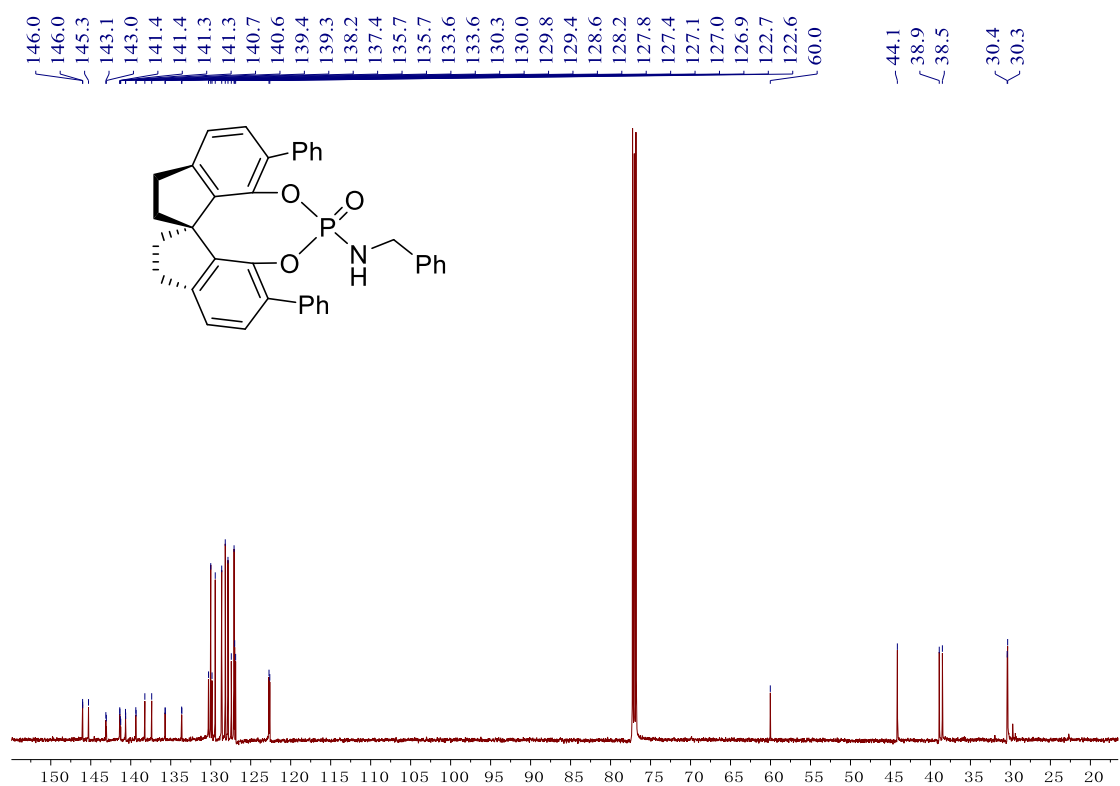
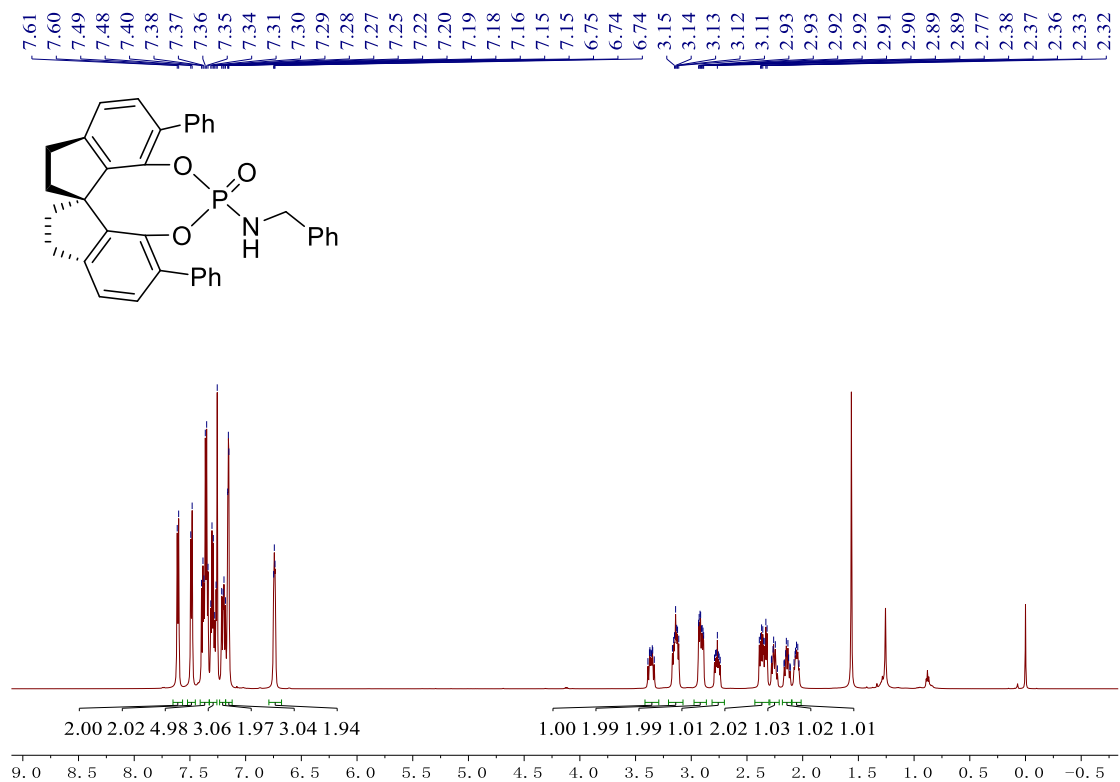
(R)-12-amino-1,10-bis(3,5-dimethylphenyl)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



(*R*)-1,10-diphenyl-12-((2,2,2-trifluoroethyl)amino)-4,5,6,7-tetrahydroindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



(*R*)-12-(benzylamino)-1,10-diphenyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



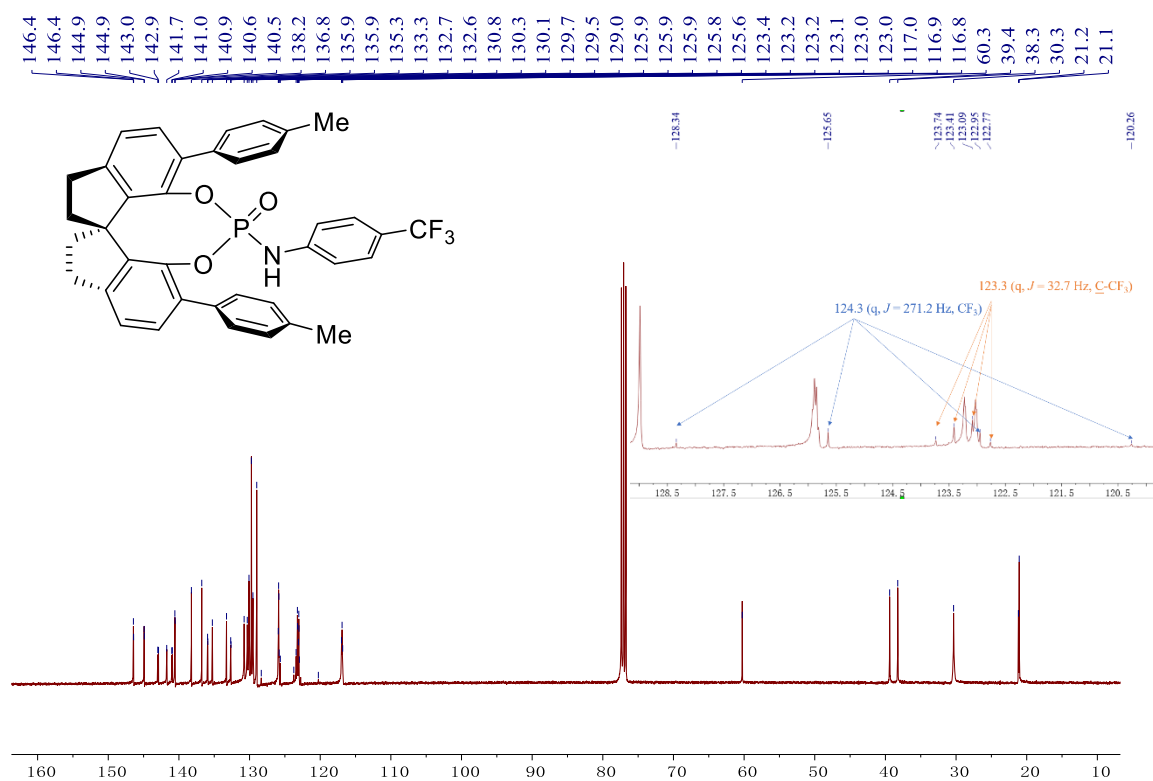
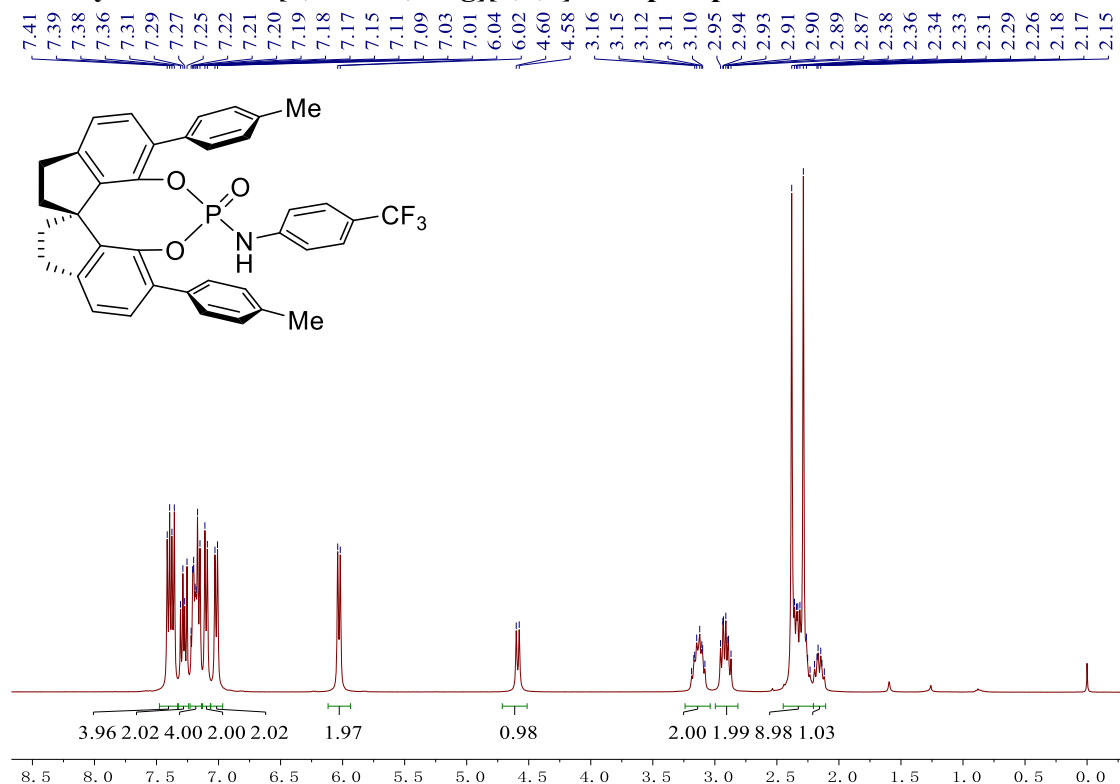
Chemical structure of (S)-1,1'-bis(2-phenylphosphoryl)spiro[3.3]heptane is shown. The ^1H NMR spectrum (400 MHz, CDCl_3) displays peaks corresponding to the structure, with chemical shifts (ppm) labeled above the peaks and integration values below the baseline.

Chemical shifts (ppm): 7.59, 7.57, 7.49, 7.47, 7.36, 7.35, 7.34, 7.32, 7.31, 7.30, 7.29, 7.28, 7.27, 7.20, 7.18, 7.16, 3.15, 3.14, 3.13, 3.12, 3.11, 3.10, 2.93, 2.92, 2.90, 2.89, 2.38, 2.37, 2.36, 2.35, 2.31, 2.30, 2.24, 2.16, 0.95, 0.94, 0.94, 0.93, 0.92, 0.92, 0.91, 0.52, 0.50, 0.49.

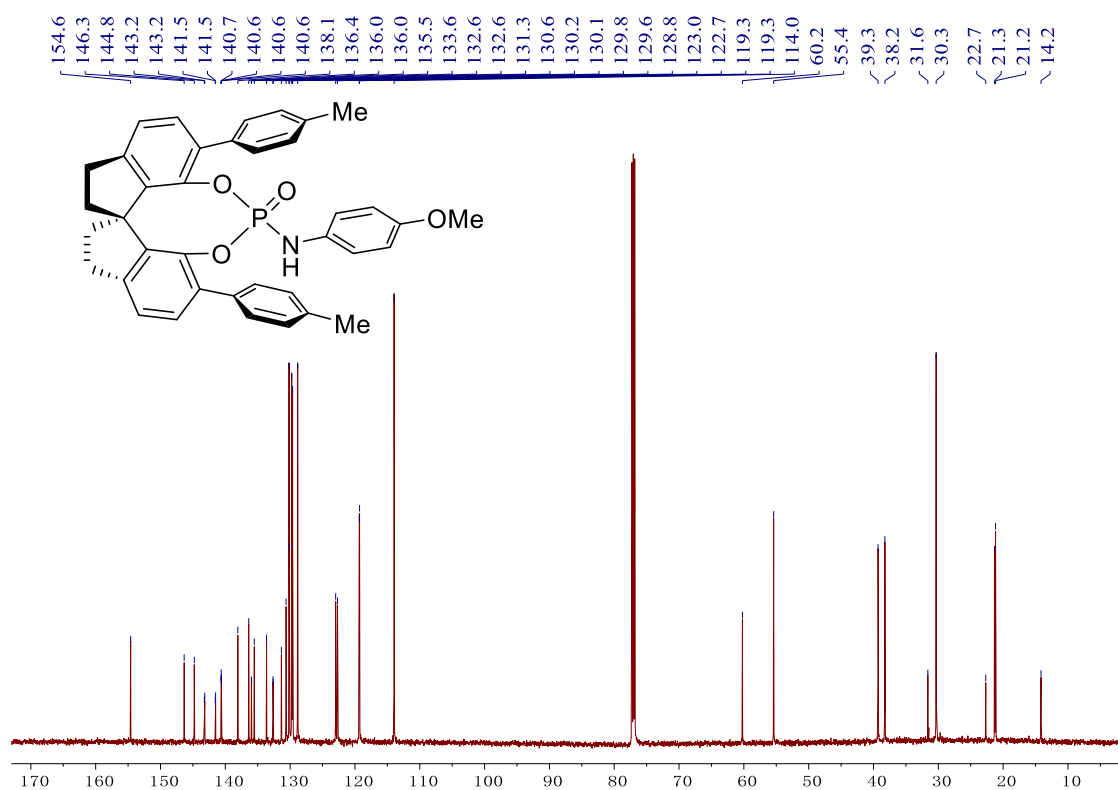
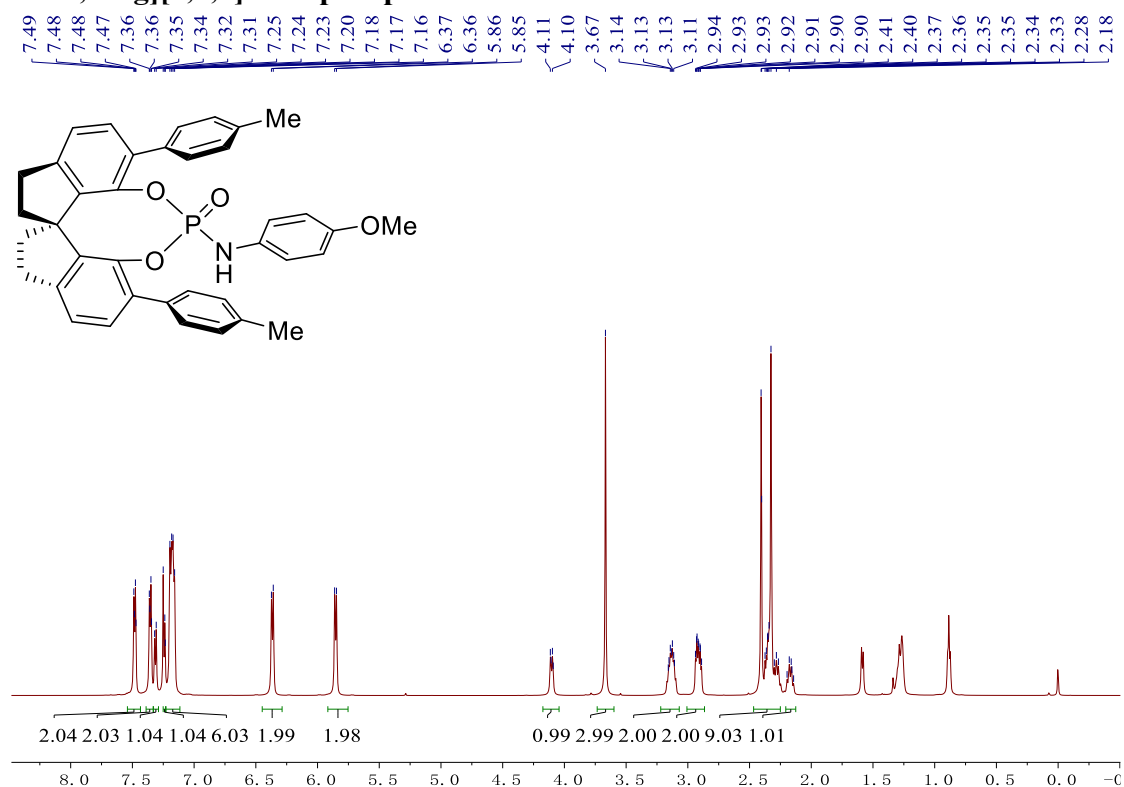
Integration values: 1.99, 1.99, 3.02, 3.00, 2.04, 1.98, 2.00, 1.99, 1.02, 1.01, 1.03, 1.02, 1.02, 1.02, 1.03, 3.06, 2.98.



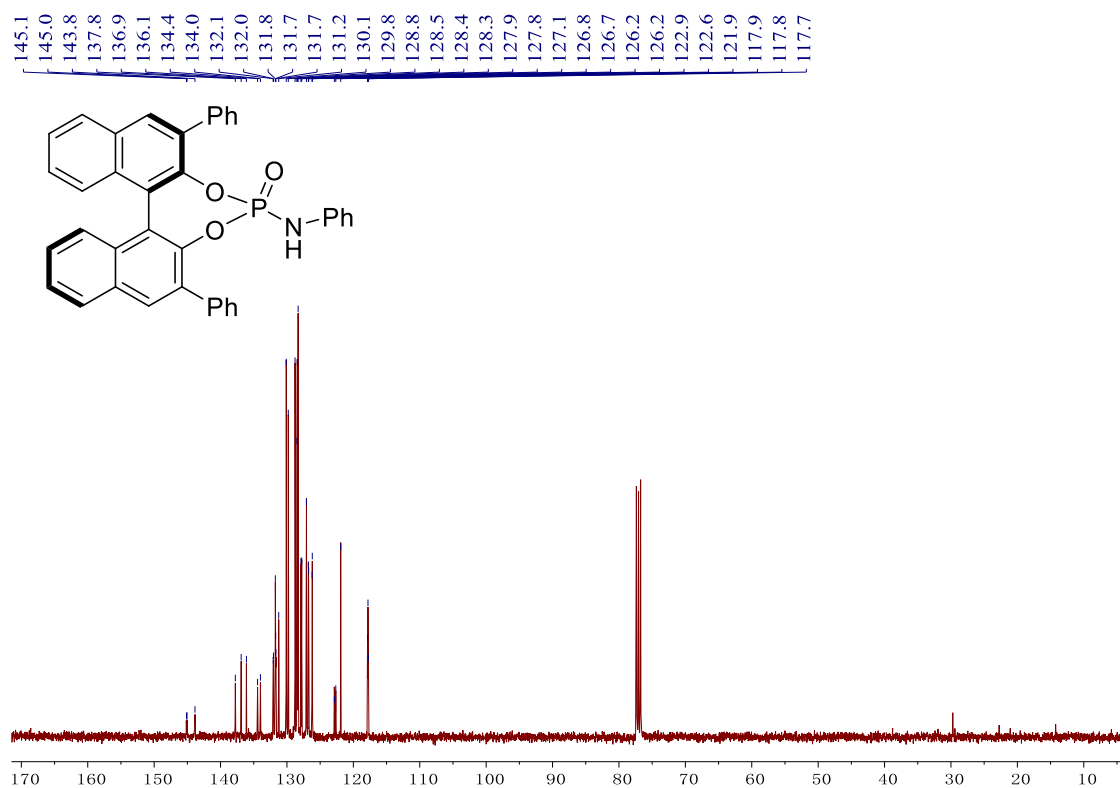
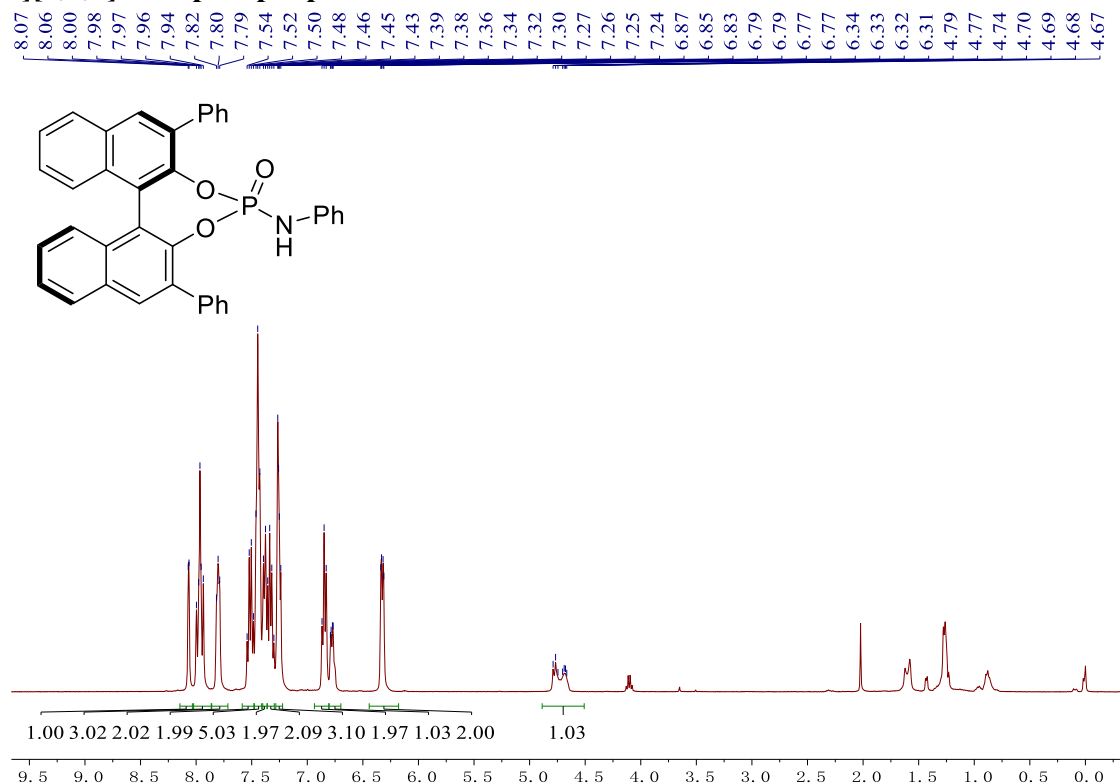
(*R*)-1,10-di-*p*-tolyl-12-((4-(trifluoromethyl)phenyl)amino)-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



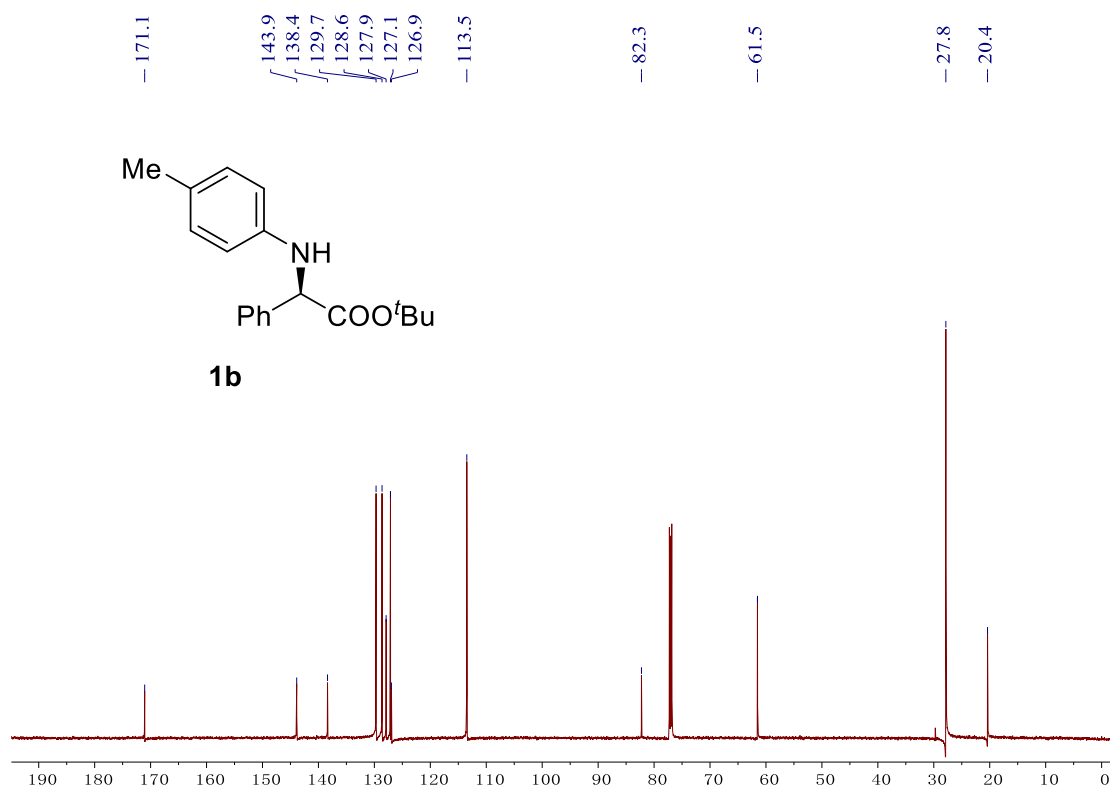
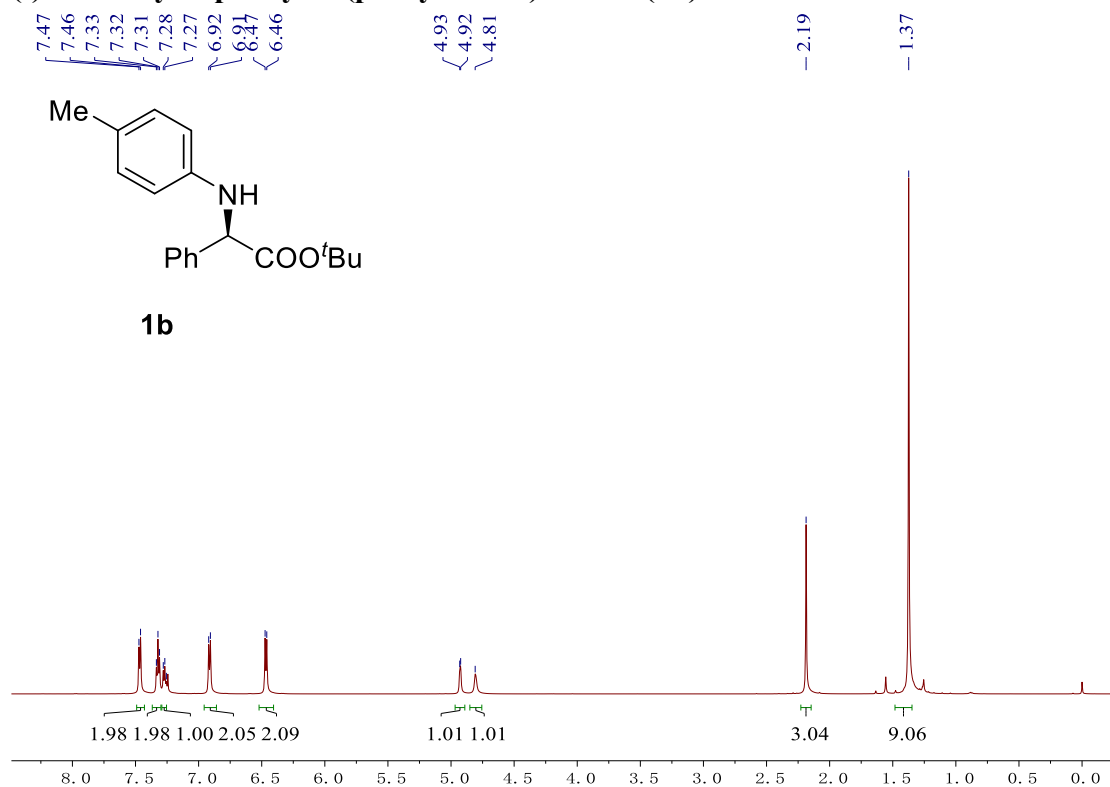
(*R*)-12-((4-methoxyphenyl)amino)-1,10-di-*p*-tolyl-4,5,6,7-tetrahydrodiindeno[7,1-de:1',7'-fg][1,3,2]dioxaphosphocine 12-oxide



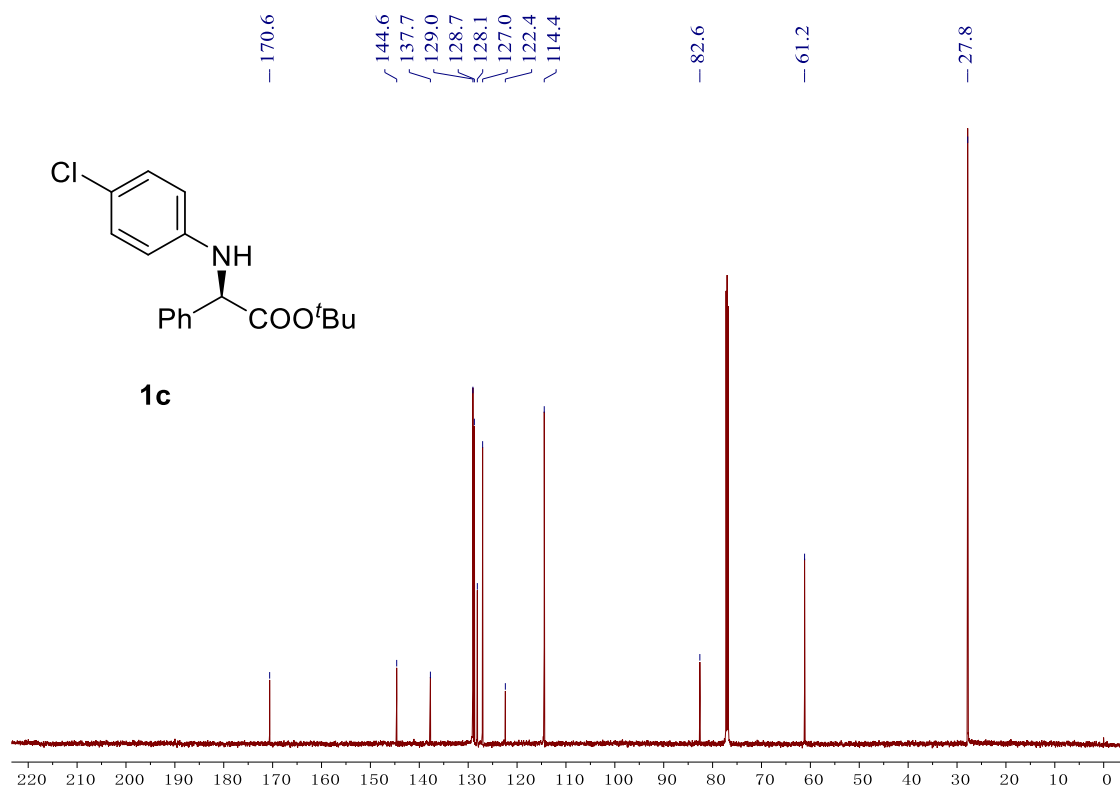
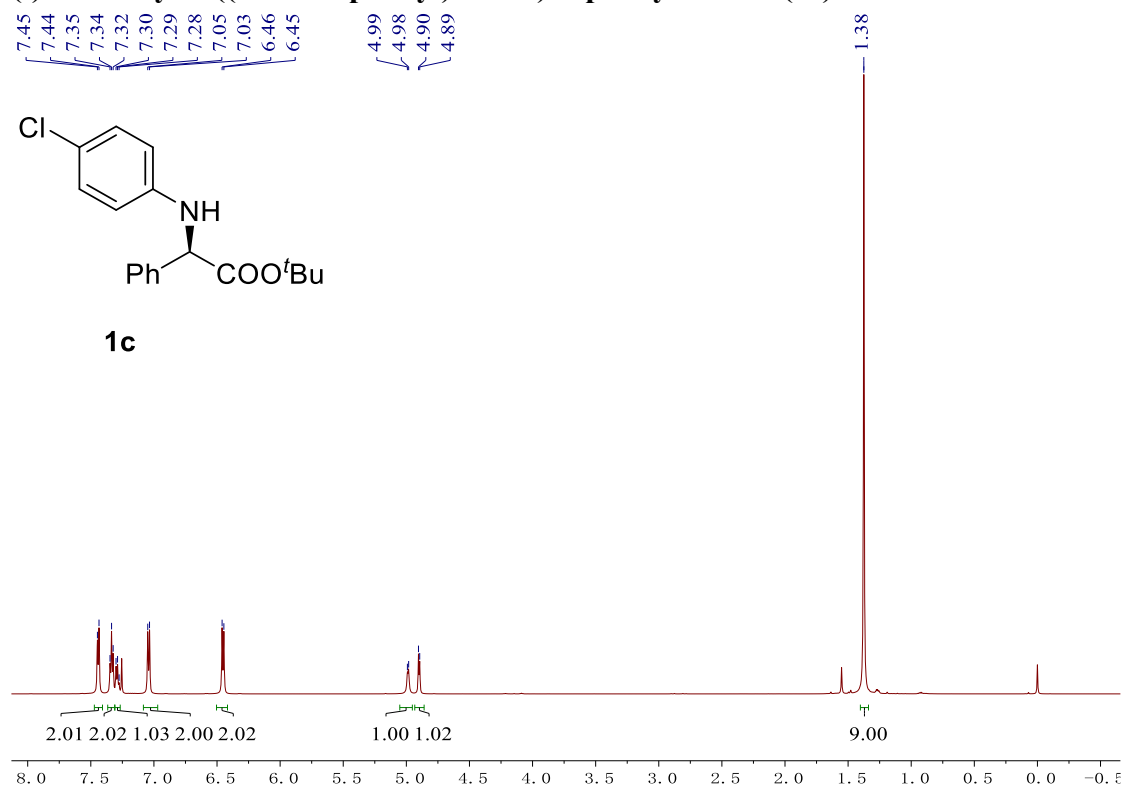
(*R*)-2,6-diphenyl-4-(phenylamino)dinaphtho[2,1-d:1',2'-f][1,3,2]dioxaphosphine 4-oxide



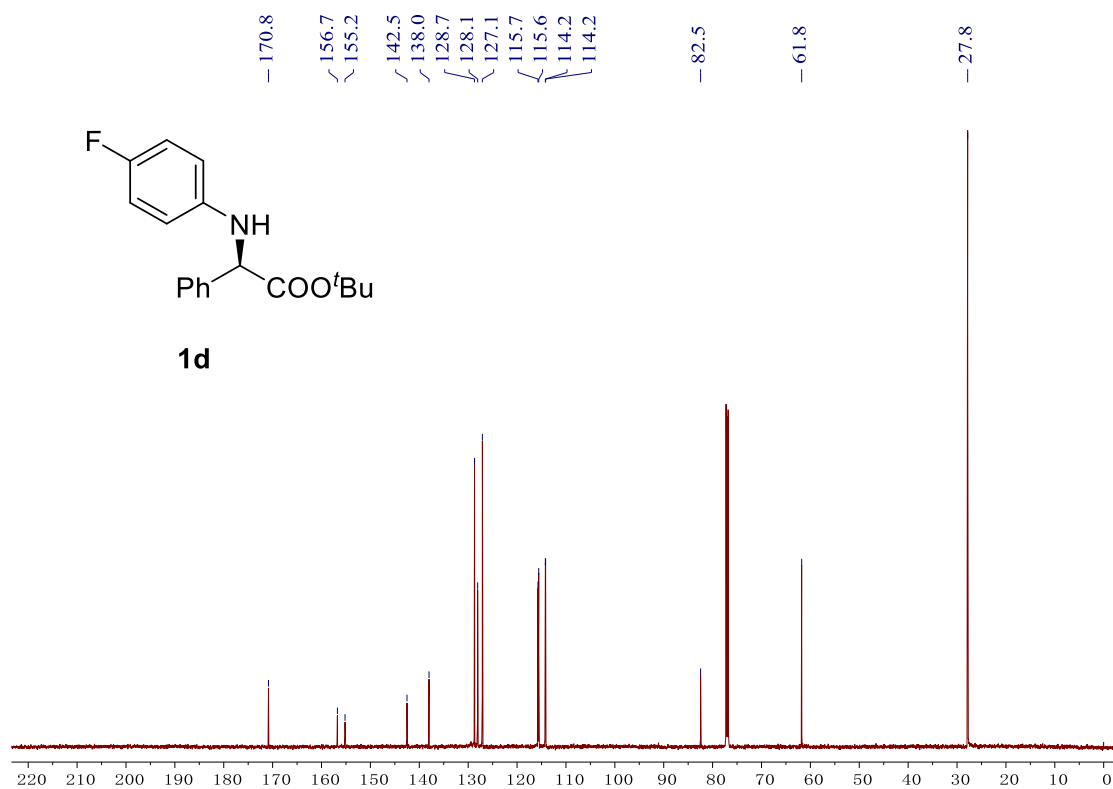
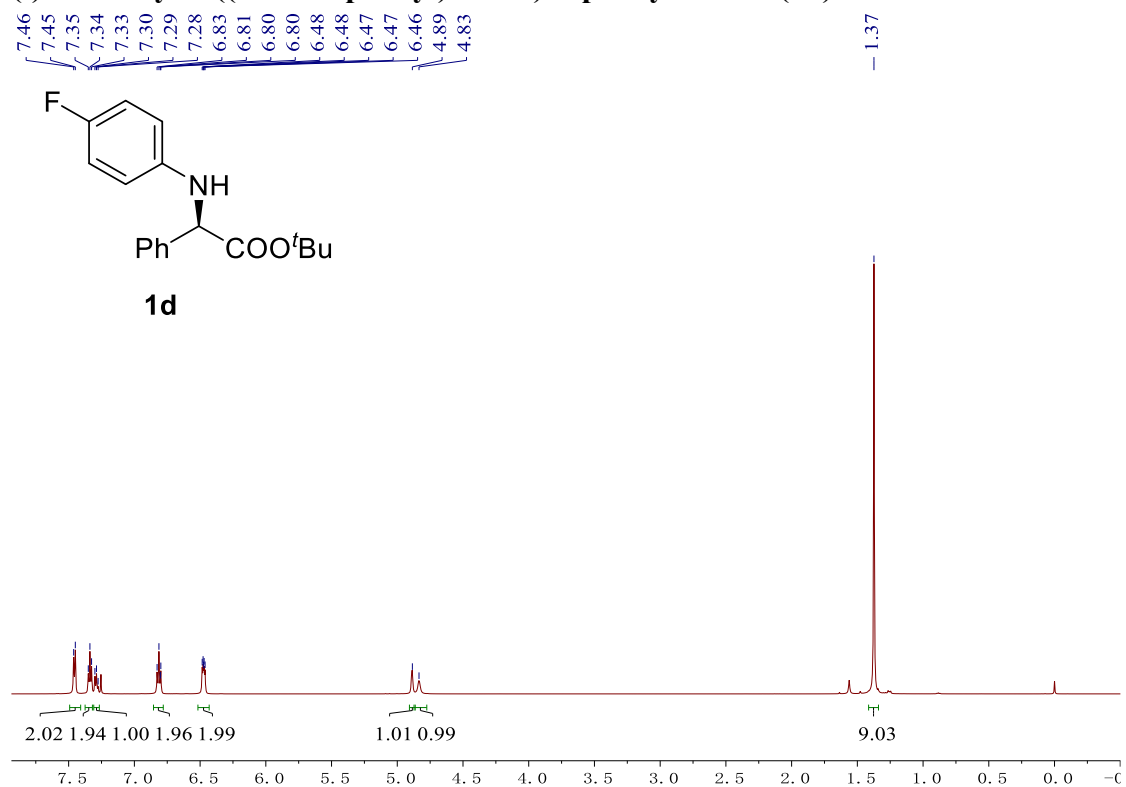
(-)-*tert*-butyl 2-phenyl-2-(*p*-tolylamino)acetate (1b**)**



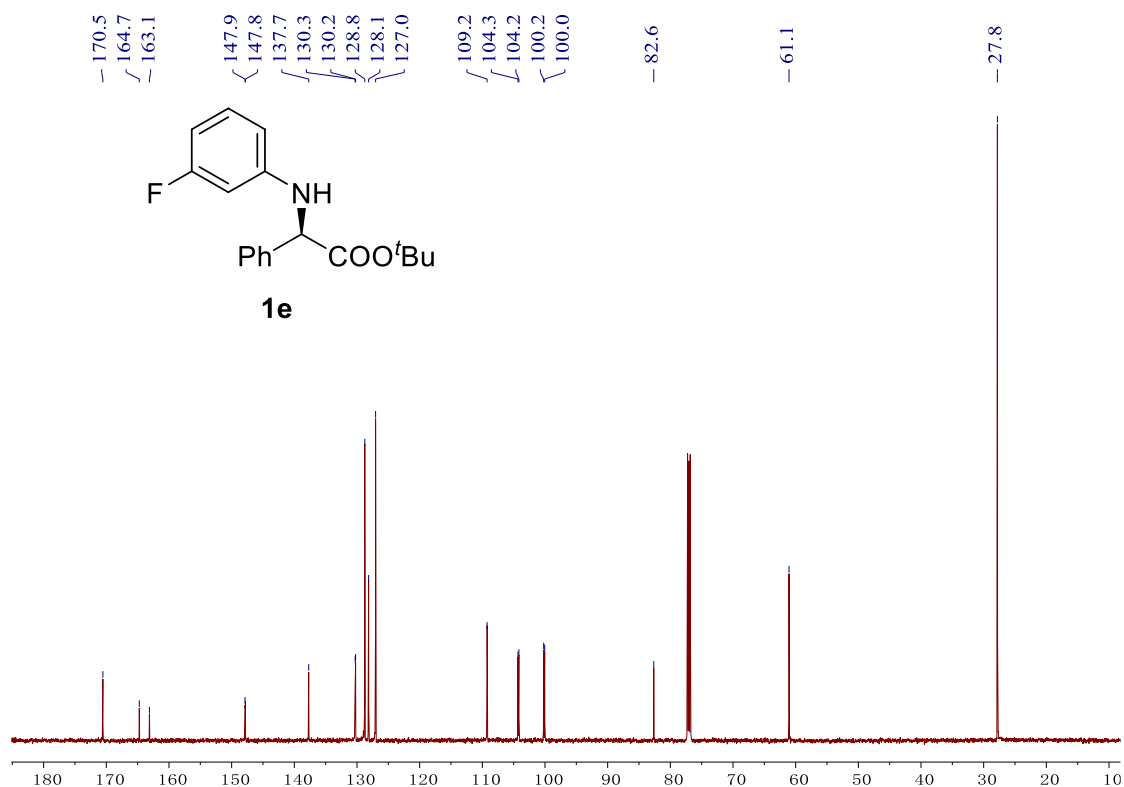
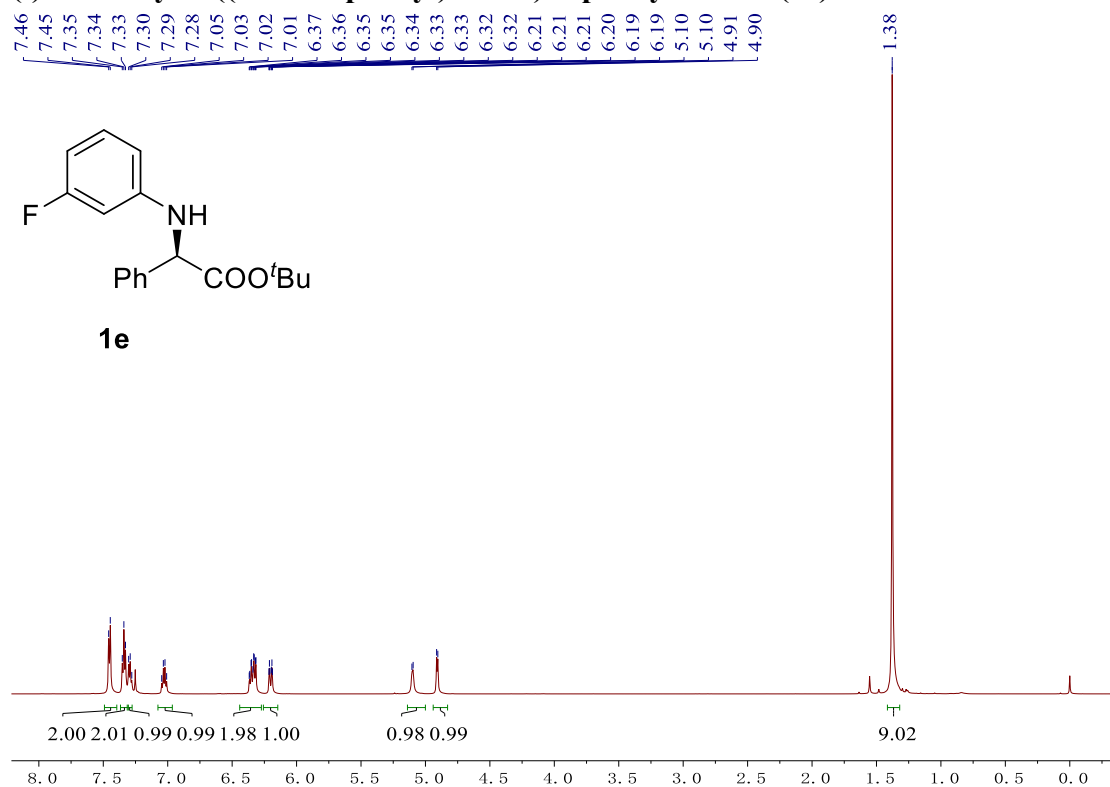
(-)-*tert*-butyl 2-((4-chlorophenyl)amino)-2-phenylacetate (1c)



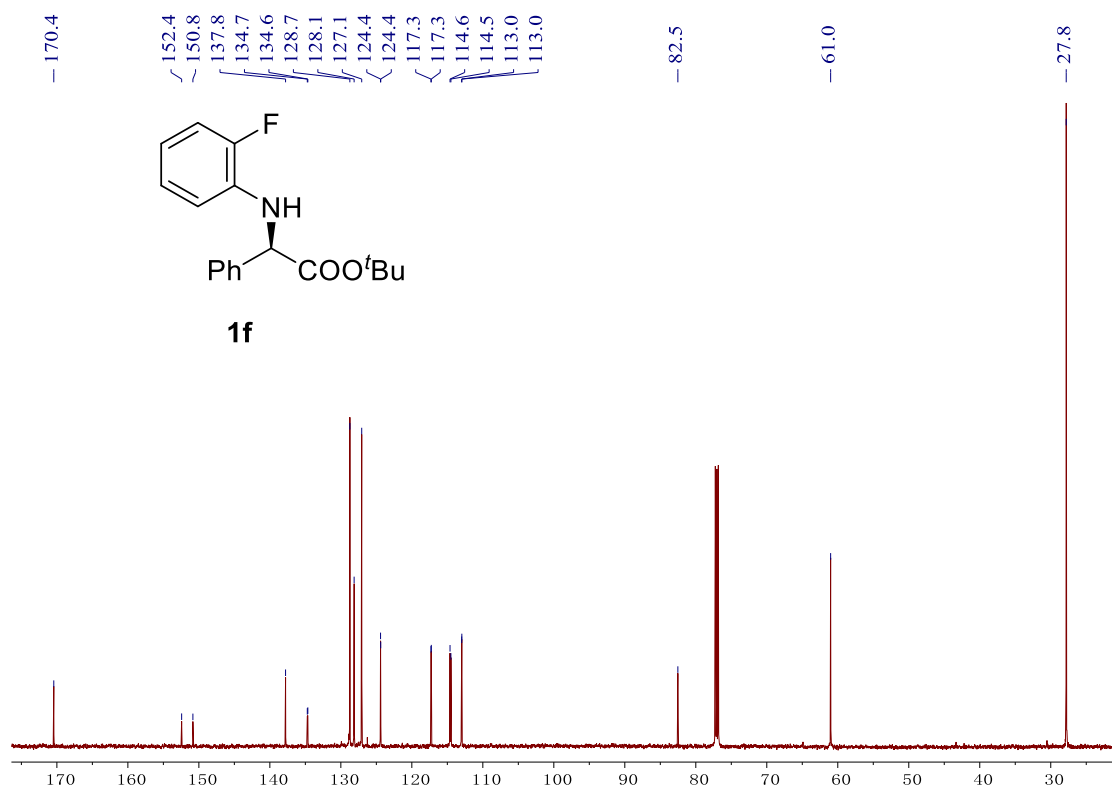
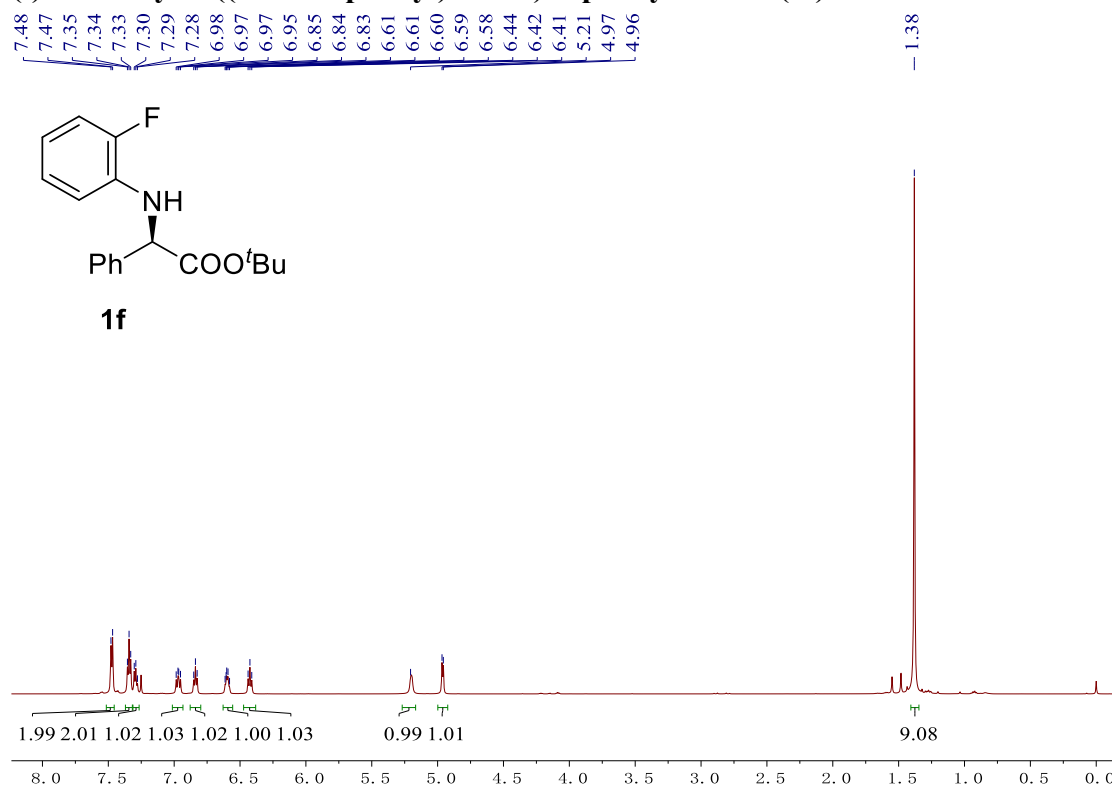
(-)-*tert*-butyl 2-((4-fluorophenyl)amino)-2-phenylacetate (1d)



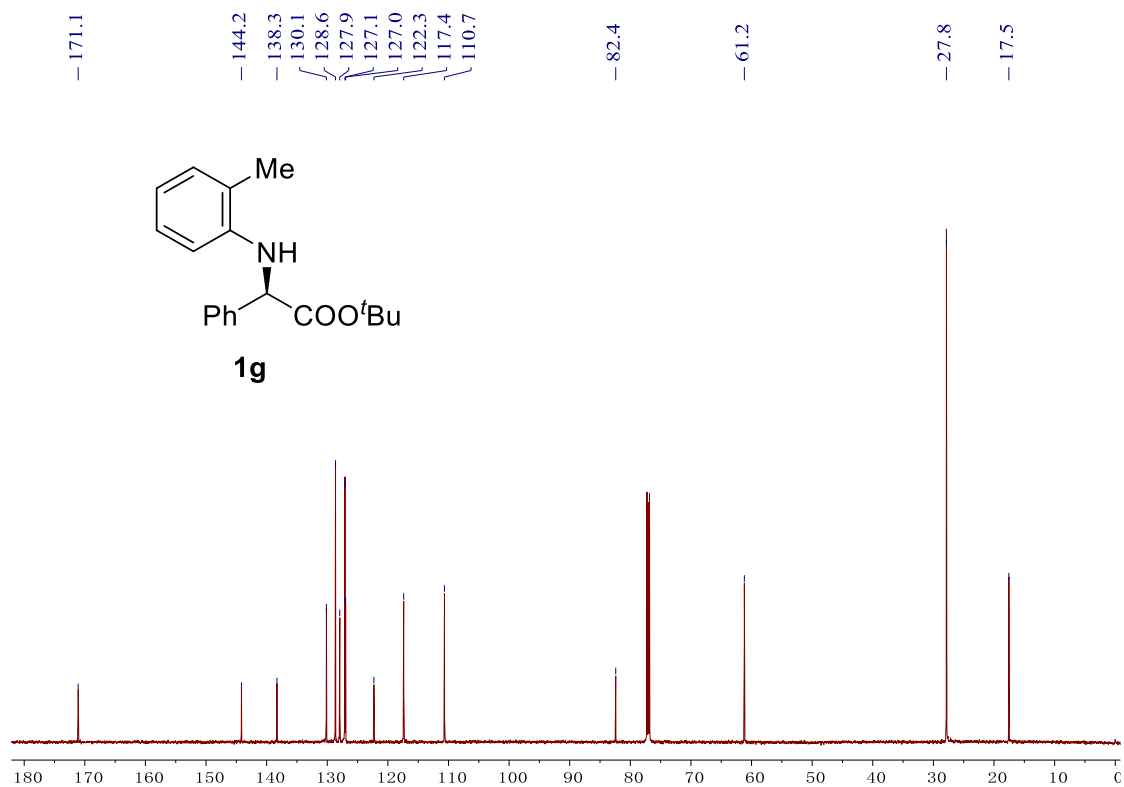
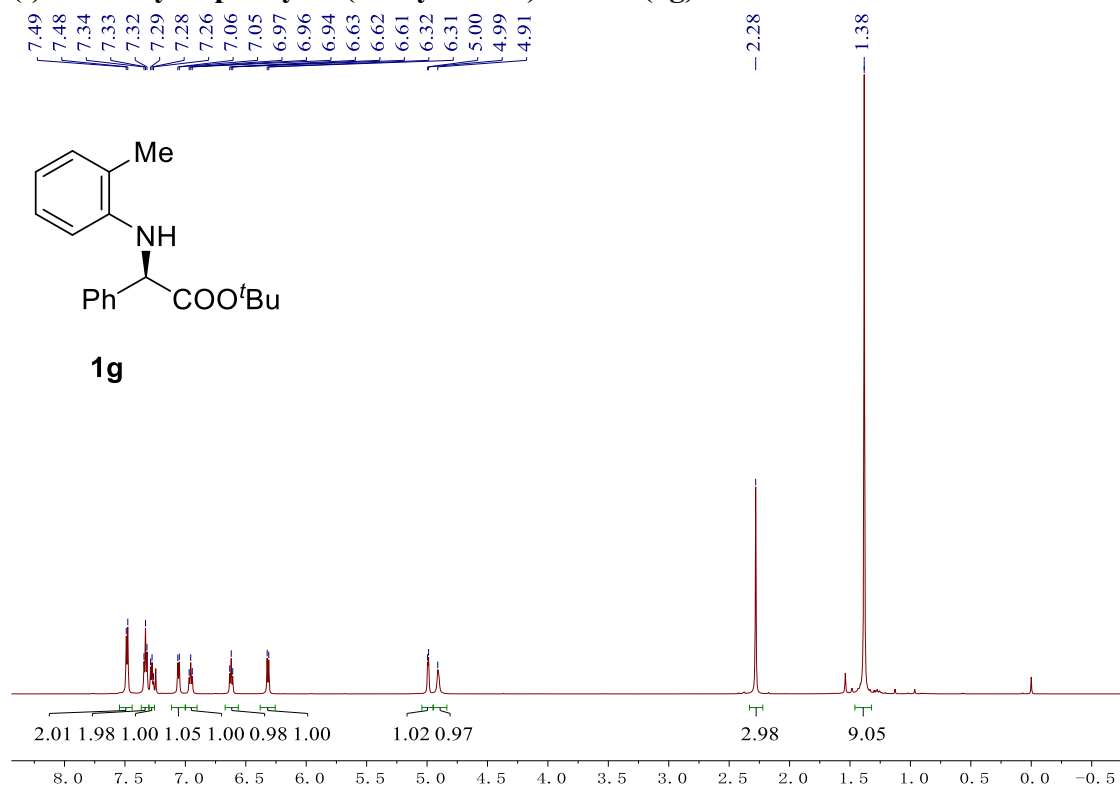
(-)-*tert*-butyl 2-((3-fluorophenyl)amino)-2-phenylacetate (1e)



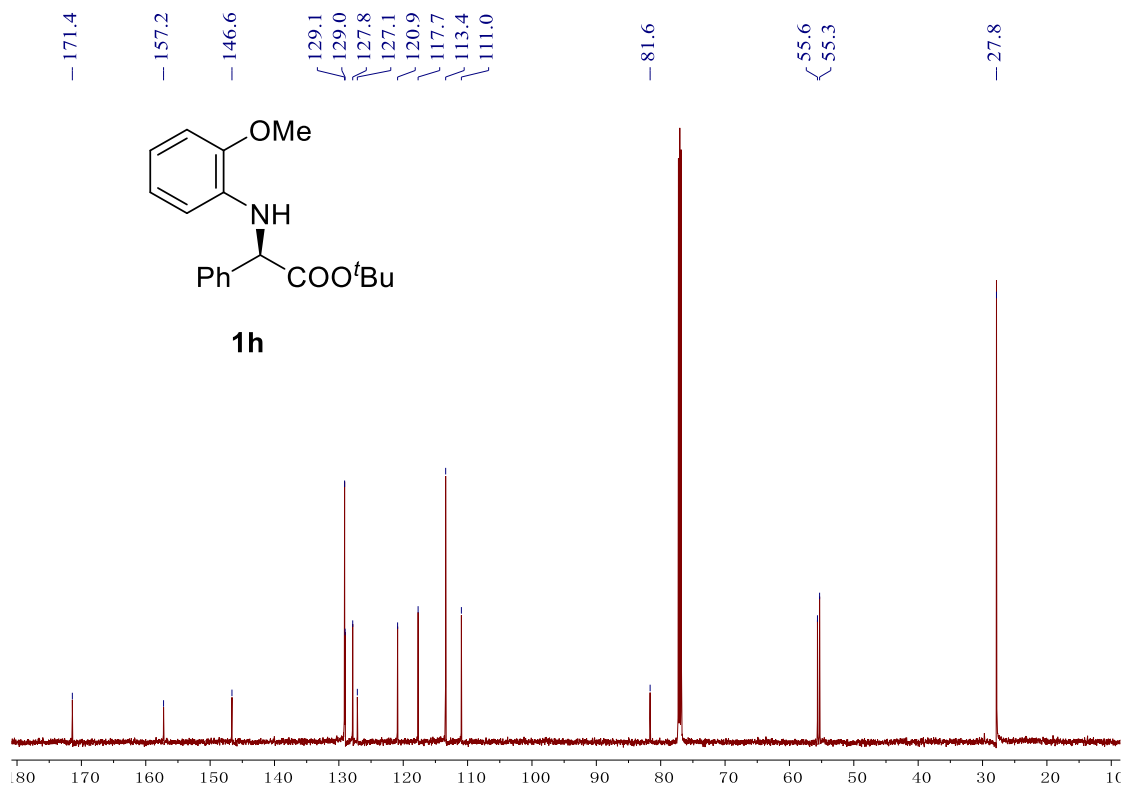
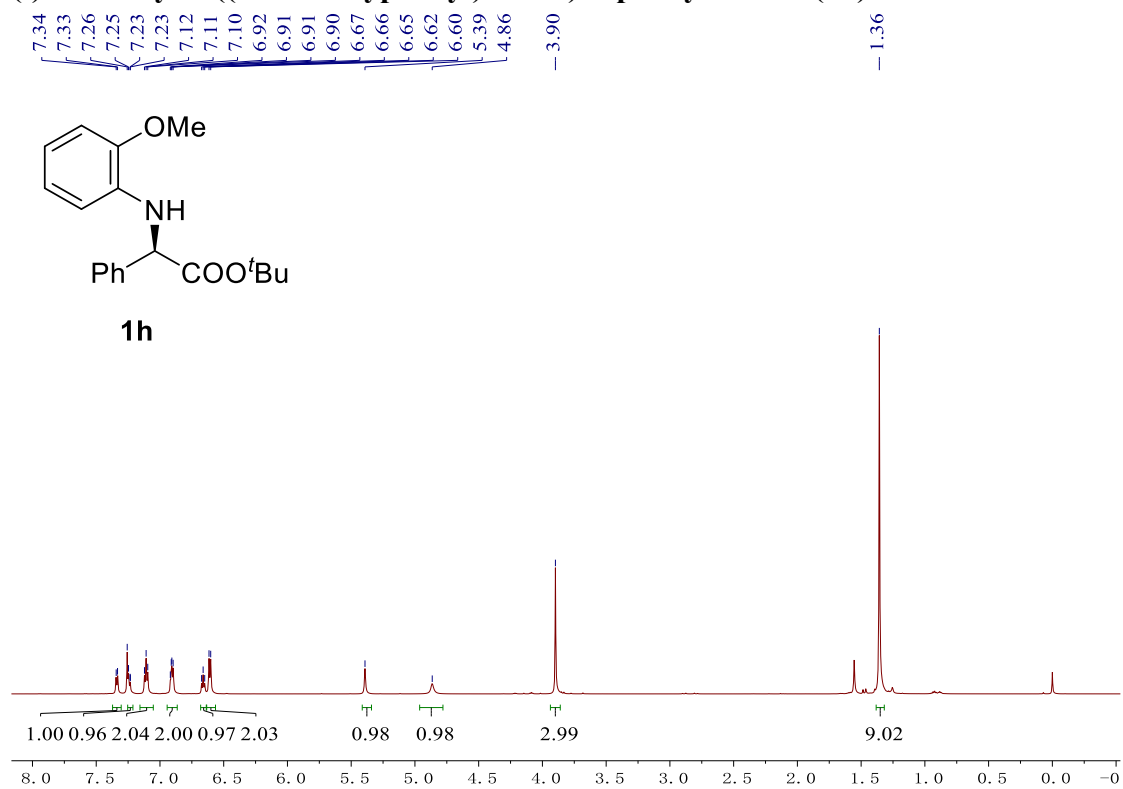
(-)-*tert*-butyl 2-((2-fluorophenyl)amino)-2-phenylacetate (1f)



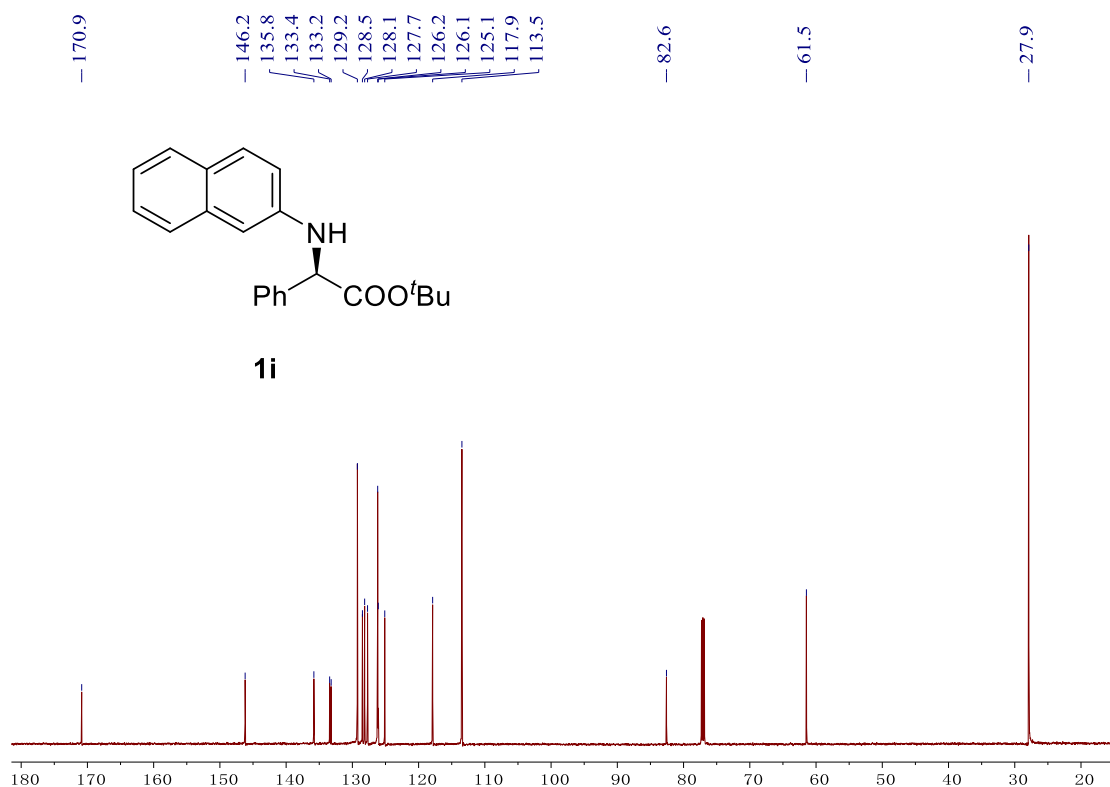
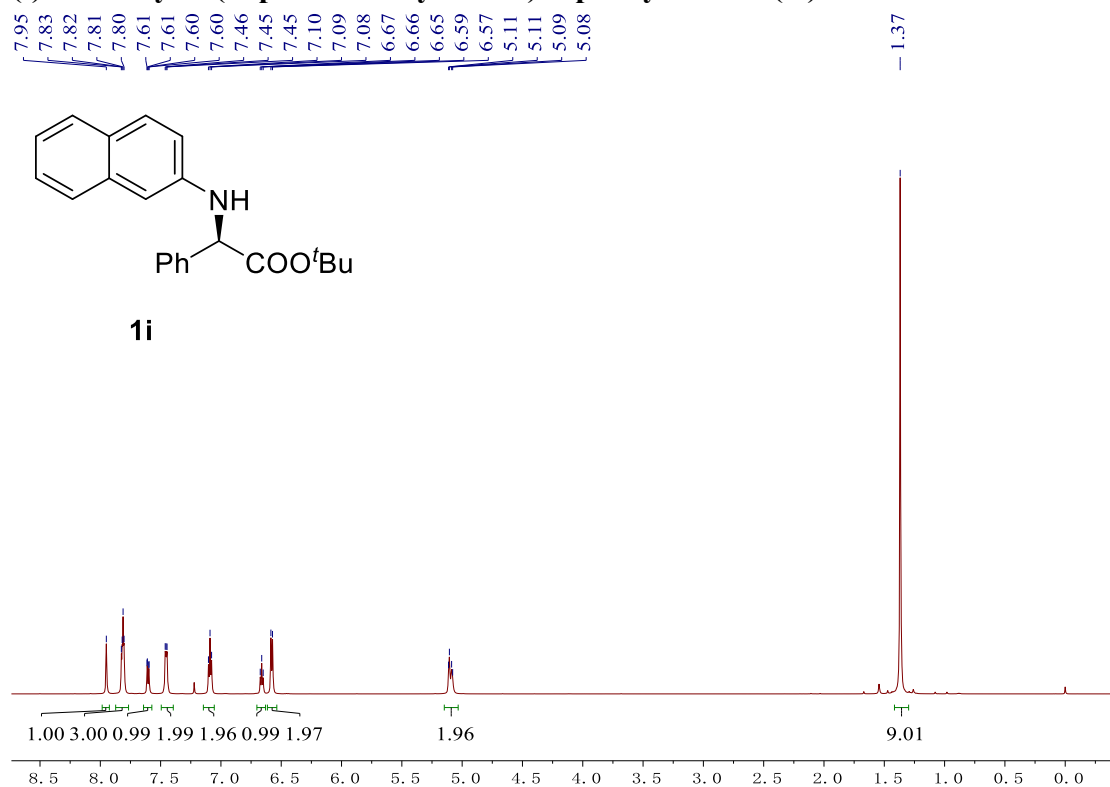
(-)-*tert*-butyl 2-phenyl-2-(*o*-tolylamino)acetate (1g)



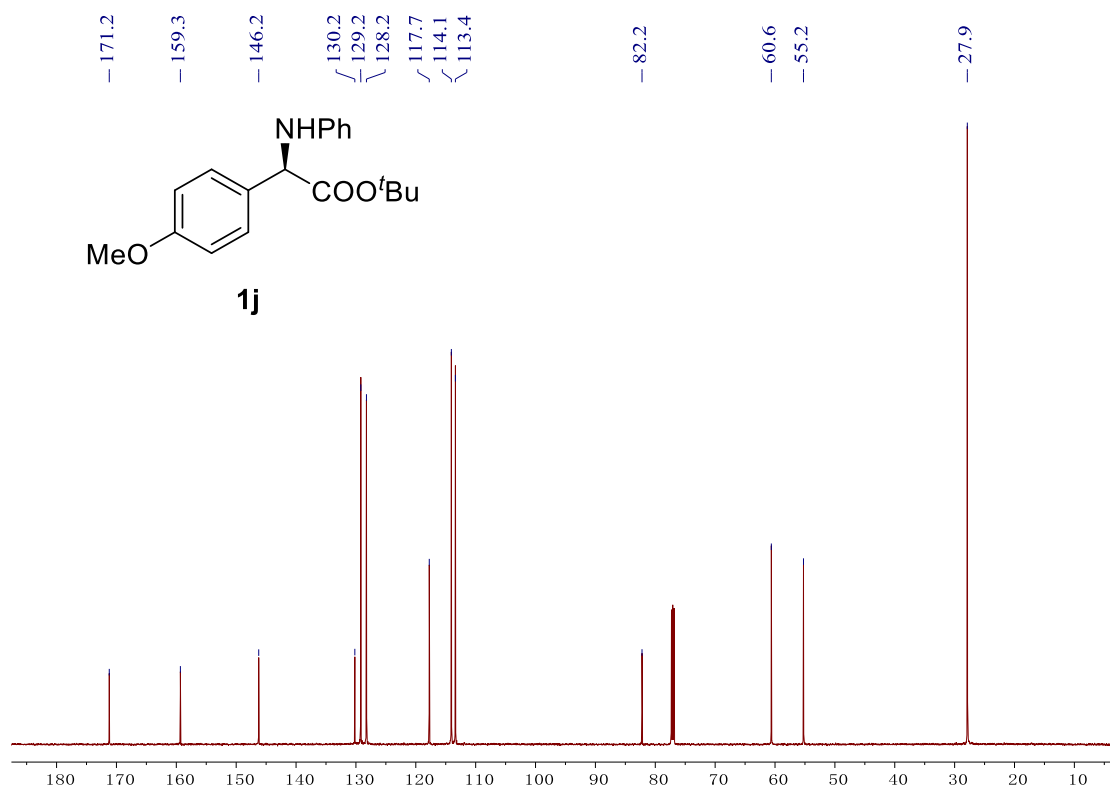
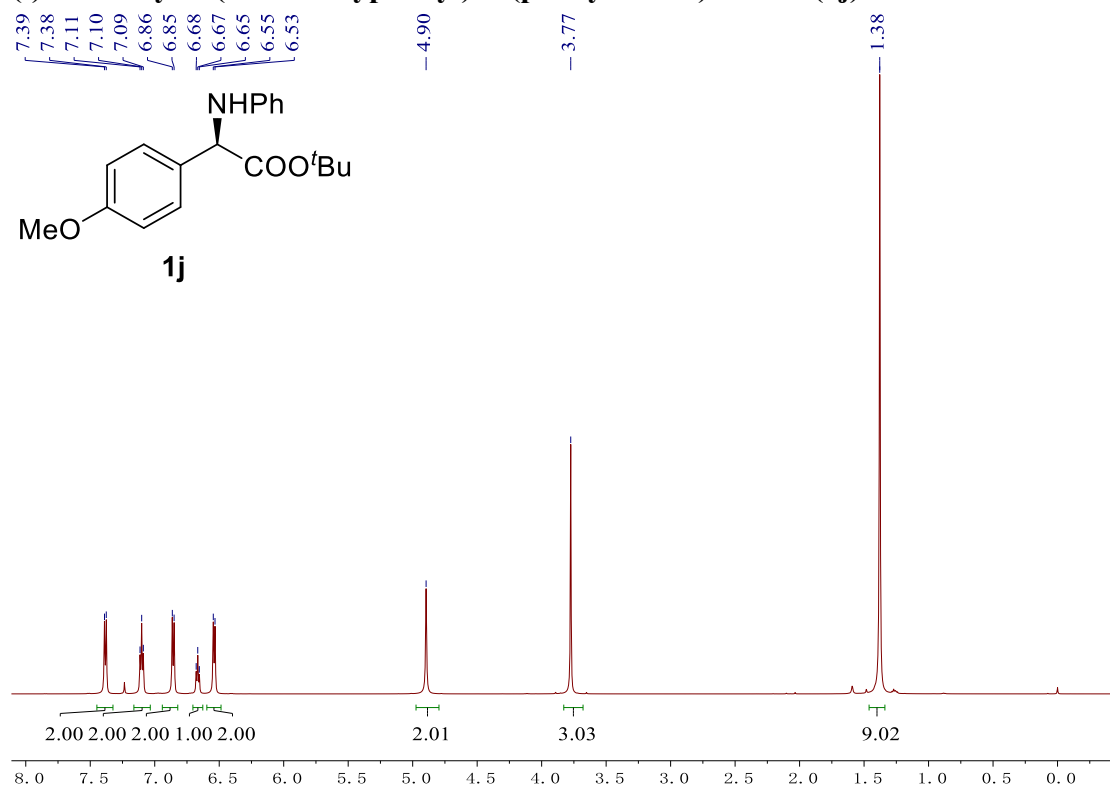
(-)-*tert*-butyl 2-((2-methoxyphenyl)amino)-2-phenylacetate (1h)



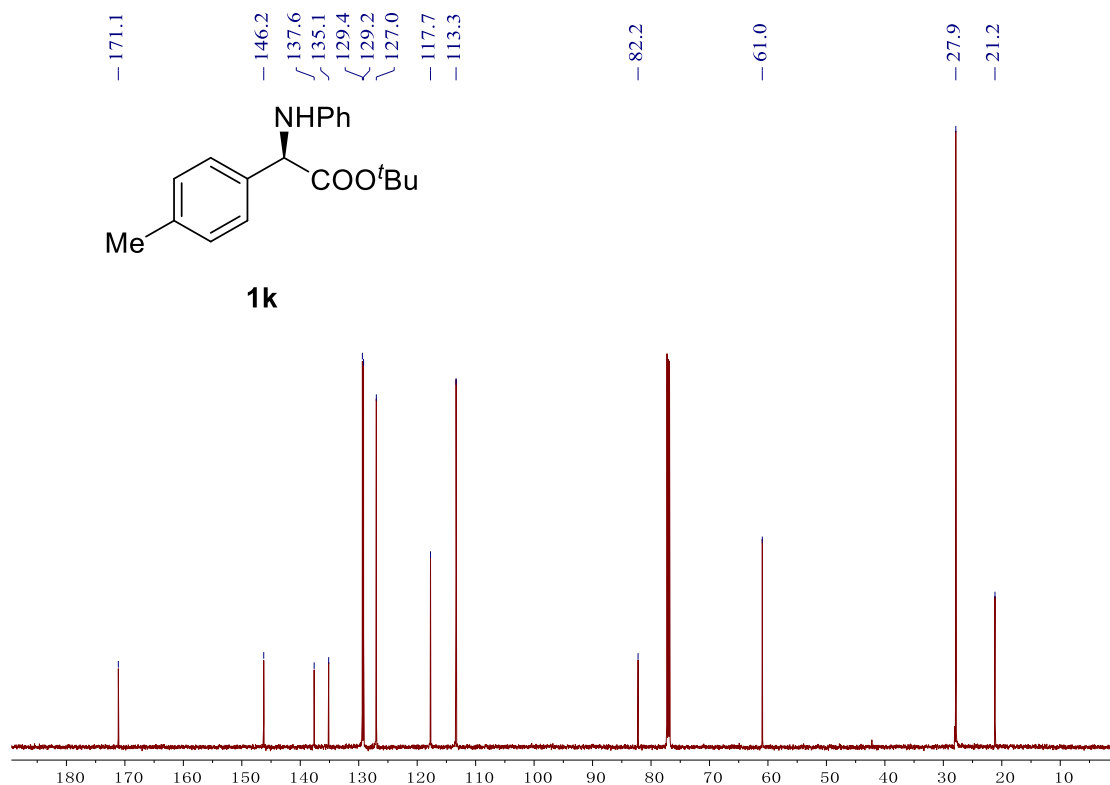
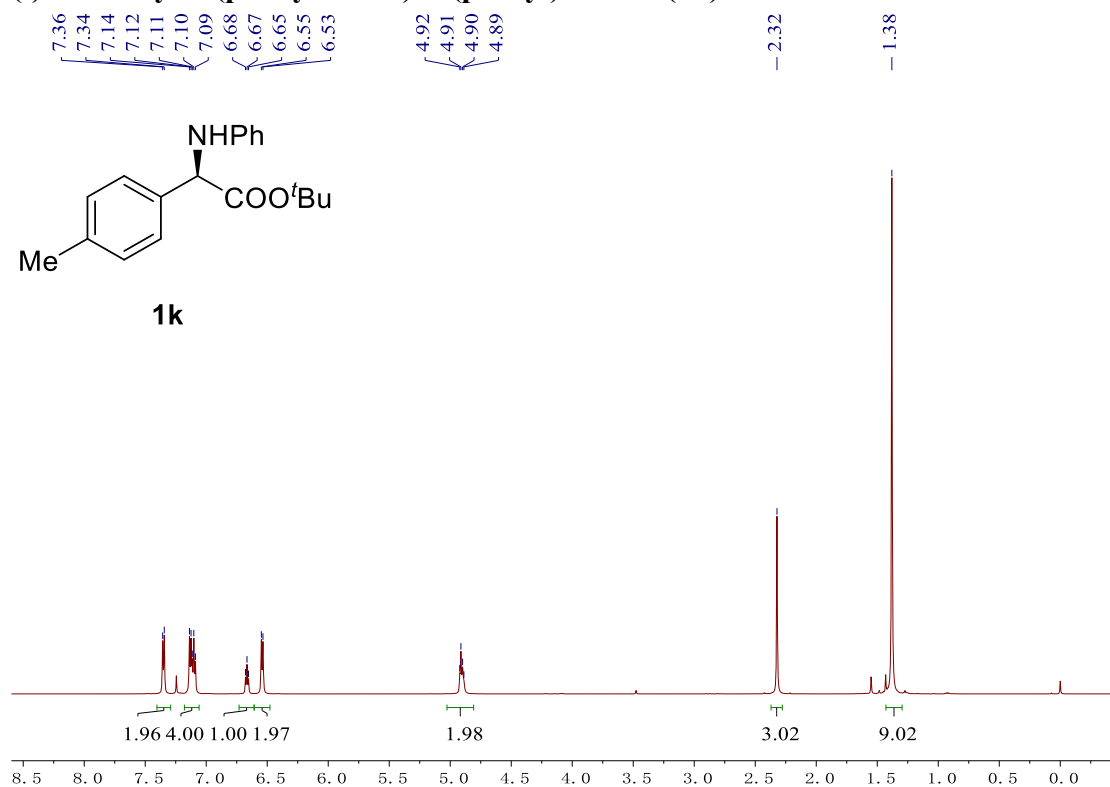
(-)-*tert*-butyl 2-(naphthalen-2-ylamino)-2-phenylacetate (1i)



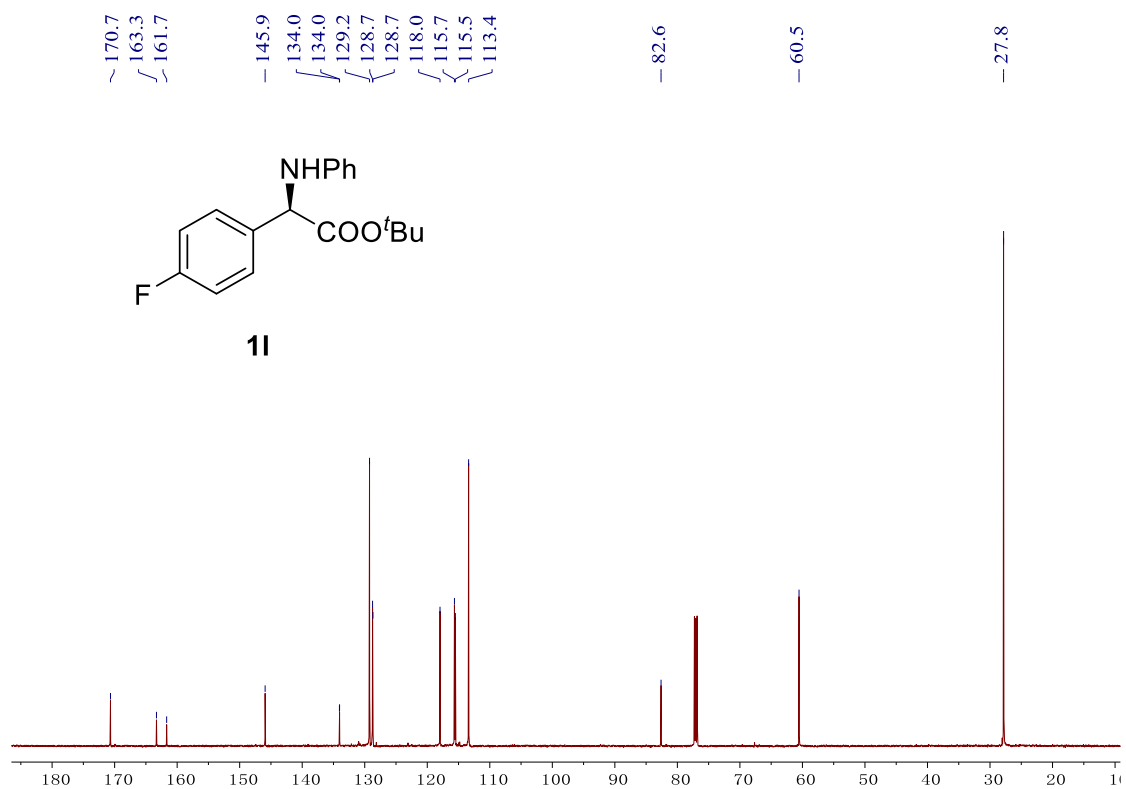
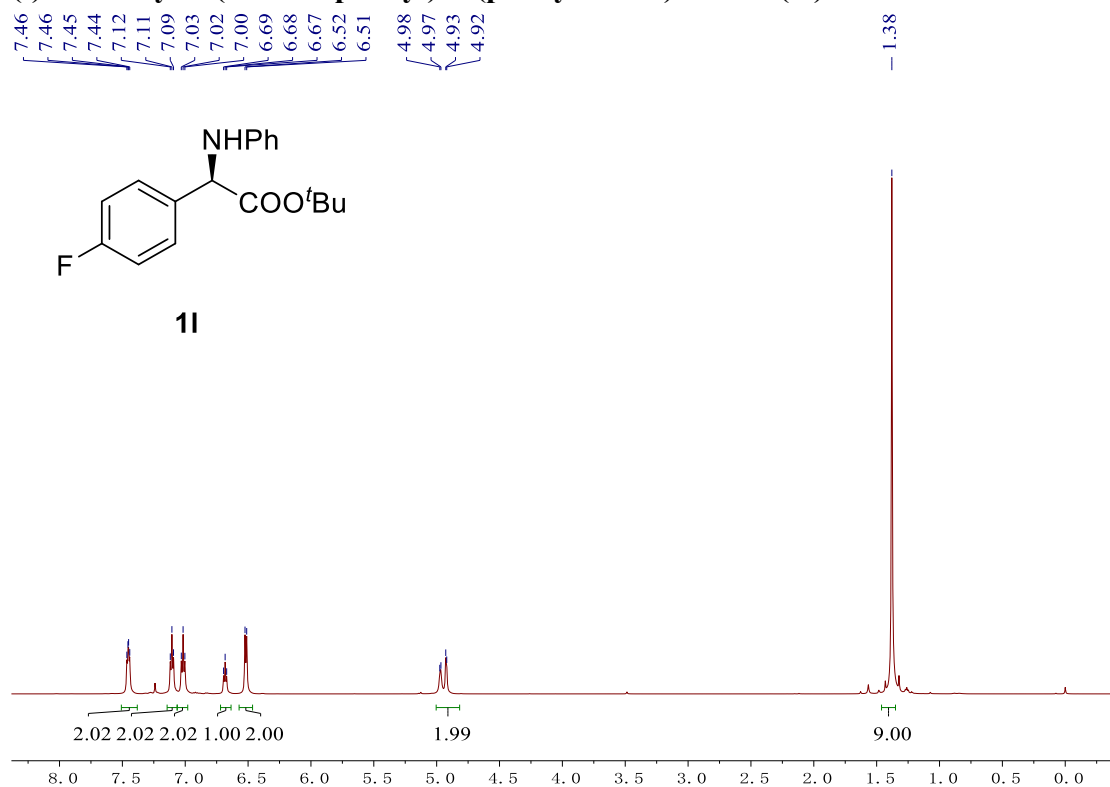
(-)-*tert*-butyl 2-(4-methoxyphenyl)-2-(phenylamino)acetate (1j)



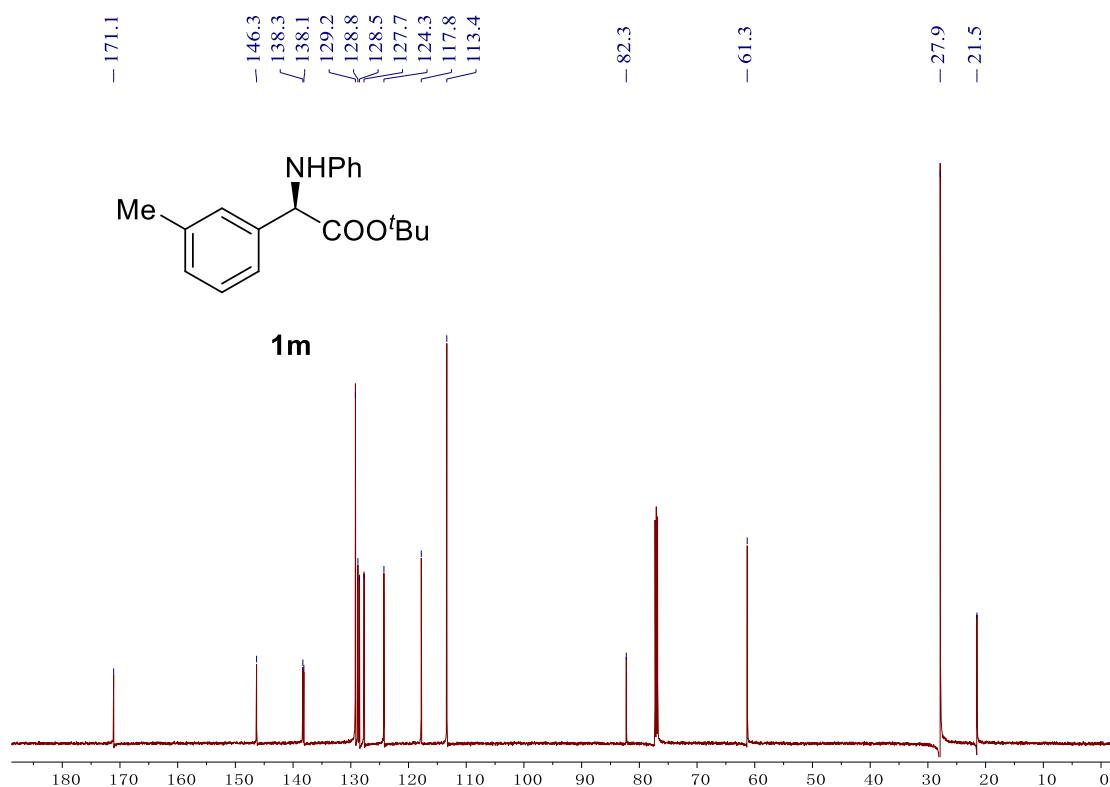
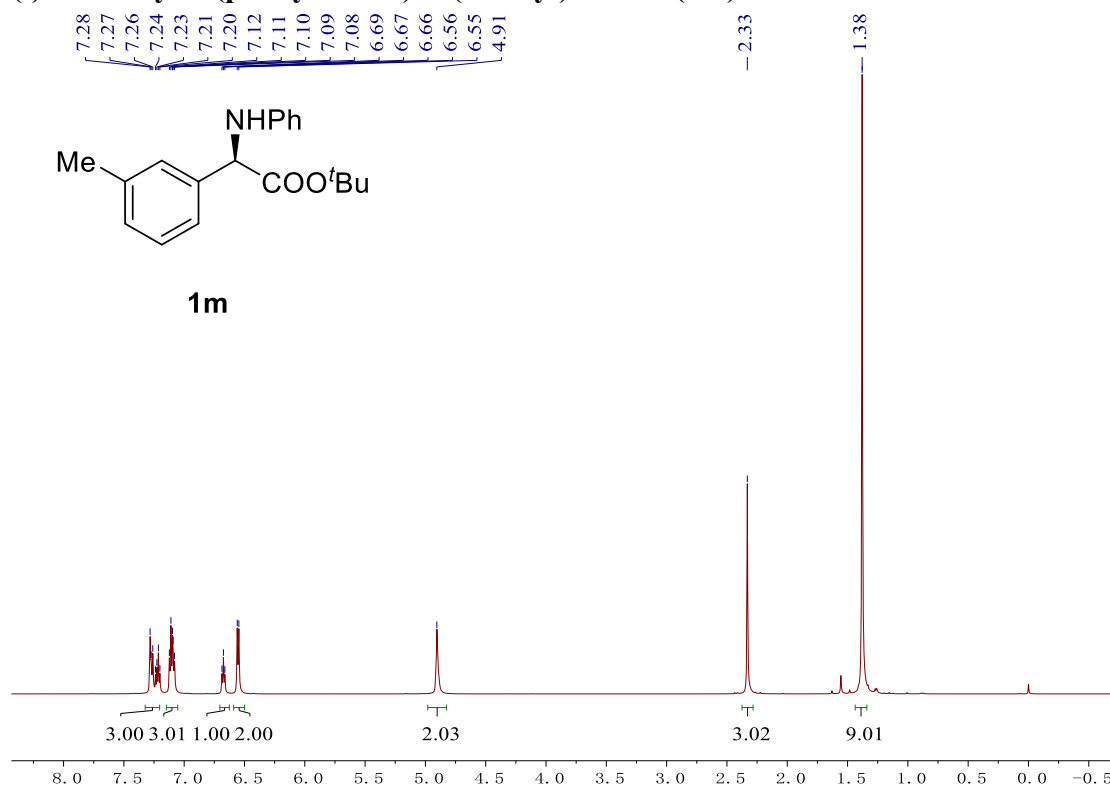
(-)-*tert*-butyl 2-(phenylamino)-2-(*p*-tolyl)acetate (1k)



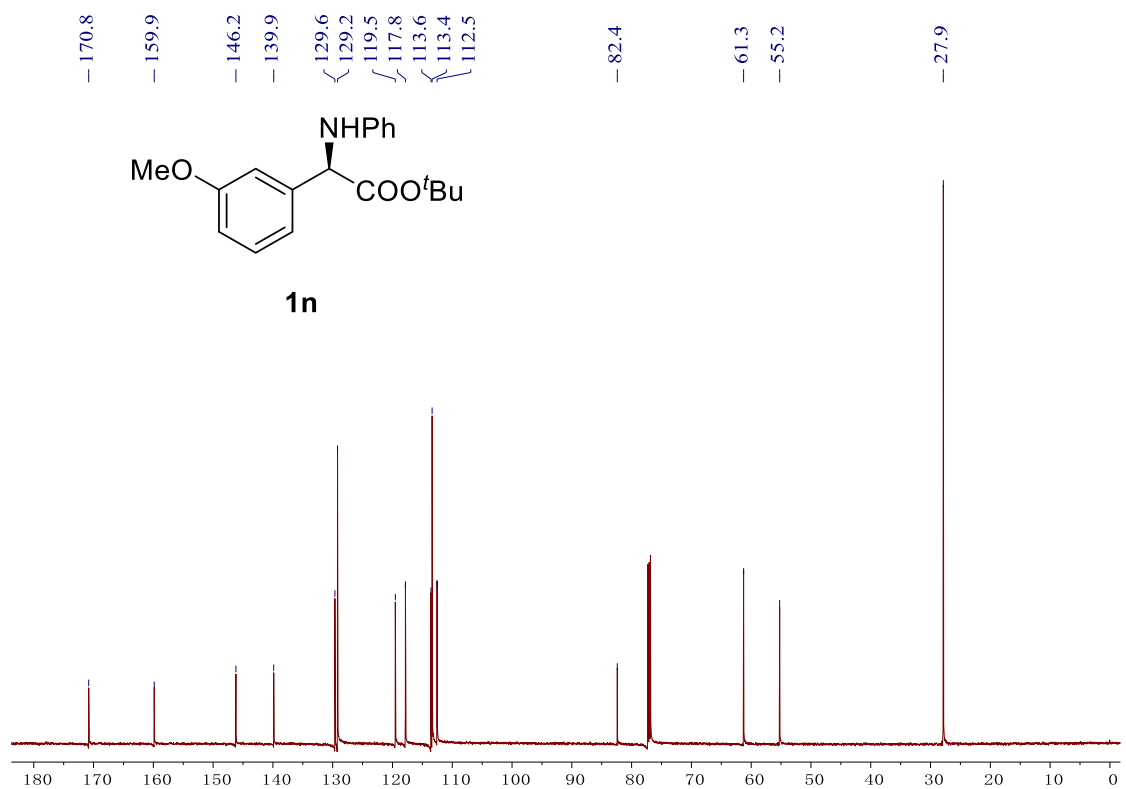
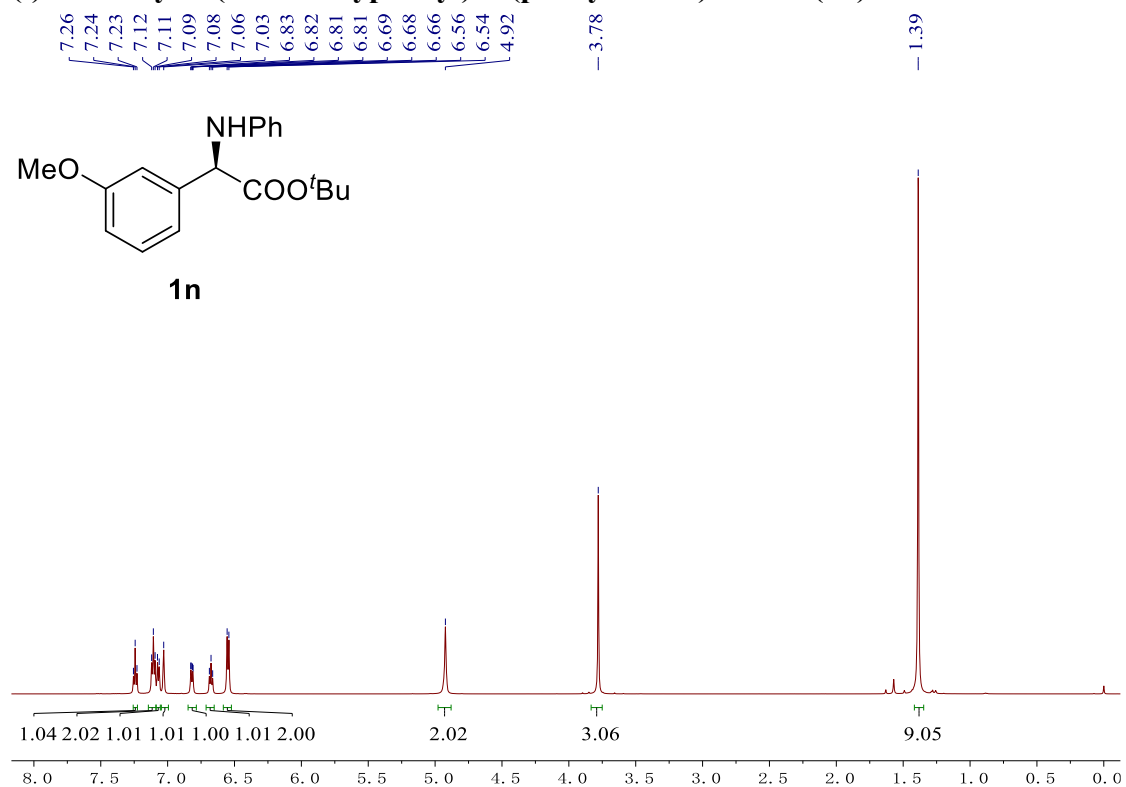
(-)-*tert*-butyl 2-(4-fluorophenyl)-2-(phenylamino)acetate (11)



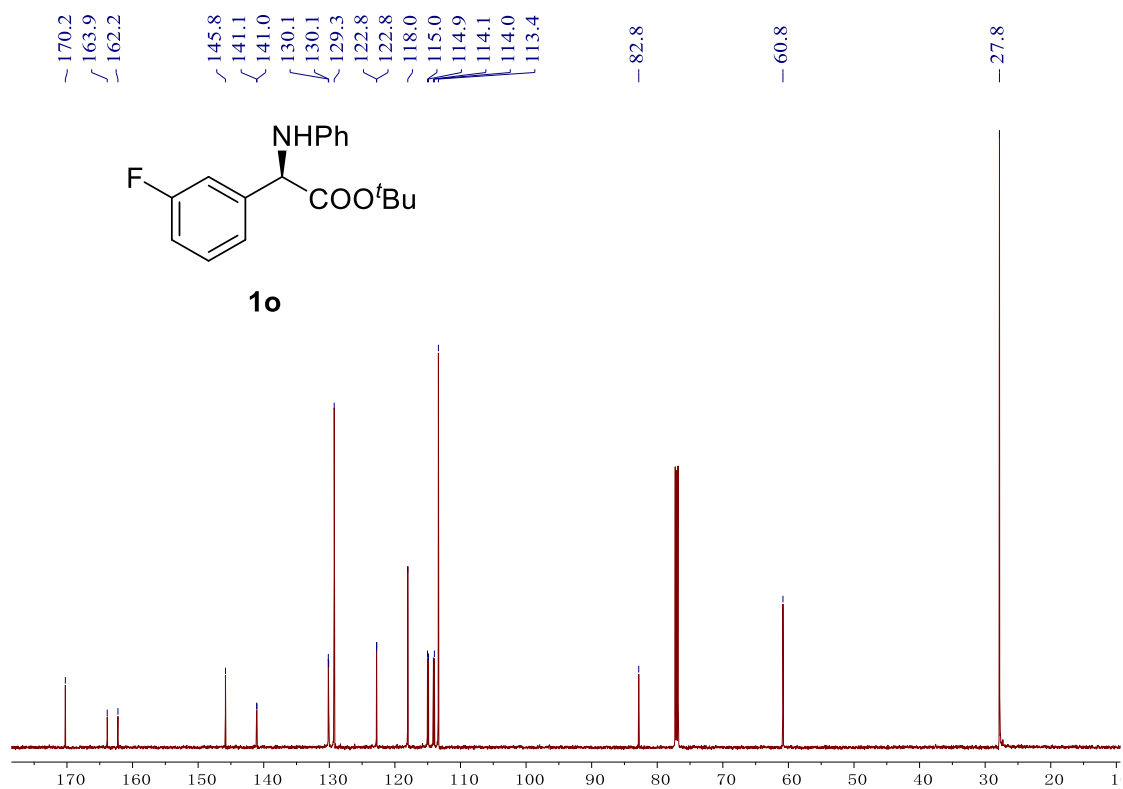
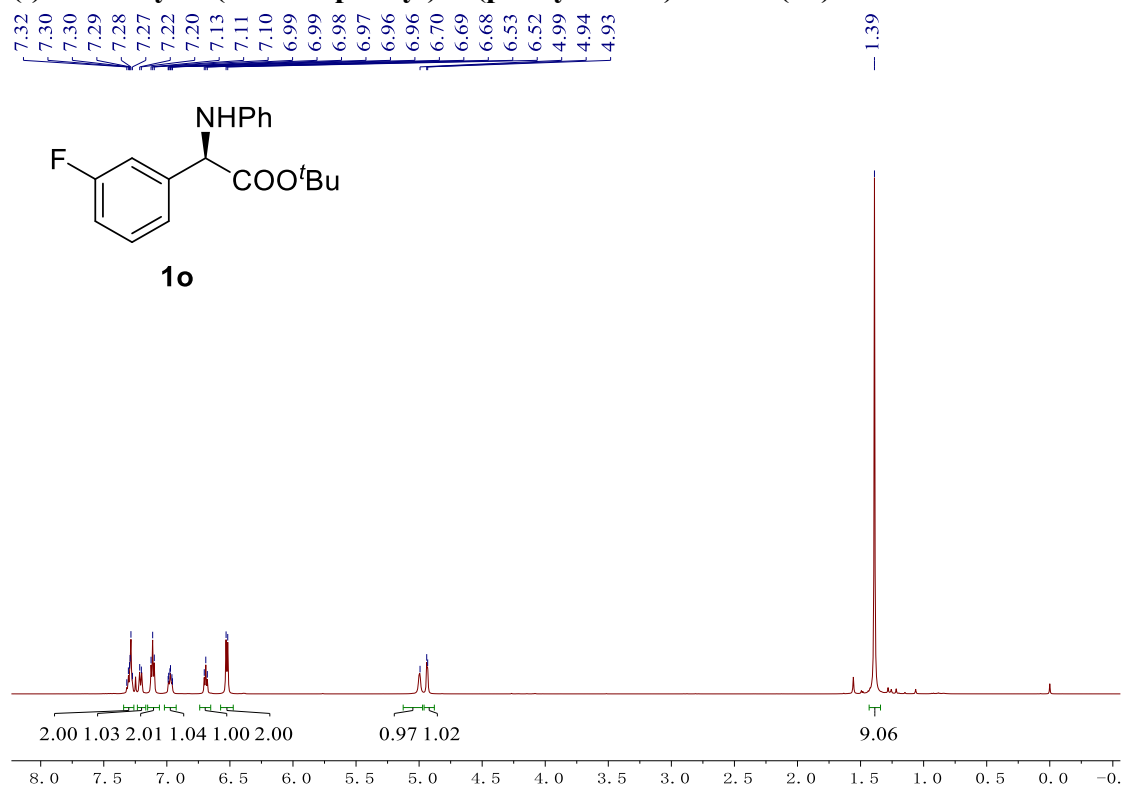
(-)-*tert*-butyl 2-(phenylamino)-2-(*m*-tolyl)acetate (1m)



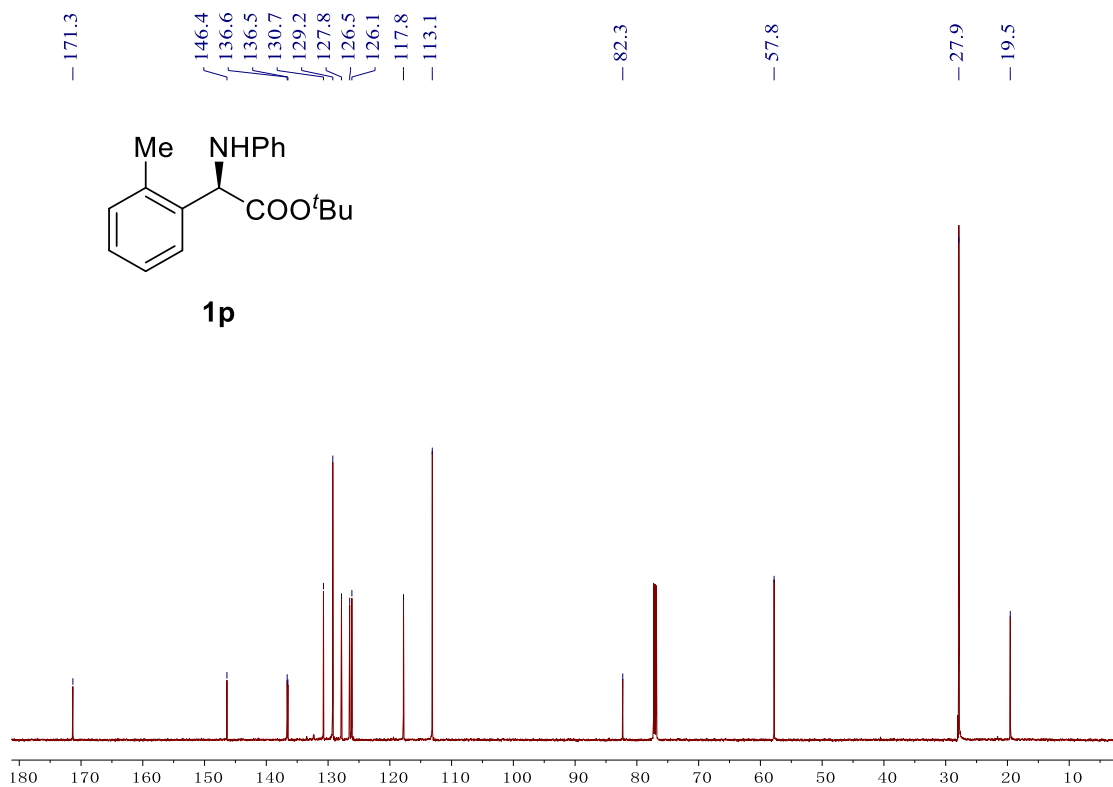
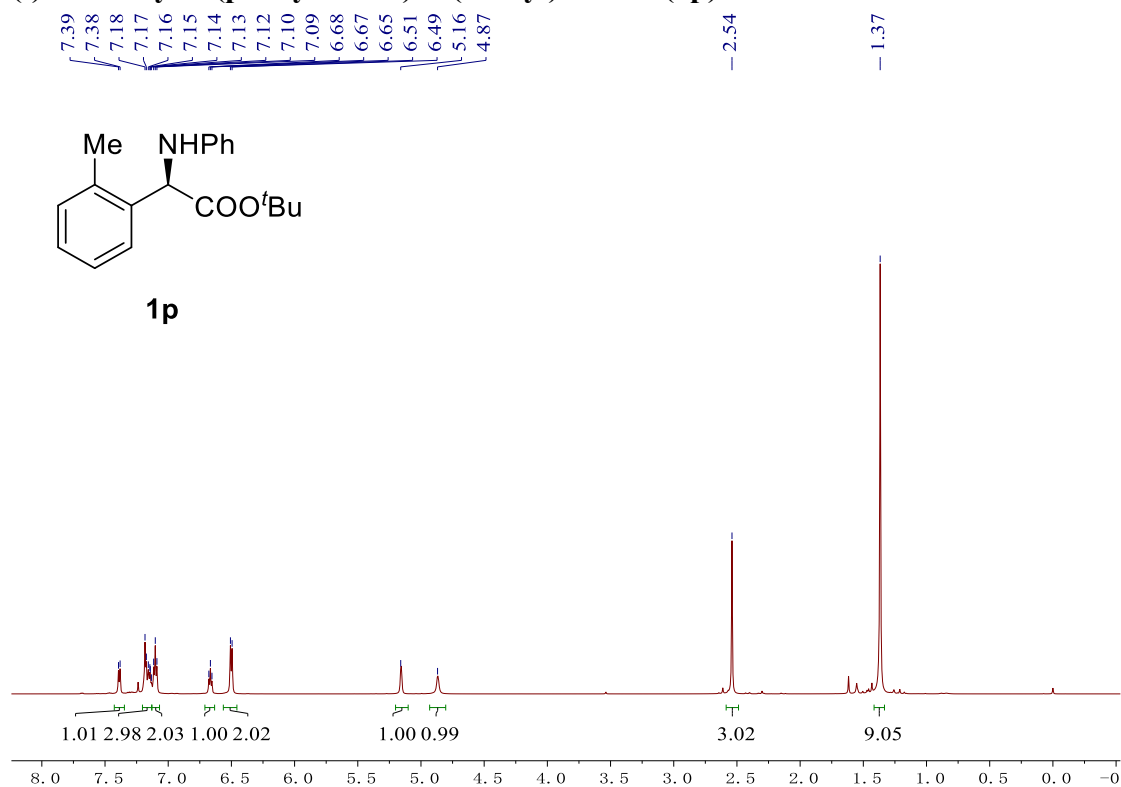
(-)-*tert*-butyl 2-(3-methoxyphenyl)-2-(phenylamino)acetate (1n)



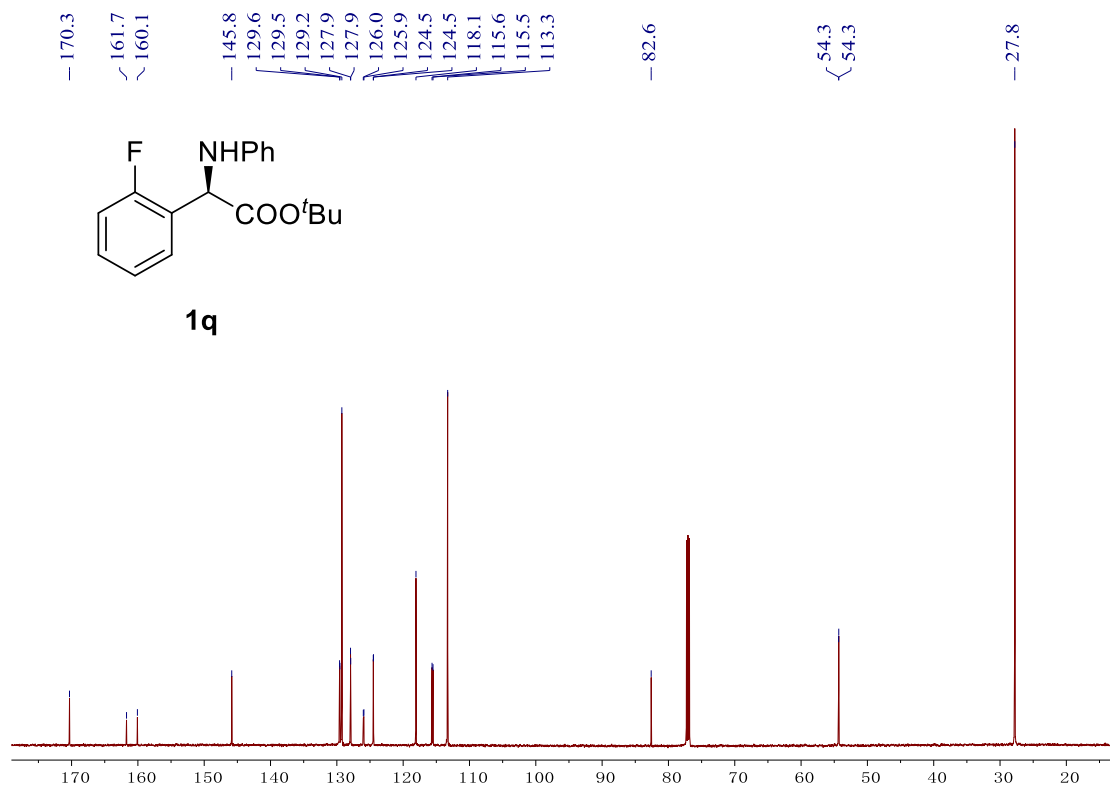
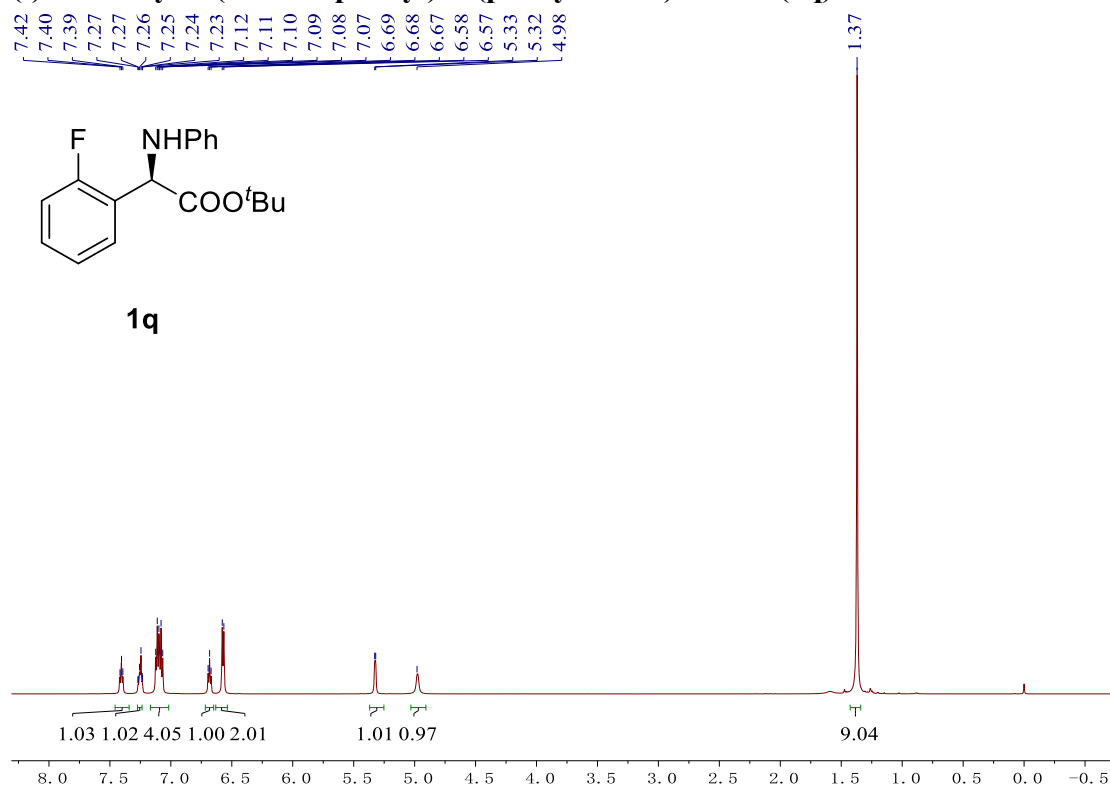
(-)-*tert*-butyl 2-(3-fluorophenyl)-2-(phenylamino)acetate (1o)



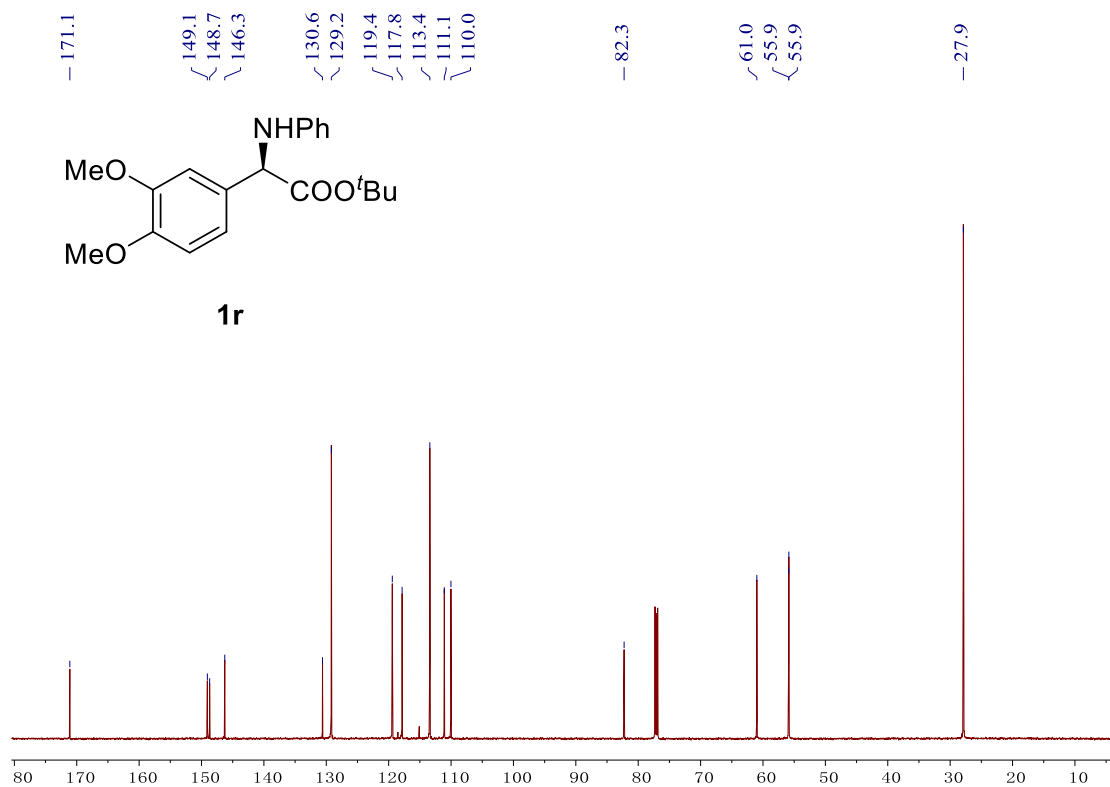
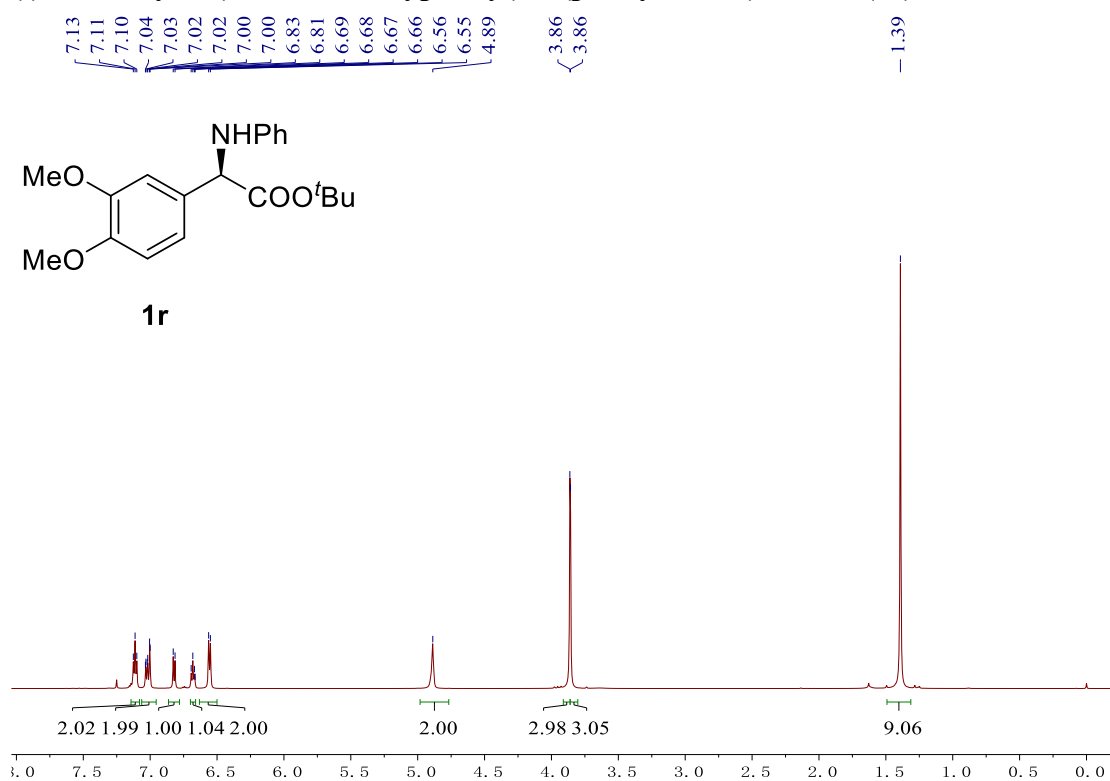
(-)-*tert*-butyl 2-(phenylamino)-2-(*o*-tolyl)acetate (1p)



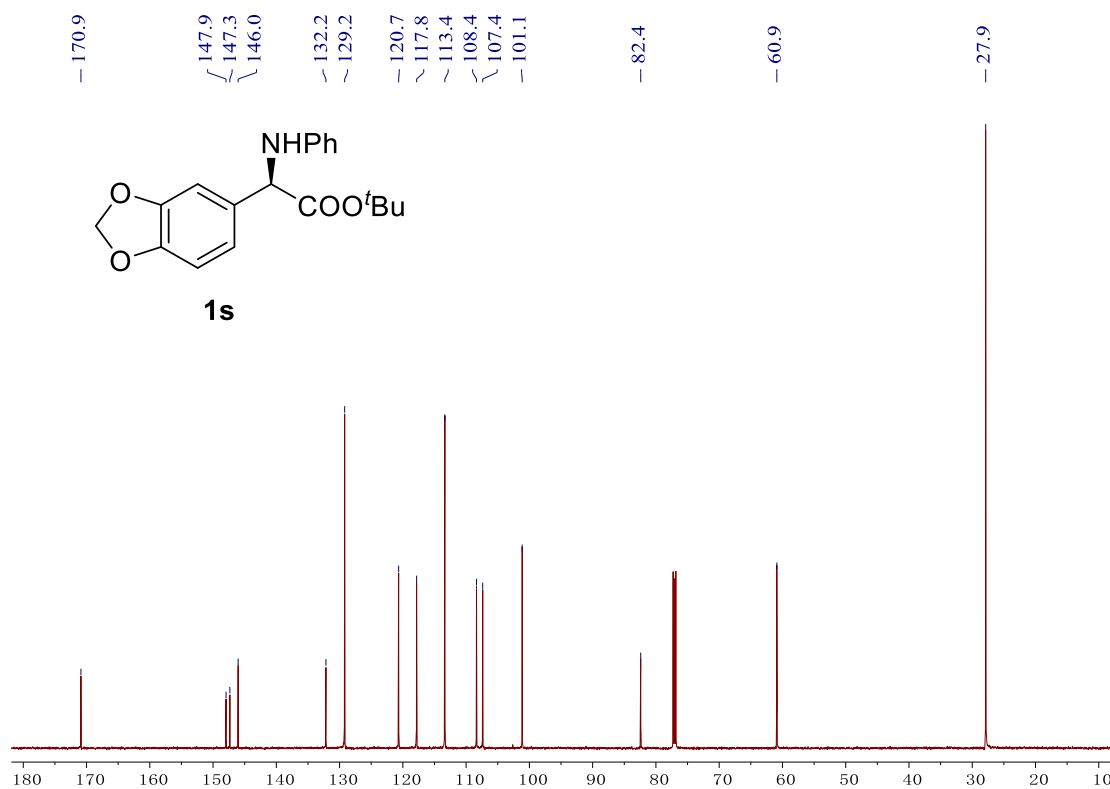
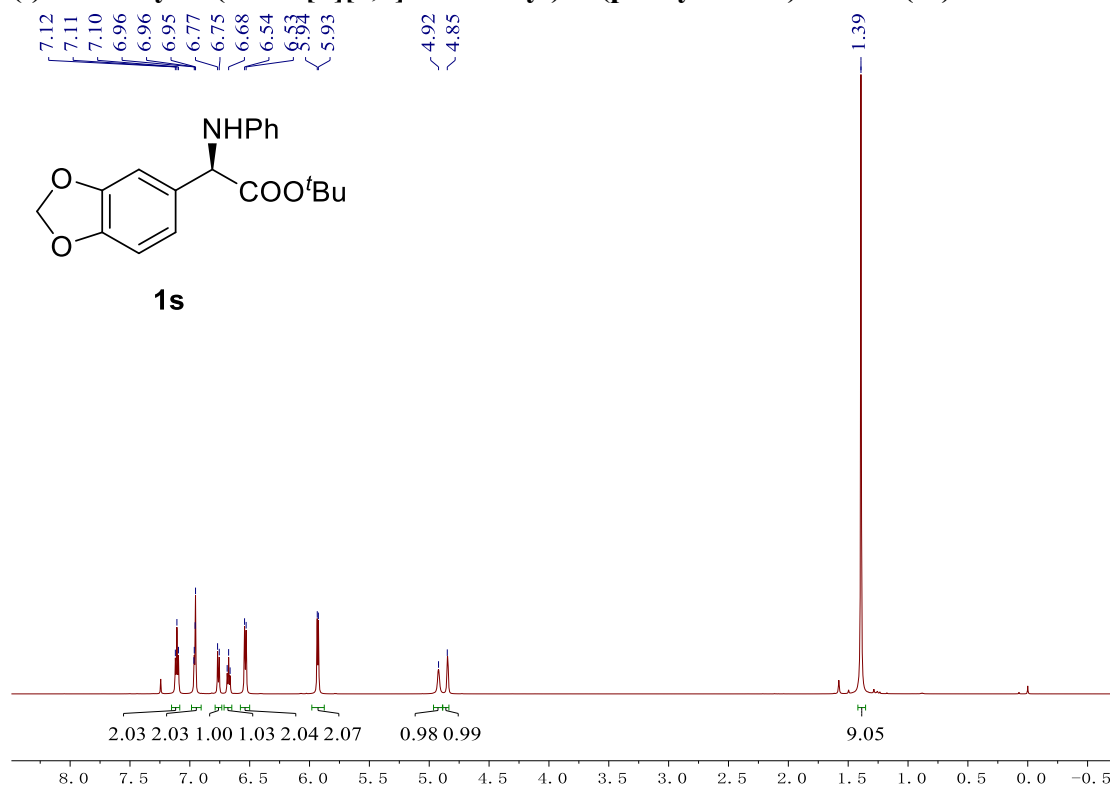
(-)-*tert*-butyl 2-(2-fluorophenyl)-2-(phenylamino)acetate (1q)



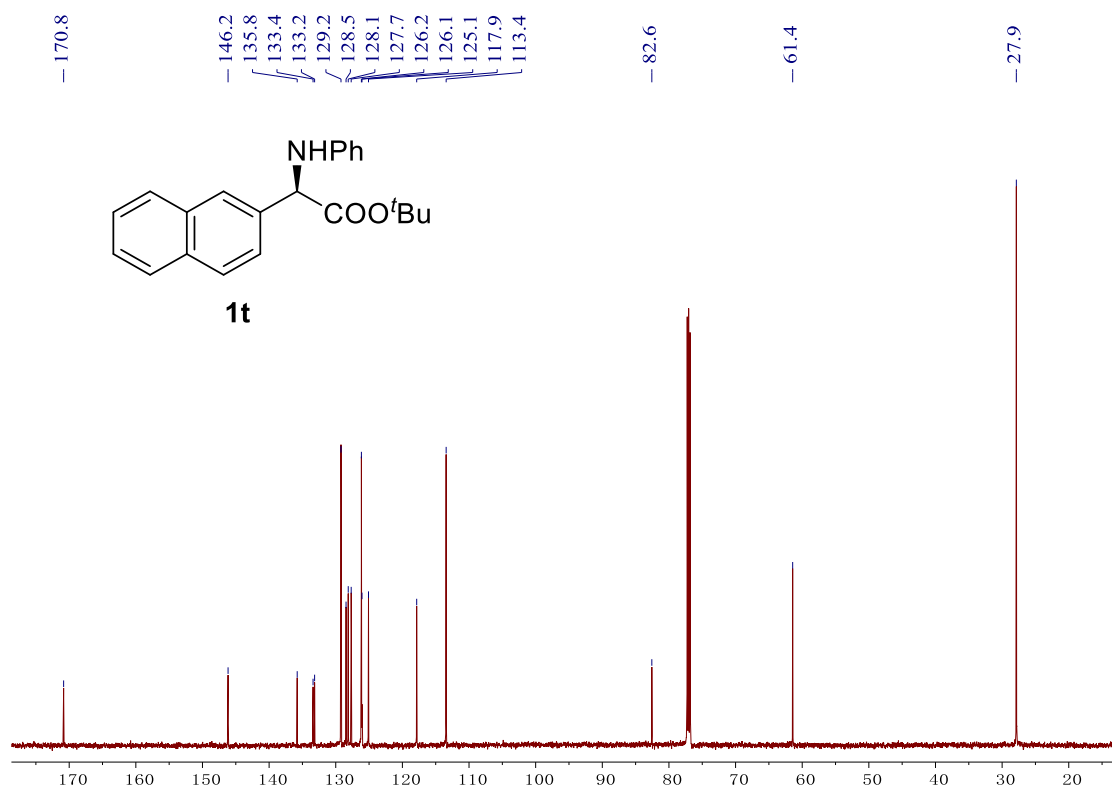
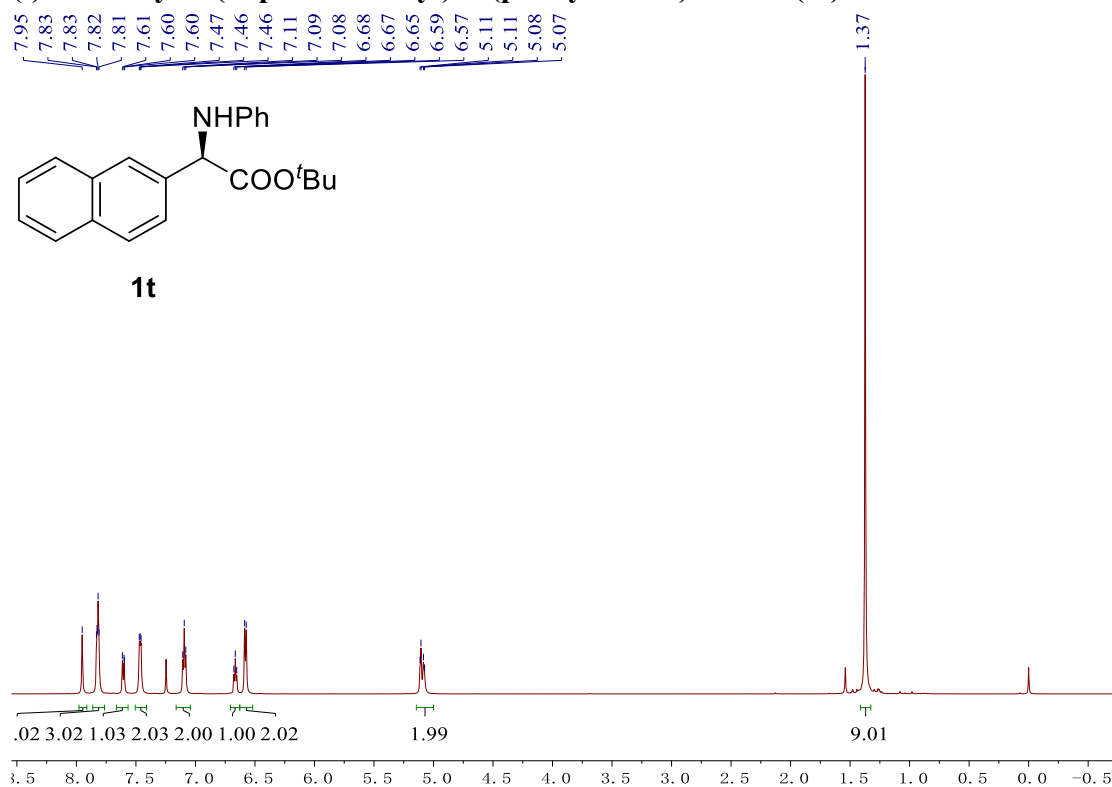
(-)-*tert*-butyl 2-(3,4-dimethoxyphenyl)-2-(phenylamino)acetate (1r)



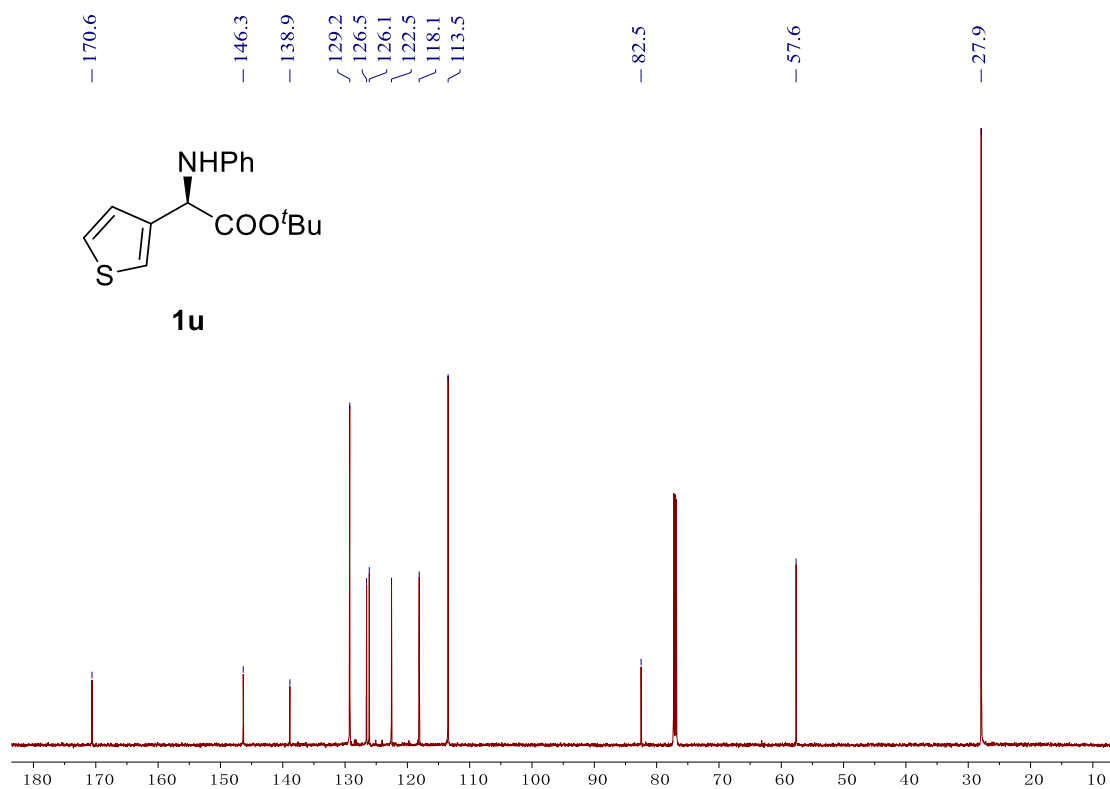
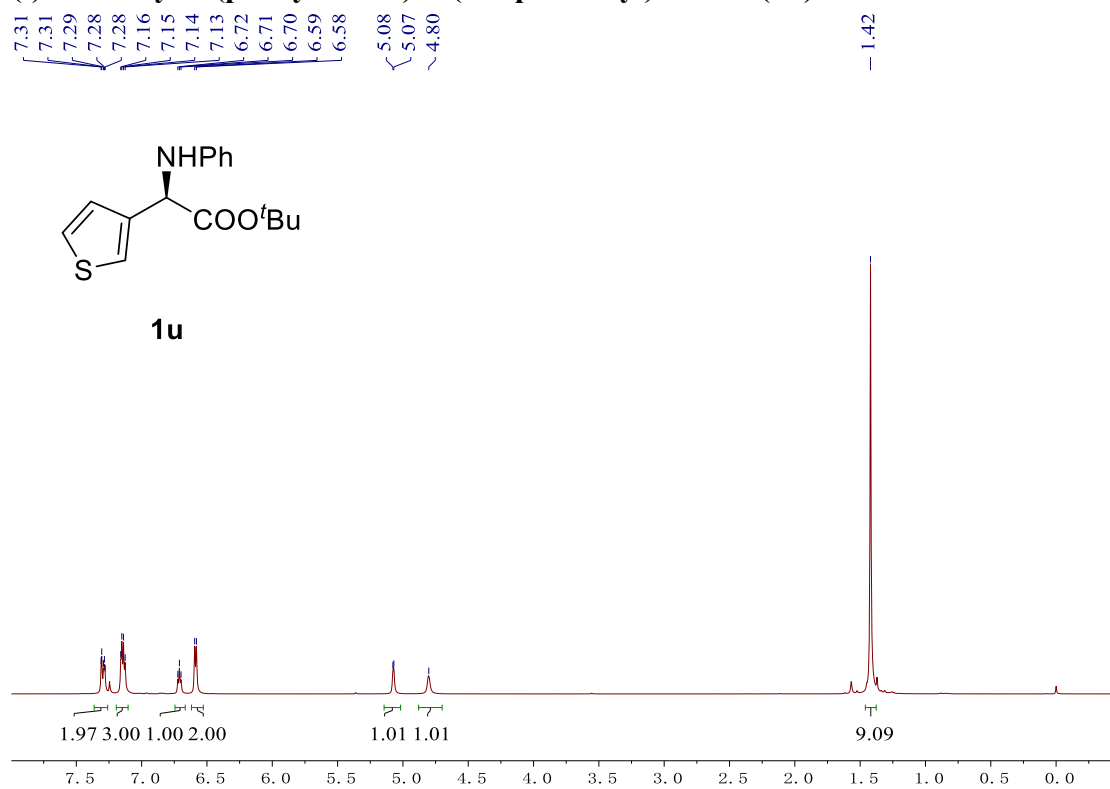
(-)-*tert*-butyl 2-(benzo[d][1,3]dioxol-5-yl)-2-(phenylamino)acetate (1s)



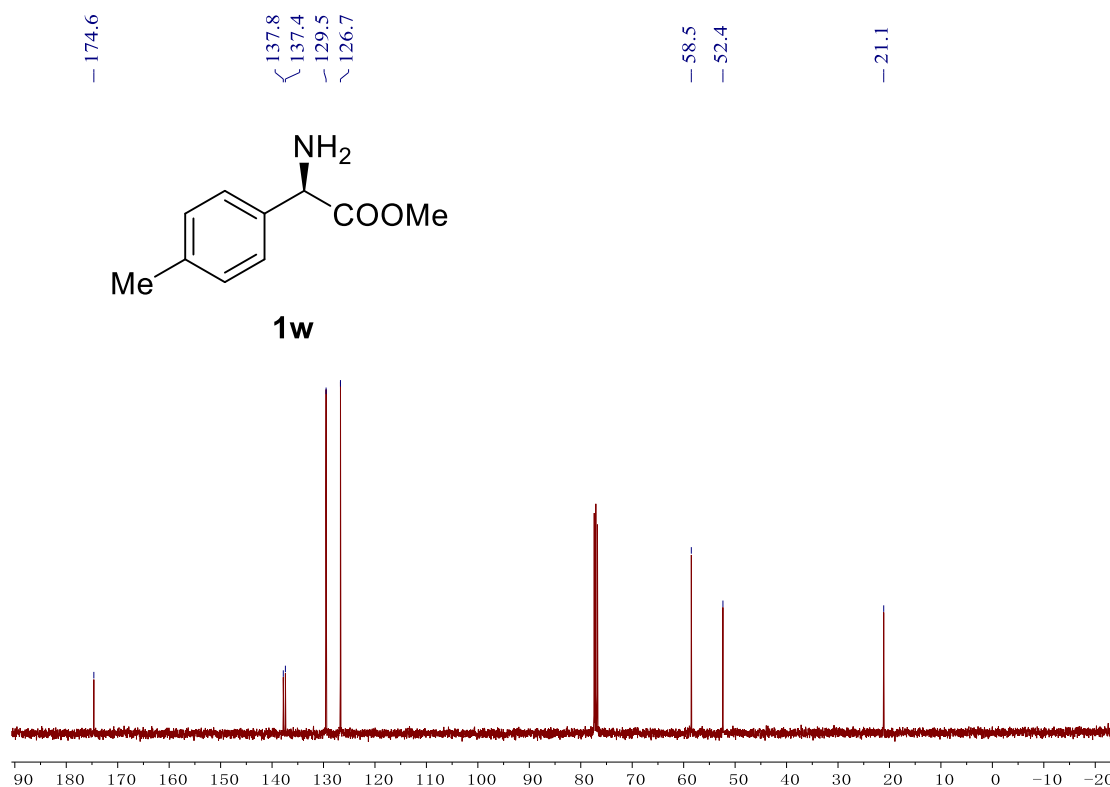
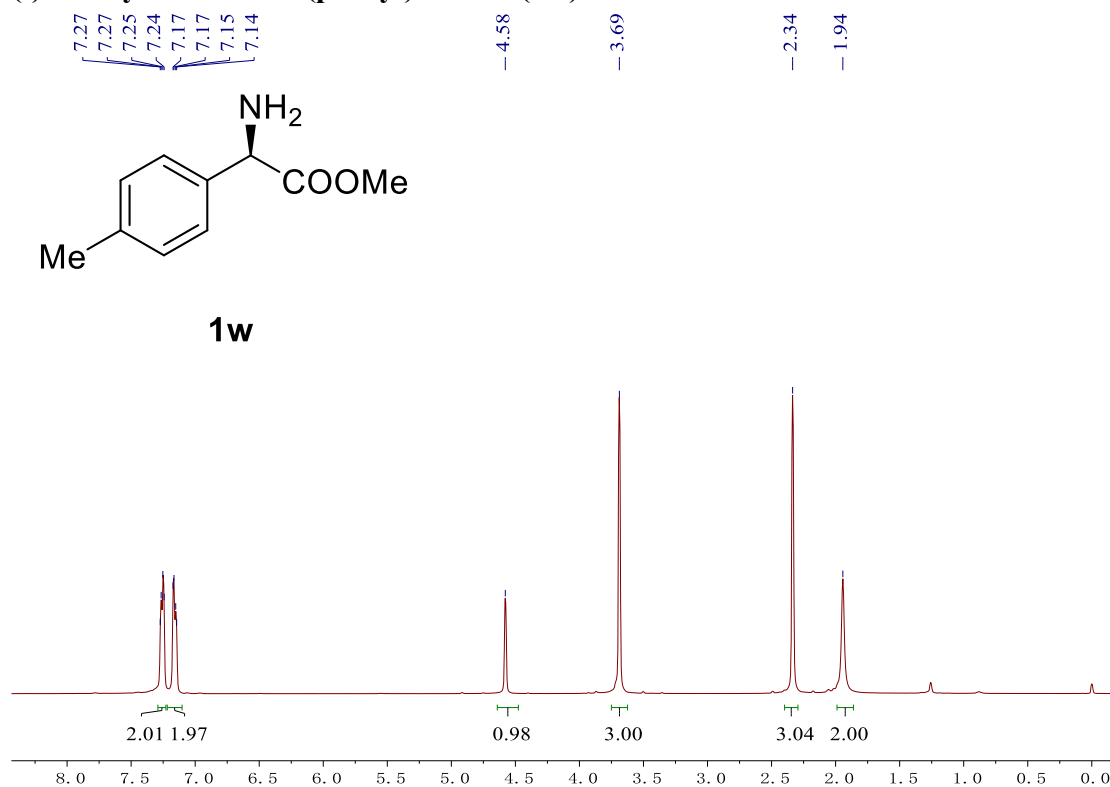
(-)-*tert*-butyl 2-(naphthalen-2-yl)-2-(phenylamino)acetate (1t)



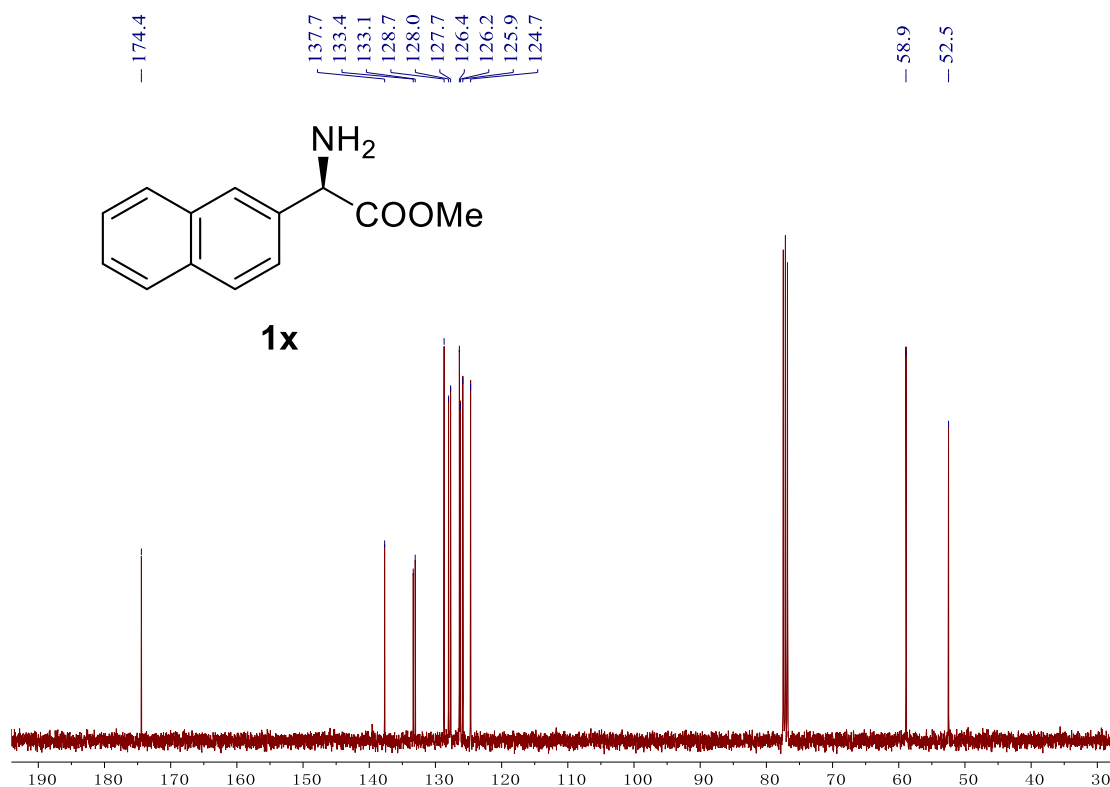
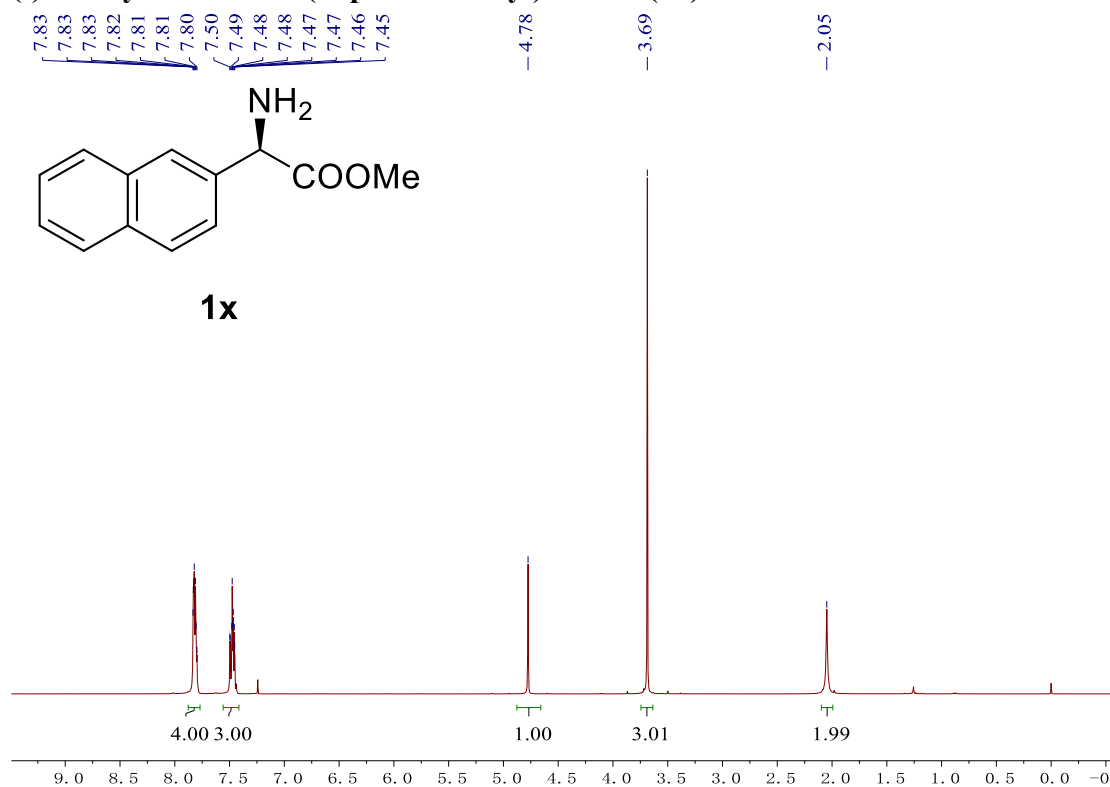
(-)-*tert*-butyl 2-(phenylamino)-2-(thiophen-3-yl)acetate (1u)



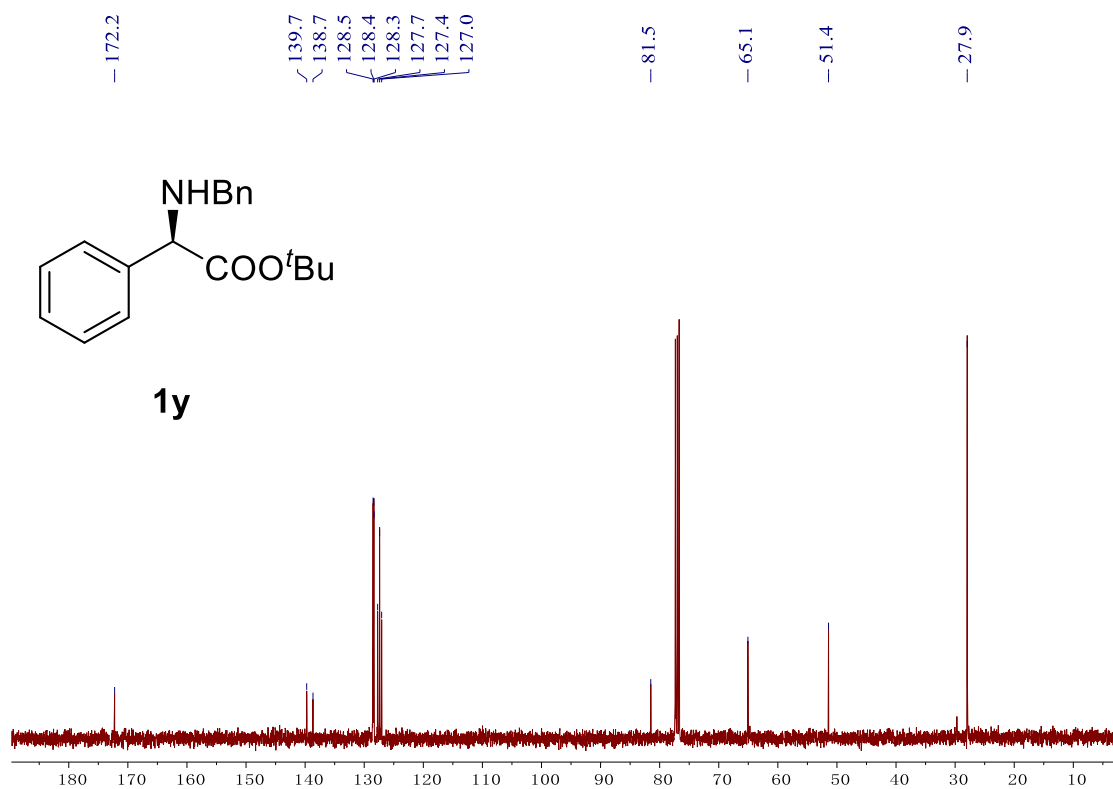
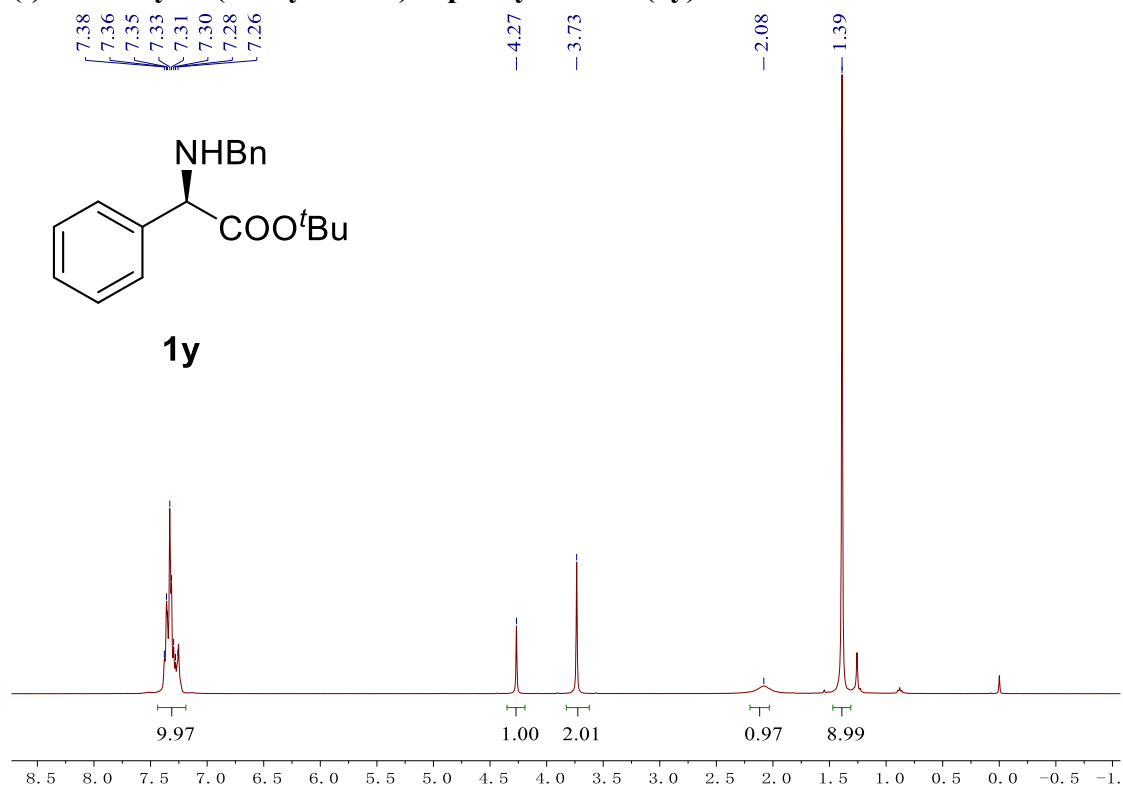
(-)-methyl 2-amino-2-(p-tolyl)acetate (1w)



(-)-methyl 2-amino-2-(naphthalen-2-yl)acetate (1x)

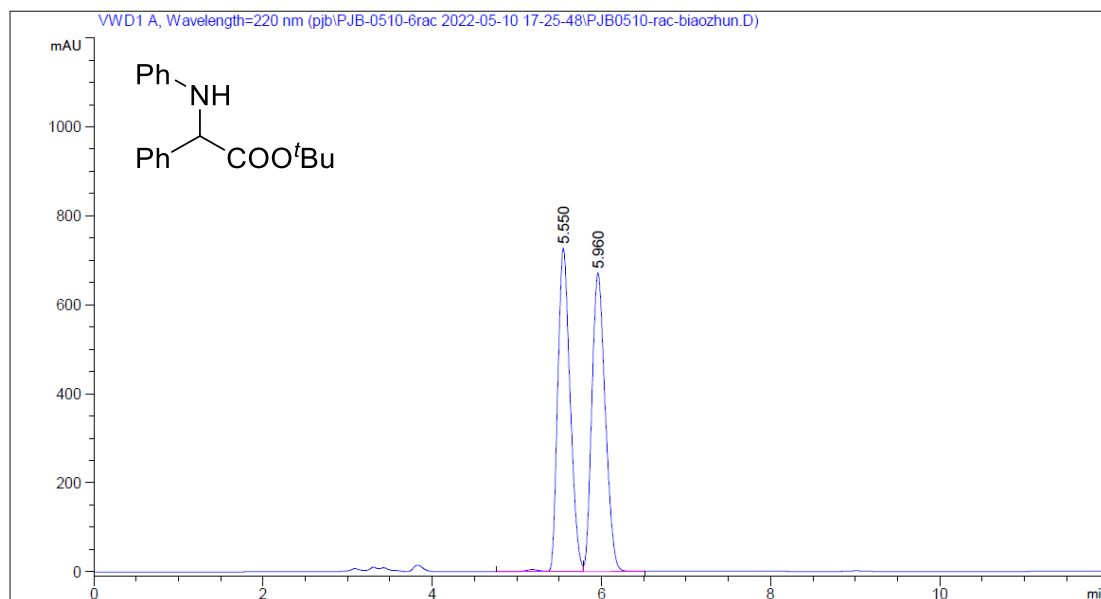


(-)-*tert*-butyl 2-(benzylamino)-2-phenylacetate (1y)

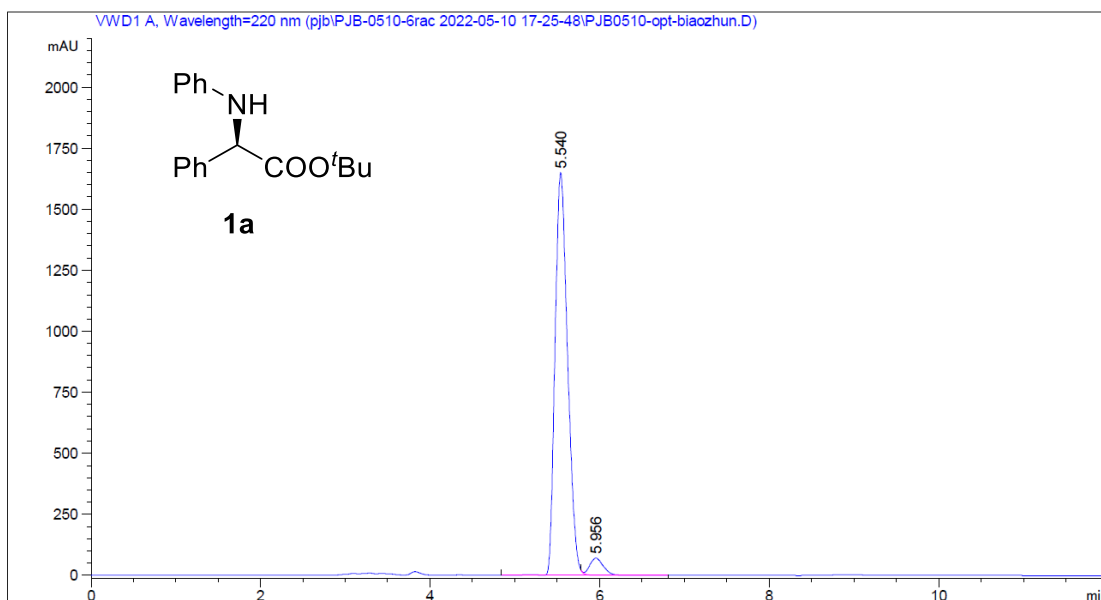


7. HPLC Spectra of Products

(*R*)-*tert*-butyl 2-phenyl-2-(phenylamino)acetate (**1a**)

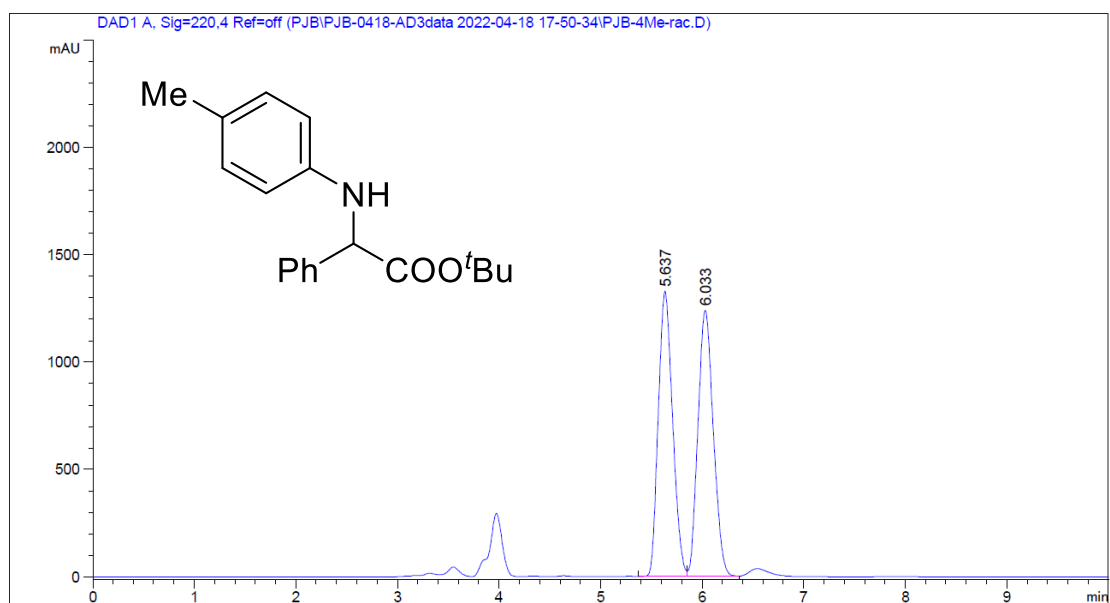


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.550	VV R	0.1570	7396.03955	726.20374	50.0627
2	5.960	VB	0.1713	7377.51953	670.80664	49.9373

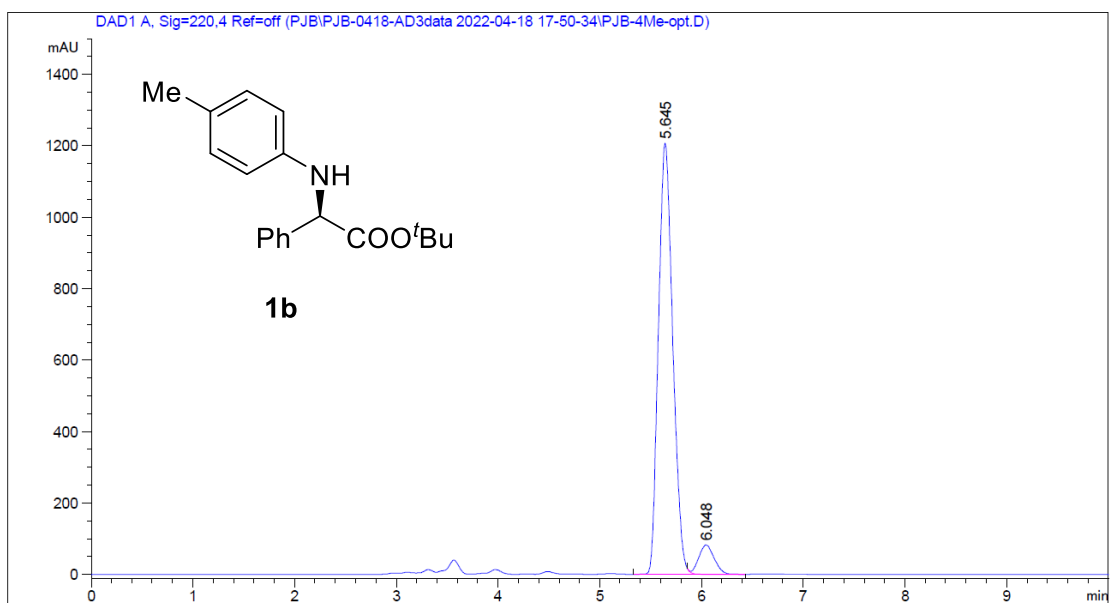


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.540	VV R	0.1588	1.69570e4	1648.67102	95.5918
2	5.956	VB E	0.1701	781.96606	70.65005	4.4082

(-)-*tert*-butyl 2-phenyl-2-(*p*-tolylamino)acetate (1b**)**

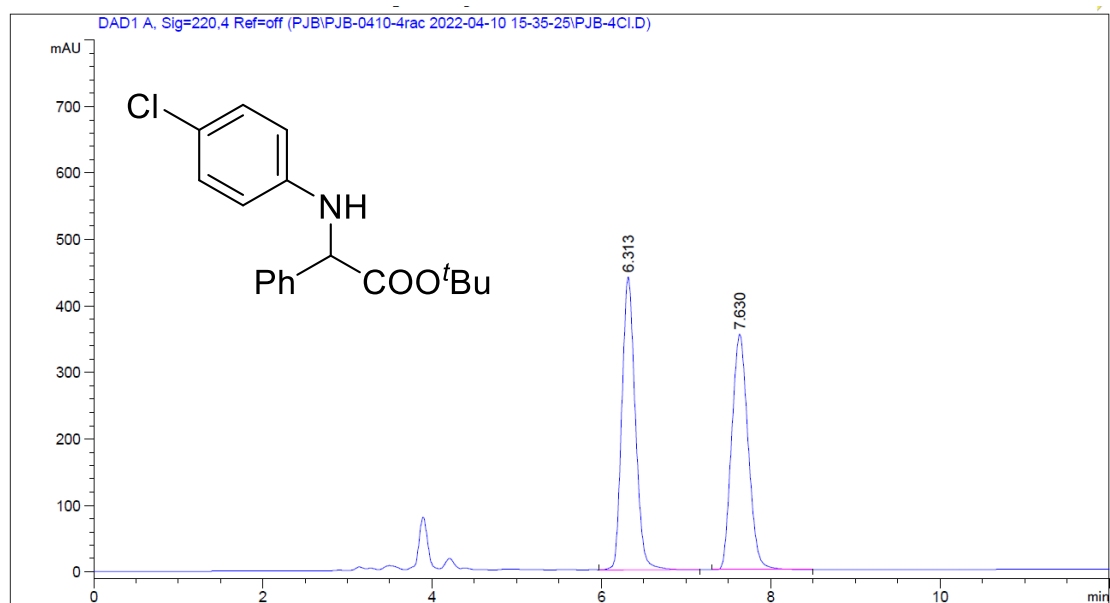


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.637	BV	0.1521	1.29850e4	1328.25122	49.8557
2	6.033	MF R	0.1755	1.30601e4	1240.49072	50.1443

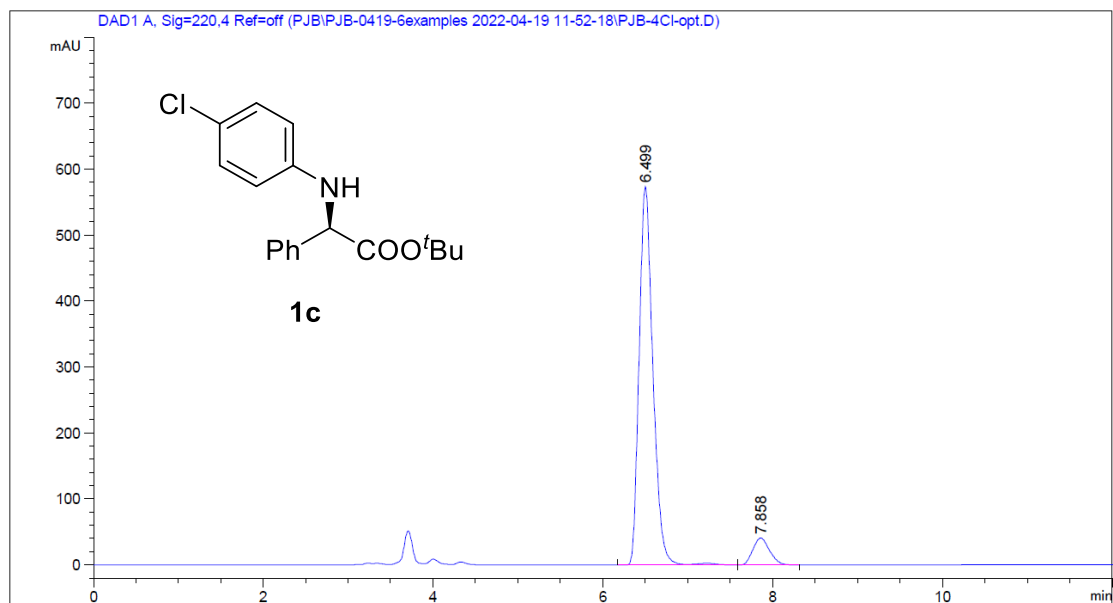


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.645	BV R	0.1523	1.18303e4	1207.83252	93.1526
2	6.048	VB E	0.1650	869.61926	82.47665	6.8474

(-)-tert-butyl 2-((4-chlorophenyl)amino)-2-phenylacetate (1c)

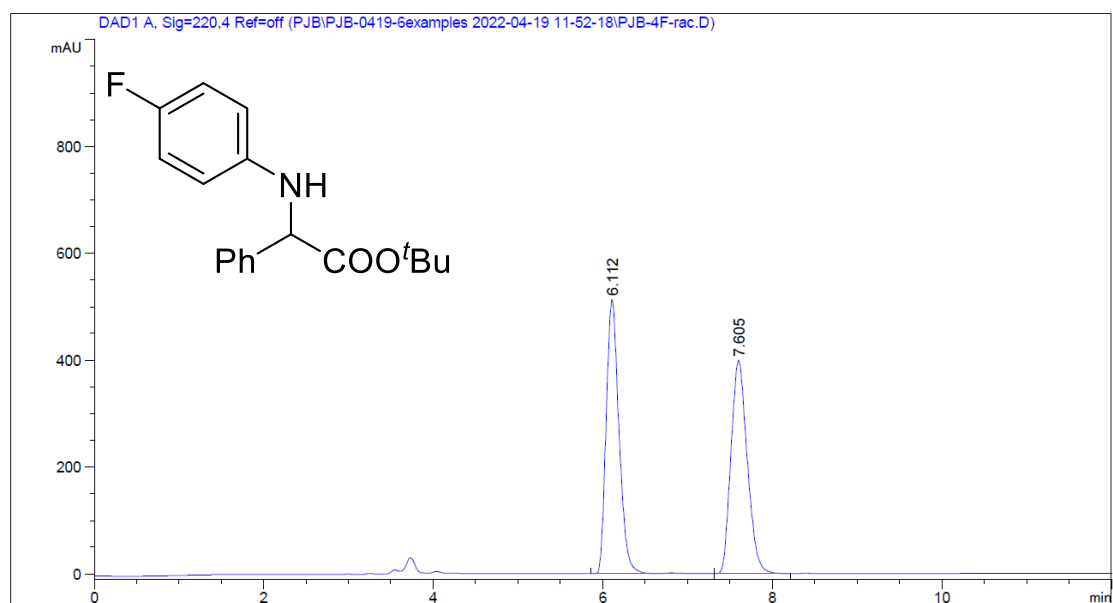


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.313	BB	0.1704	4849.78125	440.81076	50.3939
2	7.630	BB	0.2133	4773.97217	353.96808	49.6061

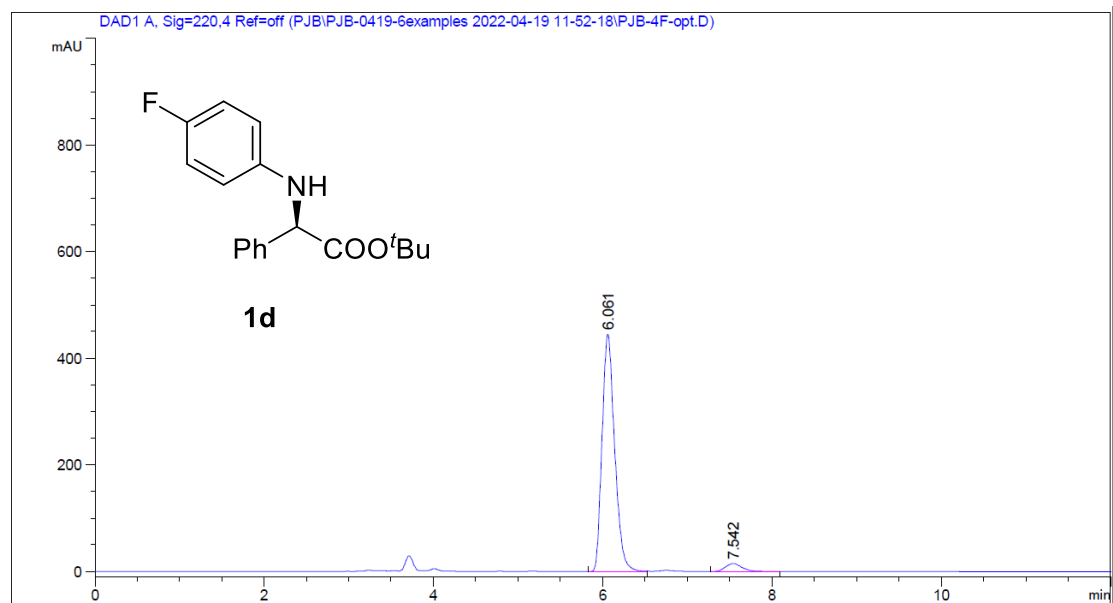


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.499	BV R	0.1704	6480.19727	574.13177	92.0192
2	7.858	VB	0.2112	562.02356	41.18242	7.9808

(-)-tert-butyl 2-((4-fluorophenyl)amino)-2-phenylacetate (1d)

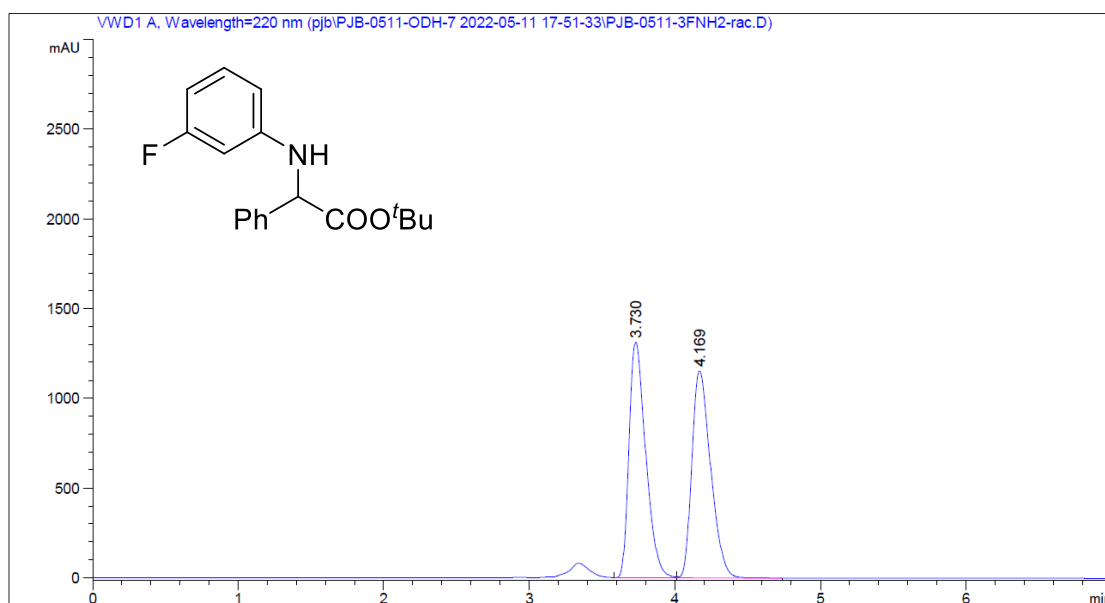


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.112	BV R	0.1620	5386.94678	513.39008	50.3046
2	7.605	BB	0.2076	5321.70215	398.94370	49.6954

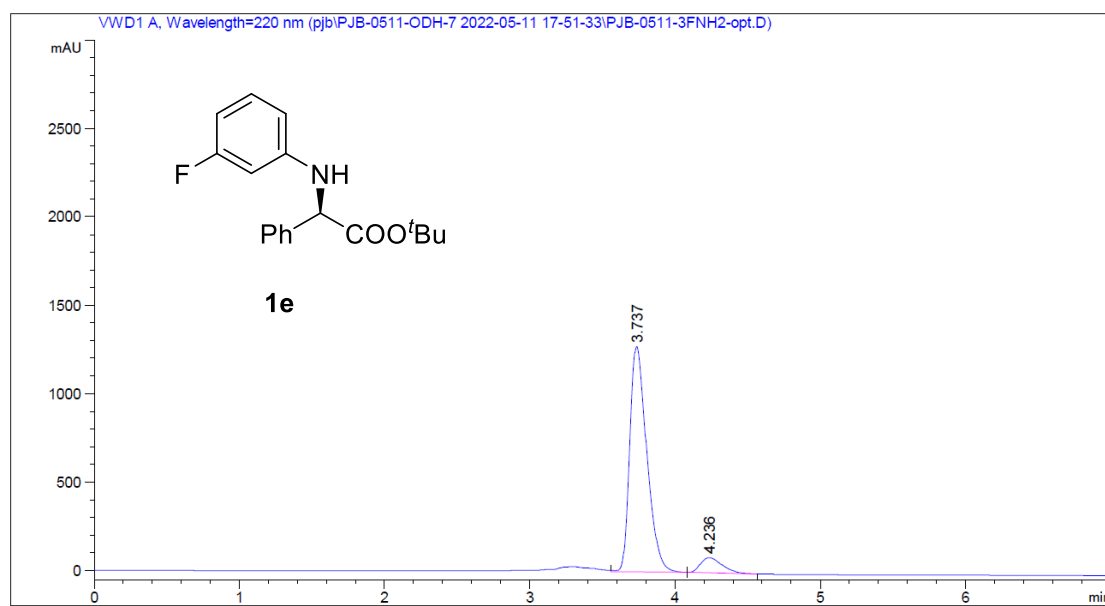


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.061	MF R	0.1722	4602.09766	445.40170	95.8400
2	7.542	BB	0.2079	199.75833	14.95093	4.1600

(-)-tert-butyl 2-((3-fluorophenyl)amino)-2-phenylacetate (1e)

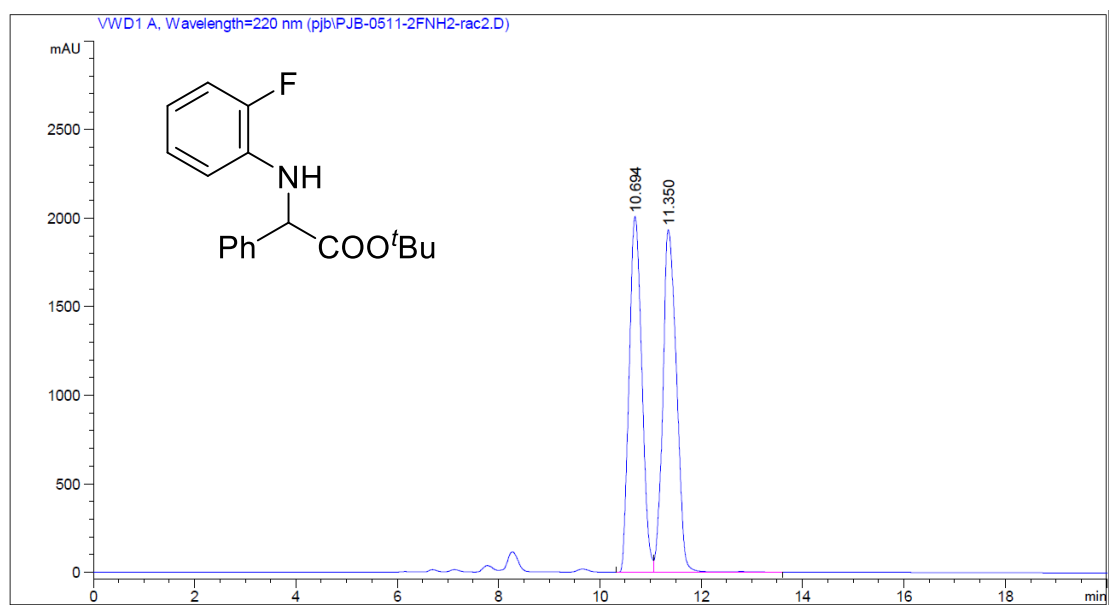


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	3.730	BV	0.1210	1.04630e4	1313.08215	50.2988
2	4.169	VB	0.1377	1.03387e4	1152.22192	49.7012

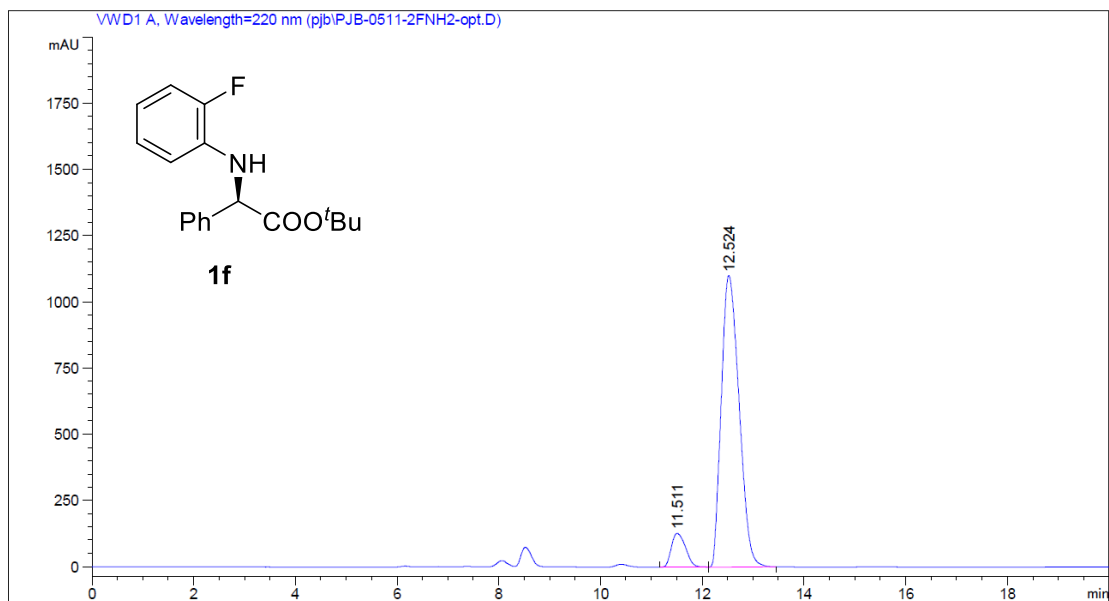


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	3.737	FM R	0.1361	1.04034e4	1274.29419	92.0834
2	4.236	BB	0.1587	894.39642	86.47750	7.9166

(-)-tert-butyl 2-((2-fluorophenyl)amino)-2-phenylacetate (1f)

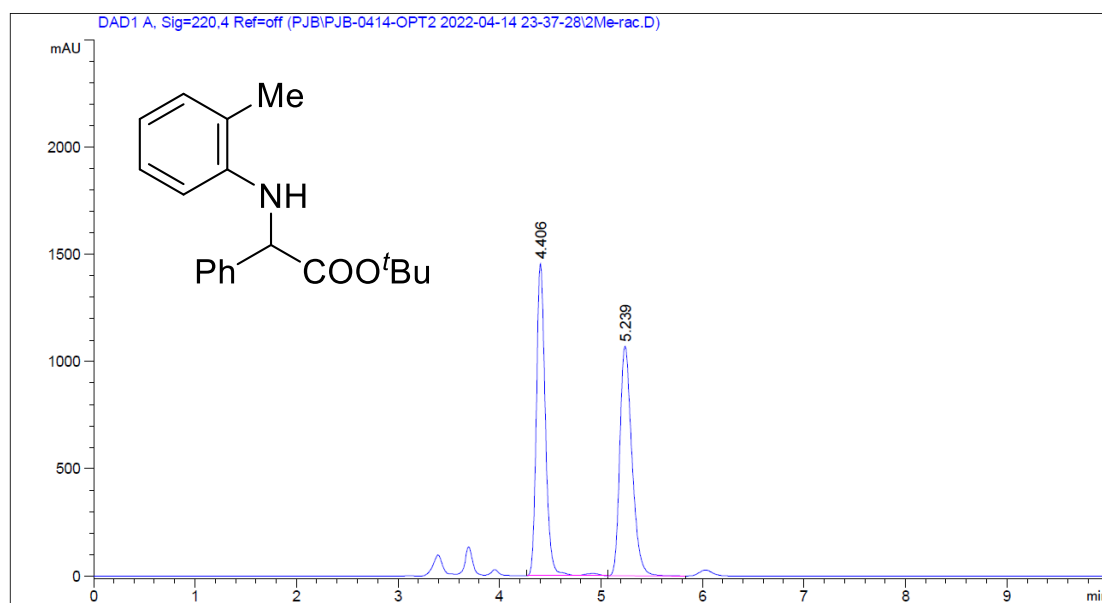


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	10.694	BV	0.2721	3.49424e4	2007.35840	49.0822
2	11.350	VV R	0.2999	3.62491e4	1933.45715	50.9178

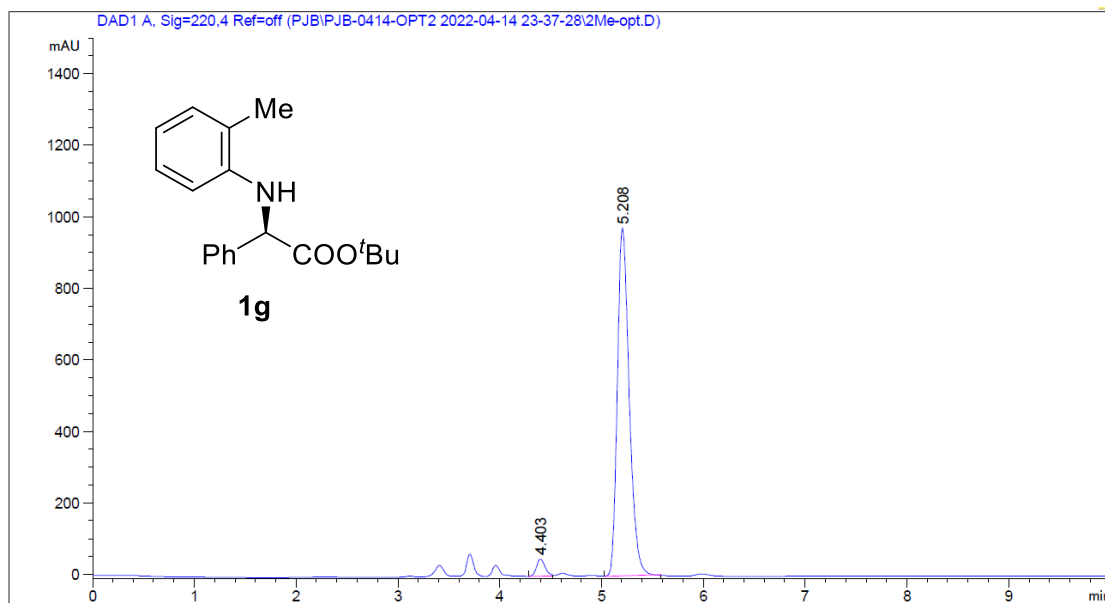


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	11.511	BB	0.3210	2469.39038	127.49702	8.4659
2	12.524	BB	0.3831	2.66994e4	1100.43201	91.5341

(-)-*tert*-butyl 2-phenyl-2-(*o*-tolylamino)acetate (1g)

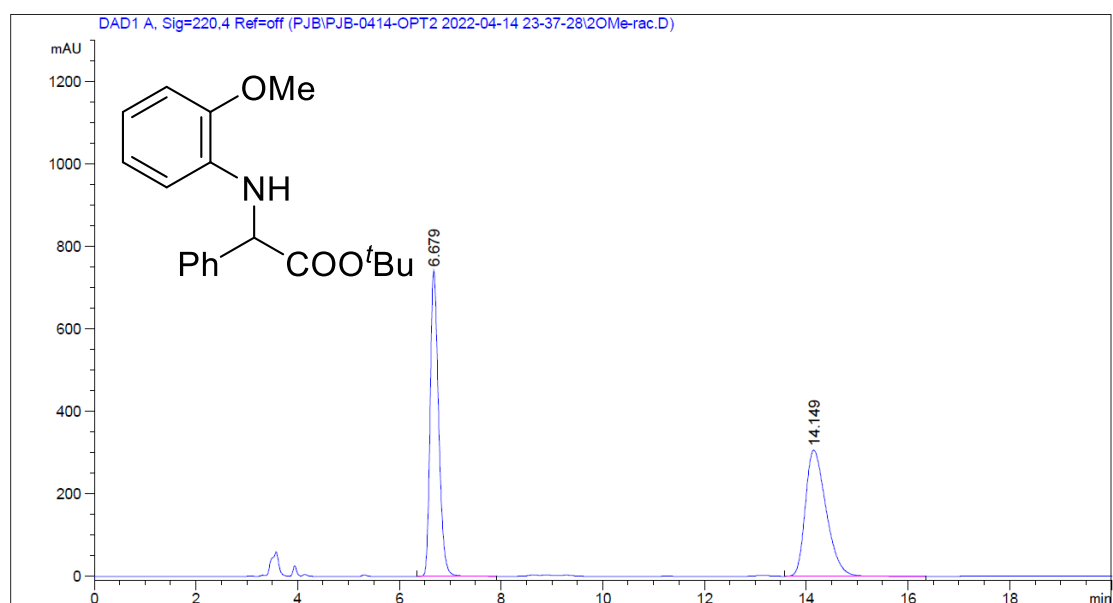


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	4.406	BV R	0.0947	8939.94238	1457.30518	50.1627
2	5.239	VB	0.1267	8881.96289	1071.47437	49.8373

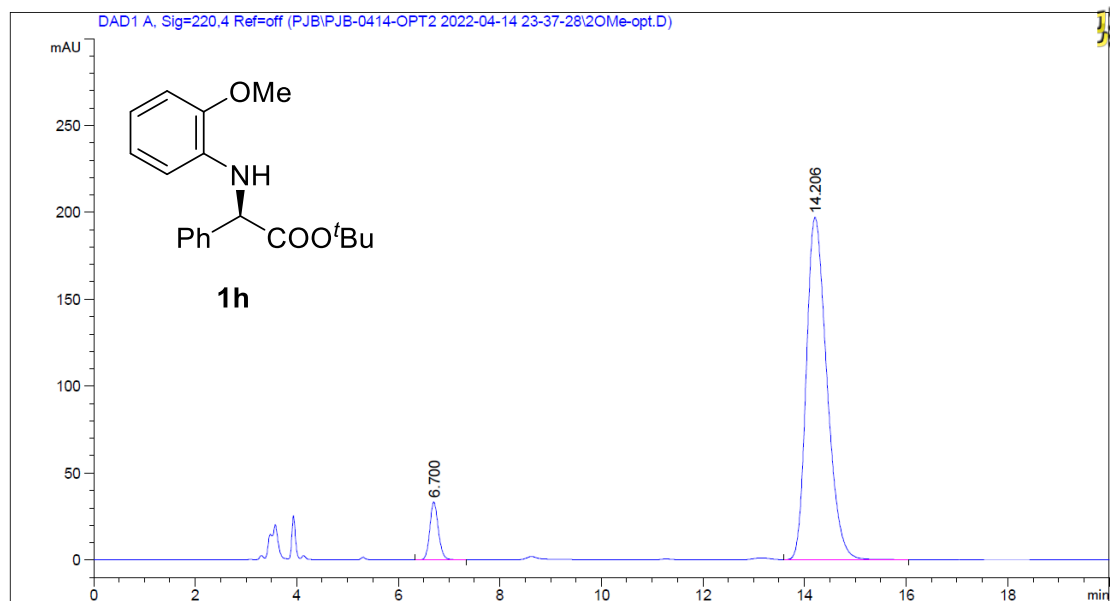


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	4.403	MF R	0.0984	284.04251	48.12967	3.4920
2	5.208	MM R	0.1346	7850.11230	972.08752	96.5080

(-)-tert-butyl 2-((2-methoxyphenyl)amino)-2-phenylacetate (1h)

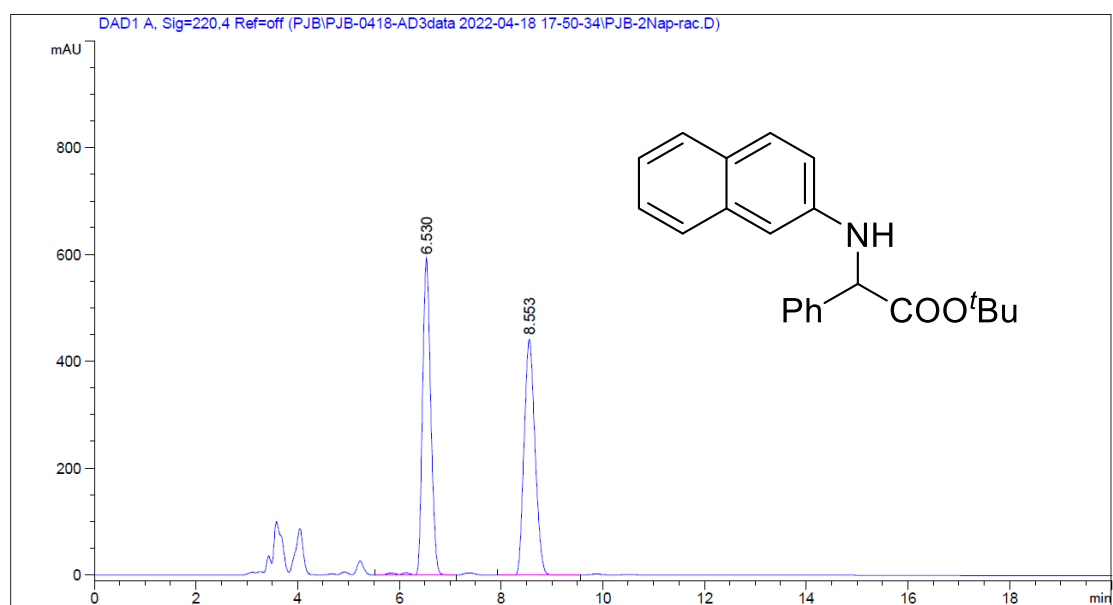


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.679	BB	0.1767	8561.54590	741.76379	49.7255
2	14.149	BB	0.4363	8656.06250	305.72040	50.2745

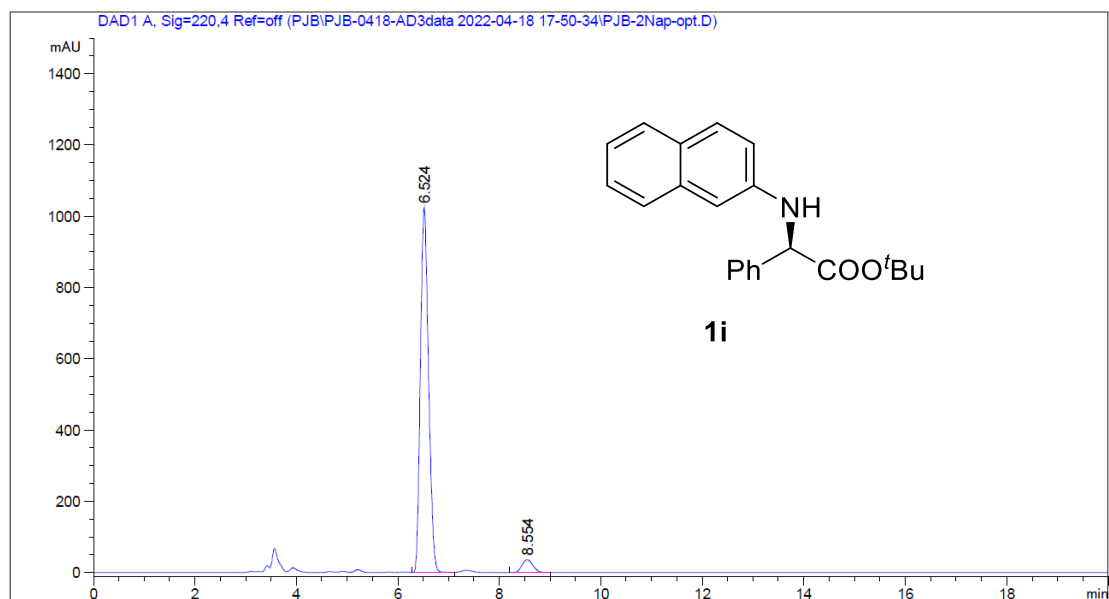


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.700	BB	0.1791	391.93268	33.34833	6.6240
2	14.206	BB	0.4329	5524.91211	197.15662	93.3760

(-)-tert-butyl 2-(naphthalen-2-ylamino)-2-phenylacetate (1i)

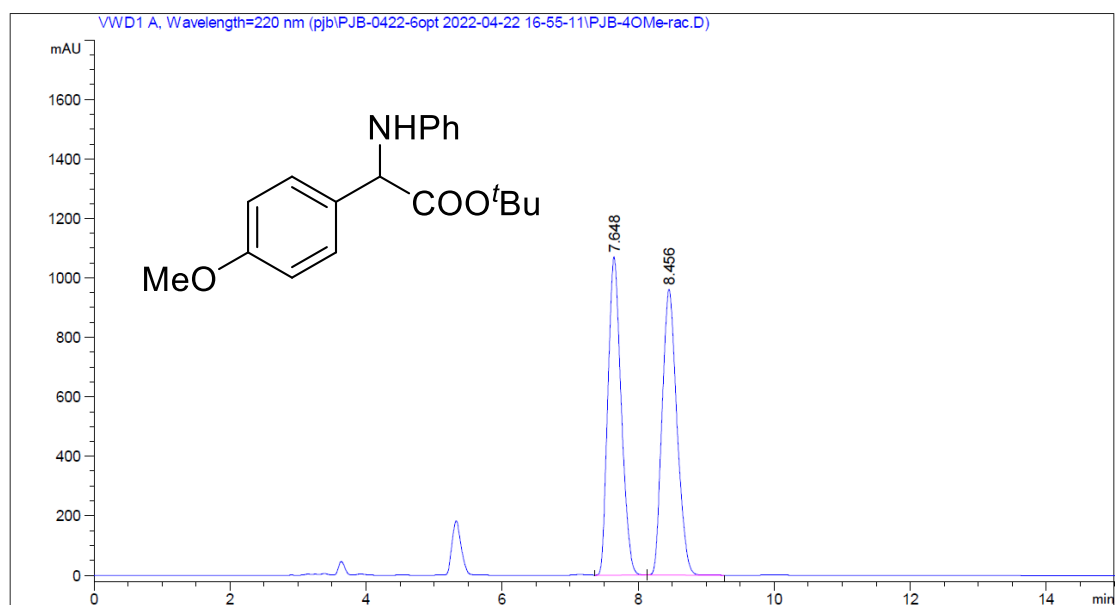


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.530	VB R	0.1718	6681.75195	592.93964	50.4754
2	8.553	BB	0.2312	6555.88574	441.57294	49.5246

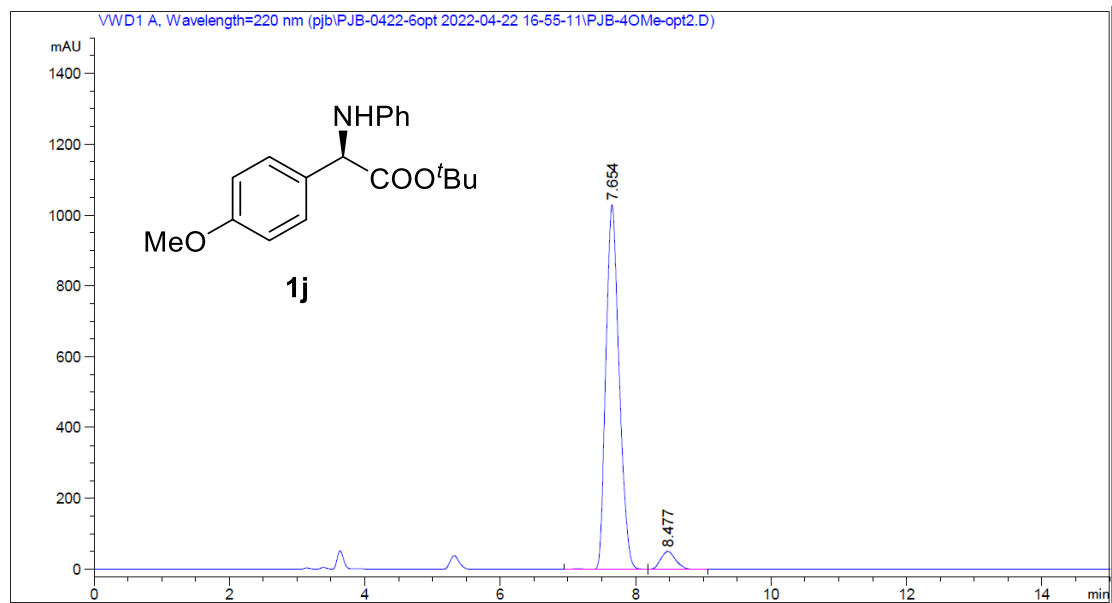


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.524	VV	0.1742	1.14362e4	1024.92065	95.4756
2	8.554	BB	0.2303	541.94458	36.69541	4.5244

(-)-tert-butyl 2-(4-methoxyphenyl)-2-(phenylamino)acetate (1j)

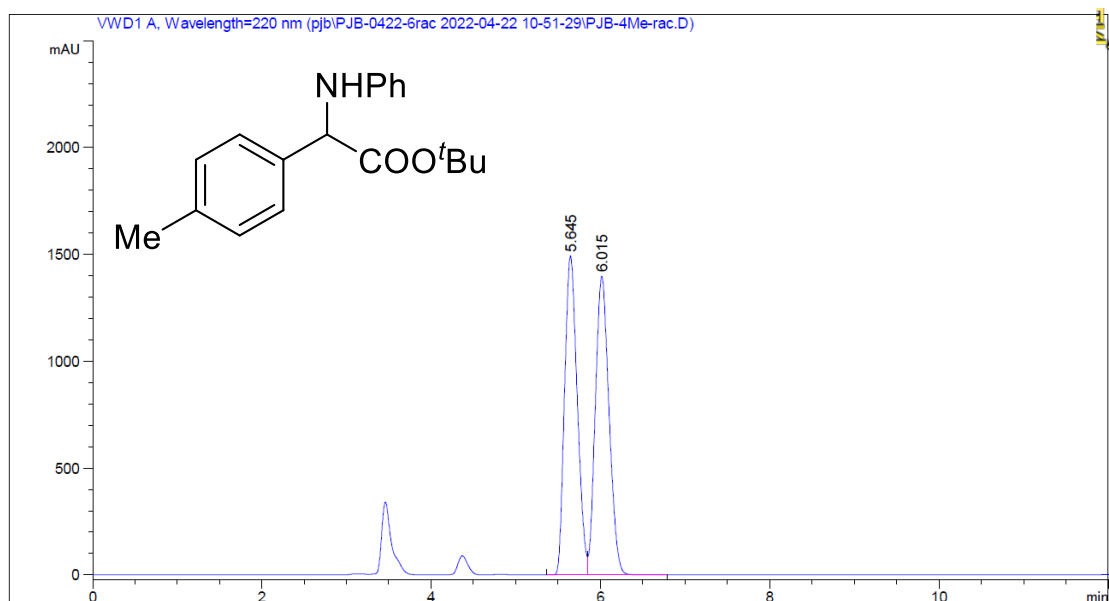


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	7.648	BB	0.2104	1.45196e4	1069.67859	49.9654
2	8.456	BB	0.2357	1.45397e4	960.13916	50.0346

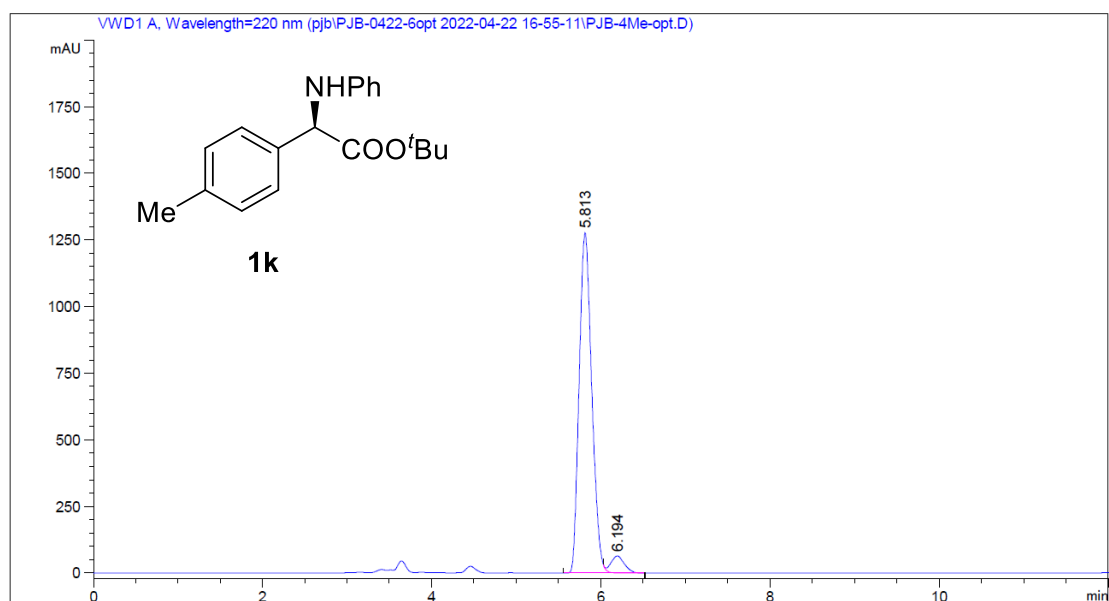


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	7.654	VV R	0.2111	1.40386e4	1029.48645	94.7762
2	8.477	VB	0.2361	773.76611	50.98281	5.2238

(-)-tert-butyl 2-(phenylamino)-2-(p-tolyl)acetate (1k)

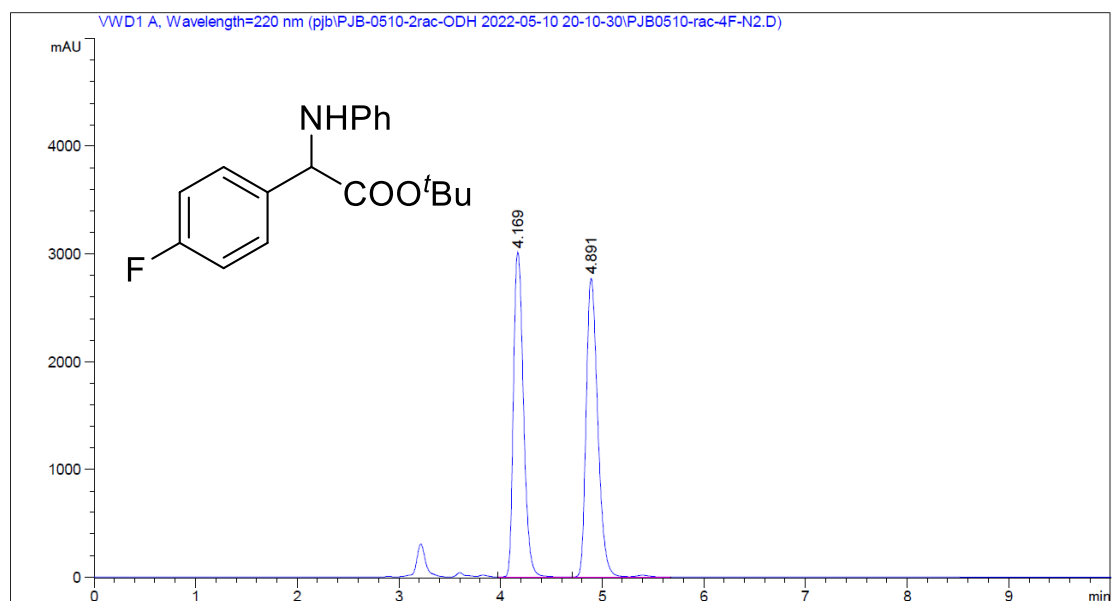


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.645	BV	0.1634	1.55141e4	1491.25940	49.6539
2	6.015	VB	0.1763	1.57303e4	1397.78125	50.3461

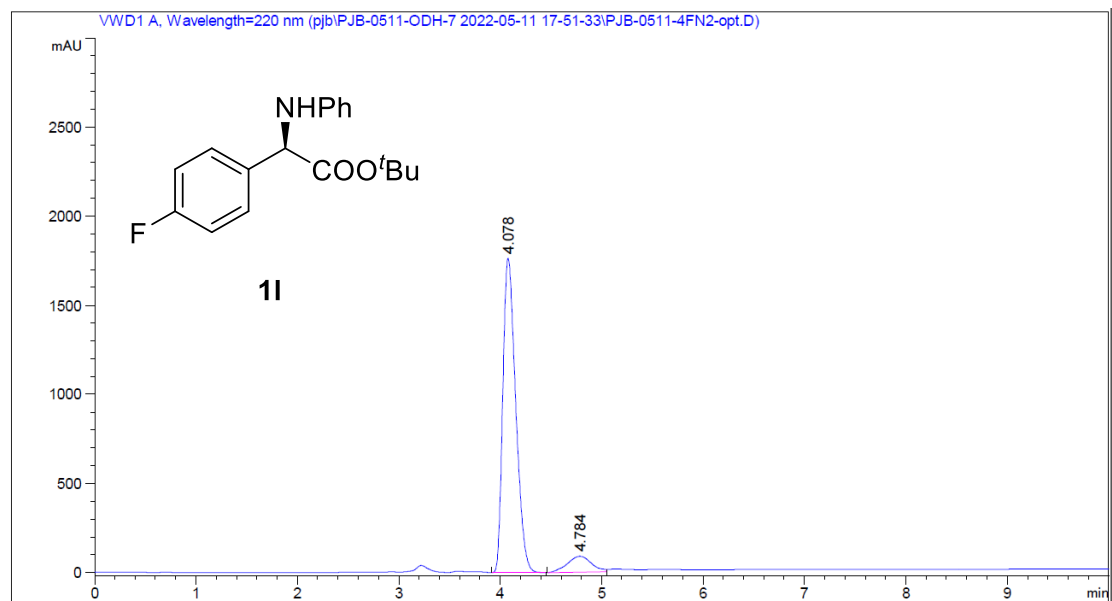


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.813	BV R	0.1607	1.32140e4	1277.61230	95.0352
2	6.194	VB E	0.1672	690.31891	63.80010	4.9648

(-)-*tert*-butyl 2-(4-fluorophenyl)-2-(phenylamino)acetate (11)

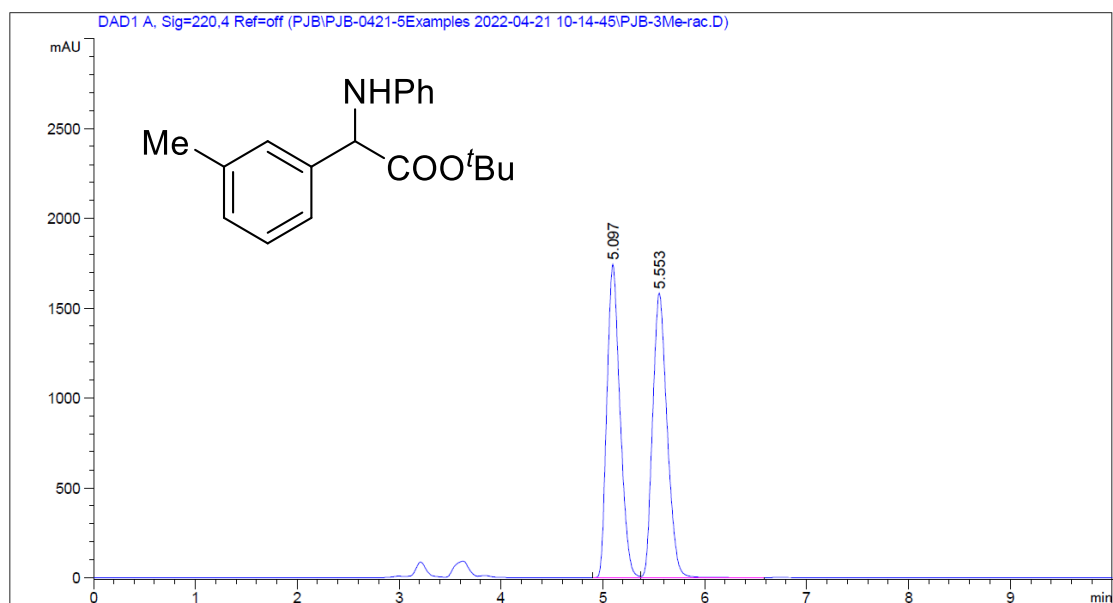


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	4.169	BV R	0.1077	2.06075e4	3016.51294	48.3452
2	4.891	BV R	0.1229	2.20182e4	2773.85913	51.6548

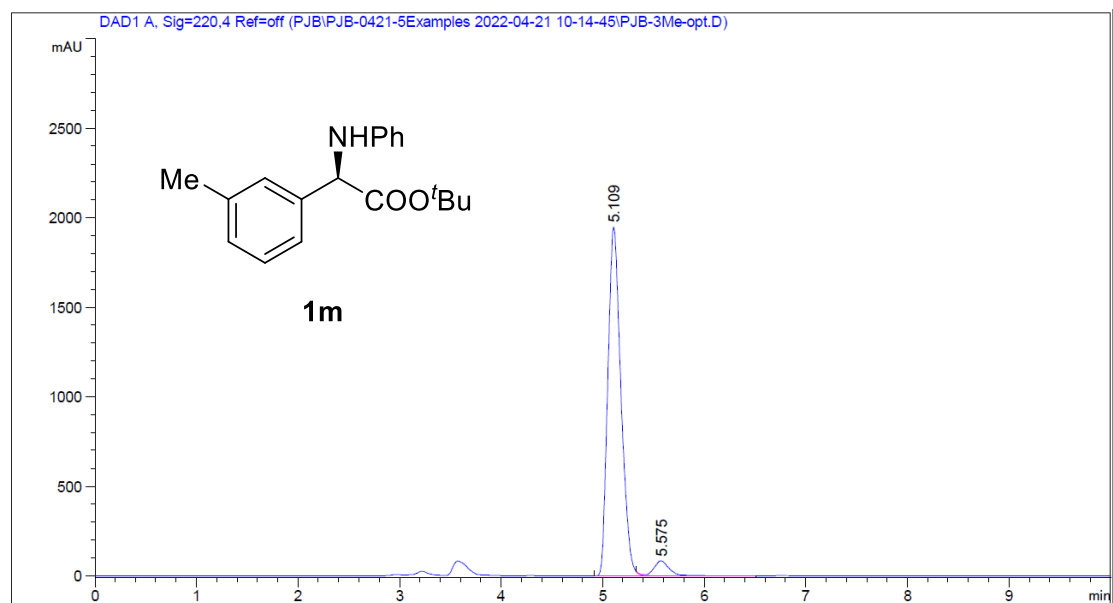


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	4.078	BB	0.1360	1.55791e4	1764.63623	91.4445
2	4.784	MF R	0.2740	1457.56848	88.64651	8.5555

(-)-tert-butyl 2-(phenylamino)-2-(m-tolyl)acetate (1m)

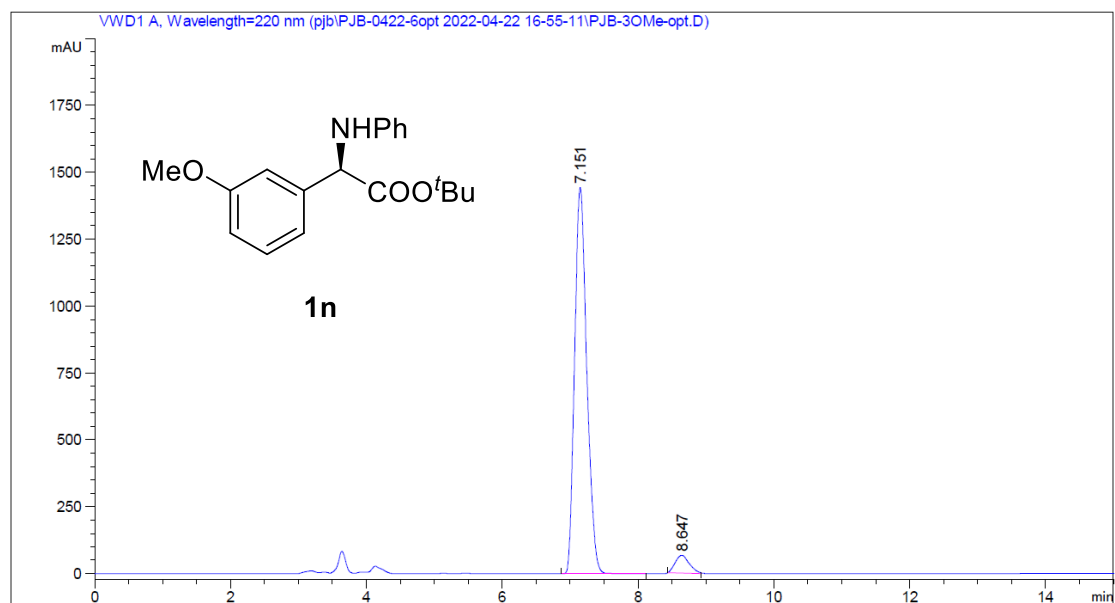
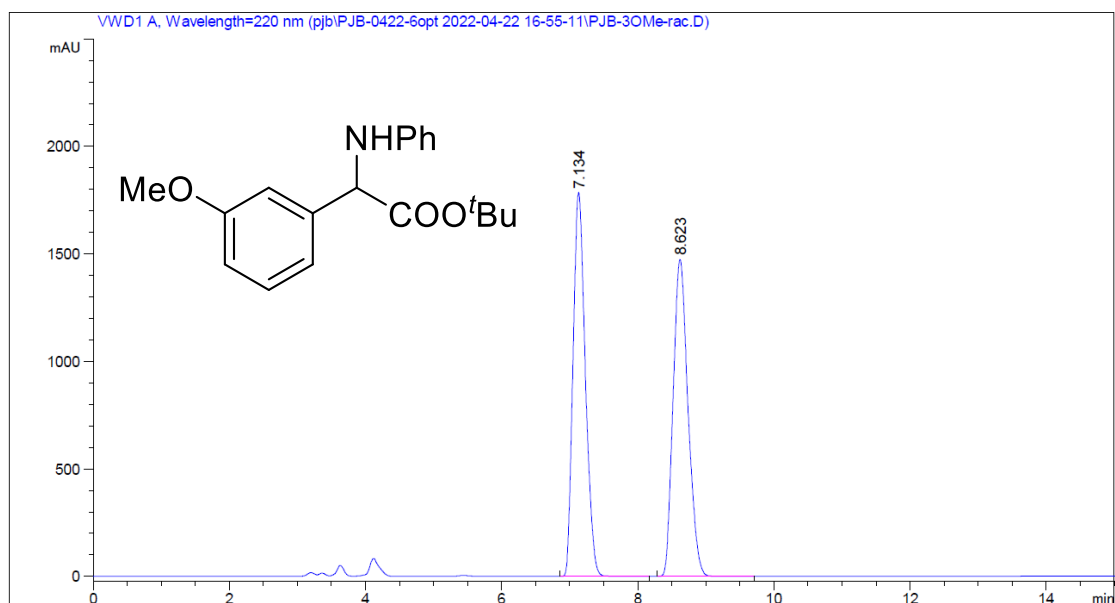


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.097	BV	0.1358	1.55252e4	1745.35999	49.8919
2	5.553	VB	0.1508	1.55925e4	1584.33472	50.1081

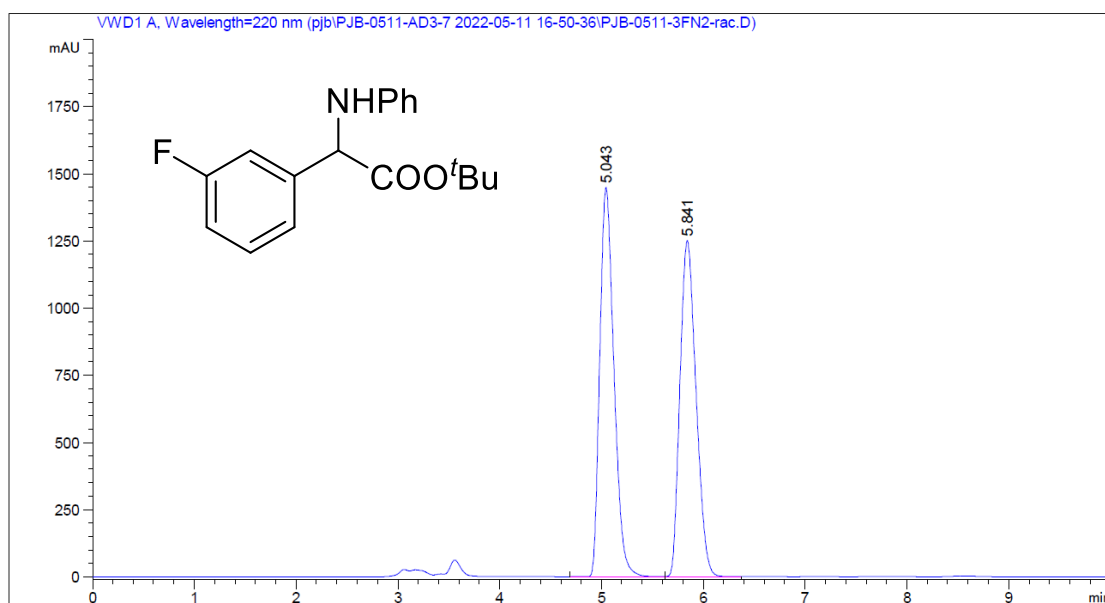


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.109	BV R	0.1382	1.74022e4	1948.37048	95.2510
2	5.575	VB E	0.1574	867.64148	83.41332	4.7490

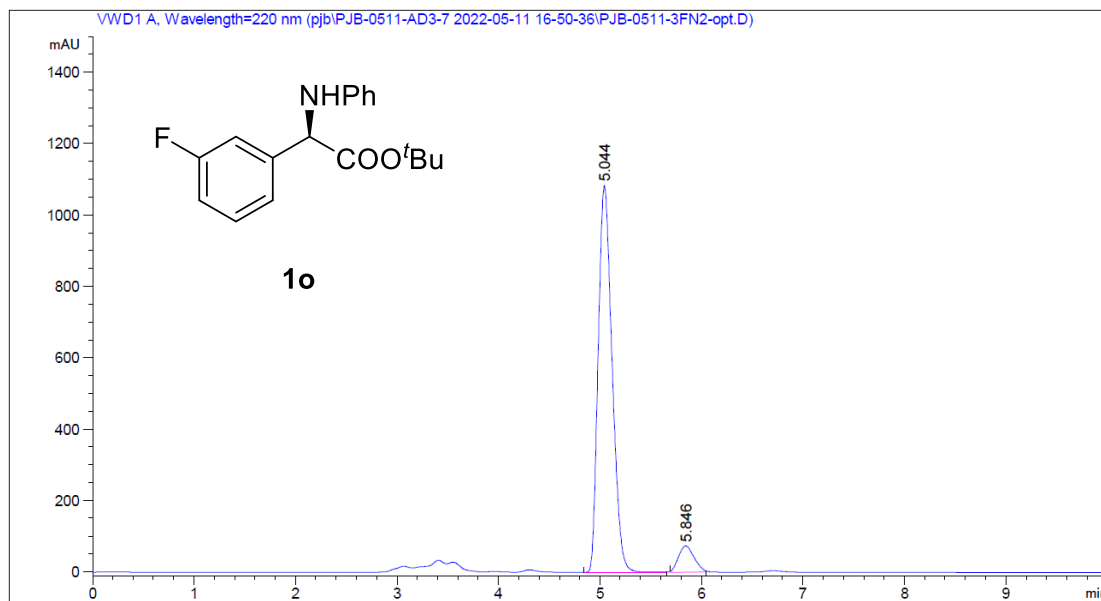
(-)-tert-butyl 2-(3-methoxyphenyl)-2-(phenylamino)acetate (1n)



(-)-tert-butyl 2-(3-fluorophenyl)-2-(phenylamino)acetate (1o)

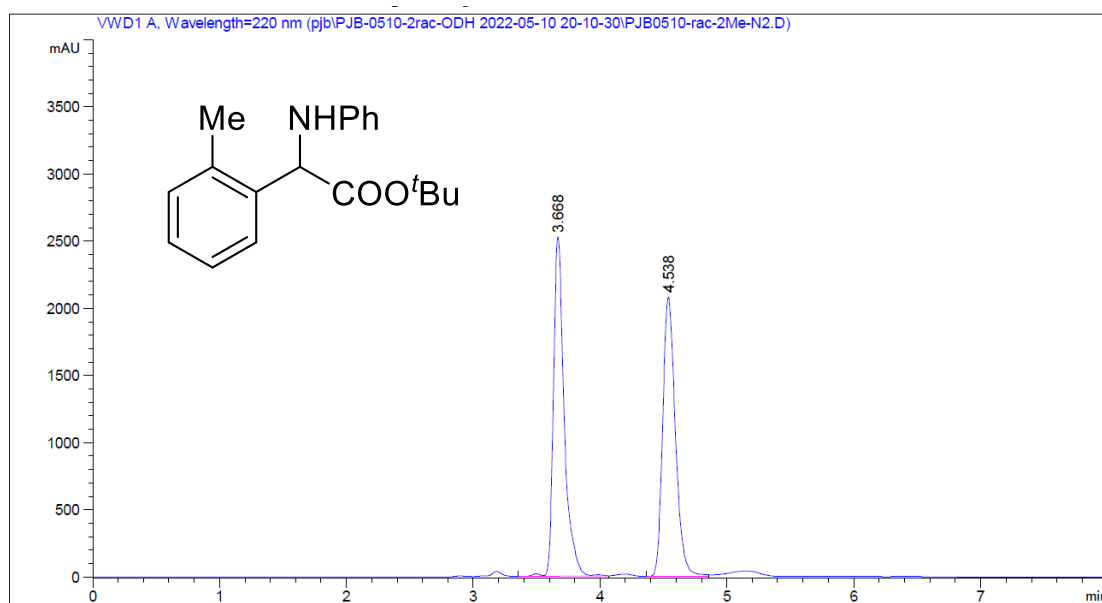


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.043	BV R	0.1496	1.39811e4	1449.06836	50.5067
2	5.841	VB	0.1717	1.37005e4	1251.56995	49.4933

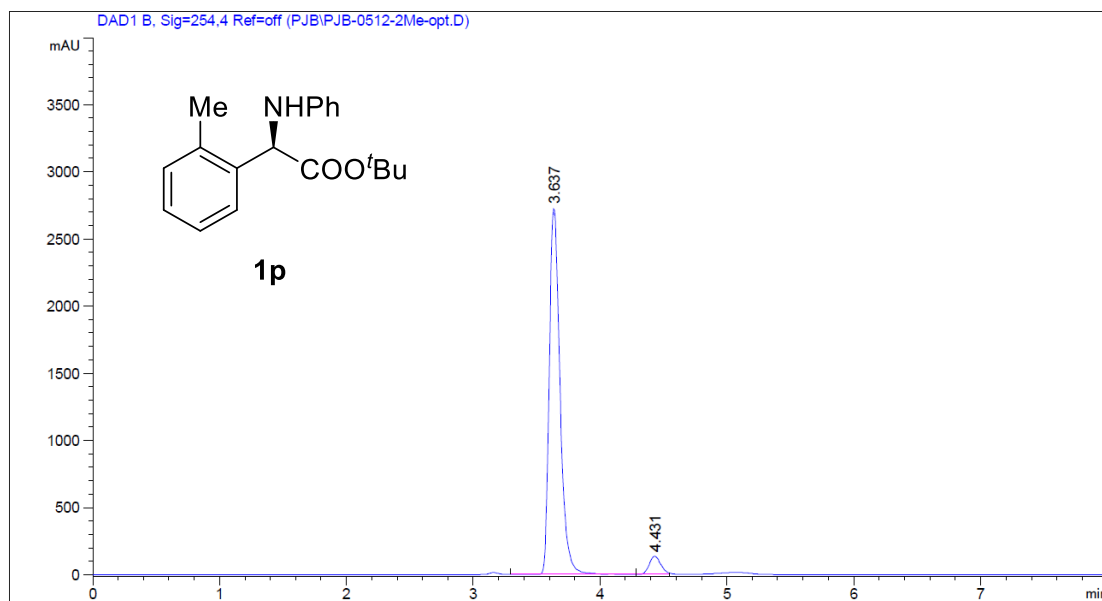


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.044	BV R	0.1468	1.02873e4	1083.28613	92.9303
2	5.846	MM R	0.1773	782.61029	73.55360	7.0697

(-)-tert-butyl 2-(phenylamino)-2-(o-tolyl)acetate (1p)

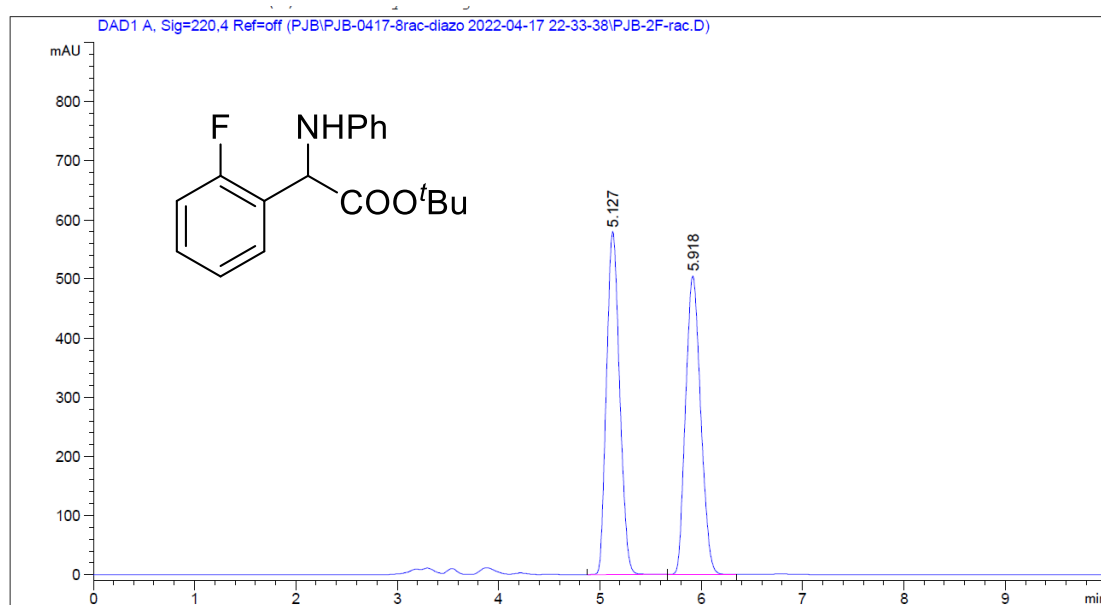


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	3.668	VV R	0.0892	1.49142e4	2529.33325	50.6368
2	4.538	MF R	0.1163	1.45391e4	2084.39746	49.3632

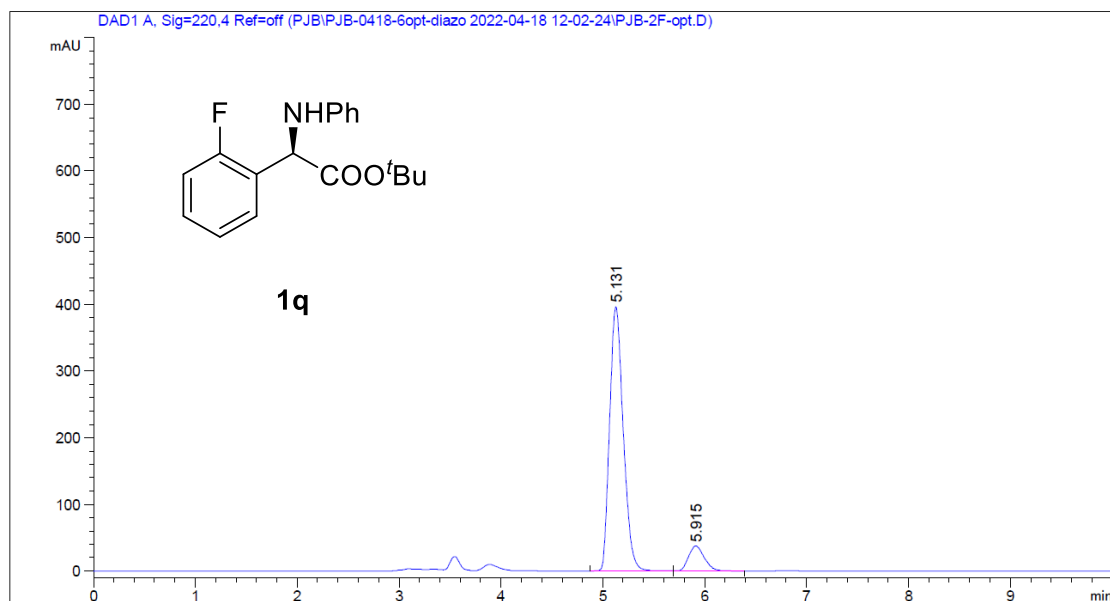


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	3.637	VV R	0.0924	1.60525e4	2731.53467	94.8444
2	4.431	MF R	0.1057	872.59406	137.62570	5.1556

(-)-tert-butyl 2-(2-fluorophenyl)-2-(phenylamino)acetate (1q)

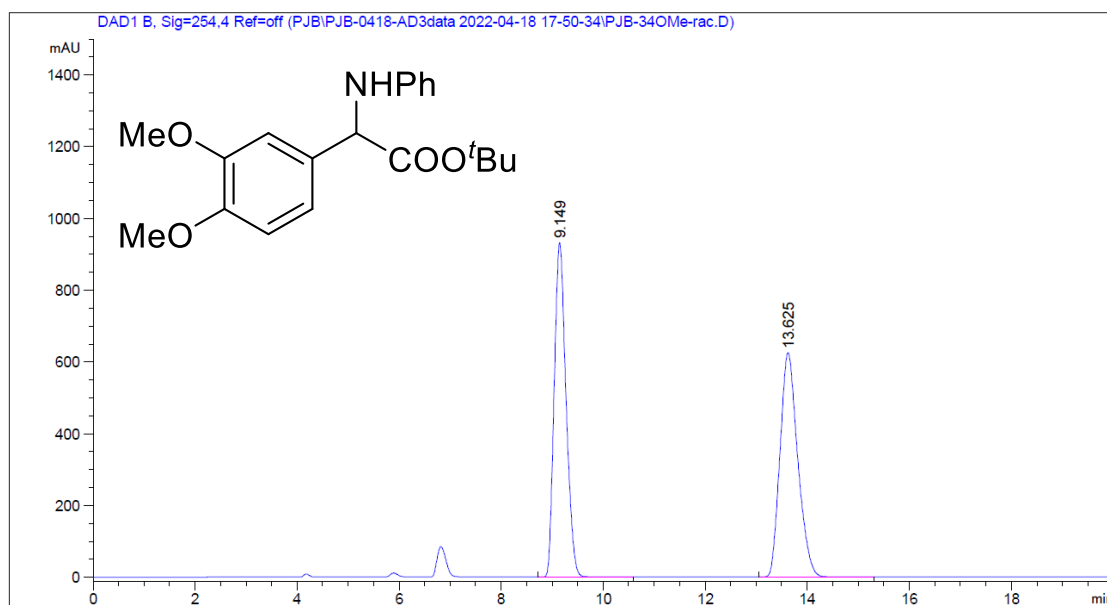


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.127	BB	0.1412	5226.06543	579.80334	50.0520
2	5.918	BB	0.1645	5215.19824	504.98361	49.9480

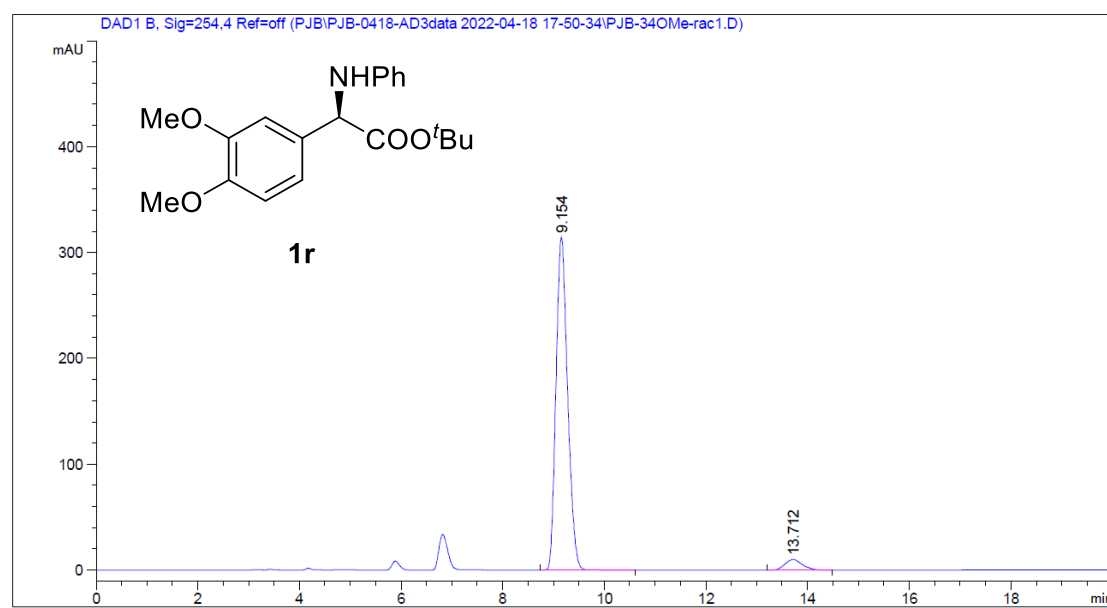


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.131	BV	0.1442	3673.96851	396.03690	90.2725
2	5.915	VB	0.1657	395.89413	37.33714	9.7275

(-)-*tert*-butyl 2-(3,4-dimethoxyphenyl)-2-(phenylamino)acetate (1r)

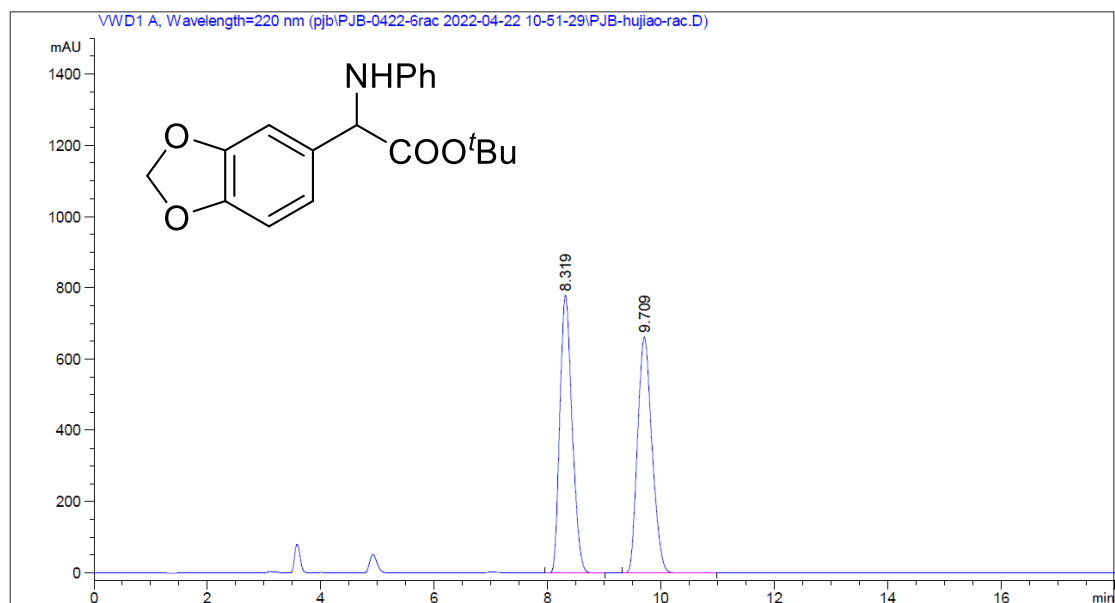


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	9.149	BB	0.2451	1.49421e4	931.90973	49.7865
2	13.625	BB	0.3613	1.50702e4	624.91559	50.2135

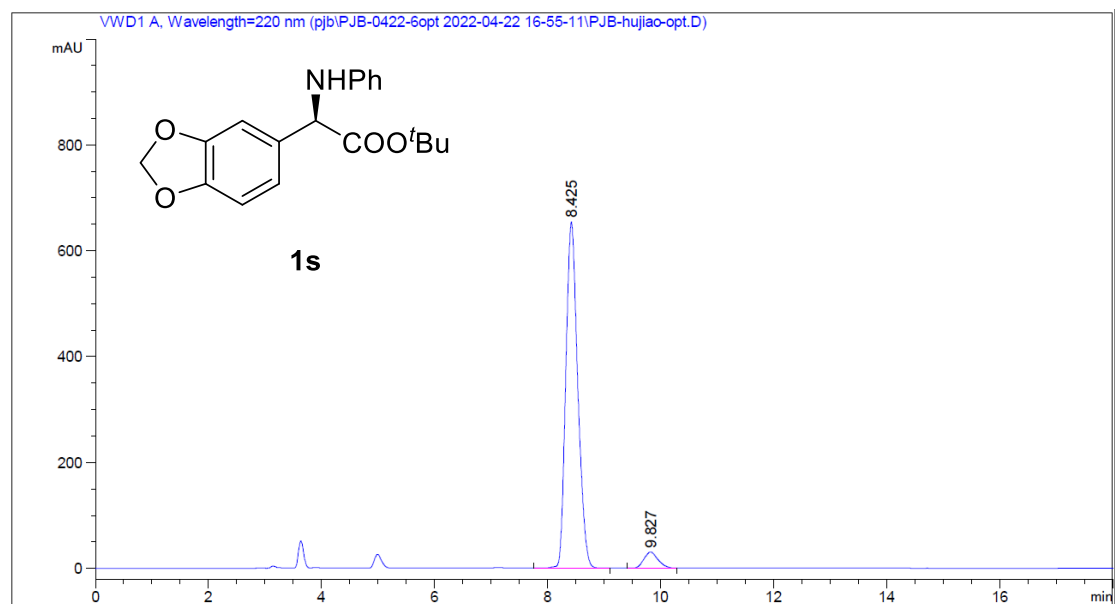


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	9.154	BB	0.2425	4973.70410	314.57571	95.5116
2	13.712	BB	0.3609	233.73279	9.92005	4.4884

(-)-tert-butyl 2-(benzo[d][1,3]dioxol-5-yl)-2-(phenylamino)acetate (1s)

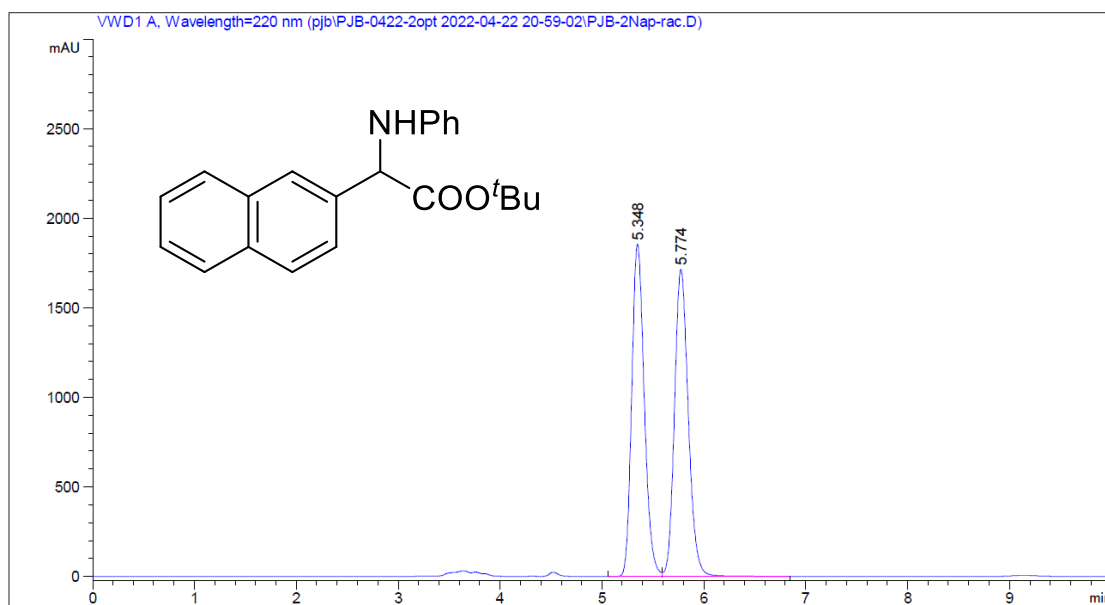


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	8.319	BB	0.2334	1.17152e4	779.36652	49.9662
2	9.709	BB	0.2756	1.17310e4	662.38708	50.0338

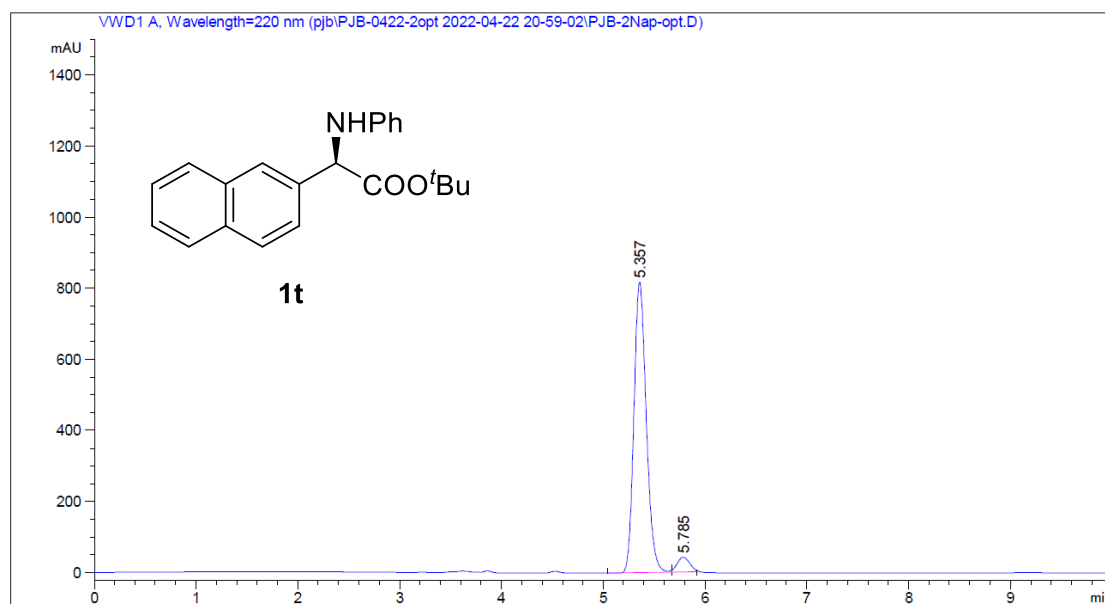


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	8.425	BB	0.2253	9602.04688	653.97711	94.9093
2	9.827	BB	0.2654	515.02899	30.12673	5.0907

(-)-tert-butyl 2-(naphthalen-2-yl)-2-(phenylamino)acetate (1t)

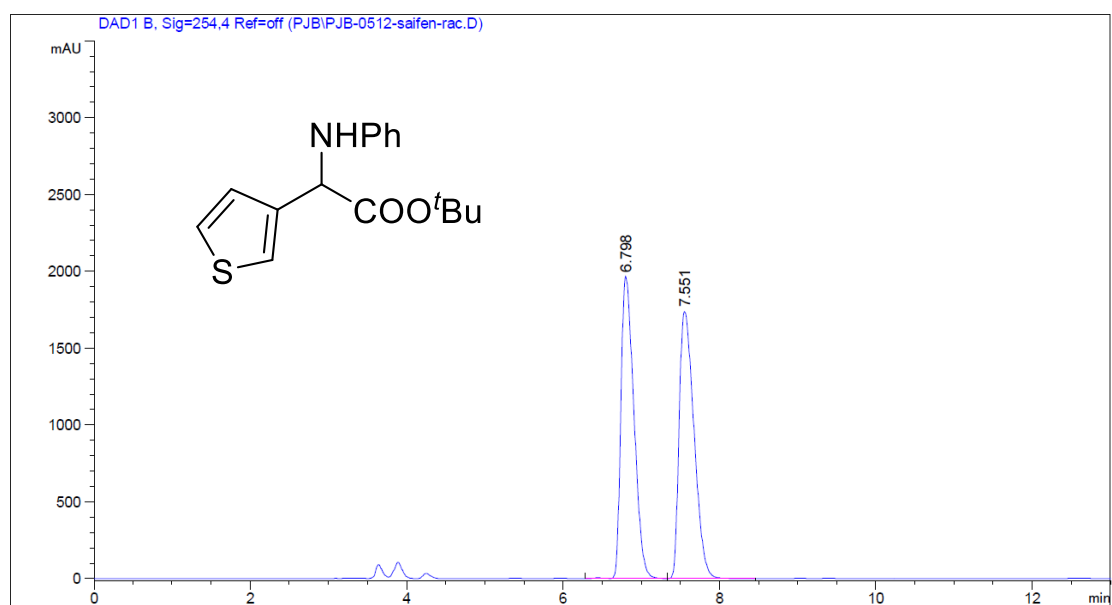


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.348	BV	0.1317	1.58633e4	1855.42639	49.5725
2	5.774	VB	0.1458	1.61369e4	1715.36108	50.4275

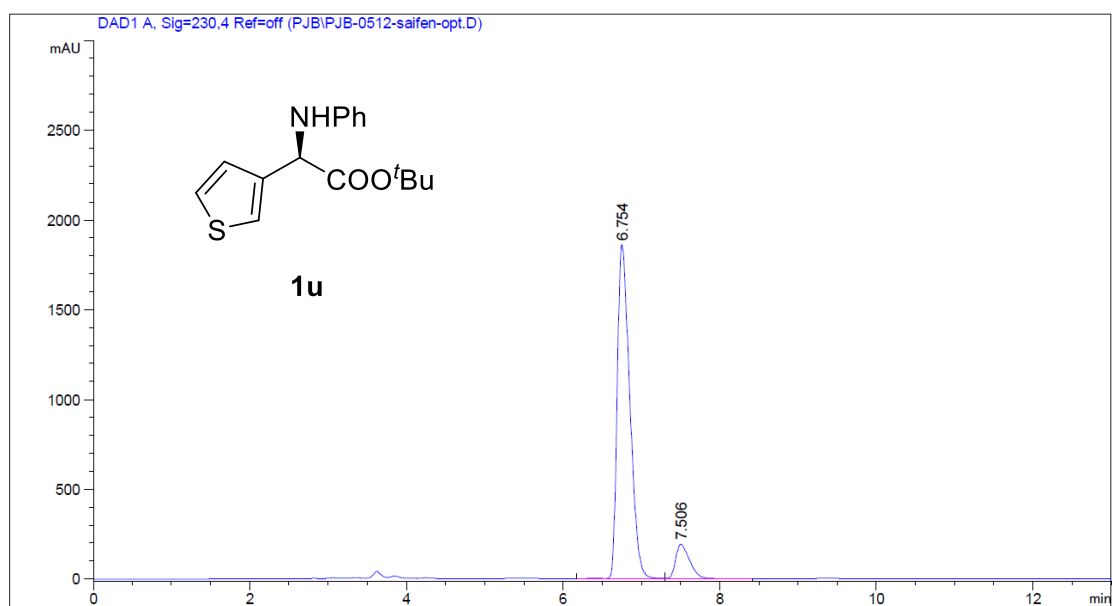


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	5.357	MF R	0.1404	6878.24756	816.34326	95.1058
2	5.785	FM R	0.1409	353.96024	41.88057	4.8942

(-)-*tert*-butyl 2-(phenylamino)-2-(thiophen-3-yl)acetate (1u)

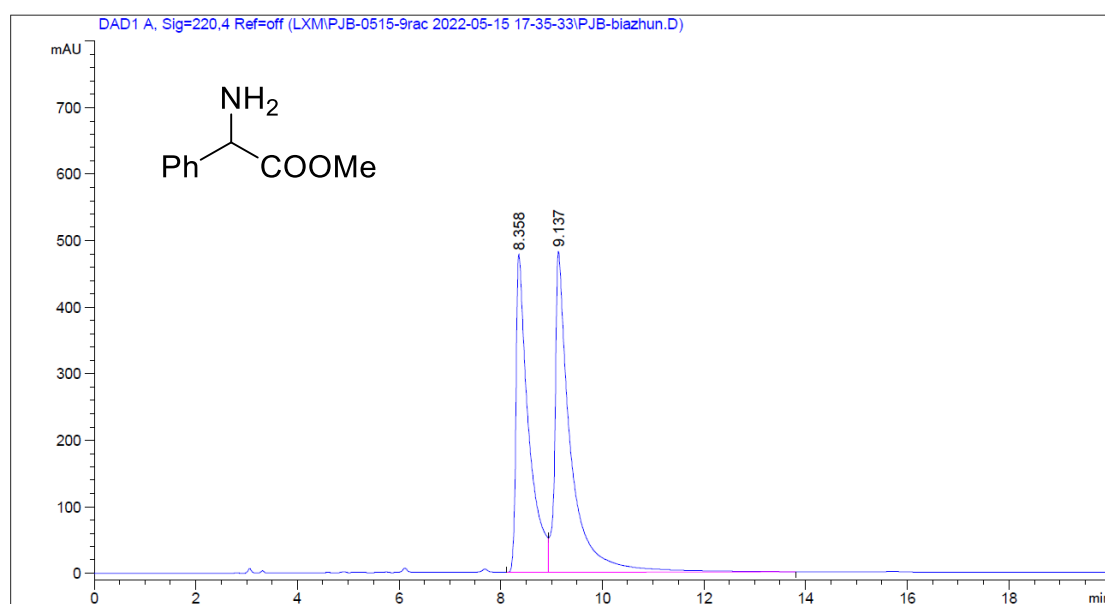


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.798	VB R	0.1841	2.26502e4	1966.27710	49.7213
2	7.551	BB	0.2098	2.29041e4	1737.88098	50.2787

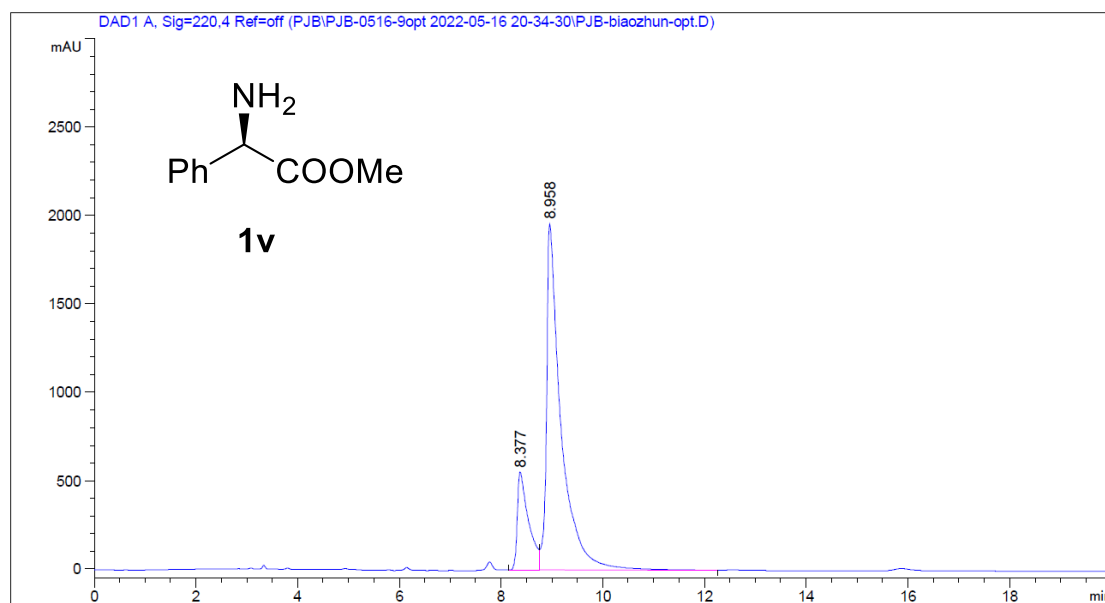


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	6.754	VV R	0.1754	2.06163e4	1858.91223	89.8581
2	7.506	VB	0.1910	2326.86841	189.78113	10.1419

(R)-methyl 2-amino-2-phenylacetate (1v)

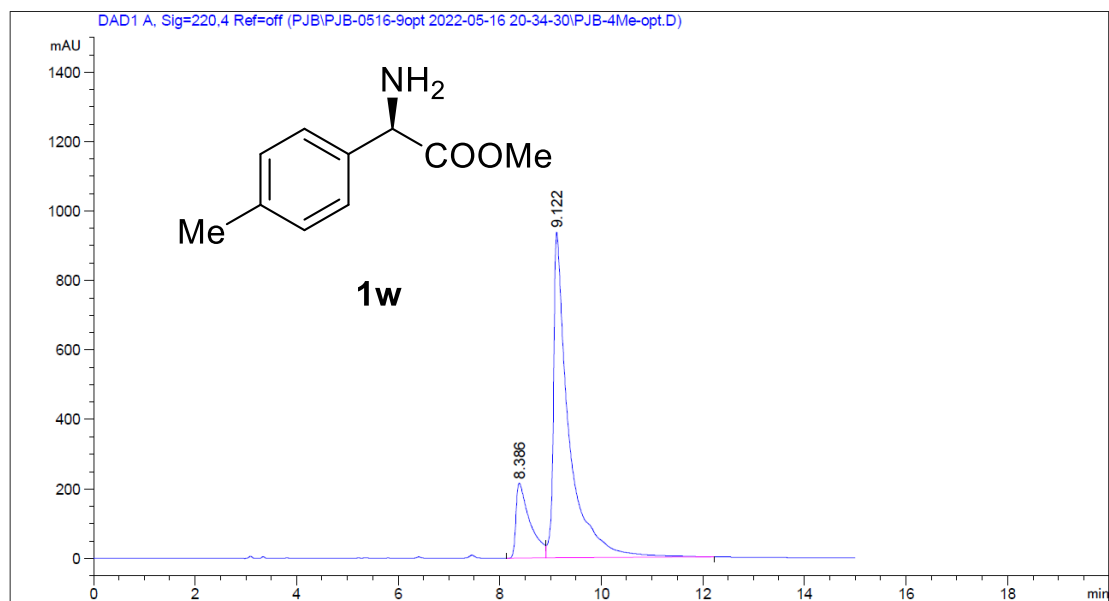
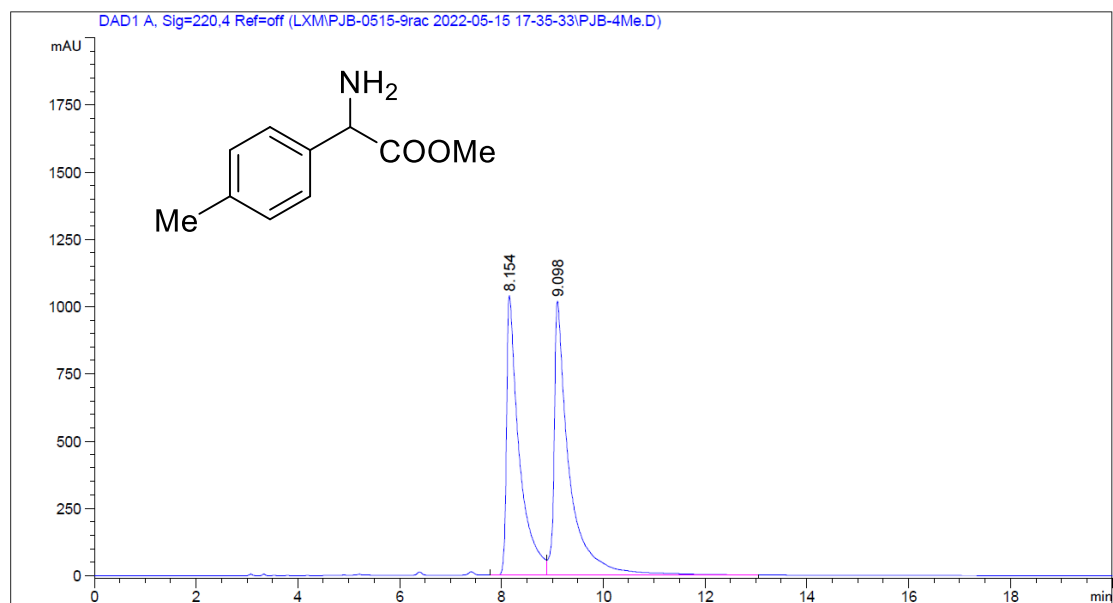


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	8.358	BV	0.2369	8228.66504	477.92642	44.3941
2	9.137	VB	0.2852	1.03068e4	482.53906	55.6059

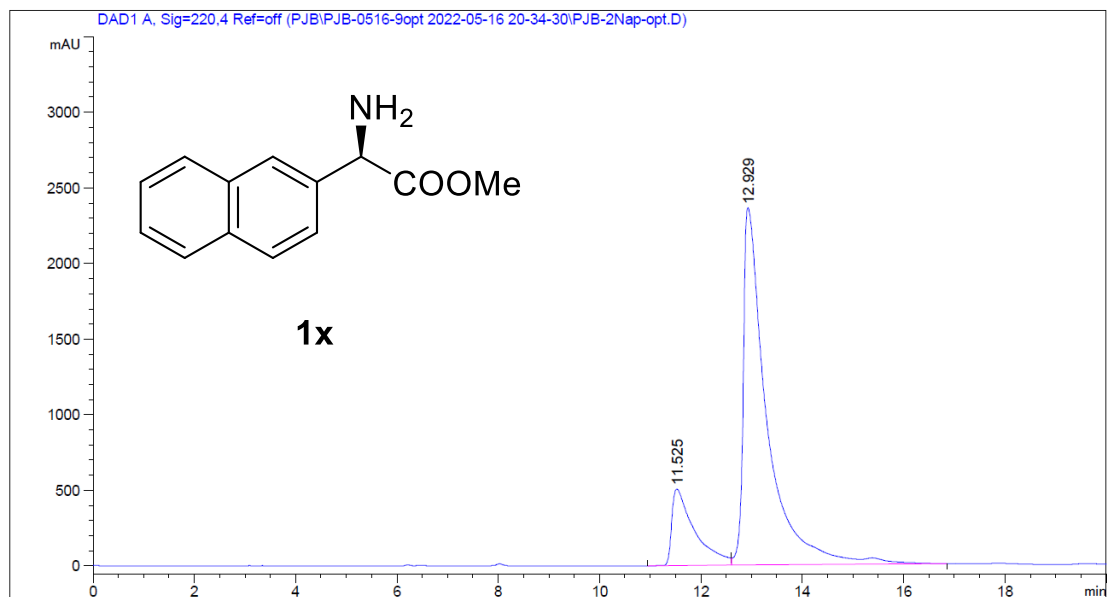
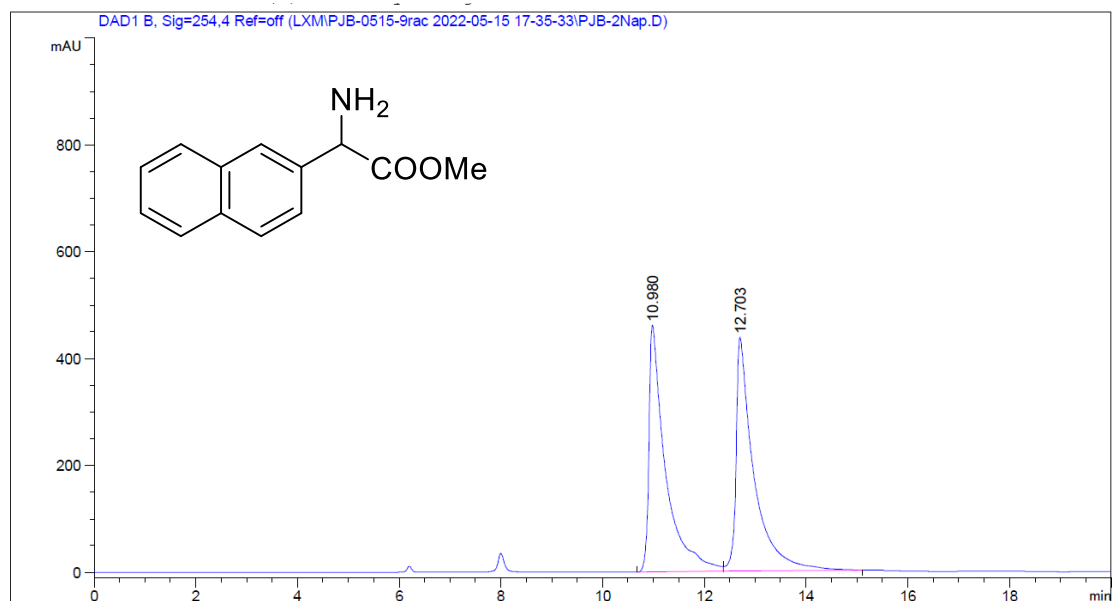


Peak #	RetTime [min]	Type	Width [min]	Area [mAU*s]	Height [mAU]	Area %
1	8.377	BV	0.2154	8528.44336	554.63654	18.0037
2	8.958	VB	0.2713	3.88421e4	1959.00464	81.9963

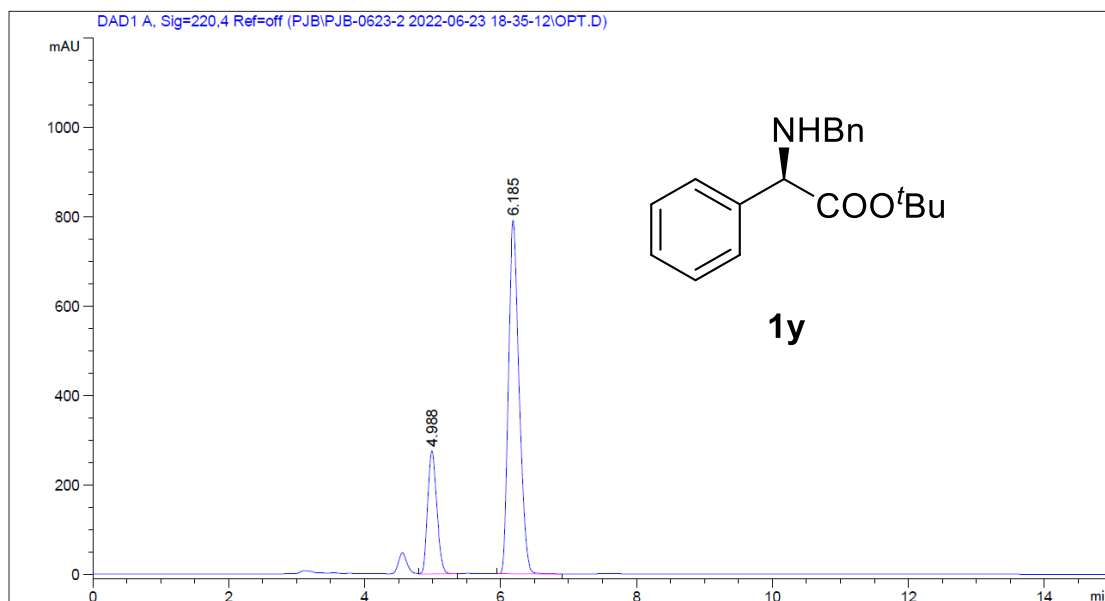
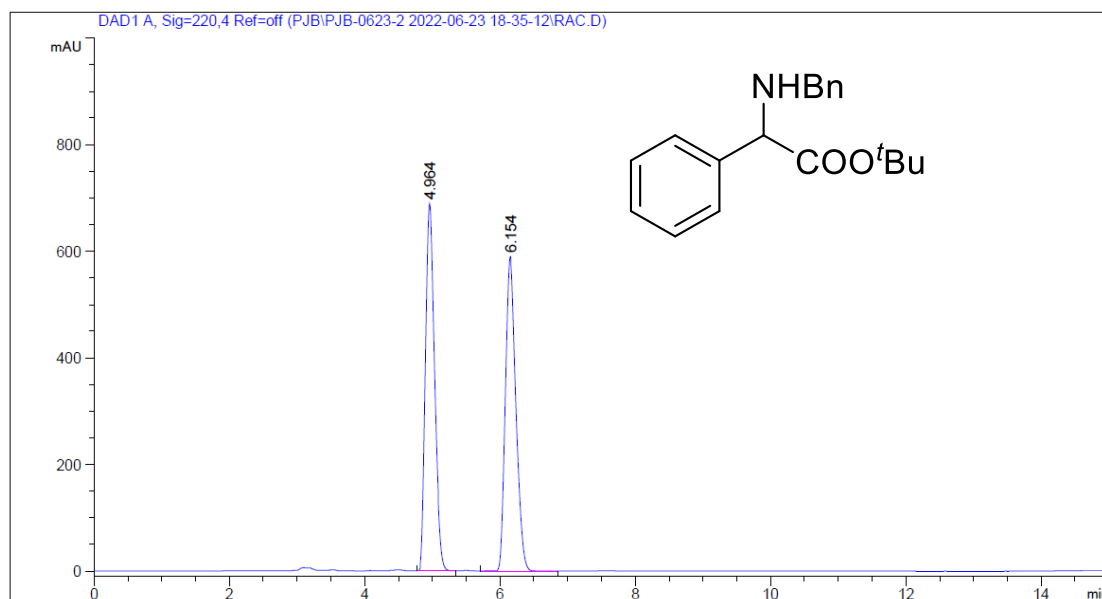
(-)-methyl 2-amino-2-(p-tolyl)acetate (1w)



(-)-methyl 2-amino-2-(naphthalen-2-yl)acetate (1x)



(-)-tert-butyl 2-(benzylamino)-2-phenylacetate (1y)



8. Cartesian coordinates (Å) for Optimized Structures

HA-1

C	0.52512700	3.69682500	0.53753900
C	-0.72477200	4.25472000	0.24115700
C	-1.70770300	3.42862200	-0.30364800
C	-1.45141500	2.07887600	-0.54830000
C	-0.18936700	1.49807100	-0.26422100
C	0.79430700	2.35417100	0.29543500
C	0.06506200	0.10660200	-0.54435000
C	1.20714200	-0.71956700	-0.47187300
O	1.17743600	-1.90976400	-0.90069200
O	2.30959900	-0.16305000	0.07946600
C	3.57061600	-0.90214400	0.20806500
H	1.30744200	4.31960900	0.96568200
H	-0.92522200	5.30461200	0.43471500
H	-2.69121200	3.82993200	-0.53719300
H	-2.26168000	1.47442300	-0.94561900
H	1.76883700	1.94855500	0.53068700
N	-1.00514600	-0.68514100	-1.17960300
H	-0.44553000	-1.56287300	-1.39284300
H	-1.31360000	-0.27399900	-2.07038400
C	-2.20016400	-1.04441400	-0.38433200
C	-3.40056200	-1.28147200	-1.04632700
C	-2.08653900	-1.16717200	0.99424100
C	-4.52317400	-1.65329800	-0.30328500
H	-3.46612700	-1.17235900	-2.12709400
C	-3.21245200	-1.54403000	1.72823800
H	-1.13571300	-0.95142300	1.46998000
C	-4.42923800	-1.78590000	1.08434700
H	-5.46702500	-1.83453300	-0.80902600
H	-3.13855300	-1.64012100	2.80744700
H	-5.30320800	-2.07157100	1.66244800
C	4.49148900	0.12942200	0.86283100
H	5.48573300	-0.30201500	1.02171300
H	4.59375600	1.01413600	0.22544900
H	4.08978000	0.44464500	1.83174500
C	3.38332100	-2.11393500	1.12494500
H	2.95235100	-1.79979400	2.08246800
H	2.72672300	-2.85502400	0.66646900
H	4.35770600	-2.57557700	1.32443000
C	4.10134900	-1.29483500	-1.17332000
H	4.16287900	-0.41150300	-1.81900200
H	5.10927300	-1.71437100	-1.07130600

H	3.45497700	-2.03549900	-1.64659200
A-1			
C	0.86180800	3.59744500	0.50049600
C	-0.24557900	4.30241400	0.00842300
C	-1.24869800	3.56863300	-0.63729500
C	-1.15010700	2.18867200	-0.78793400
C	-0.02912000	1.44886100	-0.30652200
C	0.97376800	2.21754800	0.35544400
C	0.00796300	0.01683200	-0.48166000
C	1.08623200	-0.88422700	-0.33686200
O	1.05914400	-2.09084200	-0.67307300
O	2.25424500	-0.32418100	0.18648300
C	3.48386500	-1.07613700	0.23843900
H	1.65901000	4.13450700	1.01463000
H	-0.32650700	5.38031500	0.13110800
H	-2.12796400	4.07892300	-1.02992600
H	-1.94343300	1.65074200	-1.29517600
H	1.84077900	1.71032600	0.75422500
N	-1.14570300	-0.61223400	-1.06838200
H	-0.88330500	-1.35188000	-1.71196700
C	-2.29267400	-0.90732800	-0.35444800
C	-3.24809400	-1.79859900	-0.89576700
C	-2.58454400	-0.30177100	0.88793300
C	-4.43597200	-2.07374900	-0.22530600
H	-3.03633700	-2.27592200	-1.85134300
C	-3.78182600	-0.58200600	1.54518900
H	-1.86010300	0.37672600	1.32263500
C	-4.72332500	-1.46498100	1.00438800
H	-5.14486200	-2.77359300	-0.66530800
H	-3.97968100	-0.10011400	2.50131100
H	-5.65401300	-1.67598400	1.52560500
C	4.48486900	-0.06304300	0.80846300
H	5.47862300	-0.51847600	0.90445100
H	4.55909300	0.81080600	0.15171500
H	4.15918200	0.27895200	1.79742100
C	3.35787800	-2.27627900	1.18826400
H	3.00849800	-1.93700000	2.17086600
H	2.64195700	-2.99879700	0.79365500
H	4.33691700	-2.75850900	1.31635000
C	3.92739800	-1.51107000	-1.16591400
H	3.96467800	-0.63931900	-1.83019200
H	4.93108200	-1.95559900	-1.12112100
H	3.22614000	-2.23756500	-1.57954000

HA-2

C	-2.69382200	-1.96949200	-0.00009700
C	-3.98848600	-1.43262400	0.00002000
C	-4.13200200	-0.04593000	0.00014400
C	-3.01114000	0.78666000	0.00014600
C	-1.69263200	0.26257100	0.00000200
C	-1.57012200	-1.15214800	-0.00012300
C	-0.55284500	1.13881200	-0.00011600
C	0.84607200	0.95895900	0.00032000
O	1.63173000	1.95067300	0.00006700
O	1.27313400	-0.32408400	0.00074300
C	2.69952200	-0.66828800	-0.00001100
H	-2.55703700	-3.04853200	-0.00020900
H	-4.85952800	-2.08137500	-0.00004800
H	-5.12334000	0.40133800	0.00028800
H	-3.18301300	1.86163100	0.00029700
H	-0.58122700	-1.59043300	-0.00017500
N	-0.80814500	2.59387200	-0.00019500
H	0.18351100	2.96038700	-0.00026900
H	-1.31035600	2.93468200	0.82891600
C	2.66911500	-2.19797100	-0.00029000
H	3.69045800	-2.59391000	-0.00017200
H	2.15021800	-2.57314400	0.88822600
H	2.15054700	-2.57293600	-0.88909900
C	3.37195100	-0.14431700	-1.27176500
H	2.82749900	-0.49127500	-2.15730200
H	3.40226300	0.94633300	-1.27903500
H	4.39694700	-0.52974600	-1.32810500
C	3.37293300	-0.14475800	1.27142600
H	2.82982000	-0.49294600	2.15730100
H	4.39839900	-0.52913600	1.32630200
H	3.40212900	0.94592500	1.27946300
H	-1.31044200	2.93446100	-0.82943200

A-2

C	-2.60351600	-1.92409800	-0.22891900
C	-3.89916200	-1.44105400	0.01809400
C	-4.04591900	-0.06521600	0.23668900
C	-2.95030600	0.79385200	0.20629600
C	-1.61926500	0.32804200	-0.03713500
C	-1.49934600	-1.07986000	-0.26031800
C	-0.54290000	1.27869900	-0.08003400
C	0.84241000	1.06123500	-0.18083200
O	1.71718800	1.96854300	-0.18438000

O	1.23868300	-0.28363100	-0.26608500
C	2.58001500	-0.67578800	0.07938300
H	-2.45043500	-2.98908200	-0.40676300
H	-4.75652600	-2.11082500	0.03644100
H	-5.03422900	0.34911400	0.43914100
H	-3.09307900	1.84979300	0.40952700
H	-0.52104100	-1.49267100	-0.46174300
N	-0.90995700	2.68008700	0.04773400
H	-0.03356700	3.20042700	0.07701700
C	2.51137600	-2.20866200	0.07221700
H	3.49426100	-2.63934400	0.30341900
H	1.78697700	-2.56158200	0.81452400
H	2.19412700	-2.56923000	-0.91313500
C	3.58785600	-0.19450600	-0.97608400
H	3.27918000	-0.54394400	-1.96912100
H	3.62489700	0.89582200	-0.98645400
H	4.58641400	-0.60084000	-0.76016200
C	2.95719900	-0.17860100	1.48257200
H	2.19787500	-0.49802800	2.20622400
H	3.92546400	-0.60125700	1.78414000
H	3.01729400	0.91139200	1.49687000
H	-1.40553500	2.99609000	-0.79191300

HA-3

C	-2.27279800	-3.57342600	0.22563200
C	-1.23368900	-4.49437100	0.03886300
C	0.04448600	-4.00418600	-0.22657400
C	0.28125900	-2.63069700	-0.29989300
C	-0.75722000	-1.67988700	-0.11491400
C	-2.04977700	-2.20335400	0.15384100
C	-0.49689400	-0.26544400	-0.20687100
C	-1.30928800	0.89073100	-0.18728900
O	-0.81769700	2.04165700	-0.34814700
O	-2.63311400	0.67465600	0.00978800
C	-3.60458400	1.77203900	0.01463800
H	-3.27923600	-3.92990700	0.43349000
H	-1.41846400	-5.56292500	0.09973800
H	0.87298200	-4.69220900	-0.37954200
H	1.29900100	-2.31100300	-0.51346300
H	-2.87027700	-1.51455500	0.30093300
N	0.89276400	0.16389900	-0.40408600
H	0.76863700	1.16184500	-0.68498200
H	1.36592400	-0.33717200	-1.16504100
C	-4.92760100	1.04132300	0.25388800

H	-5.75453800	1.75935800	0.27980200
H	-5.11854500	0.31871800	-0.54661200
H	-4.90370800	0.50366900	1.20785800
C	-3.31078500	2.73874400	1.16544300
H	-3.24894500	2.19023100	2.11229800
H	-2.37169700	3.26934400	1.00120500
H	-4.12450700	3.46938900	1.24638700
C	-3.61963800	2.47623100	-1.34502900
H	-3.77477700	1.74528900	-2.14669600
H	-4.44576800	3.19674300	-1.37470500
H	-2.68158700	3.00416600	-1.52361800
C	1.77519900	0.14168200	0.83421500
H	1.68079500	-0.85485700	1.26662800
H	1.32040700	0.86735200	1.51086600
C	3.19248300	0.48451000	0.47537000
C	3.58065700	1.82363500	0.32559700
C	4.12601300	-0.53009500	0.22434200
C	4.88124200	2.14227000	-0.06726700
H	2.86173700	2.61603100	0.52322100
C	5.42835000	-0.21280300	-0.16795900
H	3.83464800	-1.57081400	0.34926300
C	5.80674500	1.12397400	-0.31517300
H	5.17367700	3.18290400	-0.17562600
H	6.14624200	-1.00688200	-0.35325000
H	6.82018300	1.37222900	-0.61783800

A-3

C	2.30670400	3.45099500	0.51332200
C	1.31668900	4.41490300	0.27068700
C	0.05995300	3.96162400	-0.15120100
C	-0.19831700	2.60457000	-0.32026000
C	0.79145700	1.60179900	-0.08272600
C	2.06239600	2.09112700	0.34789600
C	0.44364100	0.21480100	-0.25840000
C	1.25730100	-0.93504600	-0.19502100
O	0.85423600	-2.11106700	-0.38032000
O	2.61413800	-0.69474200	0.06052100
C	3.59158800	-1.74655300	-0.05542200
H	3.29676300	3.76663300	0.84344600
H	1.51750300	5.47548800	0.40687100
H	-0.73531100	4.67899500	-0.35526000
H	-1.17268500	2.28110000	-0.66952400
H	2.85223400	1.38158600	0.55004200
N	-0.94673800	-0.07223700	-0.55039000

H	-0.96781300	-0.96959600	-1.03284300
C	4.91634500	-1.02138200	0.21527300
H	5.75729300	-1.72452200	0.16113200
H	5.07308000	-0.22611700	-0.52192700
H	4.90403000	-0.56572900	1.21200700
C	3.36694100	-2.83313500	1.00714400
H	3.33091500	-2.37558800	2.00337900
H	2.42321800	-3.34940900	0.82506000
H	4.19459700	-3.55600100	0.98784000
C	3.60367500	-2.33876900	-1.47243100
H	3.72459000	-1.53560800	-2.20925500
H	4.44432700	-3.03800300	-1.58048000
H	2.66710100	-2.86126300	-1.67428600
C	-1.75554100	-0.21490300	0.67005100
H	-1.67299200	0.73224200	1.22104800
H	-1.36868200	-1.00568900	1.33581800
C	-3.20964900	-0.48213900	0.35402100
C	-3.87201300	0.27167000	-0.62805300
C	-3.93173200	-1.47458700	1.02841700
C	-5.22011000	0.04862000	-0.91280000
H	-3.30506700	1.02142400	-1.17143400
C	-5.28365000	-1.70089400	0.74943400
H	-3.42347000	-2.08068300	1.77564200
C	-5.93471400	-0.93760400	-0.22288900
H	-5.71603700	0.64359600	-1.67717900
H	-5.82424500	-2.47859200	1.28519800
H	-6.98494900	-1.11270800	-0.44644100

SPAM-1

P	-0.99978200	0.75729300	0.84306500
O	-0.48539100	-0.43323200	1.83967400
O	-1.54825500	-0.06872500	-0.44776900
O	-1.90693900	1.66435500	1.56982000
C	-1.18778500	-3.02033900	0.34927500
C	0.17452600	-2.42047000	0.66336500
C	0.52390200	-1.30415800	1.41558700
C	1.87034400	-0.98865900	1.69353200
C	2.85341700	-1.83521800	1.15077600
C	2.52640400	-2.91426900	0.32997400
C	1.18674200	-3.19993100	0.08337400
C	0.58898200	-4.26856100	-0.80444400
C	-0.86466300	-3.77874500	-0.97967600
C	-2.45993100	-2.19730100	0.21973200
C	-2.66431400	-0.89943200	-0.23091800

C	-3.95209700	-0.35958700	-0.39643100
C	-5.04186300	-1.17861500	-0.05630700
C	-4.86093300	-2.46327300	0.46053900
C	-3.57072700	-2.96633800	0.60123600
C	-3.12231000	-4.29131500	1.17722200
C	-1.62232000	-4.04505700	1.44731800
C	2.26328500	0.21534400	2.46606600
C	3.40475700	0.94442300	2.09747600
C	3.79216600	2.08373400	2.80362900
C	3.05691600	2.53922500	3.90401300
C	1.91919700	1.80695800	4.27593400
C	1.52782900	0.66882500	3.57629600
C	-4.16490700	1.02749500	-0.87492200
C	-3.48365000	1.52944900	-1.99416800
C	-3.70828200	2.82989800	-2.44060700
C	-4.61201600	3.67713300	-1.78468100
C	-5.29213000	3.17192500	-0.66931500
C	-5.07265100	1.87103800	-0.21960600
H	3.89670600	-1.62252000	1.36128800
H	1.12164300	-4.36877100	-1.75668400
H	0.62117300	-5.25266600	-0.31658600
H	-1.57824000	-4.58363000	-1.17917800
H	-0.92103700	-3.06836300	-1.81307400
H	-6.04510900	-0.78428200	-0.18818500
H	-5.72213600	-3.05825900	0.75421600
H	-3.26838700	-5.10723200	0.45545700
H	-3.67566200	-4.56366200	2.08262000
H	-1.01668000	-4.95603500	1.42346500
H	-1.49337700	-3.57927700	2.43082100
H	3.97875500	0.63870700	1.22915900
H	4.67700800	2.63125000	2.48588900
H	1.32703100	2.13413500	5.12779900
H	0.64063600	0.13148300	3.88965200
H	-2.77880500	0.89521800	-2.52232300
H	-3.17299200	3.19286900	-3.31549800
H	-5.99378700	3.80953700	-0.13593100
H	-5.58585300	1.51484500	0.66910500
H	3.31415200	-3.51636700	-0.11550400
N	0.30403900	1.55401400	0.16492600
H	0.48031700	2.42539500	0.65447700
C	1.31478500	1.17343300	-0.73405100
C	2.44523600	2.00034200	-0.83836800
C	1.24210300	0.01898000	-1.52724000
C	3.49311300	1.66593000	-1.68738800

H	2. 51066700	2. 89787800	-0. 22978200
C	2. 29926900	-0. 31945000	-2. 36318900
H	0. 37207000	-0. 61838900	-1. 47800800
C	3. 43174500	0. 49534600	-2. 44960200
H	4. 36158500	2. 31347400	-1. 74981200
H	2. 23639900	-1. 22641900	-2. 95522100
C	-4. 81978400	5. 09948900	-2. 24580800
H	-5. 82747400	5. 45544700	-2. 00509900
H	-4. 10752000	5. 77905200	-1. 75837900
H	-4. 67444400	5. 19516700	-3. 32750600
C	3. 47638700	3. 76393400	4. 68004300
H	2. 63389500	4. 44924800	4. 83162100
H	3. 85396900	3. 49133500	5. 67423300
H	4. 27054200	4. 31232900	4. 16284100
C	4. 59464100	0. 08290900	-3. 29676200
F	5. 30948000	1. 14343600	-3. 75209900
F	5. 47836500	-0. 70190500	-2. 60915500
F	4. 21724100	-0. 64028000	-4. 38225200

SPAM-1-

P	-0. 89601400	0. 52981300	1. 03166300
O	-0. 51888000	-1. 02563500	1. 59854900
O	-1. 57116000	0. 08221200	-0. 46336300
O	-1. 92844100	1. 04545900	1. 96757200
C	-1. 49784500	-2. 95994500	-0. 54690100
C	-0. 07573100	-2. 59828800	-0. 14469400
C	0. 38404200	-1. 78483200	0. 89293700
C	1. 76798100	-1. 69718100	1. 17093700
C	2. 65353900	-2. 41038600	0. 34497800
C	2. 20762600	-3. 16119700	-0. 74239800
C	0. 83952200	-3. 24971300	-0. 98286600
C	0. 11935400	-3. 95556700	-2. 11107200
C	-1. 28550800	-3. 31893900	-2. 05379800
C	-2. 68575400	-2. 02812200	-0. 37213900
C	-2. 74395200	-0. 63680700	-0. 42840700
C	-3. 98793700	0. 03013100	-0. 38393600
C	-5. 14965100	-0. 74739600	-0. 24352600
C	-5. 09558900	-2. 13759800	-0. 11445700
C	-3. 85760800	-2. 77175500	-0. 17495700
C	-3. 53077000	-4. 24187500	-0. 02048600
C	-2. 00650700	-4. 21788100	0. 22856400
C	2. 30206400	-0. 84194700	2. 25763400
C	3. 48328300	-0. 11134500	2. 06338900
C	4. 01087400	0. 68544900	3. 07968900

C	3.37830500	0.78549300	4.32291400
C	2.19889400	0.05174100	4.51758900
C	1.66637800	-0.74335100	3.50716700
C	-4.07963300	1.50818600	-0.44923500
C	-3.30866000	2.25388500	-1.35555800
C	-3.42089000	3.64055100	-1.41702700
C	-4.29913100	4.33778300	-0.57614800
C	-5.06839700	3.59361100	0.32564500
C	-4.95851200	2.20480200	0.39157200
H	3.71830800	-2.34592100	0.54893600
H	0.61261400	-3.81024000	-3.07969700
H	0.07075900	-5.04087700	-1.93671600
H	-2.07936100	-3.96683400	-2.44095200
H	-1.29544500	-2.38939300	-2.63462500
H	-6.11156300	-0.24203200	-0.22034300
H	-6.00878800	-2.70997300	0.03559200
H	-3.77439000	-4.80315100	-0.93498800
H	-4.08678000	-4.71229400	0.79962200
H	-1.49504400	-5.13367100	-0.08711000
H	-1.80706200	-4.06846300	1.29575300
H	3.96977100	-0.12740400	1.09369500
H	4.91724400	1.25768300	2.88995500
H	1.67909700	0.11668800	5.47192600
H	0.73915100	-1.27836600	3.67693100
H	-2.60535300	1.74305400	-2.00359800
H	-2.80338300	4.19271800	-2.12314800
H	-5.74577500	4.10887400	1.00456300
H	-5.52954500	1.65581400	1.13485600
H	2.92264200	-3.65694800	-1.39521700
N	0.40267500	1.37236200	0.72518900
C	1.35396000	1.31017100	-0.24224000
C	2.48363400	2.16796900	-0.10847000
C	1.34261600	0.45886300	-1.38432500
C	3.53306700	2.14758900	-1.01133000
H	2.50664900	2.82780700	0.75319200
C	2.39895900	0.43495500	-2.28180500
H	0.49711500	-0.19880300	-1.53898100
C	3.51101300	1.27510400	-2.11374400
H	4.38310400	2.80878200	-0.86337000
H	2.36244800	-0.24864800	-3.12552200
C	-4.37314000	5.84597600	-0.61093800
H	-5.30834700	6.21087400	-0.17047800
H	-3.54515300	6.29509400	-0.04523900
H	-4.30959500	6.22832300	-1.63707400

C	3.91834600	1.68570200	5.40891600
H	3.32208900	2.60460200	5.49466300
H	3.89576600	1.19284800	6.38915000
H	4.95334700	1.98203400	5.20327500
C	4.67352300	1.19106500	-3.02782200
F	5.31193100	2.38852700	-3.20177800
F	5.66184400	0.33147900	-2.58938200
F	4.35002600	0.74314900	-4.27665200

TSRa-I

P	0.77851900	-1.09429200	0.48178600
O	1.98900700	-1.12471700	1.56627100
O	1.52054700	-1.69111400	-0.85116600
O	-0.30213400	-1.99165700	0.99553100
C	4.28457100	-2.53742900	0.09096900
C	4.16454500	-1.09691800	0.56339600
C	3.17706000	-0.46086900	1.31255400
C	3.33649600	0.86640100	1.74930600
C	4.50460300	1.54393200	1.37197300
C	5.47026800	0.94810000	0.56452800
C	5.29286400	-0.37129500	0.16058000
C	6.16879200	-1.21602200	-0.73390800
C	5.21995200	-2.35433400	-1.13715800
C	3.08404700	-3.39449300	-0.26109600
C	1.85302500	-3.03368700	-0.78996500
C	0.90271300	-4.00373200	-1.15477700
C	1.23773500	-5.34979700	-0.95408100
C	2.44936600	-5.72748700	-0.37482600
C	3.36690700	-4.74403600	-0.02195200
C	4.70516300	-4.87738300	0.66730900
C	4.96914100	-3.44166500	1.15391100
C	2.27330000	1.56922700	2.50918900
C	1.93257700	2.88464300	2.17006000
C	0.91972800	3.55494800	2.84880100
C	0.21927200	2.94046100	3.89144300
C	0.58490900	1.63708500	4.24763800
C	1.58682300	0.95505700	3.56517900
C	-0.42422200	-3.62614500	-1.70667200
C	-1.58344900	-4.28645500	-1.28101600
C	-2.83066500	-3.96547900	-1.81748400
C	-2.96440900	-2.97096000	-2.79207100
C	-1.80277100	-2.30136600	-3.20728400
C	-0.55655700	-2.61868600	-2.67622600
H	4.63604000	2.57377700	1.70778400

H	6.55868800	-0.65365100	-1.59373500
H	7.03942400	-1.59822300	-0.17556400
H	5.73178800	-3.28726000	-1.41268700
H	4.60579200	-2.03857000	-1.99496600
H	0.51601900	-6.11356000	-1.24899000
H	2.66502700	-6.78193000	-0.19412600
H	5.47906500	-5.20858900	-0.04483400
H	4.68450400	-5.61097400	1.48517300
H	6.03477900	-3.20320600	1.27990700
H	4.46644800	-3.27949400	2.11945000
H	2.43680300	3.37160500	1.33118000
H	0.65454600	4.57135300	2.54724900
H	0.06777700	1.14541100	5.07414400
H	1.84343500	-0.06524100	3.85240800
H	-1.51038100	-5.04458500	-0.49869900
H	-3.72259500	-4.47982300	-1.45095800
H	-1.88513100	-1.50281100	-3.94716300
H	0.32859200	-2.07991700	-3.01542900
H	6.35061400	1.51254600	0.25227000
N	0.23160100	0.36732800	0.09274100
H	-1.02893900	0.42956100	0.11591800
C	0.83622800	1.45505800	-0.53282900
C	2.07229200	1.39675000	-1.19882000
C	0.15474800	2.68767000	-0.48095300
C	2.62341400	2.54265800	-1.76327800
H	2.61402100	0.45226700	-1.26156600
C	0.70359500	3.82224300	-1.05835100
H	-0.79841600	2.74072400	0.04869400
C	1.94516300	3.75796000	-1.69691300
H	3.59430300	2.48860000	-2.25711800
H	0.18147300	4.77922800	-0.99069300
C	-4.30416700	-2.59363100	-3.36389400
H	-5.12145200	-3.08637200	-2.82019700
H	-4.44209000	-1.50399600	-3.30373800
H	-4.37372600	-2.88335500	-4.42349400
C	-0.91731200	3.65024100	4.57583300
H	-1.81769000	3.62606400	3.94172500
H	-1.17392100	3.16862200	5.52911800
H	-0.67028000	4.70333800	4.77173000
C	2.50387000	4.99207300	-2.33708900
F	3.80733500	4.87576000	-2.60364400
F	1.89317600	5.27652900	-3.49472800
F	2.34831200	6.06774700	-1.55588100
C	-4.32234100	2.04792300	2.99010500

C	-3.71345700	1.45592500	4.09616500
C	-2.69222400	0.52911500	3.89581900
C	-2.28126100	0.20021300	2.60573700
C	-2.90533700	0.76815600	1.48269700
C	-3.93100100	1.70235100	1.69830400
C	-2.48689200	0.30251900	0.12688600
C	-2.96935500	0.93926600	-1.10421900
O	-3.23831000	0.31311400	-2.11974100
O	-2.99937800	2.26942400	-1.02791100
C	-3.33632200	3.10306000	-2.17030400
H	-5.12437900	2.77460400	3.13113700
H	-4.03169000	1.71760200	5.10661900
H	-2.20487700	0.05677100	4.75050000
H	-1.47075900	-0.51962700	2.46449100
H	-4.43061400	2.16000100	0.84522000
N	-2.60831600	-1.18447300	-0.04889700
H	-2.56050500	-1.34033800	-1.07158900
C	-3.20759100	4.51345600	-1.60870400
H	-3.44770400	5.25392100	-2.38366000
H	-3.89072700	4.65183900	-0.75886400
H	-2.17982600	4.68696700	-1.26150200
C	-2.33813100	2.88549700	-3.30427100
H	-1.31120000	3.02292600	-2.93595100
H	-2.43558000	1.87995500	-3.72834100
H	-2.52402200	3.62699500	-4.09450900
C	-4.77297200	2.82963800	-2.60753900
H	-5.45593600	2.93333700	-1.75099100
H	-5.06670200	3.56415700	-3.37088000
H	-4.87187100	1.82237300	-3.02790100
C	-3.80306500	-1.84058900	0.46418300
C	-5.04921900	-1.53546600	-0.07513800
C	-3.65611900	-2.77021800	1.48764600
C	-6.17311900	-2.19259500	0.42087400
H	-5.12576500	-0.80245400	-0.87994000
C	-4.78865900	-3.41923400	1.97781000
H	-2.65858400	-2.98064500	1.88093200
C	-6.04454800	-3.13315100	1.44431300
H	-7.15592500	-1.96791100	0.00462000
H	-4.68680400	-4.14984900	2.78100700
H	-6.92885000	-3.64243300	1.82931100
H	-1.74825800	-1.64709000	0.36384300

TSRa-II

P	0.67334800	-1.14133100	0.46380200
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O	1.89566300	-1.21930100	1.53187800
O	1.34431600	-1.84731200	-0.85399000
O	-0.46282900	-1.93511500	1.02680200
C	4.03601700	-2.89551700	0.09723500
C	4.05268900	-1.43205400	0.51142700
C	3.13697900	-0.67811500	1.24291400
C	3.42834000	0.64195800	1.63243900
C	4.65410900	1.18849800	1.22590100
C	5.54952700	0.47575600	0.43295500
C	5.24120200	-0.83271000	0.07533100
C	6.02506000	-1.78805500	-0.79254400
C	4.97122000	-2.84844500	-1.14374000
C	2.76106000	-3.65672300	-0.21060900
C	1.55996700	-3.21055400	-0.74321500
C	0.52383500	-4.10748600	-1.05856000
C	0.74320800	-5.46881200	-0.80756400
C	1.92425100	-5.92805000	-0.22432700
C	2.92770400	-5.01537700	0.08122200
C	4.25585400	-5.23806300	0.76676100
C	4.64896300	-3.81271100	1.19243500
C	2.44762300	1.47608500	2.37031100
C	1.69212800	0.96713600	3.43613900
C	0.76867700	1.77118500	4.09474100
C	0.55118900	3.09872400	3.70632300
C	1.32394600	3.60821300	2.65883700
C	2.25882100	2.81390500	2.00143800
C	-0.77463500	-3.63925000	-1.60877300
C	-0.83740400	-2.66228000	-2.61580900
C	-2.06076200	-2.26448600	-3.14581000
C	-3.26767400	-2.81961000	-2.69235800
C	-3.20250000	-3.78199700	-1.67936900
C	-1.97830300	-4.18354300	-1.14397600
H	4.88883500	2.21102000	1.52572000
H	6.45481000	-1.29689700	-1.67669900
H	6.86368600	-2.22725400	-0.22720400
H	5.39207800	-3.83480000	-1.38499500
H	4.37970800	-2.51128900	-2.00921200
H	-0.04585900	-6.17755300	-1.06479900
H	2.04937000	-6.98934300	-0.00295100
H	4.99090600	-5.66455000	0.06426100
H	4.17880200	-5.93345400	1.61410800
H	5.73248100	-3.66376100	1.30248700
H	4.17143700	-3.56791300	2.15334400
H	1.83326800	-0.06756700	3.75000900

H	0.19613300	1.35934800	4.92821100
H	1.17704400	4.64117500	2.33383400
H	2.81720600	3.22681000	1.15663700
H	0.08414700	-2.21197600	-2.98555400
H	-2.08773000	-1.49193400	-3.91672800
H	-4.12830200	-4.20613900	-1.28261800
H	-1.95678300	-4.91368700	-0.33248400
H	6.47795600	0.94106800	0.09728400
N	0.22766200	0.33961500	0.02558200
H	-1.02606100	0.49844000	0.06397900
C	0.91581500	1.37377300	-0.60290900
C	0.33755300	2.65689000	-0.55573800
C	2.14249500	1.20995100	-1.27032400
C	0.99202000	3.74661300	-1.11148400
H	-0.61723200	2.78604500	-0.04204600
C	2.80179000	2.30919300	-1.80752400
H	2.59250100	0.21912200	-1.34737500
C	2.23661500	3.58131400	-1.72258400
H	0.54994800	4.74298500	-1.05044400
H	3.76694400	2.17522500	-2.29840100
C	-4.57946300	-2.35768500	-3.26657700
H	-4.68405200	-2.68050000	-4.31364800
H	-5.42709500	-2.76297100	-2.69778400
H	-4.62916700	-1.25914400	-3.24676700
C	-0.51315600	3.93659800	4.36128200
H	-0.21399200	4.99244200	4.41719000
H	-1.44953100	3.87902300	3.78337000
H	-0.73346800	3.58129300	5.37732500
C	3.00487200	4.77847500	-2.19012800
F	2.20004100	5.76155000	-2.60511800
F	3.76057300	5.29303200	-1.20770300
F	3.83460100	4.48494200	-3.19501800
C	-4.09107300	2.49275400	2.93655000
C	-3.51424700	1.88905500	4.05346900
C	-2.58777900	0.86429600	3.87018500
C	-2.23709000	0.45114700	2.58653300
C	-2.83128100	1.03224100	1.45390900
C	-3.76252200	2.06416800	1.65217400
C	-2.48355400	0.48451900	0.10903600
C	-2.93468800	1.11505400	-1.13665600
O	-3.27262900	0.47869100	-2.12457400
O	-2.85055600	2.44489500	-1.10829900
C	-3.10887600	3.26192900	-2.28268900
H	-4.81971500	3.29530100	3.06391100

H	-3.78452900	2.21640000	5.05869700
H	-2.12786500	0.38015600	4.73343600
H	-1.49848900	-0.34461600	2.45891600
H	-4.23705700	2.53395900	0.79150100
N	-2.72267200	-0.99344000	-0.00837200
H	-2.70718600	-1.19289000	-1.02516300
C	-2.85320800	4.67519800	-1.77452000
H	-3.02348600	5.40401200	-2.57847600
H	-3.52286400	4.90768900	-0.93452400
H	-1.81451100	4.76766400	-1.42897900
C	-2.13032800	2.90980000	-3.39976500
H	-1.09721700	2.96555900	-3.02671600
H	-2.31867400	1.90183800	-3.78604800
H	-2.24256300	3.63422100	-4.21916900
C	-4.56324900	3.10423700	-2.71824900
H	-5.23576500	3.30147000	-1.86995300
H	-4.78676600	3.83394200	-3.50950700
H	-4.75285300	2.09525900	-3.10155500
C	-3.95360100	-1.53362500	0.55264400
C	-5.18320100	-1.14703800	0.02784600
C	-3.85825500	-2.43764900	1.60461200
C	-6.34464200	-1.69470900	0.56820400
H	-5.21835500	-0.43752400	-0.80057500
C	-5.02787000	-2.97680800	2.13915300
H	-2.87221600	-2.71477700	1.98530000
C	-6.26853700	-2.60831800	1.62094600
H	-7.31541200	-1.40539700	0.16383400
H	-4.96672600	-3.68639900	2.96500300
H	-7.18176400	-3.03192400	2.04064800
H	-1.89158200	-1.50504100	0.40697900

TSRb-I

P	-0.49688500	1.00733000	0.32551200
O	-1.37326900	0.78993400	1.68640100
O	-1.48982100	1.93645700	-0.56989700
O	0.75480100	1.72771500	0.75189400
C	-4.11524600	1.96796000	0.96440300
C	-3.67866800	0.51990900	1.08772400
C	-2.49207500	-0.02543100	1.56307900
C	-2.37399500	-1.40678300	1.80373300
C	-3.44452700	-2.22782900	1.42232400
C	-4.59545600	-1.70688800	0.83126900
C	-4.71560100	-0.32865400	0.68539900
C	-5.83551700	0.47840300	0.06901400

C	-5.17477500	1.84989400	-0.16431200
C	-3.12560700	3.09530900	0.73585900
C	-1.95247700	3.11766700	-0.00807100
C	-1.19569700	4.29176400	-0.14237400
C	-1.64980400	5.44225800	0.51676500
C	-2.80879900	5.43309400	1.29135600
C	-3.54292900	4.25662700	1.39624900
C	-4.79370400	3.97819700	2.19503600
C	-4.78511300	2.44493400	2.28673600
C	-1.14991100	-1.96999300	2.42593300
C	-0.60768100	-3.18702600	1.99746700
C	0.54861500	-3.69975000	2.58412700
C	1.18921300	-3.02728200	3.62789600
C	0.60763700	-1.84320900	4.09947500
C	-0.53172900	-1.31443100	3.50470800
C	0.03641400	4.34122700	-0.97074900
C	0.02254800	3.91168900	-2.30195100
C	1.14959600	4.06282100	-3.10656300
C	2.32833000	4.62285500	-2.60374400
C	2.34871900	5.01451900	-1.25945200
C	1.22085000	4.88243400	-0.45591200
H	-3.35965500	-3.30365600	1.58508400
H	-6.21666400	0.02269200	-0.85605100
H	-6.68866000	0.55648700	0.76283000
H	-5.88285500	2.69092700	-0.15956500
H	-4.65369200	1.85341800	-1.13463300
H	-1.07003100	6.36092500	0.41051100
H	-3.12888200	6.33733800	1.81176300
H	-5.68383300	4.34328900	1.65600500
H	-4.78688700	4.46837200	3.17835500
H	-5.77951400	1.99948500	2.43137300
H	-4.14746800	2.13144400	3.12714100
H	-1.08316900	-3.72753500	1.17830100
H	0.97335400	-4.63076300	2.20304000
H	1.07482300	-1.31603600	4.93416300
H	-0.95428400	-0.37935800	3.87547200
H	-0.88939800	3.47847100	-2.71618300
H	1.11039600	3.74387600	-4.15094600
H	3.26372000	5.43710200	-0.83719400
H	1.25642200	5.18816200	0.59142700
H	-5.38277700	-2.37896900	0.48543700
N	-0.21653800	-0.34366500	-0.48557600
H	1.05251900	-0.77358800	-0.41740500
C	-1.11380900	-1.19546600	-1.14283800

C	-0.78946300	-2.56089700	-1.22697500
C	-2.31186700	-0.75609300	-1.73529400
C	-1.66387700	-3.46675300	-1.81512600
H	0.14985100	-2.90919500	-0.79604100
C	-3.18640100	-1.66618600	-2.31350700
H	-2.55914800	0.30362400	-1.72514800
C	-2.87839900	-3.02621900	-2.33928100
H	-1.40345200	-4.52538700	-1.85692400
H	-4.12253400	-1.31472000	-2.75232800
C	3.54639100	4.79852700	-3.47197500
H	3.89400600	5.84177500	-3.45708800
H	4.37816000	4.17224500	-3.11514400
H	3.33607100	4.52470100	-4.51452600
C	2.48981800	-3.51369200	4.20651900
H	3.29508500	-2.79269900	3.98522800
H	2.43100000	-3.60952200	5.30044300
H	2.77525800	-4.48854700	3.78877000
C	-3.88845800	-3.99913200	-2.86504000
F	-3.33870900	-5.16880500	-3.19905900
F	-4.83848900	-4.26467400	-1.95425700
F	-4.52394300	-3.52851400	-3.94306800
C	2.63266500	-2.90420800	-1.76672700
C	2.73346400	-2.46762700	-0.43625400
C	3.13687100	-3.38962100	0.53507400
C	3.44371500	-4.70630700	0.18790500
C	3.34646600	-5.12622000	-1.13643000
C	2.93745600	-4.21630100	-2.11417600
H	2.29021400	-2.20653500	-2.53461200
H	3.20042900	-3.06161000	1.57347600
H	3.76163300	-5.40738500	0.96234100
H	3.58199700	-6.15631200	-1.40758400
H	2.84514700	-4.53526800	-3.15375500
C	2.35104800	-1.06949000	-0.04442500
C	3.00579800	0.01895600	-0.74737800
O	2.76960200	1.24130200	-0.50544400
H	1.89821500	1.45550200	0.13285400
O	3.85467200	-0.29101400	-1.68227000
C	4.44133000	0.65603500	-2.63274600
N	2.19622900	-0.80232100	1.34841800
H	1.25197400	-0.73764400	1.71756100
C	3.20812600	-0.31076800	2.15541900
C	2.89311200	0.43158100	3.30760300
C	4.56243700	-0.53317100	1.84575700
C	3.90183900	0.89233200	4.14628900

H	1.84623200	0.66263800	3.51026700
C	5.56214800	-0.05228200	2.68867700
H	4.82580700	-1.10194600	0.95103000
C	5.24494000	0.65239300	3.84907900
H	3.63418900	1.46667900	5.03522400
H	6.60664000	-0.24253900	2.43449600
H	6.03263500	1.02117200	4.50636400
C	3.32669300	1.25308100	-3.48045500
H	3.76783600	1.88686900	-4.26332700
H	2.74998300	0.45357500	-3.96822500
H	2.65204800	1.86930700	-2.87331200
C	5.25512400	1.71363100	-1.89333300
H	5.90943400	1.23492200	-1.15050600
H	5.89019600	2.24152300	-2.61964500
H	4.61099100	2.43848900	-1.38217900
C	5.34787500	-0.24031200	-3.46480400
H	4.76469700	-1.05126900	-3.92204800
H	5.82733900	0.34737600	-4.25908900
H	6.12651200	-0.68762400	-2.83197900

TSSa-I

P	-0.35611500	1.03811000	0.50101100
O	-1.54777300	1.31251900	1.59147000
O	-0.77465300	2.07363600	-0.72605600
O	0.95379200	1.41058100	1.13158000
C	-3.11404400	3.68689900	0.39339300
C	-3.54015000	2.24182800	0.61085800
C	-2.89152000	1.18291400	1.24856900
C	-3.55491400	-0.03956900	1.48580000
C	-4.86843900	-0.17702700	1.00268800
C	-5.50152500	0.84049300	0.28999800
C	-4.83515000	2.04888900	0.09984700
C	-5.30288900	3.28011400	-0.63955400
C	-3.99447200	4.06397300	-0.84083500
C	-1.67274100	4.10648400	0.17032000
C	-0.62957800	3.42897200	-0.45117500
C	0.60680000	4.06420100	-0.69568500
C	0.76068500	5.39037100	-0.25541000
C	-0.25511500	6.05941900	0.43247100
C	-1.47040900	5.41247600	0.64510800
C	-2.69655700	5.89294800	1.38738300
C	-3.48491200	4.58730600	1.61403200
C	-2.90927000	-1.17511400	2.18497000
C	-3.07941400	-2.48680700	1.70992400

C	-2.52581300	-3.56941200	2.38920900
C	-1.79579200	-3.38883500	3.57308800
C	-1.63497300	-2.07863300	4.04813000
C	-2.16724200	-0.98812000	3.36206400
C	1.71954600	3.36463700	-1.38177100
C	1.49359000	2.58877400	-2.53308900
C	2.54764400	1.95077900	-3.18466000
C	3.86830700	2.05023000	-2.71346600
C	4.09297900	2.83353900	-1.57236600
C	3.04018400	3.47609300	-0.91760100
H	-5.38879100	-1.11975900	1.17906300
H	-5.81173700	3.03806700	-1.58545800
H	-6.02349100	3.85269000	-0.02885000
H	-4.13540400	5.14996900	-0.93836900
H	-3.48397300	3.70994300	-1.75032000
H	1.70751300	5.89794100	-0.45090400
H	-0.09232300	7.07613000	0.79825900
H	-3.27204000	6.60740400	0.77235300
H	-2.44539500	6.41287700	2.32481100
H	-4.56996700	4.73518900	1.71190500
H	-3.12880700	4.09085400	2.53002700
H	-3.63077900	-2.66005200	0.78444400
H	-2.65784800	-4.57696000	1.98541800
H	-1.08280200	-1.90861800	4.97599300
H	-2.02079600	0.01991700	3.75165300
H	0.47844900	2.48238700	-2.91666400
H	2.34465700	1.35116200	-4.07419800
H	5.10463400	2.91380200	-1.16845400
H	3.23995600	4.04568600	-0.00797700
H	-6.50781500	0.68957800	-0.10812400
N	-0.33878900	-0.44237700	-0.13519200
H	0.82038100	-0.94206300	-0.14516500
C	-1.28395800	-1.19604500	-0.82711500
C	-1.07445900	-2.58886400	-0.90497800
C	-2.41635700	-0.64492000	-1.45525100
C	-1.99770800	-3.40601400	-1.54420600
H	-0.19469200	-3.02325500	-0.42771200
C	-3.35043600	-1.47174300	-2.07090200
H	-2.57538200	0.43256400	-1.43765700
C	-3.15592200	-2.85692300	-2.11034000
H	-1.83200800	-4.48390600	-1.58297800
H	-4.24314500	-1.03521700	-2.52141100
C	4.98778300	1.32486000	-3.41315000
H	5.90086800	1.31123600	-2.80171800

H	4.70506100	0.28282600	-3.63282700
H	5.23045800	1.80425400	-4.37681200
C	-1.18408800	-4.56511200	4.28853000
H	-0.33160200	-4.97050500	3.71827000
H	-0.81174700	-4.28360000	5.28432200
H	-1.91120400	-5.38356200	4.41194300
C	-4.22861200	-3.75261000	-2.64899100
F	-3.73482800	-4.89301400	-3.16555200
F	-5.10118400	-4.11911400	-1.67793100
F	-4.96035400	-3.15742000	-3.61052000
C	1.82851600	-1.70105300	-4.05927100
C	2.55711500	-2.86235900	-4.32974700
C	3.15917800	-3.55318300	-3.27357900
C	3.05697800	-3.08105100	-1.96344600
C	2.34439700	-1.90121400	-1.67960600
C	1.71431600	-1.23643300	-2.74867000
C	2.21512400	-1.32075100	-0.31081800
C	2.44504100	-2.14870600	0.87232600
O	2.35680600	-3.36407700	0.90178600
O	2.66346100	-1.37239800	1.95777500
C	2.99710900	-1.91862700	3.26921100
H	1.32430900	-1.16727600	-4.86897500
H	2.64028000	-3.23624900	-5.35291400
H	3.71715500	-4.47208900	-3.47121600
H	3.51082000	-3.63683300	-1.14388900
H	1.10246300	-0.35714600	-2.54250500
N	2.81969600	0.05935300	-0.19371200
H	2.20401500	0.63064800	0.45922800
C	4.22022200	0.17687800	0.21372400
C	4.53095100	1.05204400	1.25227300
C	5.20595900	-0.56292200	-0.43412900
C	5.86139700	1.19396200	1.64759500
H	3.72877700	1.60622100	1.74222500
C	6.53618200	-0.41429800	-0.03174700
H	4.93793000	-1.24831600	-1.23794900
C	6.86630000	0.46192900	1.00609300
H	6.11336100	1.87386100	2.46444400
H	7.31713100	-0.98987700	-0.53335400
H	7.90709400	0.57189400	1.31845300
C	4.35630400	-2.61453400	3.18167900
H	4.67529300	-2.94175000	4.18344100
H	4.30004800	-3.49469900	2.52756400
H	5.11401200	-1.92162200	2.78405100
C	3.06895800	-0.67105700	4.14863700

H	3.87953700	-0.00773100	3.81467100
H	2.12118900	-0.11569200	4.08929700
H	3.25589900	-0.95238400	5.19577100
C	1.89264800	-2.85421600	3.75869200
H	1.85567400	-3.77081200	3.16010600
H	2.06851100	-3.11456000	4.81418800
H	0.91929800	-2.35188700	3.68190200
H	2.71003100	0.52357400	-1.10693700

TSSa-II

P	-0.64165400	1.13183100	0.47849900
O	-1.70121500	0.87955600	1.69521900
O	-1.55153400	2.00190200	-0.55748400
O	0.52837900	1.87269900	1.03551600
C	-4.36115100	1.93210700	0.66310100
C	-3.89849700	0.49992700	0.85871500
C	-2.74548500	0.00247200	1.45489100
C	-2.58995800	-1.37317800	1.71670600
C	-3.61982900	-2.22257400	1.27758500
C	-4.73132500	-1.75045400	0.58263500
C	-4.86990100	-0.38213100	0.37753000
C	-5.94077700	0.37900700	-0.36969800
C	-5.29215500	1.76188900	-0.56783600
C	-3.39439100	3.09595700	0.52255800
C	-2.13859300	3.16248000	-0.07193700
C	-1.42079600	4.36874700	-0.11830300
C	-2.00225800	5.50711700	0.45440800
C	-3.25009100	5.45594000	1.07256500
C	-3.93710200	4.24801500	1.10662800
C	-5.25716400	3.92781400	1.76515300
C	-5.18733700	2.40033100	1.89664700
C	-1.38464900	-1.92690100	2.39084500
C	-1.00110600	-3.25745600	2.16724400
C	0.07563100	-3.82416900	2.84270900
C	0.84475300	-3.07986600	3.74049900
C	0.46880200	-1.74976100	3.96123700
C	-0.62546700	-1.18400200	3.31345200
C	-0.07649000	4.44884100	-0.74381200
C	0.13027700	4.03155500	-2.06347500
C	1.38428800	4.15126500	-2.65354200
C	2.47867900	4.65748500	-1.94169900
C	2.26061400	5.08536700	-0.62738000
C	1.00342800	4.98569100	-0.03616100
H	-3.53847800	-3.29218800	1.47066800

H	-6.21235400	-0.10751100	-1.31749100
H	-6.86331700	0.44987700	0.22963500
H	-6.01690800	2.58357800	-0.65878500
H	-4.67159800	1.75734800	-1.47774400
H	-1.44831000	6.44667700	0.41242900
H	-3.67336800	6.34966300	1.53417900
H	-6.09639600	4.23775800	1.12033500
H	-5.38322100	4.43908500	2.72950900
H	-6.16912600	1.90897200	1.94825700
H	-4.62642800	2.13509900	2.80560900
H	-1.55003400	-3.86441900	1.44649500
H	0.32582700	-4.87144100	2.66308800
H	1.03257500	-1.14718300	4.67845800
H	-0.90303700	-0.15684100	3.54242000
H	-0.70624600	3.62381900	-2.63387700
H	1.51478600	3.85461400	-3.69630300
H	3.09220500	5.50023600	-0.05152800
H	0.85845900	5.30331900	0.99813900
H	-5.47042400	-2.45566200	0.19851200
N	-0.22424100	-0.23076500	-0.27608600
H	0.95335100	-0.52811300	-0.08559400
C	-0.99022600	-1.15533100	-0.99213800
C	-0.56308000	-2.49458800	-1.01057100
C	-2.15729200	-0.80934500	-1.69497200
C	-1.33153700	-3.47135600	-1.63176600
H	0.37114700	-2.75951200	-0.51403200
C	-2.92461200	-1.79101500	-2.30628800
H	-2.46667200	0.23361900	-1.73823300
C	-2.53075500	-3.12828000	-2.25580100
H	-1.00462000	-4.51266800	-1.62069700
H	-3.84485600	-1.51657600	-2.82589900
C	3.85125800	4.69612600	-2.55792100
H	4.47017600	5.48667500	-2.11099900
H	4.36463900	3.73213900	-2.40365400
H	3.79883800	4.86993300	-3.64177100
C	2.05114500	-3.67041100	4.41492900
H	2.95321600	-3.39348600	3.84767400
H	2.16247800	-3.29339100	5.44109300
H	1.99148300	-4.76672300	4.45199700
C	-3.44140300	-4.18457300	-2.80016700
F	-2.79522500	-5.32173500	-3.06994900
F	-4.41458200	-4.49306000	-1.92676100
F	-4.05447300	-3.79229400	-3.92121100
C	4.99213500	-0.10560900	-2.66961800

C	4.31815000	0.78536200	-3.50604700
C	3.03870800	1.21221900	-3.16211800
C	2.42587800	0.75529200	-1.99327600
C	3.08118700	-0.15475800	-1.15557300
C	4.37691300	-0.56653600	-1.51133000
C	2.44903300	-0.63787000	0.11950400
C	2.84056600	-1.97628700	0.57497500
O	3.11105500	-2.25878800	1.73417600
O	2.81314900	-2.86205000	-0.42307100
C	3.28764500	-4.22384400	-0.29092000
H	5.99987700	-0.44061000	-2.92057900
H	4.78959000	1.14406900	-4.42260000
H	2.50298600	1.90270700	-3.81302500
H	1.42466300	1.10746400	-1.73722500
H	4.91323700	-1.25921500	-0.85858600
N	2.65691200	0.30639800	1.28507600
H	2.58138900	-0.30097100	2.11580700
H	1.85287600	0.99604400	1.30143500
C	3.91899000	1.02888900	1.30338000
C	5.06751100	0.41310900	1.79066000
C	3.94591800	2.30448800	0.74850500
C	6.27687300	1.10309300	1.72214000
H	5.00514100	-0.59639900	2.20172400
C	5.16286200	2.97977000	0.68059300
H	3.02169400	2.74890100	0.36875500
C	6.32622700	2.38130600	1.16511100
H	7.18615100	0.63671300	2.10330900
H	5.19929300	3.97824200	0.24183600
H	7.27667500	2.91332200	1.10835800
C	2.42046800	-4.99532900	0.69654100
H	2.70109100	-6.05855500	0.67898600
H	2.54252800	-4.60357000	1.71320500
H	1.36117700	-4.91108400	0.41043700
C	3.12240400	-4.77747500	-1.70078300
H	2.06663900	-4.73152900	-2.00315400
H	3.70878600	-4.17938700	-2.41189300
H	3.46093100	-5.82182200	-1.74278100
C	4.75720400	-4.22888800	0.12159600
H	4.88625200	-3.80285300	1.12392400
H	5.13109000	-5.26257200	0.12663600
H	5.35538700	-3.65327200	-0.59989400

TSSa-III

P	-0.53654400	1.16150800	0.55570300
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O	-1.64369500	1.17676100	1.75147500
O	-1.30668200	2.00188400	-0.60288900
O	0.69659500	1.81751200	1.07930400
C	-4.10559800	2.47394500	0.50330900
C	-3.86017100	1.02420800	0.89297400
C	-2.79489600	0.43384300	1.56435300
C	-2.82407600	-0.92183200	1.94844000
C	-3.96383000	-1.66085300	1.59304300
C	-5.00208800	-1.10827200	0.84583200
C	-4.94307200	0.23533100	0.49095800
C	-5.88827000	1.04415100	-0.36692700
C	-5.03907500	2.27745200	-0.72315800
C	-2.98883000	3.47270700	0.23070200
C	-1.72119500	3.29171900	-0.31349700
C	-0.84112800	4.37191800	-0.51353300
C	-1.31493700	5.65212300	-0.18574200
C	-2.57291400	5.85432400	0.37651900
C	-3.39805600	4.75862200	0.60280600
C	-4.75082800	4.70676800	1.27068500
C	-4.87165800	3.21971100	1.63392000
C	-1.66827000	-1.57941400	2.61634700
C	-0.81102100	-0.89747300	3.49688500
C	0.27611700	-1.54338800	4.07950400
C	0.55074300	-2.89308500	3.82441300
C	-0.32027400	-3.57729700	2.97169400
C	-1.40058500	-2.93445300	2.37381900
C	0.55716300	4.19643400	-0.99297200
C	0.91987000	3.24446100	-1.95849200
C	2.24020500	3.12223800	-2.37991400
C	3.24866700	3.95082700	-1.87281400
C	2.88655800	4.90290300	-0.91491400
C	1.56991500	5.01470600	-0.47341200
H	-4.02202300	-2.70891600	1.88850400
H	-6.22854000	0.48621600	-1.25095400
H	-6.79008700	1.32771300	0.20017200
H	-5.63174500	3.17670900	-0.94321800
H	-4.41681500	2.06170500	-1.60610200
H	-0.66435500	6.50830300	-0.36879700
H	-2.89543000	6.86014400	0.65101300
H	-5.54434600	5.01526100	0.56984400
H	-4.81442500	5.37054400	2.14403200
H	-5.90738000	2.86637500	1.73599200
H	-4.35160900	3.02871700	2.58486100
H	-0.99894500	0.14843000	3.73476500

H	0.91887700	-0.98789900	4.76751000
H	-0.14902900	-4.63648000	2.76743100
H	-2.03470200	-3.49382200	1.68510500
H	0.16211900	2.59151500	-2.39105000
H	2.49422300	2.36558000	-3.12521100
H	3.64805000	5.56866700	-0.50001000
H	1.32763300	5.74249500	0.30266500
H	-5.83842800	-1.73484800	0.53069000
N	-0.27663100	-0.31302900	-0.04402400
H	0.90155100	-0.66330400	0.04081100
C	-1.12325500	-1.19036700	-0.72640000
C	-0.78103700	-2.55426100	-0.73046000
C	-2.26912100	-0.78474600	-1.43389100
C	-1.59919600	-3.49076200	-1.35031000
H	0.13186900	-2.86768400	-0.22320700
C	-3.09151600	-1.72611500	-2.03726400
H	-2.51821300	0.27251000	-1.49431100
C	-2.77347500	-3.08351700	-1.98245500
H	-1.33055600	-4.54853700	-1.33290300
H	-3.99235200	-1.40151900	-2.56186700
C	4.67518000	3.79190900	-2.32543000
H	5.26096900	4.70147200	-2.13193200
H	5.15738300	2.95594500	-1.79317800
H	4.72643700	3.56465300	-3.39956000
C	1.74938600	-3.57390500	4.42549600
H	1.58721500	-4.65674300	4.51614000
H	2.62482200	-3.41164500	3.77812000
H	1.97927400	-3.17278100	5.42224600
C	-3.73016900	-4.08829800	-2.54766200
F	-3.13940600	-5.25819000	-2.80508100
F	-4.73796900	-4.34215900	-1.69960500
F	-4.29164400	-3.66100800	-3.68298800
C	4.54777400	0.25863500	-2.91948500
C	3.64357600	0.08598900	-3.96644500
C	2.34438300	-0.32720900	-3.67947200
C	1.95517200	-0.55281600	-2.35988700
C	2.85207900	-0.39440700	-1.29143000
C	4.16253800	0.01313600	-1.60171000
C	2.39967100	-0.81212700	0.08404000
C	2.73333400	-2.17387700	0.50655100
O	3.01346400	-2.47549400	1.66046300
O	2.62600900	-3.05369000	-0.48826900
C	2.92870600	-4.46136900	-0.32513900
H	5.57421600	0.56620700	-3.12793300

H	3.95271800	0.26389300	-4.99779500
H	1.62367100	-0.47857900	-4.48459400
H	0.93880700	-0.88900600	-2.15984800
H	4.91253500	0.11082100	-0.81739800
N	2.67159500	0.09814800	1.24932300
H	2.55004500	-0.52396600	2.06378300
H	1.90061100	0.83111800	1.27769300
C	3.94634600	0.78995700	1.36224700
C	5.07533900	0.11041100	1.81268900
C	3.98972900	2.12574100	0.97927900
C	6.28901800	0.79325100	1.85940000
H	4.99280100	-0.93928700	2.10089400
C	5.21122100	2.79612400	1.03311600
H	3.07755900	2.62212700	0.63599100
C	6.35882900	2.13151200	1.46513400
H	7.18554200	0.27760500	2.20578800
H	5.25988400	3.84318800	0.72929200
H	7.31270000	2.65923700	1.50250000
C	1.97768500	-5.10132900	0.68243000
H	2.15569800	-6.18610100	0.71187100
H	2.12728800	-4.68410300	1.68535000
H	0.93254500	-4.93491100	0.37963500
C	2.68696000	-5.02206000	-1.72102200
H	1.64681300	-4.83777900	-2.02544700
H	3.34936500	-4.52979900	-2.44617500
H	2.87751600	-6.10390800	-1.73587000
C	4.38931800	-4.63368900	0.08277300
H	4.56220700	-4.25066100	1.09549200
H	4.65459300	-5.70009700	0.05465900
H	5.03884500	-4.09558700	-0.62284800

TSSa-IV

P	-0.37124200	1.16608800	0.60472500
O	-1.67582800	1.33925900	1.58166000
O	-0.70047200	2.25888400	-0.60008800
O	0.86011600	1.53406100	1.37508000
C	-3.10323800	3.82416100	0.43505300
C	-3.55955600	2.37230900	0.49616900
C	-2.97960700	1.25400500	1.10049300
C	-3.67554600	0.02786800	1.17828000
C	-4.94435300	-0.04311000	0.57391200
C	-5.50204800	1.03860300	-0.10472100
C	-4.80804300	2.24533000	-0.13708100
C	-5.20311800	3.54124200	-0.80362700

C	-3.87628900	4.31731200	-0.83027700
C	-1.64709900	4.23992400	0.35324600
C	-0.56769000	3.59559800	-0.23892600
C	0.68521100	4.23402500	-0.35816500
C	0.81241700	5.52736700	0.17899500
C	-0.24319700	6.15573700	0.84522700
C	-1.47283300	5.50667600	0.93254500
C	-2.74508200	5.94176900	1.62421500
C	-3.55739300	4.63179600	1.69125300
C	-3.13422700	-1.17905900	1.84834300
C	-2.47633500	-1.10891100	3.08685600
C	-2.09921200	-2.27124000	3.75970900
C	-2.34302100	-3.54140500	3.21701800
C	-2.95914300	-3.60582000	1.95831200
C	-3.35173700	-2.45068200	1.28790000
C	1.83825400	3.57329700	-1.01497200
C	1.67669800	2.83222900	-2.20050400
C	2.77090100	2.24434900	-2.83229000
C	4.07106400	2.36670700	-2.31225300
C	4.23056300	3.10346600	-1.12975300
C	3.13650700	3.69128200	-0.49118400
H	-5.49304900	-0.98414300	0.63238800
H	-5.63572000	3.38834600	-1.80452600
H	-5.96584500	4.07081800	-0.20528900
H	-3.99777200	5.40981400	-0.84651400
H	-3.29586700	4.03224000	-1.72182100
H	1.77062100	6.04129200	0.08014200
H	-0.09965400	7.14309900	1.29077700
H	-3.27144500	6.71051400	1.03108000
H	-2.55597000	6.38176500	2.61560600
H	-4.64566200	4.78581500	1.72027600
H	-3.27349400	4.06191300	2.58953000
H	-2.27762500	-0.13610900	3.53739300
H	-1.60769500	-2.18912700	4.73263000
H	-3.13805000	-4.57905900	1.49257200
H	-3.82027800	-2.53853500	0.30690800
H	0.68119000	2.71603500	-2.62888400
H	2.61676400	1.67395500	-3.75090300
H	5.22296500	3.19113600	-0.68187400
H	3.28787900	4.22449700	0.44914200
H	-6.47352300	0.93975200	-0.59498900
N	-0.24848700	-0.27897200	-0.09890300
H	0.92352200	-0.71484400	0.01016600
C	-1.11135700	-1.03211500	-0.88919900

C	-0.86992400	-2.41875800	-0.98045900
C	-2.19117200	-0.48282100	-1.60643000
C	-1.71600500	-3.23418200	-1.72068000
H	-0.02698600	-2.84431600	-0.43437700
C	-3.04838400	-1.30856000	-2.32647600
H	-2.37063300	0.59157700	-1.57718700
C	-2.82577000	-2.68931800	-2.38026000
H	-1.52985000	-4.30846100	-1.76823800
H	-3.90339500	-0.87531200	-2.84789600
C	5.23811900	1.72215200	-3.01285200
H	6.13510800	1.71753800	-2.37832600
H	5.00626100	0.68181400	-3.29224200
H	5.48409400	2.25968200	-3.94451100
C	-1.96411900	-4.80145700	3.95166000
H	-2.85663900	-5.40403800	4.18952700
H	-1.29978100	-5.43533100	3.34258900
H	-1.44559300	-4.57919800	4.89516300
C	-3.83011000	-3.58714100	-3.03373500
F	-3.27693400	-4.71720000	-3.51158700
F	-4.79170000	-3.97409200	-2.15753600
F	-4.46980900	-2.98696900	-4.05534900
C	2.38283700	-1.52872000	-3.73667800
C	3.22671200	-2.63147300	-3.89361300
C	3.77468100	-3.24234800	-2.76092000
C	3.49841200	-2.74540000	-1.48623500
C	2.65700700	-1.63118800	-1.31199100
C	2.09304200	-1.04457000	-2.46013200
C	2.35463300	-1.03505200	0.02255400
C	2.48299900	-1.79687900	1.26281800
O	2.87382100	-1.33615400	2.32123200
O	2.01269600	-3.05452500	1.10402100
C	2.04115600	-4.03556900	2.18555800
H	1.92761600	-1.05685600	-4.61112200
H	3.44693100	-3.02033300	-4.89052300
H	4.43049200	-4.10967900	-2.87197400
H	3.93792700	-3.22482800	-0.61106400
H	1.39967000	-0.21035200	-2.34426100
N	2.93569600	0.34447600	0.20315000
H	2.26556700	0.87582700	0.83034600
C	4.31453900	0.48209800	0.67635700
C	4.54390900	1.18431900	1.85670200
C	5.36301500	-0.05291400	-0.06782900
C	5.85625700	1.37142000	2.29119900
H	3.69635900	1.56873000	2.42614800

C	6.67421900	0.13591100	0.37679200
H	5.15987600	-0.60530100	-0.98453700
C	6.92336500	0.85079500	1.55147000
H	6.04562300	1.91960800	3.21665900
H	7.50358900	-0.27658600	-0.20212900
H	7.94979500	0.99744100	1.89507300
C	1.30008700	-3.51646500	3.41808100
H	1.14551600	-4.34467200	4.12657000
H	1.85848900	-2.71428200	3.91190700
H	0.31717000	-3.12570800	3.12442600
C	1.30110500	-5.22681200	1.57885000
H	0.26248900	-4.94917300	1.34356900
H	1.79138900	-5.55334900	0.64960200
H	1.28818900	-6.06961900	2.28618200
C	3.49701400	-4.38659600	2.49780700
H	4.03636500	-3.50260200	2.86308300
H	3.53697000	-5.16674900	3.27359000
H	4.00404700	-4.77111100	1.59890200
H	2.85887200	0.82968600	-0.70113900

TSSa-V

P	0.21945900	1.13118300	-0.56050300
O	1.46080900	1.25888300	-1.62572000
O	0.62504500	2.28750600	0.55266100
O	-1.05609400	1.45193900	-1.28170500
C	3.27795100	3.42477800	-0.38935700
C	3.49828500	1.92466300	-0.54639800
C	2.76139700	0.95646400	-1.22972900
C	3.28417000	-0.33909700	-1.45913400
C	4.52764000	-0.63798100	-0.86867500
C	5.21487300	0.27715500	-0.07327600
C	4.70419500	1.56350000	0.07575900
C	5.25439600	2.71881400	0.87934400
C	4.06036500	3.69109200	0.93378400
C	1.90671000	4.07779100	-0.37273100
C	0.69793300	3.60343700	0.12528500
C	-0.46307700	4.40537000	0.11876300
C	-0.35369300	5.70562100	-0.40653800
C	0.84400800	6.18485600	-0.94242500
C	1.96970500	5.36388400	-0.93254400
C	3.34720400	5.62046200	-1.49728000
C	3.94318400	4.20206800	-1.57246100
C	2.59300900	-1.36303000	-2.28409000
C	1.74521800	-1.01880700	-3.35526400

C	1. 18786700	-1. 99739600	-4. 17479100
C	1. 45195200	-3. 36000000	-3. 97449600
C	2. 28229400	-3. 70589500	-2. 90037300
C	2. 83343700	-2. 73248700	-2. 06881700
C	-1. 76985200	3. 89763500	0. 60405200
C	-1. 88683200	3. 13925100	1. 77978500
C	-3. 13372800	2. 68974500	2. 22368400
C	-4. 30952700	2. 97319300	1. 50932600
C	-4. 18802900	3. 72894300	0. 33230100
C	-2. 94758400	4. 18049400	-0. 11085900
H	4. 95694800	-1. 62808500	-1. 01847200
H	5. 60209300	2. 40739800	1. 87687600
H	6. 12195300	3. 17561200	0. 37114400
H	4. 35040700	4. 74605500	1. 04370300
H	3. 40563500	3. 43466800	1. 78164900
H	-1. 23707700	6. 34725000	-0. 40040400
H	0. 88950400	7. 18845500	-1. 37222400
H	3. 93338400	6. 26636500	-0. 81944000
H	3. 31766400	6. 12881900	-2. 47339600
H	5. 04113900	4. 17759500	-1. 52447900
H	3. 63823000	3. 72085100	-2. 51465400
H	1. 52450900	0. 02779000	-3. 55564200
H	0. 53414300	-1. 69170100	-4. 99638300
H	2. 49351200	-4. 76031100	-2. 70194000
H	3. 45126600	-3. 04652800	-1. 22781100
H	-0. 99488900	2. 89767700	2. 35744900
H	-3. 19123400	2. 11173400	3. 14826200
H	-5. 08298300	3. 95246600	-0. 25384500
H	-2. 88150100	4. 74036300	-1. 04580000
H	6. 14170600	-0. 02208800	0. 42122100
N	0. 17987700	-0. 26125500	0. 24522000
H	-1. 01991100	-0. 70012400	0. 36274400
C	1. 20584800	-0. 90582200	0. 94454200
C	1. 31169400	-2. 30558100	0. 84068300
C	2. 13594000	-0. 21667400	1. 74745200
C	2. 36312400	-2. 98365100	1. 44895800
H	0. 58153800	-2. 84109500	0. 23457500
C	3. 19627600	-0. 89414100	2. 33617900
H	2. 03640300	0. 86052700	1. 86860600
C	3. 33035400	-2. 27843700	2. 17397700
H	2. 45353000	-4. 06518000	1. 33278900
H	3. 93664200	-0. 34297900	2. 91787500
C	-5. 65657700	2. 47211400	1. 96319500
H	-6. 05500200	1. 71813500	1. 26337200

H	-5.60258000	2.01433000	2.96106600
H	-6.39240500	3.29098100	2.00474700
C	0.86259300	-4.40400200	-4.88658500
H	1.30570000	-4.34589000	-5.89544400
H	1.03717200	-5.42020100	-4.50364900
H	-0.22401400	-4.26560600	-5.00256500
C	4.56545500	-2.96995300	2.66177800
F	4.36262700	-4.27744100	2.90745200
F	5.56220000	-2.90377800	1.74196700
F	5.05511200	-2.41754900	3.78756700
C	-2.08436000	-2.39280500	3.99362800
C	-3.12196800	-1.74239300	4.67124600
C	-3.94647800	-0.86911800	3.96246800
C	-3.74069500	-0.65507900	2.59634400
C	-2.70036300	-1.29790200	1.89977600
C	-1.87096100	-2.16990100	2.63545200
C	-2.41350700	-1.03667800	0.46195800
C	-2.43922600	-2.09679700	-0.55793200
O	-2.65446900	-1.87956100	-1.73725600
O	-2.12353500	-3.29802300	-0.04186100
C	-2.10435200	-4.51494500	-0.85170200
H	-1.41981700	-3.07187800	4.53358900
H	-3.28034700	-1.91248100	5.73855300
H	-4.76595400	-0.35155900	4.46828600
H	-4.41813400	0.02067700	2.07649600
H	-1.04400400	-2.66695500	2.13277900
N	-3.04460500	0.19996500	-0.08989800
H	-2.34500900	0.63656400	-0.76579600
C	-4.37144000	0.15980100	-0.71785400
C	-5.32292600	-0.78651300	-0.34614200
C	-4.65049400	1.12984800	-1.67863200
C	-6.58912600	-0.74092000	-0.93429400
H	-5.07734100	-1.55754600	0.38356600
C	-5.91897200	1.16910300	-2.25965900
H	-3.87362000	1.83998000	-1.96588200
C	-6.89235200	0.23861700	-1.88408300
H	-7.34039600	-1.48075800	-0.64966300
H	-6.14353200	1.92541000	-3.01500500
H	-7.88391600	0.26887600	-2.34103400
C	-1.04171100	-4.40346000	-1.94211700
H	-0.96503300	-5.35761400	-2.48537600
H	-1.28409900	-3.60513600	-2.65257700
H	-0.05802100	-4.18268800	-1.50551300
C	-1.72950200	-5.59173600	0.16475900

H	-0.74060300	-5.38266800	0.60003400
H	-2.46505000	-5.62329900	0.98210900
H	-1.69585100	-6.57870900	-0.32038000
C	-3.49924100	-4.76936900	-1.42683400
H	-3.77596700	-3.98652700	-2.14378100
H	-3.51704400	-5.74379200	-1.93883800
H	-4.24563300	-4.79279000	-0.61754000
H	-3.06739700	0.90106900	0.66808100

TSSb-I

P	-0.57595200	1.09266700	0.45464800
O	-1.67300000	1.04012600	1.65721000
O	-1.35368600	1.99957200	-0.65036100
O	0.63297000	1.78824400	1.02080100
C	-4.18410000	2.28739800	0.41334300
C	-3.86632900	0.83354500	0.72625500
C	-2.79885900	0.26391600	1.41366100
C	-2.79139800	-1.10390700	1.75283300
C	-3.86770300	-1.88268900	1.29861200
C	-4.89087000	-1.34945200	0.51779500
C	-4.88881200	0.01266800	0.23831800
C	-5.84391000	0.81681300	-0.61290000
C	-5.05888600	2.11798800	-0.85787900
C	-3.10749300	3.34662500	0.25563600
C	-1.82178600	3.24786000	-0.26121500
C	-0.96788900	4.36297300	-0.31638000
C	-1.46630700	5.59099300	0.13856500
C	-2.75319100	5.71036500	0.66133700
C	-3.56337600	4.58204600	0.73069000
C	-4.94278000	4.43189400	1.32623600
C	-5.02746800	2.91642800	1.56145000
C	-1.67391000	-1.72584700	2.51219400
C	-1.33465700	-3.07032500	2.29855600
C	-0.30757400	-3.67869800	3.01315300
C	0.42881400	-2.96983300	3.96744400
C	0.07446900	-1.63605600	4.20038200
C	-0.95294100	-1.02154800	3.49002200
C	0.44436000	4.21699000	-0.74981300
C	0.78428800	3.55681900	-1.93377300
C	2.11974900	3.33448100	-2.26133600
C	3.15305400	3.74245400	-1.41332200
C	2.80724500	4.43671700	-0.24691600
C	1.47706100	4.67589000	0.07858200
H	-3.88969500	-2.94381900	1.54888900

H	-6.11190100	0.29431100	-1.54231500
H	-6.78435300	1.01039600	-0.07130700
H	-5.69673800	2.99399700	-1.04311500
H	-4.39323200	1.99585500	-1.72679700
H	-0.81118600	6.46281300	0.09166500
H	-3.10816200	6.67319600	1.03289900
H	-5.71249400	4.77067400	0.61284900
H	-5.07292400	5.01850300	2.24612300
H	-6.05378800	2.52368800	1.58521000
H	-4.54580000	2.66600300	2.51877800
H	-1.86865800	-3.64773100	1.54380000
H	-0.06922800	-4.72618400	2.81524400
H	0.60550500	-1.06516400	4.96601800
H	-1.20400900	0.01575300	3.70629300
H	-0.00696200	3.20330900	-2.59735500
H	2.37355000	2.82092200	-3.18943900
H	3.59670800	4.77078600	0.43109100
H	1.22586800	5.17518100	1.01660100
H	-5.67074200	-2.00297700	0.12301100
N	-0.24202800	-0.32813100	-0.18790600
H	1.07704800	-0.58736800	-0.01904000
C	-1.06043900	-1.21525000	-0.90060100
C	-2.13571400	-0.78559200	-1.70135700
C	-0.79183800	-2.59228000	-0.82245500
C	-2.96222700	-1.70852500	-2.32633400
H	-2.32886200	0.28039900	-1.81125700
C	-1.61675000	-3.51216600	-1.46027100
H	0.05351300	-2.93529700	-0.22557800
C	-2.71980000	-3.07591100	-2.19223300
H	-3.80765400	-1.36331700	-2.92506800
H	-1.40579100	-4.57918700	-1.37558400
C	4.59091700	3.42249500	-1.72183700
H	5.20085200	4.33678600	-1.77611000
H	5.01405700	2.78400900	-0.93080900
H	4.68775900	2.87756500	-2.67134400
C	1.58848500	-3.60605000	4.68653300
H	2.52104500	-3.47949000	4.11018800
H	1.74749600	-3.15135900	5.67385700
H	1.43314500	-4.68513800	4.82278600
C	-3.69342000	-4.05293700	-2.77627200
F	-4.78705100	-4.17624500	-2.00669900
F	-4.12522000	-3.67099700	-3.98242400
F	-3.17195700	-5.27496400	-2.90245300
C	2.41274800	-2.91932300	0.94064300

C	2.58399500	-2.21741300	-0.26342600
C	2.77273200	-2.95119700	-1.43950800
C	2.81133100	-4.34583300	-1.41000000
C	2.66102700	-5.03237000	-0.20731800
C	2.46008000	-4.30924700	0.97042900
H	2.20079300	-2.36635600	1.85988400
H	2.87523100	-2.42845600	-2.39092600
H	2.95723200	-4.89770900	-2.33991000
H	2.69079700	-6.12261500	-0.18653000
H	2.32186700	-4.83294900	1.91886800
C	2.40029200	-0.72674000	-0.29402600
C	3.12014100	0.05892900	0.69741300
O	2.95862600	1.30729200	0.83832100
H	1.89638900	1.61063300	0.74310000
O	3.97320300	-0.55928200	1.46153800
C	4.51010600	-0.00139800	2.70772200
N	2.35819300	-0.08574300	-1.56786500
C	5.38815800	1.21007300	2.41403100
H	5.95541800	1.46422800	3.32062900
H	6.10642700	0.96942200	1.61669500
H	4.79362600	2.07767700	2.10882900
C	3.34732700	0.31515100	3.64147000
H	2.74352700	-0.58944200	3.80868400
H	3.74318700	0.65225300	4.60975500
H	2.69991200	1.10191600	3.23428100
C	5.34278700	-1.15570100	3.24747800
H	6.15925100	-1.39589700	2.55290400
H	5.77167900	-0.88475100	4.22151200
H	4.71667000	-2.05060000	3.36770300
C	3.45576600	0.07411900	-2.39220000
C	4.76686800	-0.15722800	-1.93802100
C	3.27484700	0.46829500	-3.73199900
C	5.85132000	0.04854800	-2.78799500
H	4.93930200	-0.51854700	-0.92329100
C	4.36650500	0.67096100	-4.56801600
H	2.25939600	0.61098300	-4.10920500
C	5.66845400	0.47385400	-4.10258100
H	6.85899800	-0.13492300	-2.41054300
H	4.19581700	0.98168900	-5.60040000
H	6.52325900	0.63289700	-4.75985700
H	1.44111600	0.07015000	-1.97236100

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